

Scuola Superiore Sant'Anna
BIONICS ENGINEERING
ISTITUTO DI BIOBOTICA

LECTURE NOTES

Neuromorphic Engineering

– THEORY PART –

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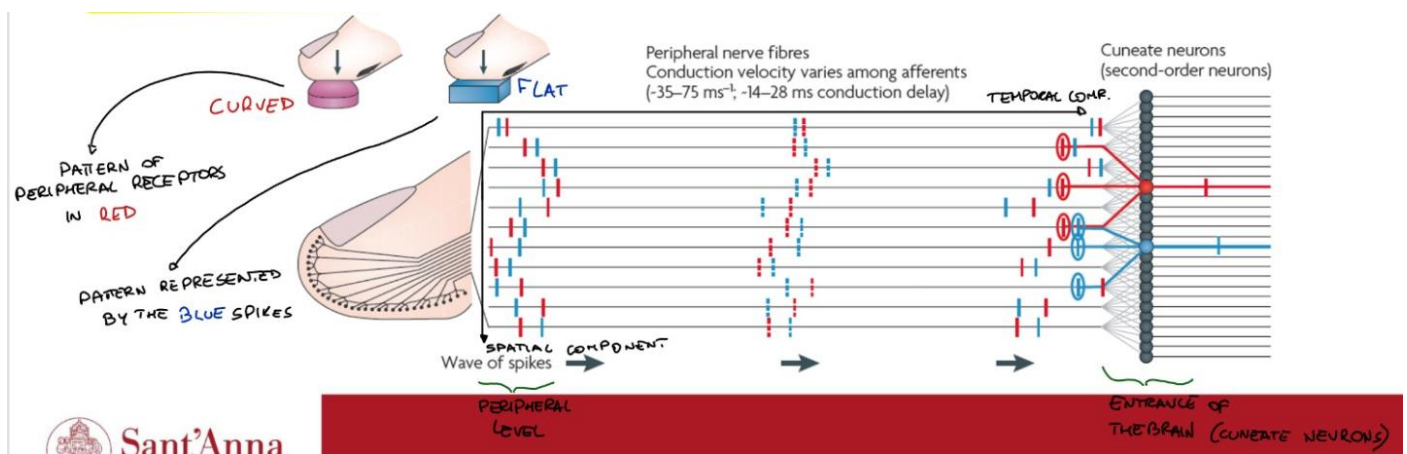
Neuromorphic Engineering

Course presentation

Neuro-robotics aims at developing hybrid bionic systems (robotic systems operating in close interaction with humans), by integrating research methods and skills which are peculiar of neurosciences and robotics. It can be categorized into three main sub - areas:

1. Neuro – robotics 1: “developing mechatronic systems for neuroscientific measurements and analyses”;
2. Neuro-robotics II: “modeling neuroscientific phenomena and behaviors”;
3. Neuro-robotics III: “developing hybrid- bionic systems”;

Neuromorphic engineering looks at how biological systems perform computations. In the human tactile system, the processing of sensory information does not happen only in the brain; instead, a crucial pre-processing occurs directly in the "hardware" (the nerve fibers) while the signal is traveling. When we touch an object (e.g., a curved or flat surface), peripheral receptors in the fingertip generate a spatiotemporal pattern of spikes. Notably, the receptors located more centrally relative to the stimulus generate spikes with a shorter delay compared to the more peripheral ones. Notice that the relative delay between two stimuli captured by the same channel is preserved from the periphery to the cuneate neuron (ex. In the first channel we'll always have first the blue stimulus and after the red one, regardless of the level of the channel. If we now compare two channels, we'll see that when the two couples of stimuli are travelling towards the brain, the latency between them varies because of the different delays of the two channels (ex. If we compare the first two channels, at the peripheral level the first channel captures the couple of stimuli before than the second one but, at the cuneate neurons, we can see that the second channel is able to convey that couple of stimuli earlier than the first channel, thanks to the fact that it has a higher conduction velocity).



The fundamental problem is how the brain rapidly decodes these complex timing patterns. The brain cannot rely on an independent, precise internal clock to measure spike latencies, because it cannot predict contact events with millisecond precision.

To solve this, the nervous system relies on the physical properties of the peripheral nerve fibers. These afferent fibers have varying axonal conduction velocities, introducing transmission delays that range from ~14 to ~28 ms over a distance of roughly 1 meter. This creates a natural **spatiotemporal filter**. Because different channels have different, characteristic delays introduced at the peripheral level, the system possesses "**memory by wires.**" This means the neural architecture acts as a hardware filter, eliminating the need to actively store or save data in a separate memory center. Mathematically, this spatiotemporal filtering can be modeled as:

$$y(t) = \sum_j a_j x_j(t + \Delta T_j)$$

The differential delays along the converging neural pathways serve a very specific purpose: **re-synchronization**. Spikes that are fired asynchronously and uncoordinated at the peripheral level (the fingertip) travel at different speeds. Thanks to the specific delays of each channel, these spikes arrive *simultaneously* at the second-order neurons (cuneate neurons at the entrance of the brain). These cuneate neurons are programmed to act as **coincidence detectors**: they only fire when they receive highly synchronous inputs from multiple sources. Without the differential delays introduced by the peripheral fibers, this filtering and synchronization would be impossible. So if the red cuneate neuron fires, it means that we have touched the red object (same for the blue). So we can say that the activation of the specified neuron is obtained by a precise pattern of spikes (unsynchronized) at the peripheral level that, thanks to the differential delay introduced at this level, will result in a synchronized pattern of spikes at the cuneate nucleus (causing the firing of it).

Ultimately, this mechanism allows for a **temporal-to-spatial conversion**. Because of the massive divergence and convergence of primary afferents onto cuneate neurons, a specific temporal pattern of spikes at the fingertip (representing a specific shape, like a flat edge) is delayed precisely so that it triggers a specific *spatial* set of cuneate neurons. In other words, relative timing information is converted into a "place code." Different tactile stimuli will perfectly synchronize at different second-order neurons, allowing the brain to rapidly classify and decode tactile information during object manipulation without needing an internal clock.

Thanks to this example, we can say that neuromorphic engineering takes inspiration from biology, physics, mathematics, computer science and electrical engineering to give answers about the neurobiological circuits, new efficient computational paradigms, low power hardware solutions and new technologies with robustness to damage and learning capabilities.

Possible Questions at the exam:

- How the tactile sensors can be intended in a neuromorphic manner and in a physiological manner and explain how the physiological ones work.
- How bioinspired architecture can elaborate the data, thanks to multi-stage neurons of tactile information (Jeffress Models).
- Explain different models, illustrate how those the principles they support and how they can be implemented in technical solutions.
- The variation of the Q factor in the Viktor Purpura Distance (all the cases)

Somatosensory Physiology and experimental technologies

Robotics is the field concerned with the connection of perception to action. So the passive walker is not a robot. Anyway, current robotics is based on mechatronics (plus more, including forms of artificial intelligence). Mechatronics systems generally have sensors (these are the core of any neuromorphic system).

All the senses share three common steps:

1. A physical/chemical stimulus.
2. A set of events transforming the stimulus into nerve impulses (transduction).
3. A response to this signal in the form of a perception or conscious experience of sensation.

Neurons from the periphery bring information to the brain thanks to synapses.

All sensory systems convey four basic types of information about the stimulus:

- Modality: type of energy transduced. Labelled line code: the axon of the receptor is a modality-specific communication pathway.
- Location: set of activated receptors. Each receptor responds only to stimulation within its receptive field. We have a variable density of receptors. Hearing, taste and smell receptors are spatially organized according to sensitivity.

- Intensity: Amount of energy transduced. The discharge frequency of an afferent fiber modulates with the stimulus intensity: temporal code. Stronger stimuli also activate a greater number of receptors: population code.
- Timing: dynamics of receptor signalling.

We also divide the senses based on the source of energy of the stimulus:

- Chemical: olfaction and taste;
- Physical: vision, audition and touch.

Neuromorphic engineering grounds on the understanding of physiology. The way we are designing our forms starts from physiology before electronics, computer science, etc. First of all, we have to understand how physiology works and then we are able to integrate those principles into artifacts and machines. An example is related to the Odorant Receptors and the Organization of the Olfactory Systems, presented by Richard Axel and Linda Buck.

Although the rose and rancid differs only from a double bond with oxygen from a chemical point of view, we can see that there is an amplification of the difference in the response in terms of receptors (some in common and others that are different).

This means that very minimal change from the chemical viewpoint can produce a slightly amplified but still minimal change from the set of receptors that is activated. And then this is even more amplified as much as you go from the peripheral level up to the central level. However, we can say that we have some functions that are innate (everyone prefers rose rather than rancid) and others that are acquired through sensory experiences (for example whether we prefer meat or fish).

The Jeffress model is a neurocomputational model that explains how auditory systems can register and analyze small differences in the arrival time of sounds at the two ears in order to estimate the direction of sound sources in the azimuthal plane (even without using vision). This model is only based on trigonometric calculations.

Based on this introduction, we want to focus on the understanding of audition and vision senses to biomedical and consumer applications. The most diffused and advanced consumer and medical devices ground on science – technology synergies. This is in fact the era of physical human – machine interaction. Intelligent machines need a sense of touch to interact physically with humans and the environment. So the tactile sensors cover an important role that is still not used in all systems. This is because nowadays robots can only work in structured environments (industry) but not in the unstructured one (more social, urban, changeable environment). In fact we can focus on tactile sensing to enable interaction abilities in robots.

The goal of this course is to design tactile sensors inspired by the physiology of touch. Some examples can be using piezoelectric sensors. These allow to capture some strain at the level of the receptor on the membrane, which results in modifying the flow of ions through the channel (changing the permeability of the membrane to specified ions) and therefore causing a depolarization that leads to an action potential generation.

Another important discovery was the analogy of receptors activated when eating spicy and feeling hot or mint and fresh. This confirms that there's a cross modal integration across different sensory modalities.

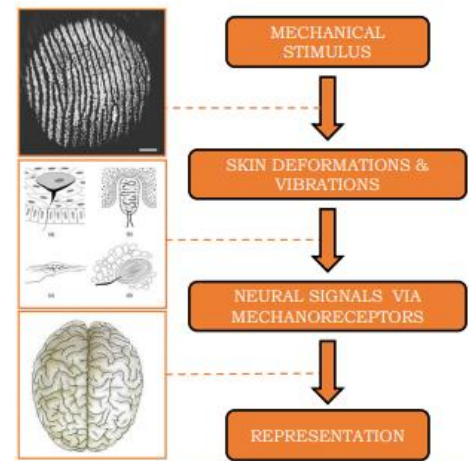
These are some sources of inspiration for building bioinspired machines.

Touch Physiology

In the context of Neuromorphic Engineering, understanding human touch physiology is fundamental for designing artificial tactile sensors. Like all sensory systems, the tactile transduction process can be generalized into three main sequential steps: stimulus, transduction and perception. When interacting with the external environment, our somatosensory system processes mechanical stimuli to extract four critical pieces of information (modality, location, timing and intensity). This is perfectly illustrated by classical models where a stimulus (e.g., heat or touch) triggers a specific sensory pathway.

Focusing specifically on the fingers (digital tactile perception), the biological process of converting a physical touch into a mental representation follows a precise pathway:

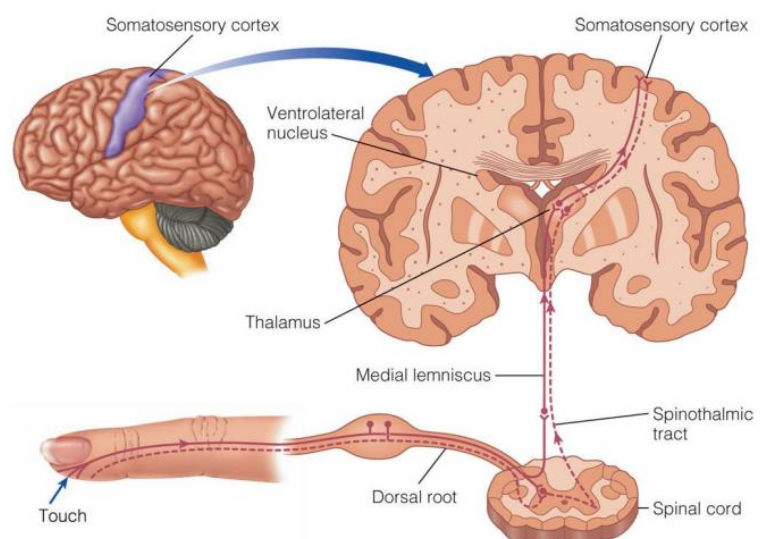
1. **Mechanical Stimulus:** The process begins with the administration of a mechanical stimulus to the cutaneous tissues (the skin of the fingers).
2. **Skin Deformations & Vibrations:** The physical contact forces the skin tissue out of its original resting state. This generates mechanical deformations and traveling vibration waves that propagate through the skin tissue.
3. **Neural Signals via Mechanoreceptors:** Specialized mechanosensors (transducers) embedded within the skin layers detect these traveling waves. They translate these mechanical tissue deformations into a sequence of electrical spikes (action potentials).
4. **Representation & Central Processing:** These spikes travel along the afferent somatosensory pathways to create a representation of the stimulus. The neural signals are elaborated through distinct processing stages in the central nervous system, primarily passing through:
 - a. The Cuneate Nucleus (in the brainstem)
 - b. The Thalamus
 - c. The Somatosensory Cortex
5. **Active Perception:** The final elaboration at the cortical level gives rise to our conscious, active perception of touch.



Somatosensory pathways

Once tactile information is transduced into neural spikes, nerve fibers travel in bundles (peripheral nerves) toward the central nervous system. Within the spinal cord, the architecture of the somatosensory system is organized into two major distinct ascending pathways. They differ significantly in both function and neuroanatomy:

- **The Dorsal Column-Medial Lemniscal Pathway:**
 - **Structure:** Consists of large-diameter nerve fibers.
 - **Function:** Primarily carries information related to fine touch and proprioception (the sense of body position).
- **The Spinothalamic Pathway:**
 - **Structure:** Consists of smaller-diameter nerve fibers.
 - **Function:** Primarily carries information regarding temperature and pain.



A fundamental principle of human neurophysiology is that sensory processing is **contralateral**. If you touch an object with your right hand, the sensory information is projected to and processed by the somatosensory cortex in the left hemisphere of the brain, and vice versa.

To achieve this, the neural pathways must cross the midline of the body (a process called decussation) at some point between the peripheral skin receptors and the cortex. Eventually, information is shared with the ipsilateral (same-side) hemisphere to allow for complex bimanual coordination and sensory-motor control, but the primary projection remains contralateral.

While both pathways eventually cross over to the opposite hemisphere, **they do so at different anatomical levels.**

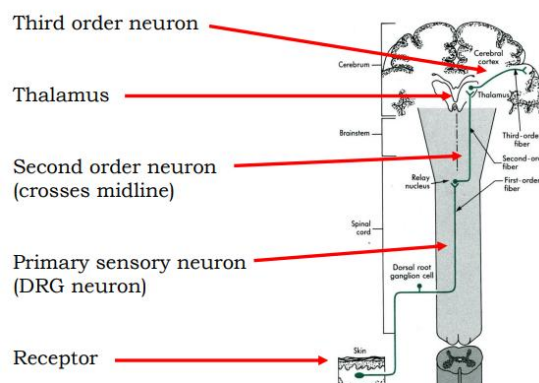
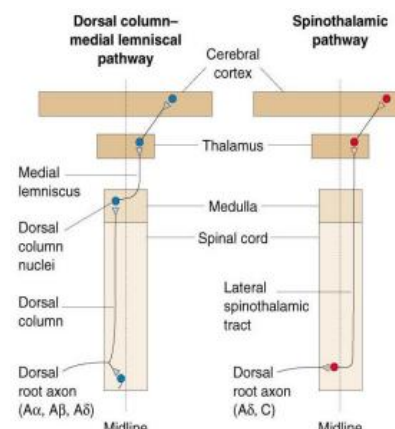
- The **Spinothalamic tract** crosses over almost immediately upon entering the spinal cord.
- The **Dorsal Column-Medial Lemniscal pathway** travels up the same side of the spinal cord and only crosses over much higher up, in the brainstem (medulla).

This anatomical difference becomes highly visible in pathology, specifically in the case of a **partial spinal cord injury** (resulting in dissociated sensory loss).

- *Example:* If a patient suffers a unilateral (one-sided) lesion on the right side of the spinal cord, they will experience a specific dissociation of senses. They will lose touch and proprioception on the **right side** (because those fibers haven't crossed yet), but they will lose temperature and pain sensation on the **left side** (because those fibers crossed immediately at the spinal entry level below the injury).

Regardless of the specific tract, the flow of somatosensory information generally follows a three-neuron chain architecture:

1. **Receptors (Skin):** Detect the physical stimulus.
2. **Primary Sensory Neuron (1st Order):** These are the **DRG (Dorsal Root Ganglion)** neurons. A ganglion is a cluster of neural bodies located just outside the spinal cord. They receive the signal from the skin and carry it into the central nervous system.
3. **Second-Order Neuron:** Located in the spinal cord or brainstem (depending on the pathway). This is the neuron that crosses the midline (decussates) and travels up toward the brain.
4. **Thalamus:** Acts as the primary relay station. Here, the second-order neurons synapse with the next stage.
5. **Third-Order Neuron:** Carries the signal from the thalamus to the final destination in the cerebral cortex.
6. **Cerebral Cortex:** The area where central processing occurs, leading to conscious tactile perception.



Nerve Fibers

Information travels from the peripheral receptors to the central nervous system through different types of nerve fibers. These fibers are classified based on their diameter and whether they possess a **myelin sheath** (an insulating layer that significantly increases the speed of electrical conduction). The human somatosensory system relies on four main classes of nerve fibers:

- **Aα (A-alpha) Fibers:**
 - **Structure:** Very large diameter (13–20 μm) and highly myelinated.
 - **Speed:** Extremely fast (80–120 m/s).
 - **Function:** Proprioceptors of skeletal muscles (e.g., muscle spindles). They are crucial for immediate motor control and reflexes.

- **A β (A-beta) Fibers:**
 - **Structure:** Intermediate diameter (6–12 μm) and myelinated.
 - **Speed:** Fast (35–75 m/s).
 - **Function:** **These are the primary fibers of interest for tactile sensors.** They carry information from the skin's mechanoreceptors. Because of their speed, tactile feedback (e.g., the pressure exerted to grasp an object) reaches the brain with a latency of only about **30 milliseconds**.
- **A δ (A-delta) Fibers:**
 - **Structure:** Small diameter (1–5 μm) and thinly myelinated.
 - **Speed:** Moderate (5–30 m/s).
 - **Function:** Carry information about sharp, immediate pain and temperature.
- **C Fibers:**
 - **Structure:** Very small diameter (0.02–1.5 μm) and **unmyelinated**.
 - **Speed:** Very slow (0.5–2 m/s).
 - **Function:** Carry deep, throbbing pain, temperature, and itch. Due to their lack of myelin, a signal traveling from the hand to the brain can take **1 to 2 seconds** to arrive.

A critical consequence of these different fiber architectures is that **sensory information does not arrive at the brain all at once**. Imagine accidentally touching a burning hot stove.

1. **Immediate Reaction (Spinal Reflex & A-fibers):** The tactile contact (A β) and the sharp, initial warning of extreme heat/pain (A δ) travel rapidly. This triggers a spinal reflex to instantly retract your hand, protecting the tissue from further severe damage before the brain fully processes the event.
2. **Delayed Sensation (C-fibers):** One to two seconds *after* you have safely pulled your hand away, the deep, agonizing sensation of the burn reaches your brain via the slow C fibers. This delayed signal serves a different purpose: it tells your conscious brain that an injury has occurred, prompting you to seek medical care and protect the wound while it heals.

From an engineering and physiological standpoint, this staggered arrival of information is a brilliant optimization of the system's computational workload.

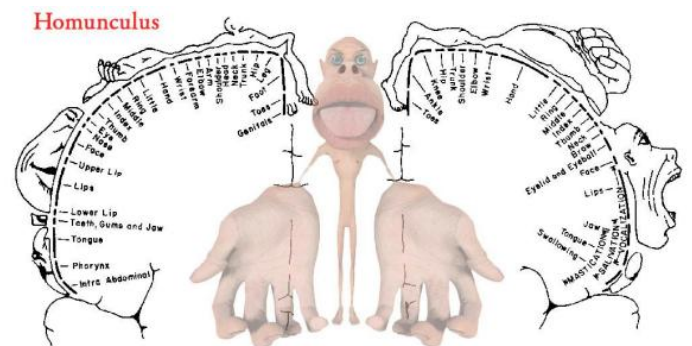
- **Priority-Based Processing:** The brain is not immediately overloaded with the complex processing of deep pain when its absolute first priority must be motor control (moving the hand away).
- **Hardware-Level Spatiotemporal Filtering:** The physical properties of the nerves themselves (their diameter and myelination) act as a built-in delay filter. **The intelligence is embedded in the body's hardware**, not just in the central "processor" (the brain).

When designing biorobotic systems or artificial sensory architectures, we do not need to send all raw data instantly to the central processing unit. By taking inspiration from human physiology, we can embed "intelligence" directly into the physical design of the artificial nervous system—using hardware delays and specialized pathways to filter information and trigger low-level reflexes autonomously.

Cortical Projections and Homunculus

Once tactile information reaches the brain via the pathways discussed previously, it is projected onto the **Primary Somatosensory Cortex**. A fundamental characteristic of this central processing is that different regions of our body innervate specific, distinct areas in the cortex. However, this mapping is not proportional to the actual physical size of the body parts.

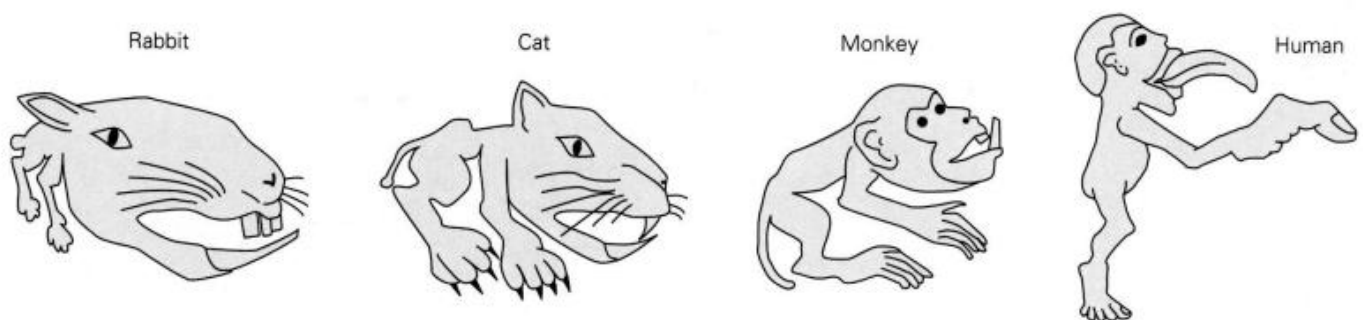
If we map the dimensions of the cortical areas dedicated to different body parts and morph a human body to match these proportions, we get a representation known as the **Sensory Homunculus**.



- **Cortical Overrepresentation:** Certain body parts are massively "overrepresented" in the cortex compared to their real anatomical geometry. The **hands, lips, and face** take up a disproportionately large area of the brain.
- **The "Why":** This overrepresentation directly correlates with receptor density. Regions like our fingertips and lips have a vastly higher number of mechanoreceptors integrated into the skin to support fine discrimination and complex interactions, thus requiring more "computational space" (cortical area) in the brain to process the incoming signals.

Looking at the cortical mapping across different species provides fascinating insights into how sensory needs drive neurophysiology. If we apply the "homunculus" morphing principle to animals, we see clear evolutionary shifts:

- **Lower Mammals (e.g., Rabbit, Cat):** Their cortical representation is heavily dominated by the face and snout (particularly the whiskers), which are their primary organs for exploring the immediate environment. The paws have a relatively small representation.
- **Primates (e.g., Monkeys, Humans):** As we move to non-human primates and humans, there is a massive shift in importance toward the **hands**.
- **The Human Hand:** In humans, the hand occupies a substantial fraction of the entire somatosensory cortex. This is a direct consequence of our evolutionary reliance on complex manipulation, tool use, and the exceptionally high density of mechanoreceptors in our digital skin.



Somatosensory system: mechanoreceptors

The human hand is an incredibly dense sensory instrument, containing approximately **17,000 sensory receptors**. The somatosensory system uses these receptors to transmit information about four major modalities: Touch, Proprioception (awareness of limb position), Pain, and Temperature. Proprioception provides information on the orientation of our limbs with respect to one another. Exteroception provides information on the interaction between our limb and the external world.

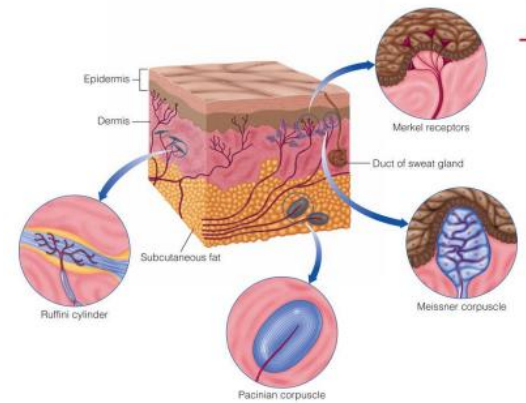
These receptors are embedded within the layers of the skin:

- **Epidermis:** The outer layer, primarily acting as a protective barrier.
- **Dermis:** The layer just below the epidermis, which houses the vast majority of the mechanoreceptors responsible for detecting pressure, stretch, and vibration.

The skin protects the organism by keeping damaging agents from penetrating the body.

The human tactile system relies on four primary types of mechanoreceptors:

- Merkel receptor - disk-shaped receptor located near the border between the epidermis and dermis;
- Meissner corpuscle - stack of flattened disks in the dermis just below epidermis;
- Ruffini cylinder - branched fibers inside a cylindrical capsule;
- Pacinian corpuscle - onion-like capsule located deep in the skin.



Their specific functions are determined by their **anatomy (morphology)** and their **location (depth)** in the skin. We can classify them along two main axes: Spatial Resolution (Type 1 vs. Type 2) and Temporal Adaptation (Slow vs. Fast).

Spatial Resolution: Depth and Receptive fields

The depth at which a receptor is located determines the size of its "receptive field" (the area of skin it monitors).

- **Type 1 (Superficial):** Located very close to the skin surface (epidermis/dermis border). Because they are shallow, they have a **small, precise, and highly localized receptive border**. They are excellent for detecting fine details and edges.
- **Type 2 (Deep):** Located deeper in the dermis or subcutaneous fat. Because they are deep, the mechanical signals spread out before reaching them, giving them a **large, broad, and blurred receptive field**. They integrate information over a wider area.

Temporal Resolution: Adaption Rates

Receptors also differ in how they fire in response to a stimulus over time.

- **Slowly Adapting (SA):** These receptors respond to **sustained (static) stimulation**. If you press on the skin and hold it, these receptors will continue firing action potentials for the duration of the contact (though the rate may slowly decrease over time). They tell the brain, *"I am still touching something."*
- **Fast Adapting (FA):** These receptors respond only to **dynamic changes (transients)**. They fire when a stimulus is first applied and when it is removed, but they *stop firing* during steady, static pressure. They tell the brain, *"Something just changed or vibrated."*

Combining these spatial and temporal properties gives us the four specific end-organs:

	Slowly Adapting (SA)(Responds to sustained stim.)	Fast Adapting (FA)(Responds to dynamic changes)	Receptive Field
Type 1 (Superficial)	SA1: Merkel Receptors	FA1: Meissner Corpuscles	Precise, small, localized border.
Type 2 (Deep)	SA2: Ruffini Cylinders	FA2: Pacinian Corpuscles	Large, broad, blurred border.

Once again, a crucial lesson for bio-inspired robotics: **specialization happens at the periphery, not just in the brain.** Nature did not create one generic tactile sensor and leave it to the brain to sort out the math. Instead, it evolved four distinct hardware transducers, each pre-filtered for a specific type of mechanical event (edges vs. vibrations, static vs. dynamic). By mimicking this in artificial neuromorphic systems, engineers can capture highly specialized data right at the contact point, drastically reducing the computational burden on the central processor.

Microneurography

To understand the precise firing patterns (spatial and temporal properties) of the mechanoreceptors we just discussed, researchers rely on a specific electrophysiological technique called **Microneurography**.

- **What it is:** It is an acute (temporary), minimally invasive method used to record the electrical action potentials (spikes) flowing directly from peripheral mechanoreceptors through human nerves.
- **How it works:** A very thin, acupuncture-like tungsten microelectrode is inserted percutaneously (through the skin) directly into a peripheral nerve (e.g., the median nerve at the wrist or the peroneal nerve). The electrode penetrates the nerve fascicle to contact a single sensory axon, allowing the recording of isolated "single-unit" activity.
- **Anatomical Constraint:** This technique can only be performed at specific anatomical sites where the nerve runs very superficially (close to the skin surface) and is stabilized by bone landmarks (e.g., the elbow or the wrist). This prevents the electrode from losing contact with the nerve during the recording session.

One of the most powerful aspects of the microneurographic electrode is its bidirectionality. Once the electrode is successfully impaled into the nerve, it can be used for two distinct purposes:

1. **Recording (Afferent Pathway):** While a mechatronic platform administers precise tactile stimuli to the skin (e.g., the fingertips), the electrode records the exact train of spikes generated by the mechanoreceptors in response to that specific stimulus.
2. **Stimulation (Efferent/Sensory Feedback):** The current can be reversed. Instead of recording, the electrode can *inject* electrical pulses directly into the nerve. By stimulating the nerve with specific spiking patterns, researchers can artificially induce a conscious tactile sensation in the patient's brain.

Microneurography is not used for chronic (long-term) applications because the patient must remain completely still (e.g., in a dentist's chair) while the needle is inserted. However, it is an invaluable tool for acute research and clinical testing:

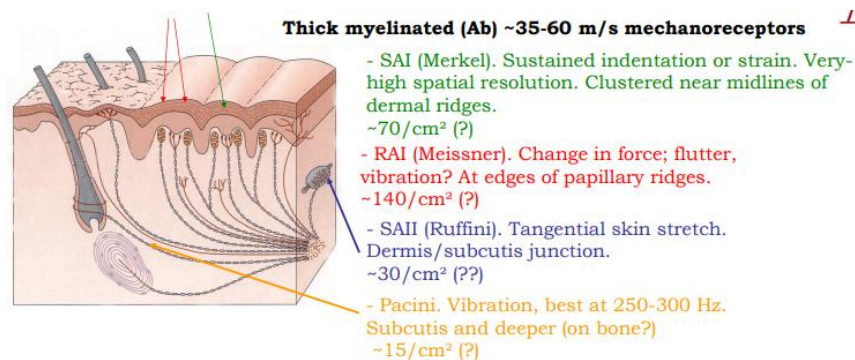
- **Biologically Accurate Neuromorphic Modeling**
This technique provides the "ground truth" data required to build biomimetic artificial sensors. By recording the natural spiking activity of human mechanoreceptors (the clean, single-unit signals), engineers can build computational models that replicate these exact firing characteristics. The result reflects in artificial neuromorphic tactile sensors that encode tactile data with highly fidelic, bio-similar spike trains, perfectly mimicking human neurophysiology.
- **Pre – surgical screening for amputees**
Before subjecting an amputee to invasive surgery for a chronic neural implant (e.g., for a bionic prosthesis with sensory feedback), microneurography is used as a personalized screening tool.
 - Researchers use neuromorphic sensors to trigger a neural stimulator, injecting pulses through the temporary microneurography electrode.
 - If the amputee successfully perceives the artificial tactile sensation (meaning their neural pathways are responsive), they are confirmed as a viable candidate for chronic surgical implantation.

Density – Receptive Field size trade off

Building on the data obtained from microneurography, we can look closer at the specific quantitative properties of our four mechanoreceptor families.

In neurophysiology, **Fast Adapting (FA)** receptors are often also referred to as **Rapidly Adapting (RA)** receptors. Therefore, Meissner corpuscles can be labeled as FA1 or RA1, and Pacinian corpuscles as FA2 or RA2.

The density of mechanoreceptors is not uniform across the body. The fingertips represent the area of maximum density ("maximal regions") for tactile sensing. However, even though this density is high for the somatosensory system, it is still tremendously lower than the density of photoreceptors in the human eye.



The most critical "take-home message" from this data is the inverse relationship between the number of receptors (Density) and the area they monitor (Receptive Field).

To ensure our nervous system can feel stimuli anywhere on the skin without having "blind spots" (or black spots), nature follows an optimization rule: **The product of Density and Receptive Field Area remains approximately constant.**

- **The Rule of Full Coverage:**

- **Smaller Receptive Fields require Higher Density:** Type 1 receptors (Merkel, Meissner) are located very close to the surface. Because they only sense a tiny, precise local area, you need a massive amount of them packed tightly together (~70-140 units/cm²) to cover the fingertip without leaving gaps.
- **Larger Receptive Fields require Lower Density:** Type 2 receptors (Pacinian, Ruffini) are located deeper in the tissue. Because their depth gives them a very broad, blurred receptive field (e.g., ~60 mm² for Ruffini, up to ~100 mm² or more for Pacinian), a single receptor covers a lot of ground. Therefore, you need far fewer of them (~15-30 units/cm²) to ensure total coverage.

To summarize the spatial design of the human tactile system:

- **Depth dictates the Receptive Field:** Surface = Small/Precise; Deep = Large/Broad.
- **Receptive Field dictates the required Density:** Small Field = High Density needed; Large Field = Low Density needed.

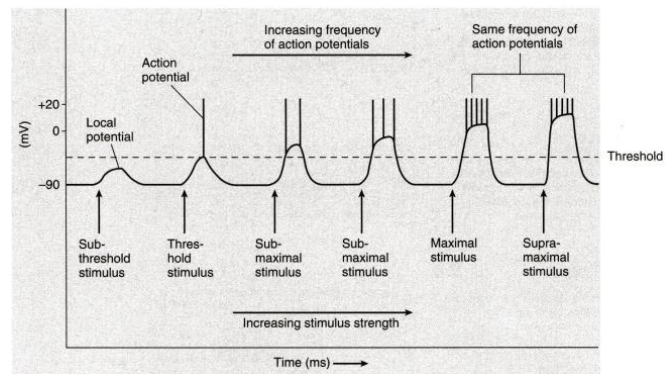


	FA1 (RA)	SAI	SAIL	PC (FAII)
Termination	Meissner corpuscle	Merkel cell	Ruffini organ	Pacini Corpuscle
Termination depth (mm)	0.3	0.5	1	2
Receptive field (mm²)	10	10	60	100
Density (units/cm²)	140	70	30	15
Adaptation	Fast (fmax=50Hz)	slow	slow	fast (fmax=250Hz)

Local potential at human tactile receptors

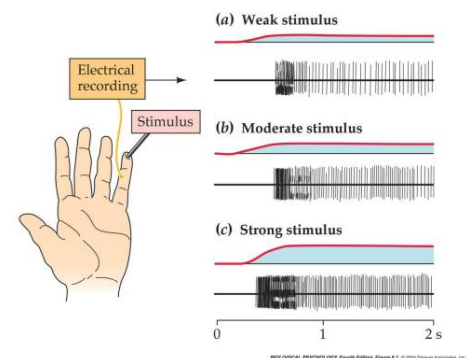
To engineer biomimetic tactile sensors, we must understand how the human nervous system encodes the *intensity* (force/pressure) of a physical stimulus into a language the brain can understand (spikes).

- **Firing Rate Proportionality:** The most basic rule is that the intensity of a stimulus is encoded by the **firing rate** (the frequency of action potentials). A weak stimulus generates a low firing rate, while a strong stimulus generates a high firing rate.
- **Thresholds:**
 - **Sub-threshold:** If the pressure is too light, no spikes are generated. The brain perceives nothing.
 - **Threshold:** The minimum force required to trigger the first action potentials.
- **Saturation:** A single mechanoreceptor cannot increase its firing rate infinitely. If you press harder and harder, the individual receptor will eventually hit its maximum firing rate and "saturate".
- **Population Coding (Recruitment):** If a single sensor saturates, how does the brain know you are pressing even harder? Through *recruitment*. As the force increases, a larger population of surrounding receptors is activated.
 - *Neuromorphic Implication:* By playing with stimulation rate (frequency) and amplitude (recruiting a larger population of fibers), engineers can create tactile illusions in amputees, tricking the brain into feeling different levels of force.



By looking at a graph of a receptor's spike train over time (during the application, holding, and release of a stimulus), we can identify its biological family.

- **Transients vs. Steady State:** Most mechanoreceptors fire at very high rates during the "transients" (when the force is changing—either pressing down or letting go).
- **Identifying Adaptation (SA vs. FA):**
 - If you see spikes occurring continuously during the "plateau" (the static phase where the stimulus is held constant), it is a **Slowly Adapting (SA)** receptor.
 - If the receptor fires only during the transients (onset/offset) and is completely silent during the static plateau, it is a **Fast Adapting (FA)** receptor.
- **The Missing Spatial Data:** You cannot tell if a receptor is Type 1 (small field) or Type 2 (large field) just by looking at a single static pressure graph. To determine the receptive field, you would have to move the stimulus across the skin to see where the firing stops.



How do we translate these biological properties (like FA and SA behaviors) into artificial robotic sensors? There are two main engineering paradigms:

A. "Soft" Neuromorphic Sensors (Computational/Software)

In this approach, the physical sensor simply measures pressure, and the "spiking" behavior is simulated computationally (e.g., using differential equations like the Euler method in LabVIEW).

- **To simulate SA (Slowly Adapting):** You provide a fully proportional input to the artificial neuron model. The output directly tracks the sustained input.

- **To simulate FA (Fast Adapting):** You introduce a **mathematical derivative component** between the sensor output and the neuron. This filters out the DC (static) component and only passes the AC (time-varying/dynamic) component, mimicking how FA receptors ignore static pressure.

B. "Hard" Neuromorphic Sensors (Physical/Hardware)

In this paradigm, the spiking behavior and filtering properties are integrated directly into the **physics and hardware** of the sensor itself, exactly as nature does.

- **The Biological Masterpiece (Pacinian Corpuscle):** The Pacinian corpuscle (an FA2 receptor) is the perfect example of biological "hard" neuromorphics. It has a characteristic "onion-like" layered capsule surrounding the nerve ending. This physical, mechanical structure acts as a hardware filter: it absorbs and dissipates static pressure (DC component) so it never reaches the nerve, but it perfectly transmits fast vibrations (AC component).
- **The Engineering Goal:** Instead of writing software to filter signals, "hard neuromorphic" engineers design the physical casing and materials of artificial sensors to naturally filter out specific physical phenomena before they even become electrical signals.

To truly understand how biological "hardware" filtering works, neurophysiologists conducted a classic experiment on the Pacinian corpuscle, observing how it generates local (receptor) potentials under different physical conditions.

The core discovery is that the **axon itself (the nerve ending inside the receptor) is inherently mechanosensitive**, but its raw behavior is completely altered by the surrounding tissue.

- **The Intact Corpuscle (Fast Adapting / FA):** when a mechanical stimulus (pressure) is applied to an intact Pacinian corpuscle, the layers of tissue (the "onion-like" lamellae) act as a mechanical high-pass filter. The physical deformation is transmitted to the axon only during the dynamic phases (when the pressure is changing: pressing down or releasing). During the static plateau (when the pressure is held constant), the fluid and connective tissue layers shift and absorb the steady force. The membrane potential depolarizes and triggers spikes *only* during the transients. This gives the Pacinian corpuscle its characteristic **Fast Adapting (FA)** behavior.
- **The Decapsulated Axon (Slowly Adapting / SA):** without the mechanical filter, the sustained static pressure acts directly on the mechanosensitive axon. The membrane undergoes a continuous, static depolarization for as long as the stimulus is applied. It loses its transient-only spiking behavior and essentially behaves like a **Slowly Adapting (SA)** receptor.

The computational property (adaptation) does not happen in the brain, nor does it happen through complex ion-channel math in the nerve; it happens because of the physical morphology of the receptor's capsule. If you "remove the onion," you completely change the sensor's temporal encoding.

Braille Code Experiment: spatial encoding

To visualize how the brain "sees" through touch, neurophysiologists study the spiking patterns of receptors when a complex geometric stimulus—like Braille text—is swept across the skin.

- **Type 1 Receptors (Merkel & Meissner):** Because they have a high density and tiny, precise receptive fields, their spiking activity acts like a high-resolution camera. The firing pattern perfectly reflects the spatial geometry of the Braille dots. If there are two horizontal dots, the spike train shows two distinct bursts of activity. They map the topography of what you are touching.
- **Type 2 Receptors (Ruffini & Pacinian):** Because they integrate information over a wide, deep area (large receptive field), they completely lose this geometric resolution. Their firing patterns blur the dots together, meaning they cannot tell the brain about the shape or texture of the object.

Each family brings a specific, specialized piece of data to the central nervous system:

- **Merkel (SA1) & Meissner (FA1):** Encode **Topography and Texture**. They tell the brain about the shape, edges, and fine spatial details of the object in contact.
- **Ruffini (SA2):** Encodes **Overall Force and Posture**. It acts as a tension/stretch sensor. The harder you press on an object, the higher its firing rate. It tells the brain the magnitude of the interaction force.
- **Pacinian (FA2):** Encodes **Slip and Velocity**. It fires when there is lateral movement or high-frequency vibration between the skin and the object. It tells the brain, *"The object is moving across the skin."*

The ability of the **Pacinian** (and to some extent, Ruffini) receptors to detect slippage is what gives humans our spectacular, almost unconscious grip control. Imagine holding a heavy glass and slowly opening your hand to let it slide down, even with your eyes closed. You don't drop it. Why? Because the moment the glass begins to fall ("incipient slip"), the Pacinian corpuscles detect the micro-vibrations of the sliding texture and instantly fire. This triggers a lightning-fast reflex to slightly increase the grip force, allowing you to regulate the fall in a perfectly controlled manner.

To close this section, we have to recap the complete anatomical pathway—the "highway" that tactile information takes from the fingertip to conscious perception in the brain.

Here is the exact sequence of the neural relay:

1. **Peripheral Receptors:** Transduce the mechanical stimulus into spikes.
2. **1st Order Neurons:** Located in the **Dorsal Root Ganglion** (just outside the spinal cord).
3. **2nd Order Neurons:** Located in the **Cuneate Nucleus** (in the brainstem/medulla). Here, the signal crosses over to the opposite side of the body.
4. **3rd Order Neurons:** Located in the **Thalamus** (the brain's sensory relay station).
5. **Destination:** The **Primary Somatosensory Cortex** on the **contralateral** (opposite) hemisphere of the brain.

Psychophysics methods and the Psychometric Curve

While microneurography tells us *how* sensors fire (the electrical code), **Psychophysics** tells us *what* the person actually feels (the perceptual reality). It studies the mathematical relationship between the physical intensity of a stimulus and the subjective perception of that stimulus.

- **Historic Laws:** Scientists like Weber, Fechner, and Stevens proposed different scaling laws (logarithmic, linear, power-law) to model this relationship. For example, our perception of sound intensity is logarithmic (which is why we use decibels).
- **Application in Bionics:** For neuromorphic engineering, psychophysics is the ultimate test. If we inject a synthetic code of spikes into a patient's nerve via a neural interface, we must use psychophysical methods to measure if the code is actually "informative"—meaning, does the patient correctly recognize the physical properties of the virtual object?
-

To evaluate how well a human (or a bionic prosthesis) can discriminate between tactile textures, researchers use the **Method of Constant Stimuli** to generate a **Psychometric Curve**.

The Experiment (Grating Discrimination): Imagine touching two different grooved surfaces (gratings).

- λ_1 : The spatial period (distance between grooves) of the first grating.
- λ_2 : The spatial period of the second grating.
- $\Delta\lambda = \lambda_2 - \lambda_1$: The difference between the two.

The subject is asked: *"Is the second grating coarser (wider grooves) than the first?"* We then plot the probability of the subject answering "Yes" against the physical difference ($\Delta\lambda$).

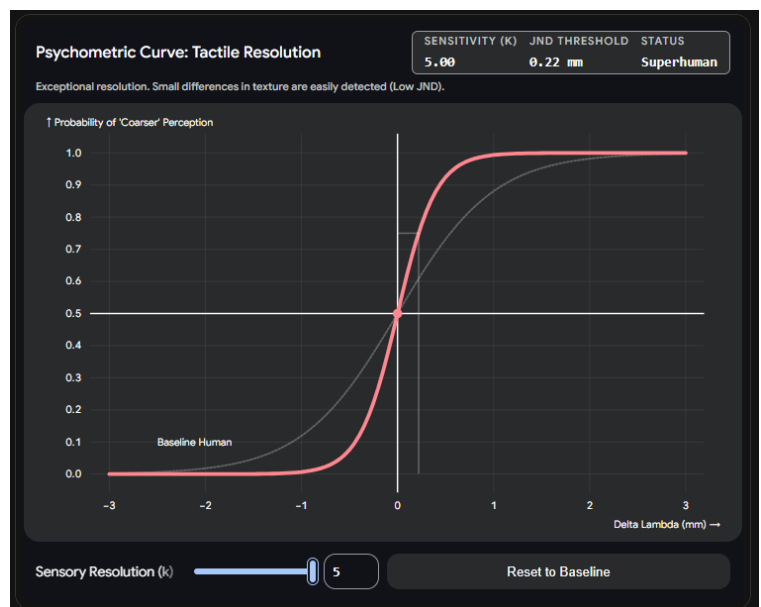
The standard human psychometric curve is a logistic (S-shaped) function:

- If $\Delta\lambda = 0$ (**Both gratings are identical**): The subject has to guess. The probability of answering "Yes" is **0.5 (50%)**.
- If $\Delta\lambda \gg 0$ (**Second is clearly coarser**): The probability approaches **1.0 (100%)**.
- If $\Delta\lambda \ll 0$ (**First is clearly coarser**): The probability approaches **0.0 (0%)**.

By comparing the psychometric curve of an amputee using a neuromorphic prosthesis against a healthy human baseline (the "White Curve" on the blackboard), we can assess the quality of the artificial sensory feedback.

- **The Baseline (White Curve)**: Represents normal human physiological resolution. The goal of a prosthesis is to match this curve perfectly.
- **Subhuman Performance (Blue Curve - Flatter)**: A flatter curve means the subject needs a much larger physical difference $\Delta\lambda$ to feel which texture is coarser. Their perceptual threshold is high; their sensory resolution is poor.
- **Superhuman Performance (Red Curve - Stiffer/Steeper)**: A steeper curve means the subject can correctly identify tiny, microscopic differences in texture that a normal human finger could not detect. Their perceptual threshold is extremely low, giving them "superhuman" tactile acuity.

The overarching conclusion of this entire physiological module is a profound lesson for engineering and robotics: **Computation does not only happen in the brain; it happens in the body.** Our peripheral sensory system is not just a passive wire sending raw data. The mechanoreceptors in our skin act as complex "smart sensors." Through their specific morphologies and adaptation rates, they perform massive pre-processing: spatial integration, temporal differentiation, and mechanical filtering. By distributing this computational load between the body (peripheral nerves) and the brain (cortex), the human nervous system achieves incredible efficiency. In Neuromorphic Engineering, the ultimate goal is to replicate this elegant architecture—building artificial sensors that pre-process data locally, relieving the central robotic processor from the burden of raw computation.

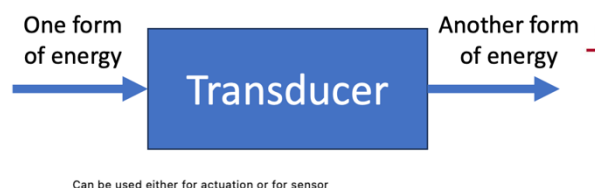


Measurements, instrumentation, sensors and transducers

Introduction:

The idea is to learn how to feed neural models, with data coming from sensor.

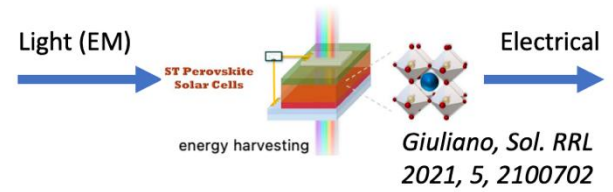
How we want to connect an input, like the current in the Izkivech to a sensor to generate real time data.



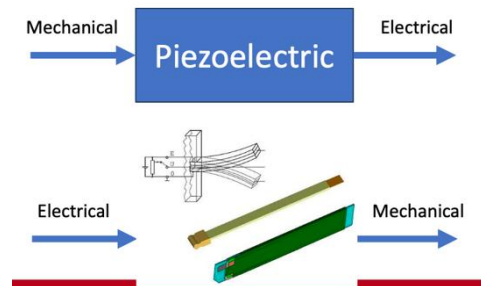
Transducer:

- Transducers are devices which transform one form of energy into another form.

So, it's related to the concept of energy transformation, that can be used either for sensor or actuation, this difference depend to the direction of conversion. For example, you can have a photovoltaic energy harvester in which you use light, electromagnetic source converted into electrical. A piezoelectric can be a bidirectional actuator or sensor that transform mechanical to electrical or vice versa.



Example: Ultrasound, piezoelectric actuator that can be used for that. If the actuator is tuned at order of magnitude relative to MHz or GHz can create oscillation and the actuator is going to generate wave. Another example with piezoelectric can be a microphone, that work with pressure that change the electrical resistance.



Not always we have piezoelectric because in the regimen of audible frequency (low one) they do not work well with piezoelectric, so we prefer electromagnetic.

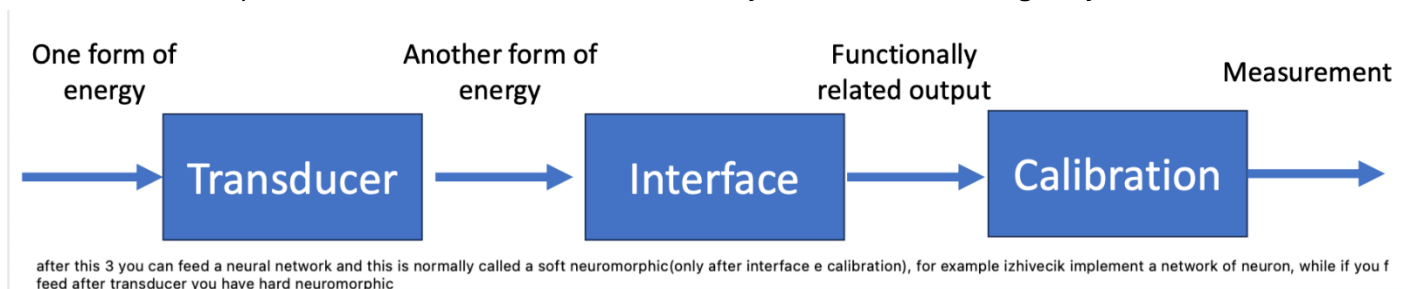
Sensors:

It's related to the concept of transducer, but it needs the introduction of calibration. The sensor is a calibrated actuator and normally the actuator is any port that transform a source of energy into another. The sensor is that port with the reverse model of the transducer, starting from the end we can reobtain the starting energy.

DEFINITION:

A Sensor is "a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an instrument. A sensor is a device, which responds to an input quantity by generating a functionally related output usually in the form of an electrical or optical signal. (US National Institute of Standards and Technology, Special Publication 800-82)

Functionally related output: remind the concept of invertibility of the function. This is not always the case, because, for example, if the function is not monotonic and you don't know the regime you cannot invert it.



Example: Transducer that has an electrical signal rather than optical, if you have weighting system (bilancia) and it is digital, you can read it optically but there is a electrical signal inside that convert the output of the strain gauge into a signal, and then you know the calibration, than the microcontroller read the signal and generate an output on the LCD screen.

The same item can be fully optical, for example a dynamometer or a fully mechanical bilancia. In this case the system is fully analog and for that reason you don't have an electrical output. Another example is a thermometer; in the same way they can be fully analog (fluid that react to the different temperature) or digital, in which you

have electrical measurement of voltage, you read the electrical signal and after the controller inverse the function and obtain the temperature.

RIASSUNTO: inutile

Different way of feeding the NN:

In neuromorphic engineering, you can feed the neural networks after the *calibration*, in this way you give an input adapted to the range of the input layer or directly after the *interface*, for example when you have a Wheatstone bridge, you can modify a bit this value like adding a derivative for the offset or scaling the amplitude; the last case is directly after the *Transducer*.

If we feed the NN after the interface or the calibration (sono sicuro 100% solo dopo la calibration) we are in the case of **Soft Neuromorphic**, this because the normal case is to have a neuronal network implemented digitally.

Example: Izhikevich equation implementing a network of neurons, so you discretize such differential equations and you have the networks implemented with the Euler method.

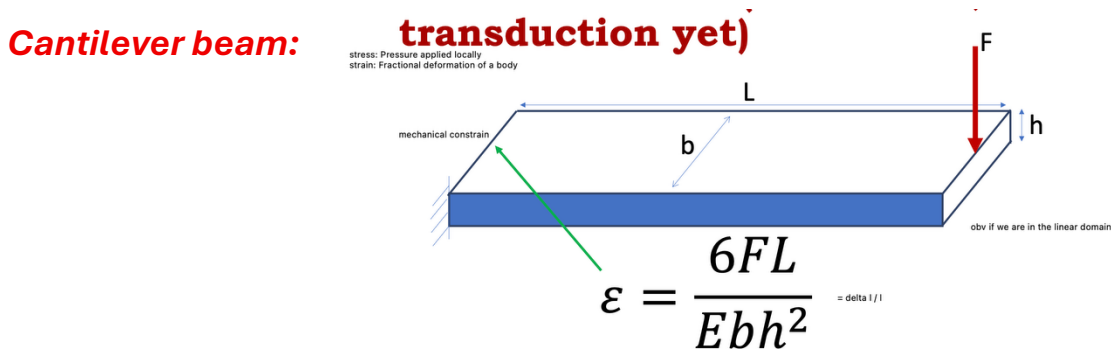
If we feed the NN after the transducer we are in the case of **Hard Neuromorphic**, in which the physics of the system is already producing some spiking implementation.

QUESTA PARTE LA AGGIUNGO ALTRIMENTI NON SI CAPISCE: L'idea è la seguente.

- **Soft:** refers to software; in other words, one works directly with clean, calibrated signals, and therefore, as stated by Oddo, the implementation is generally digital.
- **Hard:** refers to hardware; analog signals are generally used, or they are translated directly into spikes. In this case, it is closer to direct biological functioning: biological receptors pass impulses directly to the nervous system.

In this course we focus on Soft and sensor & actuators.

To implement at hardware level, you need to go on the material science and condensing matter side, to implement the physics principles that produce spiking activity.



It's a structure very important in mechanics that is very useful to give rise to tactile sensor integrating strain gauges into and onto the covered cantilever beam.

A cantilever is a trampoline, so it's a structure that typically has a constrain in one end and after you apply force on the other end.

- **Stress:** pressure applied locally [Pa = N/m²].
- **Strain:** Fractional deformation of a body is defined as $\epsilon = \frac{\Delta l}{l}$.

In the linear regimen, the strain is proportional (for the cantilever) to the force, obviously based on the upper formula if you change the thickness, length ecc, the relation stress-force is going to change.

Another important value is the *young module* that is a quantity related to the material that you are considering, in general it follows the idea that soft = smaller and hard = higher.

Table 1. Typical Equations to Calculate Bending Moment

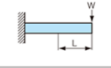
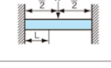
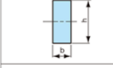
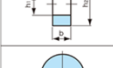
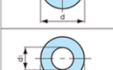
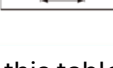
Shape of Beam	Bending Moment M
	$M = Wl$
	$0 \leq x \leq \frac{l}{2} \rightarrow M = \frac{Wx}{2} \left(\frac{l}{2} - \frac{x}{2} \right)$ $x = \frac{l}{2} \rightarrow M = \frac{Wl}{8}$ $\frac{l}{2} \leq x \leq l \rightarrow M = \frac{Wx}{2} \left(\frac{l}{2} - \frac{x}{2} \right)$
	$0 \leq x \leq \frac{d}{2} \rightarrow M = -\frac{Wx}{2}$ $x = \frac{d}{2} \rightarrow M = -\frac{Wd}{4}$ $\frac{d}{2} \leq x \leq l \rightarrow M = -\frac{W(d-x)}{2}$
	$0 \leq x \leq d_1 \rightarrow M = Wx$ $d_1 \leq x \leq (d_1 + d_2) \rightarrow M = Wd_1$

Table 2. Typical Equations to Calculate Section Modulus

Cross Section	Section Modulus Z
	$\frac{1}{6}bh^2$
	$\frac{1}{6} \cdot \frac{b(h_1^3 - h_2^3)}{h_1}$
	$\frac{\pi}{32}d^3$
	$\frac{\pi}{32} \cdot \frac{d_o^4 - d_i^4}{d_o}$

$$\epsilon = \frac{M}{ZE}$$

where, M: Bending moment (See Table 1)
 Z: Section modulus (See Table 2)
 E: Young's modulus (See table "Mechanical Properties of Industrial Materials.")

It's important to be aware of the existence of this table.

Strain gauges: $R = \rho \frac{l}{S}$

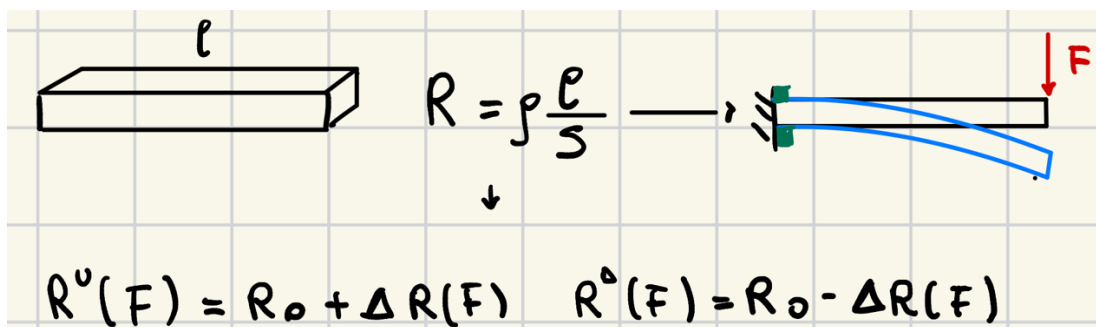
- In strain gauges resistance changes are dominated by geometric variations of the resistor due to applied strain.
- In piezo resistor major effects are due to resistivity changes due to the applied strain.

It's important to focus on the physical mean of the formula, the impedance is directly proportional to the length and inversely to the area, the idea is that if you have a lot of space you are going to have less friction, because the different molecules are not very close to each other (low impedance), on the other hand if you have a narrow space, the particles are going to encounter more friction and so the impedance is going to increase (this for what it concerns the S value).

The impedance is directly proportional to length because if you have to travel a long distance the R is going to be somehow higher (I believe the mirror reasoning is simple: if you have a higher number of interactions within a restricted space, you have an inverse relationship—the larger the 'l', the more interactions you will encounter, and therefore the impedance will increase.).

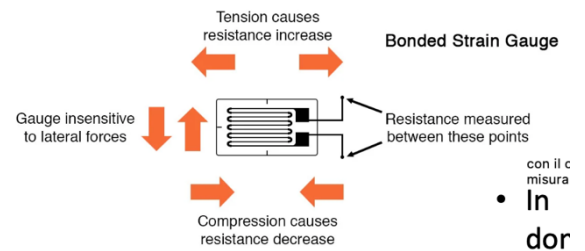
The question is how you use the resistor.

- the idea is to put it at level of the constrain in a cantilever. On the surface you have traction, while below you have compression.



The item on top (green) increases the length while the one below is the opposite, but there is also a decrease in the area when length increases; for this in general the R stays constant.

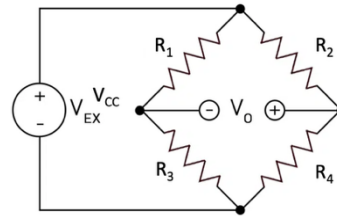
How we find the change in the resistance? The answer is the Wheatstone bridge.



- The idea is that knowing V_0 and at least three out of four the different resistance, we can find one of them, that it is what we want.

If the four resistances are equal, V_0 is equal to 0.

The bridge is **balanced**.



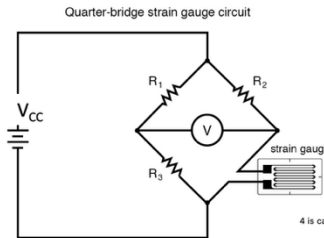
$$V_0 = V_{CC} \left(\frac{R_4}{R_2 + R_4} - \frac{R_3}{R_3 + R_1} \right)$$

What happens if we add a strain to one of the four resistances?

- Only one of the four is on the cantilever. In this case that is even you have a simplification of $2\Delta R$, this is called **quarter bridge**.

If you do the same calculation and you add another transducer in counter phase (1-3,2-4), the simplification that was done before is not anymore needed, so you have a perfect calculation.

They must be on the two different side of the cantilever.

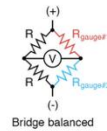
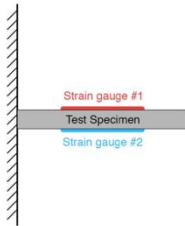
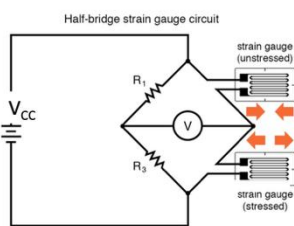


$$R_1 = R_2 = R_3 = R_0$$

$$R_4 = R_0 + \Delta R$$

$$V_0 \approx V_{CC} \frac{\Delta R}{4R_0} \approx V_{CC} \frac{G\epsilon}{4}$$

Where G: Gauge Factor



$$R_1 = R_3 = R_0$$

$$R_2 = R_0 - \Delta R$$

$$R_4 = R_0 + \Delta R$$

$$V_0 = V_{CC} \frac{\Delta R}{2R_0} = V_{CC} \frac{G\epsilon}{2}$$

If you add also on the other side is possible to obtain a **full bridge** and at the denominator you will not have a coefficient like 4 or 2.

This is a definition: $\frac{\Delta R}{R} = G\epsilon$, where G is called Gauge factor.

In the full bridge you have a *chiastic approach*, so the one in diagonal are working in the same way (compression or extension).

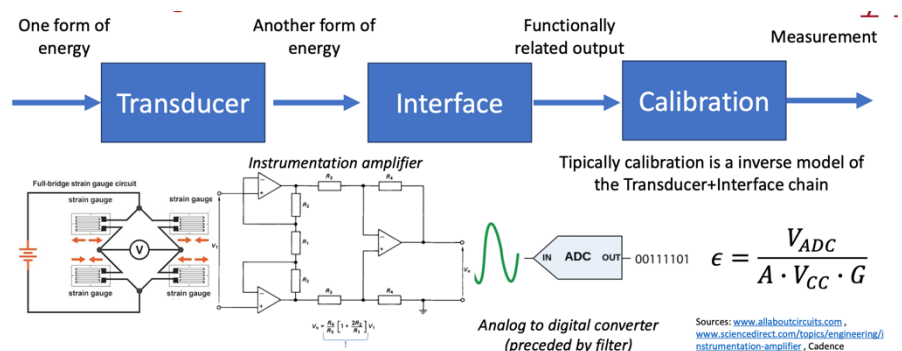
This is a **front-end** example of soft neuromorphic application.

What we do is feed the NN with the output of the ADC and knowing the relation we can obtain the deformation.

An example are the **load cells**, that measure three force and torque components. They are based on strain gauges.

- Collaborative robotics:** Force / Torque and tactile sensor can be applied in industrial robotics in order to have machine aware of the surrounding and that can react/stop when they enter in contact with different people, so increase safety.

Sensory modelling and substitutions:



Prosthetics:

The loss of the upper limb is a traumatic event that changes the quality of life radically.

Phantom limb pain and impairment of:

- Ability in **grasping** and **manipulation**.
- Ability in **sensing** through touch.
- **Gesture** (communication)

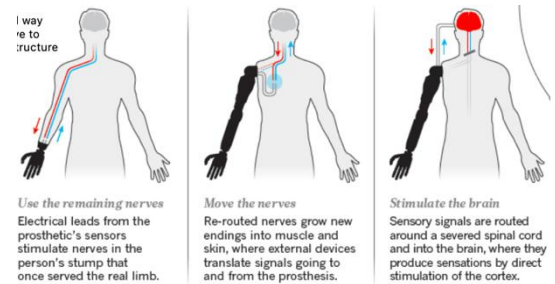
In prosthetics the problem is the bidirectional communication between the human and the system.

Ideally the objective is to restore bidirectional sensation:

- **Motor control** that is called *efference* system.
- **Sensory feedback** is called *afference* system.

Different name:

- The first is **trans radial** amputation so we use only the periphery.
- The second one is an amputation at the level of the shoulder, in which we **targeted muscle innervation**.
- The third one, is when there are SPIs, so we stimulate directly the brain.

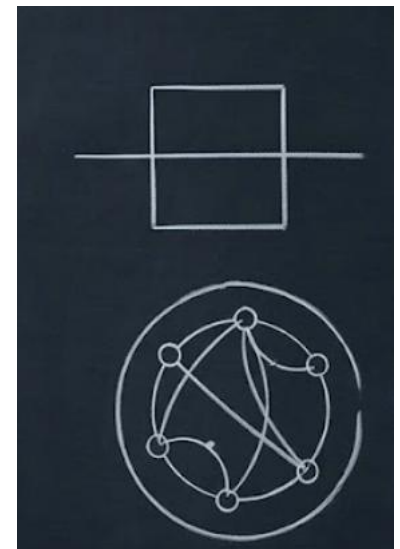


The common aspect between in the 3 cases is that it's better to stimulate as much peripheral possible, exploiting the presence of some nervous structure that are still there and, in the model, you need to consider this thing. If you stimulate peripherally you have to recreate only the mechanoreceptor.

Sensory feedback:

Different paper:

- *Dillon & Horch*: Demonstrated the possibility that a person with an amputation at level of the elbow, with neural feedback was capable to encode real time the motion of a robotic arm. This motion was measured by an electrode inserted in the median and ulnar nerve, that provide sensory feedback about the kinematics of the arm.
- *Tan et al.*: they used peripheral feedback, some used **cuff electrodes**. This type of electrode is wrapped around the nerve, like a cylinder but it's not entering inside the nerve. It can be used to measure or to stimulate, there are different electrode in the cylinder that allow to inject a current, inside the nerve. These however are not selective because you should deal with the different path of the current, that flows between the different electrodes (Essenzialmente la corrente devia il suo flusso creando diversi path tra i vari electrode, quindi non hai completa selettività). This study was an important study to demonstrate the capability to restore sensory feedback.
- *Ortiz-Catalan et al.*: They used a permanent implantation, the technological leap was this so-called osteo-integrated fixture, like the one done in dental prosthesis. You have a mechanical fixture with wires capable of passing in and out the fixture to stimulate the nerve.



In these examples they were modulating different things, like the pulse width or the amplitude but the general approach was not Neuromorphic.

Short-term implant of Transversal Intrafascicular Multichannel Electrodes (TIMEs) in an amputee:

The idea was to stimulate neurally the fascicles with pulses that give rise to a sense of touch.

The ethical approval was for only one month, so after that period the electrodes had to be removed. This because it was impossible to be sure about the safety of the electrodes after this period, at the end of the experiment they have seen that the electrodes were fine, but they had to remove them.

Description:

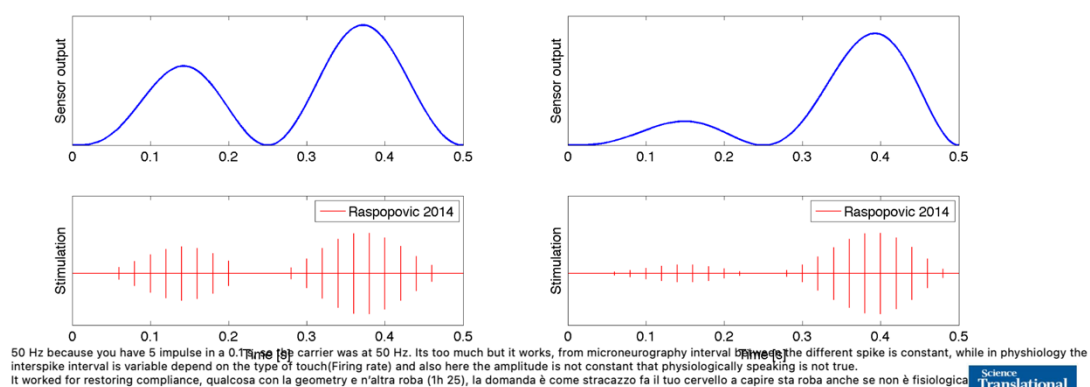
- 35-year-old man, from Denmark.
- Trans-radial amputation in 2004 (fireworks accident, NAPOLI88888).
- Subject resistant to pharmacological therapy and with no neuropathies (evaluated by Electroneurography) or other systemic diseases affecting brain/spinal cord/nerves.
- Subject with no neuropsychiatric disorders, evaluated by neuropsychological and psychiatric tests.
- Four-week implant
- Implanted nerve: Median and Ulnar, with two electrodes for nerve. These two nerves were chosen because they are associated with motor control and they provide sensory feedback, so they are bidirectional. It was possible to position the electrodes in different position, in order to measure only the signals that were ascending or descending. Other reason related to neuroanatomy, the *median nerve* innervate the first digits of the hand, in particular the first three fingers and the internal part of the fourth finger(the one between the third and the fourth finger), while the *ulnar nerve* innervate the fifth finger and the internal part of the fourth(between the fourth and the fifth). In *physiology* those two nerves innervate the territories of the palmar side of the hand. In neuroprosthetic with the median you associate the first fingers to the phantom limbs(?), the ulnar in the last(ripetizione).

Surgical technique:

- General anesthesia
- Skin incision (medial nerve of the biceps muscle).
- Exposition of the median and ulnar nerve.
- Epineural microdissection.
- TIME electrodes inserted under surgical microscope using a guiding needle.
- 8-0 suture to fix the electrodes to the epineurium.
- Subcutaneous pockets.

Also, this was not a Neuromorphic approach.

Encoding strategies:



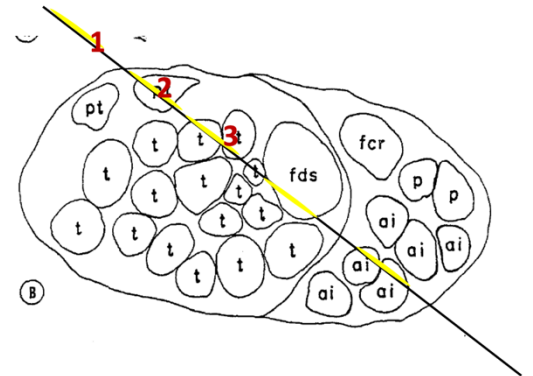
The principle of this encoding strategy is the following; Based on sensor output you are going to modulate the current delivered per each pulse proportionally to the sensor output itself, minus the threshold of what you consider as the ground price (da per scontato ci sia sempre una soglia di rumore minima, sotto la quale non dai stimolazione). The pulses are delivered at fixed frequency, in this case 50 Hz (5 pulses each 0.1s). This is not physiological, in fact it's too much from a micro neurographic perspective and also the interval it's fixed. In physiology, the firing rate change based on the intensity of the applied force (for the classical mechanoreceptor) and also the amplitude of the spike here is variable, while in the body is constant (AP, all-or-none phenomena). In the body the amplitude is not a carrier of information, but the firing rate is.

So even with this all violations, **how this method works?**

The picture is about the structure of the peripheral nervous system. Fascicles are packed, in groups some are motor and other sensory, they are all closed to each other with an organization that is locally somatotopic.

Fatto un esempio pratico: the more you add pressure when touching something the bigger is the receptive field that you are using, in the nerves this mean that you are using more fascicles.

This can be exploited by neuroengineering, if you increase the *current*, the stimulated area will be bigger. (Ha disegnato intorno alla linea gialla, che è una specie di needle, un cerchio che aumenta di grandezza con l'aumentare della corrente).



COSA SIGNIFICA STA ROBA: The sensor perceives an high input that normally would have been an high pressure applied, for that reason while physiologically we would have an high quantity of fascicles activated (bigger receptive field), here we increase the current (bigger stimulation) so we are going to increase the area of stimulation, the one that surround the needle.

IN ITALIANO CHE FA SEMPRE BENE: Fisiologicamente quando applichiamo tanta pressione, aumentiamo il firing rate ma anche il numero di recettori usati, come possiamo nella nostra neuroprotesi modellare questo aumento di recettori? Sfruttando il collegamento diretto recettori-fasci, più recettori vuol dire un maggior numero di fasci nervosi; quindi, quando il sensore della protesi misura un input elevato io aumento la corrente nel needle, che aumenterà l'area di fasci nervosi interessata.

What was obtained:

- Restored force level.
- Restore the capability of perceive the grosser shape of the object.
- Compliance, so understand if the object is soft or hard.

How the Patient Perceives "Compliance" (Stiffness vs. Softness)

Even though the sensors on the robotic hand only measure raw pressure (force) and not "softness" directly, the patient was still able to distinguish between rigid and compliant objects. This perception relies on a clever interaction between the movement of the robotic hand and the object's physical resistance over time.

Here is how the encoding strategy makes this possible:

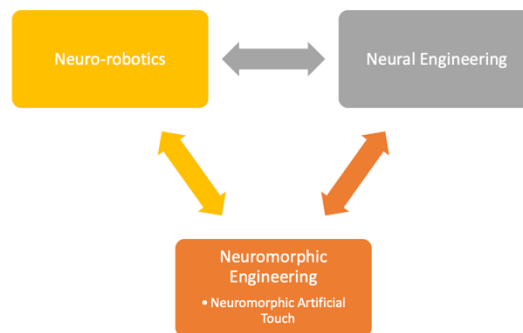
- **The Constant Speed Constraint:** The robotic hand is programmed to close its fingers at a constant speed.
- **The Temporal Profile of Force (The "Derivative"):** The key to feeling compliance is not the absolute amount of force, but *how fast* that force changes over time as the hand interacts with the object. In physics, this is the temporal derivative of force (dF/dt).

- **Rigid Objects (e.g., a wooden block):** When the hand grasps a hard object, the material does not yield. Because the hand is trying to close at a constant speed against a rigid surface, the sensor registers an almost instantaneous, sharp spike in force. The encoding algorithm translates this into a sudden, brutal increase in the amplitude of the electrical pulses.
- **Compliant Objects (e.g., a mandarin or sponge):** When grasping a soft object, the material squishes and yields. The sensor registers a much more gradual buildup of force. The encoding algorithm translates this into a slow, gentle ramp-up in the amplitude of the electrical pulses

Neuromorphic artificial touch case study

Introduction:

The idea is to have the output of a sensor that is feeding a neural model that modulate the spike interval. So instead of having a fixed interval, like before the interval is variable during time and we are exploiting this variability to carry information.

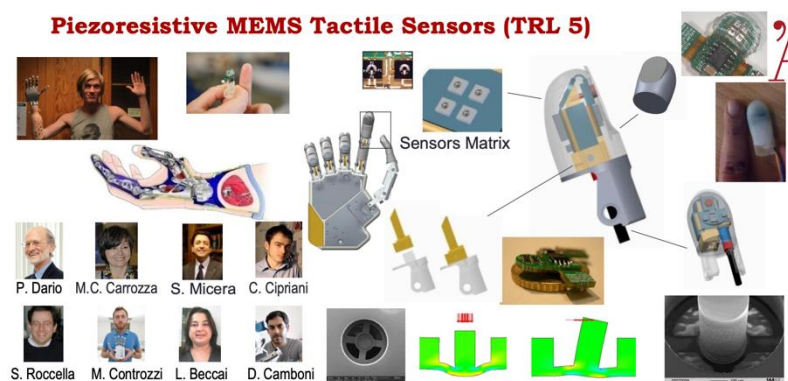


It was discovered by investigating the human sense of touch so human neurophysiology, using micro neurography and EEG, that we could create a benchmark of how the nerve and the receptor encode tactile information to reproduce them with a pattern of neural activities

Piezoresistive MEMS tactile sensors (TRL 5):

This study is how we can use an artificial finger to implement tactile sensor with electronics and then transport into inputs to activate a neural model, to create spike and then trigger the neural stimulation.

In the sensors that are integrated in the artificial fingers, the Wheatstone bridge and the cantilever are used in the fingertips.



The piezoresistive sensor are like the strain gauges and other sensor families like photonic and others are not so distant from them, so the principle of strain applied and read by the transducer is pretty much the same.

Focus on the fingertips:

Inside them there are electronics that allow to read the output of MEMS sensors, integrated in the fingertips and on top of them the MEMS and if you zoom you will find miniaturized **load cells** (quelle Verdi nella foto sopra). This load cells have a pillar structure, like mesa (that mean “altopiano”) and they exploit the principle of the cantilever (parte in giallo).

There are four cantilevers:

- *Vertical force*: All of them bends like in common mode (all together).
- *Lateral force*: Two cantilever deform with a complex compression-traction way while the other two orthogonal to the plane they just rotate (Torsion), so it's like an X(?). The two working with compression-traction have a very high change in resistance while the other working with torsion has small changes in resistance.

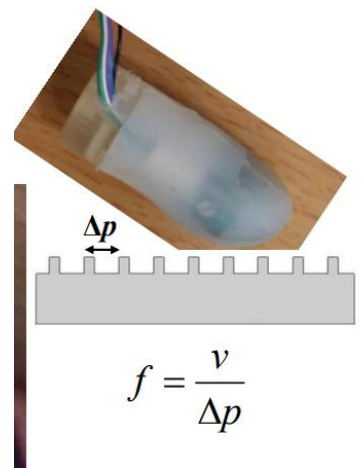
As shown in the image above, in the study it was used a robotic fingertip equipped with **MEMS (Micro-Electromechanical Systems)** tactile sensors slides across a surface. The idea is that when the artificial fingertip presses against and slides over an object, the physical structure of the sensor deforms (strain). By measuring this deformation at specific points (e.g., the roots of the sensor pillars), the system can extract two critical pieces of analog data:

1. Normal Force: The perpendicular pressure against the surface.
2. Shear (Tangential) Force: The friction/drag force as the finger slides across the surface.

These analog force signals are fed into a **Spiking Neural Network (SNN)**, which translates the continuous force values into trains of discrete spikes.

However, to understand how tactile sensors perceive complex everyday surfaces, we must first study how they perceive a very basic, fundamental shape. For this purpose, the experimental stimulus was a **grating**. A grating is a highly regular physical pattern composed of alternating **valleys and ridges**. The defining characteristic of this grating is its spatial period, denoted as Δp (the physical distance from one ridge to the next).

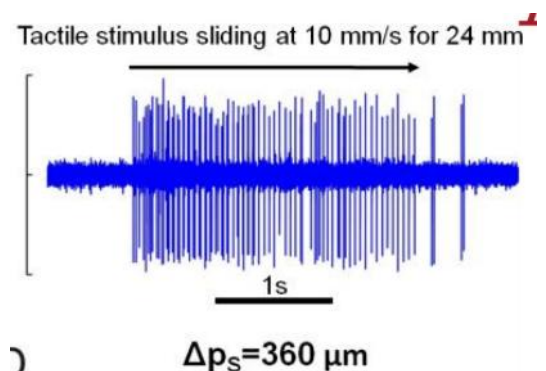
Any complex, irregular sound wave can be mathematically decomposed into a linear combination of simple, periodic sine waves (harmonics) using a Temporal Fourier Transform. Any complex, real-world texture (both its macroscopic and microscopic topography) can be decomposed into a linear combination of simple, periodic gratings with different spatial periods (Δp) and weighted by different coefficients. This is the **Spatial Fourier Transform** of a stimulus. So basically, by understanding exactly how our artificial sensory system (and the human brain!) responds to this basic "atomic" grating stimulus, we can mathematically reconstruct and understand how it will respond to *any* complex surface. This relies on the principle of **spatial summation**.



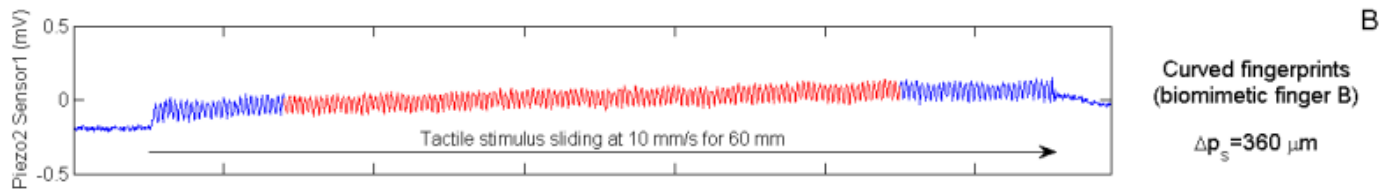
Grating Decoding

When an artificial finger and a human finger are subjected to the exact same physical stimulus (sliding over a grating at a controlled velocity of 10 mm/s), the raw signals recorded look completely different:

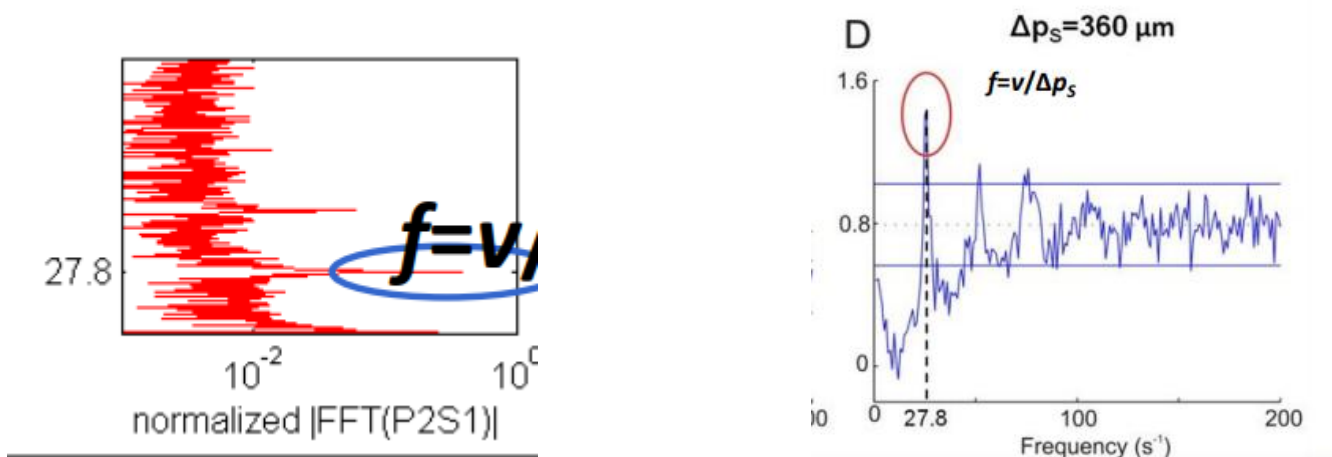
- Human Response (Microneurography): human mechanoreceptors output discrete spikes (action potentials).



- Artificial Response (MEMS Sensor): the artificial sensor outputs continuous vibrations (analog strain/voltage signals).



However, it is convenient to analyze the signals in the spectral domain using tools like **Fast Fourier Transform (FFT)**, **Short-Time Fourier Transform (STFT)**, or **Wavelet analysis**. Whether you analyze the human spike trains or the artificial continuous vibrations, **they both exhibit the exact same peak frequencies** in the spectral domain. This shared spectral signature proves that both systems are capturing the exact same physical reality.



When you slide a finger across a regular grating, the frequency of the resulting vibration depends purely on two physical parameters:

- Sliding Velocity (v):** How fast the finger moves (e.g., **10 mm/s**).
- Spatial Period (Δp):** The physical distance between two ridges on the grating.

The time (Δt) it takes too move from one ridge to the next is: $\Delta t = \frac{\Delta p}{v}$

Since the frequency (f) is the inverse of time, the fundamental frequency generated by the sliding motion is: $f = \frac{v}{\Delta p}$.

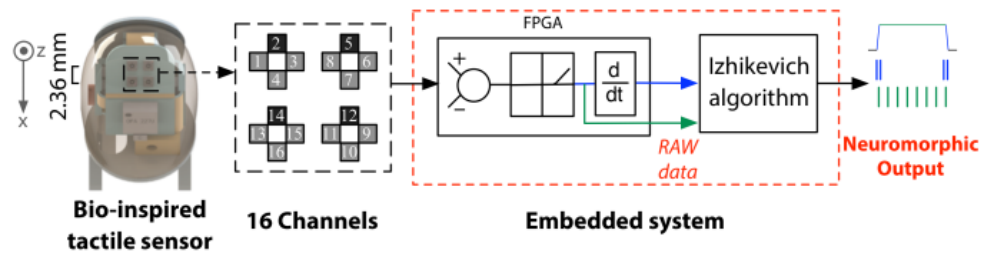
If we now use this mathematical derivation, we can compute all the peaks already observed in the plots.

- If $v = 10 \text{ mm/s}$ and $\Delta p = 0.44 \text{ mm} \rightarrow f = 10 : 0.44 = 22.7 \text{ Hz}$
- If $v = 10 \text{ mm/s}$ and $\Delta p = 0.36 \text{ mm} \rightarrow f = 10 : 0.36 = 27.8 \text{ Hz}$
- If $v = 10 \text{ mm/s}$ and $\Delta p = 0.28 \text{ mm} \rightarrow f = 10 : 0.28 = 35.7 \text{ Hz}$

Now that we know the artificial sensor contains the correct frequency information, how do we translate its continuous vibrations into the spiking language of the human nervous system?

We use the **Izhikevich artificial spiking neuron model** (discretized using the Euler method for real-time application). The input of this model will be The analog vibration from the sensor is converted into the input current (I) for the Izhikevich model. You must carefully scale/amplify the sensor's voltage before feeding it as current to the neuron. If you amplify the signal too much (e.g., multiplying a 1 mV signal by a million), the massive input current will cause the artificial neuron to enter **full saturation**. It will just fire at its absolute maximum rate continuously, completely losing the delicate frequency modulation (the 22.7 Hz or 27.8 Hz peaks) of the original

stimulus. With proper scaling, the Izhikevich model will output a spike train that mathematically mimics the biological microneurography recordings.



This neuromorphic approach can be used for different applications:

- **Bionics & Neuroprosthetics:** If we want to connect a robotic hand to a human amputee, the artificial system *must* speak the language of the human peripheral nervous system (spikes).
- **Neurorobotics (Pure AI):** Even in fully artificial systems (no human involved), biological nervous systems have evolved to process sensory data with incredible efficiency (optimality). By mimicking biological architectures (connectomes and spiking networks), we can build smarter, more energy-efficient robots.

Complex Decoding

In the previous steps, the team used simple, periodic gratings to perfectly dimension and scale the analog-to-spike conversion. Once they verified that the sensor and the Izhikevich model could accurately capture and translate spatial frequencies, they moved to **non-grating, everyday materials**. The artificial sensor was dragged across various surfaces at a controlled velocity and force. The textures included:

- Different textiles
- Velvet
- Glass
- Paper
- Bioskin (a synthetic skin-like material)
- **The "Wii" Cover:** a student cut up the silicone protective cover of a Nintendo Wii remote because it possesses a very unique, smooth, and highly lubricated texture (low friction), making it an excellent edge-case material to test!



The real-time data flow happening during the experiment was:

- **Sensor:** The piezoresistors inside the tactile sensor physically deform.
- **ADC:** An Analog-to-Digital Converter reads this continuous voltage.
- **Processing:** The signal is scaled (amplified) and the offset is subtracted.
- **SNN:** The scaled current is fed into the real-time Izhikevich spiking neuron model.
- **Output:** A train of discrete electrical spikes is generated.

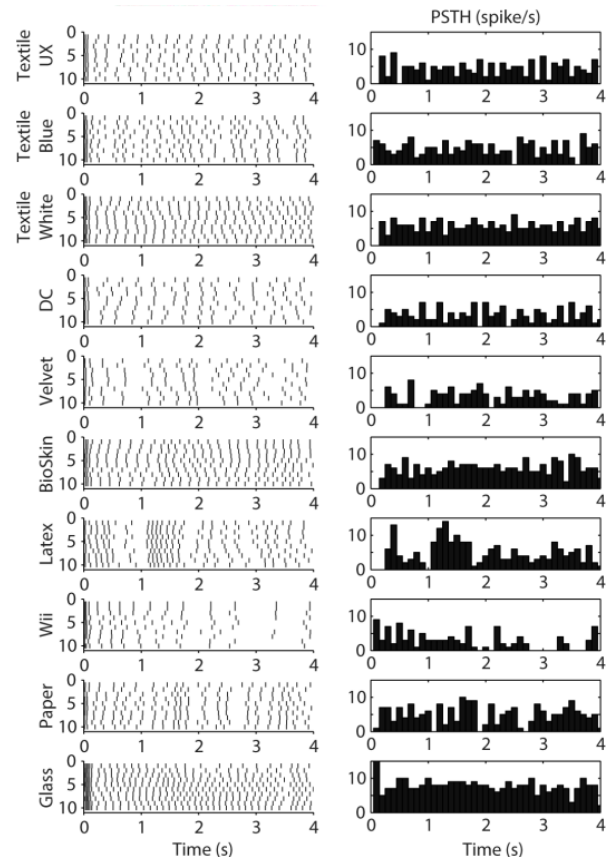
Raster Plot

To analyze the spikes produced by these different materials, neuroscientists use a specific visualization tool called a **Raster Plot**.

- **General Definition:** Typically, a raster plot puts **Time** on the x-axis and different **Channels/Neurons** on the y-axis. Every time a neuron fires, a small vertical tick mark is drawn. It allows you to see the firing activity of an entire network of neurons at a glance.
- Instead of multiple channels, the y-axis represents **Replicas (Trials)** of a *single* channel.
 - For each material (e.g., Glass), they performed **10 identical experiments** (replicas 1 through 10).
 - Each row block in the image (Glass, Paper, Wii, etc.) shows these 10 repeated trials stacked on top of each other.
 - Since there are 10 materials and 10 replicas per material, the study displays exactly **100 experiments**.

If you look closely at the raster plots, a vital pattern emerges:

- **Intra-Stimulus Consistency (Self-Similarity):** Look at the 10 replicas for a single material (e.g., the 10 rows under "Textile"). The spike patterns are highly consistent across every trial. The vertical lines almost perfectly align. If you touch the same object 10 times, the sensor produces the same "acoustic signature" or pattern of spikes every time.
- **Inter-Stimulus Variability:** Now compare the block of lines for "Textile" to the block for "Velvet". The patterns are completely different. Textile generates a very regular, periodic spiking pattern (due to its woven nature), while velvet generates a much more irregular, distinct pattern.



Because the spike patterns are **repeatable** for the same object, and **distinct** between different objects, we can build an algorithm to recognize them. We need to extract mathematical **features** from these spike patterns to build a **Classifier**—a decoding algorithm that can look at a blind spike train and confidently say, "The robot finger is currently touching glass," or "The robot finger is touching velvet."

Decoding Strategies

To classify objects based on spiking data, there are three main approaches:

1. **Feature-based decoding**
2. **Distance-based decoding**
3. **Architecture-based decoding**

1. Feature-based decoding

To classify the different materials (glass, velvet, Wii cover, textile, etc.) from the raster plots, we need to extract mathematical "features" from the spike trains.

The simplest feature is to just count how many spikes happen over a certain period, which coincides with the **average firing rate**. You count the total number of spikes in a trial and divide it by the duration of the time window. (Note: This is *not* the instantaneous firing rate, which looks at the time between individual spikes, but an overall average).

In this experiment, the duration of the time window was 4 seconds. So, if we count the spikes for each stimulus and then divide by 4, we get the AFR per each stimulus (Wii has 3.5 spikes/seconds, Velvet 3.5 spikes/seconds and Textile white has 6 spikes/seconds).

If we only look at the Average Firing Rate, we have a major problem. "Wii" and "Velvet" both produce ~3.5 spikes/second. On a 1D graph, their data points would perfectly overlap, making it impossible for an algorithm to tell them apart. We **need a second dimension**.

The "Wii" and "Velvet" stimuli have the same number of spikes, but differ in regularity. Velvet produces a very regular, steady rhythm of spikes. The Wii cover produces a more irregular, clustered pattern. To measure this mathematically, we use the **Inter-Spike Interval (ISI)**. The ISI is simply the time that passes between two consecutive spikes. In code, it's like taking the discrete derivative (the diff operator in MATLAB/Python) of the timestamp vector. To obtain the feature, we calculate the **standard deviation** of these intervals (often normalized as the Coefficient of Variation).

- A low standard deviation = highly regular firing (e.g., Velvet).
- A high standard deviation = irregular firing (e.g., Wii).

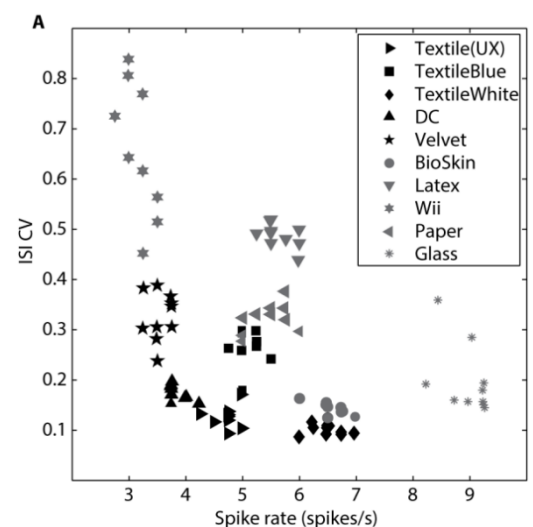
Look at the scatter plot. We now have a 2D space:

- **X-Axis:** Average Firing Rate
- **Y-Axis:** ISI Coefficient of Variation

By plotting the 100 experiments (10 materials x 10 replicas) on this graph, we see that the different materials separate into somewhat distinct clusters!

Because the data points are separated, we can use a basic Machine Learning algorithm like **K-Nearest Neighbors (KNN)** to classify them.

- **The Result:** The KNN algorithm achieves an accuracy of **78%**.
- **Context:** Since there are 10 different objects, pure random guessing (chance level) would give a 10% accuracy. So, 78% is actually a very good performance!



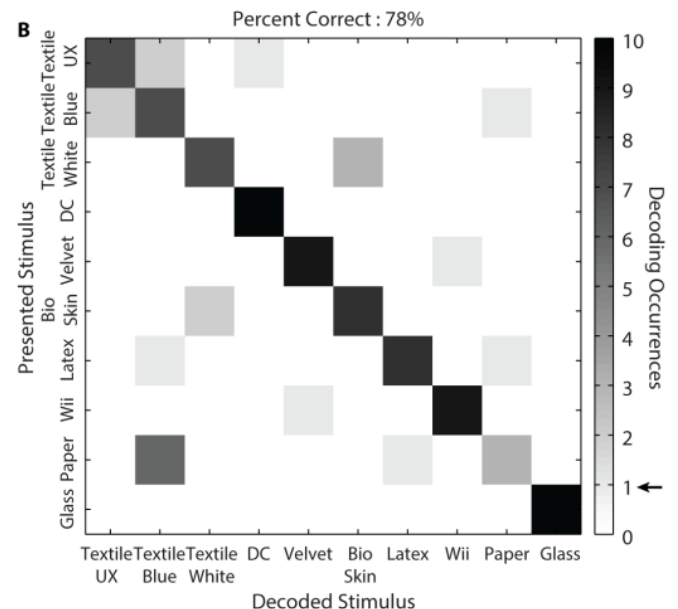
While 78% is good, if you look closely at the 2D plot, some clusters (like the triangles and squares) are still overlapping. Why can't we just write a more complex algorithm to push the accuracy to 100%?

2. **Danger of overfitting:** If you use a highly complex algorithm to force the overlapping points apart, you will likely **overfit** the model. This means the algorithm just memorizes the specific noise of this specific dataset and will fail completely when it encounters new, unseen data in the real world. The algorithm isn't the problem; the *features* are the problem.
3. **Loss of temporal richness:** Imagine two different spike patterns that happen over the same duration (Pattern A: 3 spikes... then 4 spikes... then 3 spikes. Pattern B: 3 spikes... then 3 spikes... then 4 spikes.) If you compute the features for these patterns, you get that both the Average Firing Rate and the ISICV are almost identical.

In our 2D feature space, Pattern A and Pattern B will be plotted in the exact same spot. However, biologically and physically, a sequence of "3-4-3" is very different from "3-3-4". By taking averages and standard deviations over a large 4-second window, **we lose the dynamic temporal richness of the signal**. To solve this, we cannot just extract basic statistical features. We must find a way to compare the actual timeline of the spikes. This directly leads to the next lesson: **Distance-Based Decoding**.

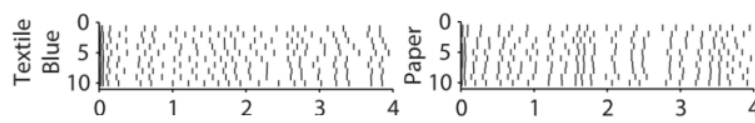
Now that we understand the limits of extracting simple statistical features (like average firing rate and ISI variation), we need a new way to classify our tactile stimuli. The goal is to evaluate the **entire temporal structure** of the spike trains.

To truly see where the Feature-Based approach (with the KNN classifier) fails, we look at a **Confusion Matrix**. A confusion matrix is a matrix where the rows represent the *actual/true* stimulus applied, and the columns represent the *predicted* stimulus decoded by the algorithm. A perfect classifier would only have high values on the main diagonal (meaning True Stimulus = Predicted Stimulus) and zeros everywhere else (off-diagonal). What we can notice is that while the diagonal is generally strong, there are significant off-diagonal errors. The most notable one is the confusion between **Paper** and **Textile Blue**.



Why are Paper and Textile Blue being confused? If we look back at the 2D feature space in the previous page, we can notice that the data points for Paper (triangles) and Textile Blue (squares) are severely overlapping. Another important thing to note is that Paper is usually misclassified as Textile Blue and not the opposite. This happens because of how the K-Nearest Neighbors (KNN) algorithm handles data dispersion. The **Textile Blue** data points are highly concentrated in a specific area (they have a compact support/centroid). The **Paper** data points are much more *dispersed*. Because Paper is dispersed, its points "invade" the compact territory of Textile Blue. When the KNN algorithm looks for the nearest neighbors of an invading Paper data point, it is surrounded by concentrated Textile Blue points. Therefore, Textile Blue "wins" as the attractor, causing the misclassification. If the 2D features of Paper and Textile Blue overlap, does that mean their actual spike trains are identical? **Absolutely not.**

- If you look at the raw raster plots (2 pages back) for Paper and Textile Blue, they look visibly different to the human eye.
- For example, the Paper raster plot has distinct "blank regions" (periods with zero spikes), whereas the Textile Blue plot has continuous spiking in those exact same time windows.
- Our previous features completely failed to capture this obvious structural difference.

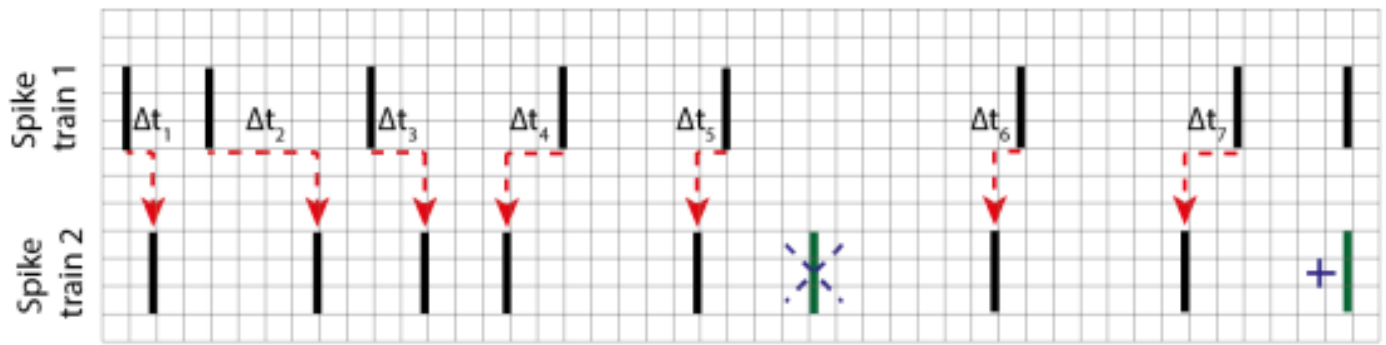


2. Distance-Based Decoding

Instead of extracting average numbers from the spike trains, what if we just compare the spike trains directly? We need to define a mathematical **Distance** between Stimulus 1 and Stimulus 2. We also want that our distance is based on the similarity definition, in order to have a lower distance if the two spike patterns are highly similar, and a higher distance if two spike patterns are very different.

We need a mathematical function that compares two entire spike trains and tells us how similar they are. This function must respect the mathematical properties of a metric distance (e.g., symmetry, triangle inequality).

The solution proposed in this paper is the **Victor-Purpura Distance (VPD)**.



The VPD is defined as the *minimum* cost required to transform Spike Train 1 (S_1) into Spike Train 2 (S_2) using a specific set of operations.

You have two allowable operations:

- **Add or Remove a spike:** Cost = 1
- **Shift a spike:** Cost = $q * \Delta t$ (where Δt is the time shift, and q is a tuning parameter).

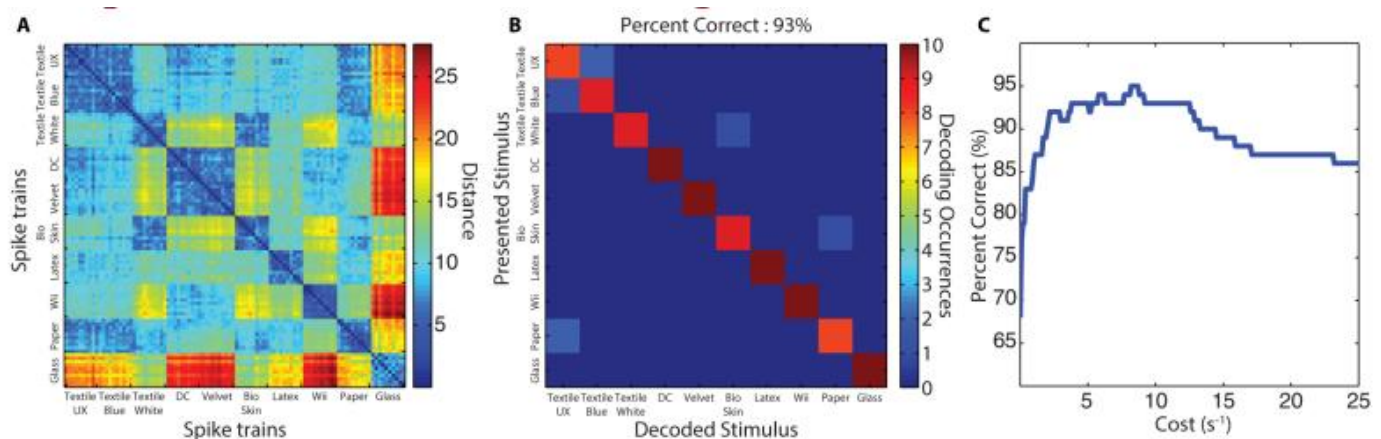
The parameter q represents the **weight given to temporal precision**.

- If q is high, you demand strict temporal precision (shifting is expensive).
- If q is low, you are more tolerant of temporal shifts (shifting is cheap).

What happens to the VPD when $q \rightarrow 0$ and when $q \rightarrow \infty$?

Let's assume S_1 has N spikes and S_2 has M spikes.

- **Case 1: $q \rightarrow 0$ (Temporal precision doesn't matter)**
 - **Policy:** Because the cost of shifting is practically zero, you can freely slide all the spikes of S_1 to perfectly align with S_2 . Once aligned, you simply add or remove the remaining unmatched spikes.
 - **Distance:** The cost is purely the difference in the total number of spikes. Distance = $|N - M|$. (If they have the same number of spikes, the distance is 0).
- **Case 2: $q \rightarrow \infty$ (Strict temporal precision)**
 - **Policy:** Shifting a spike even a tiny bit costs infinitely too much. The only mathematically viable option is to completely delete all spikes in S_1 and insert all new spikes for S_2 (unless spikes happen to fall at the exact same mathematical instant).
 - **Distance:** You pay a cost of 1 to remove each of the N spikes, and a cost of 1 to add each of the M spikes. Distance = $N + M$ (minus any naturally overlapping spikes).



To use this for Machine Learning, the researchers calculated a massive **Cross-Distance Matrix**. They compared every single one of the 100 experimental trials against each other (resulting in a 100 x 100 matrix with 10,000 distance values). The diagonal is purely 0 because the distance from a vector to itself is zero.

Using a KNN classifier on this distance matrix yields fascinating results depending on the value of q :

- **Optimal Range (q between 3 and 15):** The classification accuracy jumps to **93% - 95%**! This massive improvement proves that the VPD captures the rich temporal patterns of the spikes (like the "3-4-3" vs "3-3-4" problem discussed earlier).
- **High Range ($q \rightarrow \infty$):** Accuracy settles around **85%**.

As we can see also from the graph, if you set q near 0, the accuracy crashes to roughly **65% - 70%**. This is even worse than the Feature-Based Method. Because $q \rightarrow 0$ means the distance is *only* based on the difference in the total number of spikes. This is mathematically equivalent to looking *only* at the Average Firing Rate (the X-axis from the previous lesson). By setting $q = 0$, you are throwing away the Y-axis (the ISICV variation). You reduce a 2D feature space into a 1D feature space, completely confounding materials like "Wii" and "Velvet" that share the same firing rate.

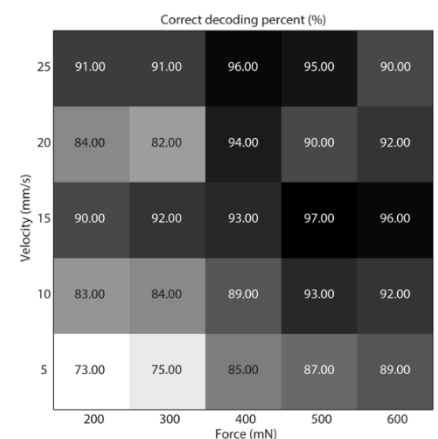
Generalization of stimuli decoding

Before diving into the solution for generalization, it is important to distinguish between the two main approaches to neural decoding discussed in this module:

- **Algorithmic Decoding (Current Focus):** Methods like the Victor Purpura Distance (VPD) rely on algorithms. This approach is non-physiological because it requires collecting a fixed "window" of data over time, and then applying mathematical computations (like a matrix) to that static window.
- **Architectural Decoding (Future Focus):** The biological brain does not wait to collect data windows. It relies on its neural architecture (starting with the cuneate nucleus). It processes a continuous streaming of data from the periphery, allowing for real-time perception and classification.

To decode sensory modalities effectively, we must first understand how physical quantities alter the spike patterns. In tactile sensing, within a specific sensory workspace, there is an almost **linear correlation** between the firing rate and both the applied force and the sliding velocity.

- **Force:** Pressing with a higher degree of intensity increases the steady-state rate of firing and enhances signal modulation. Looking at the data plots horizontally, as the contact force increases, the overall firing rate consistently increases.
- **Velocity:** As is typical in many sensory modalities, a faster phenomenon yields a higher rate of firing. Looking at the data plots vertically, as the sliding velocity increases, the number of spikes generated proportionally increases.

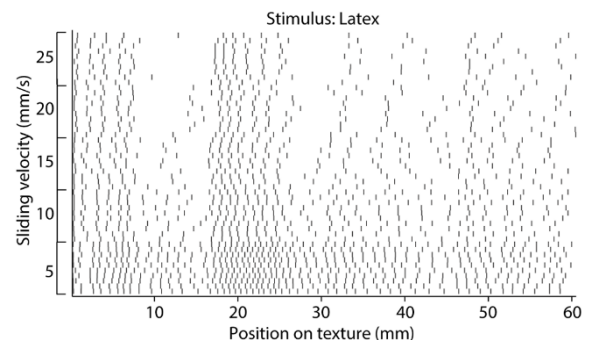


In previous models, decoding was done at a fixed velocity. The goal of generalization is to accurately classify the identity of a tactile stimulus (what we are touching) *regardless* of the exploratory sliding velocity.

The physical topography of an object remains constant. If you slide a sensor across a textured surface, the spatial layout of that texture does not change, even if you speed up or slow down your hand.

Instead of calculating the Victor Purpura Distance (VPD) as a function of time, it was implemented in the **space domain**.

- By converting the data so that the x-axis represents *Space (mm)* instead of time, the spike output becomes invariant.
- As shown in the raster plots, the spatial patterns of the spikes align beautifully across different velocities (10 mm/s, 20 mm/s, 40 mm/s) for the same texture.



While spatial representation works perfectly for velocities of 10 mm/s and above, an anomaly occurs at very slow velocities (e.g., 5 mm/s). At 5 mm/s, the spatial pattern of spikes appears slightly different, becoming denser and showing spurious spike clusters. This is caused by a motion biomechanics phenomenon known as **stick-slip**. At very low velocities, the coefficient of friction causes the motion at the interface between the stimulus and the fingertip to behave erratically. Instead of a smooth, continuous sliding motion, the movement breaks down into microscopic steps of starting and stopping. This mechanical stuttering disrupts the otherwise invariant spatial spike pattern.

To represent neural spikes in the space domain rather than the time domain, there is a fundamental prerequisite: the system must be capable of independently measuring or estimating the explorative sliding velocity. Whether this velocity estimation comes from other sensory inputs (like proprioception or vision) or direct measurement, knowing the velocity allows you to normalize the sensory data. This normalization shifts the analysis from a Temporal Victor Purpura Distance to a **Spatial Victor Purpura Distance**.

When shifting domains, the representation of the data and the mathematical units fundamentally change:

- **Temporal VPD:** The parameter Q represents a *time shifting cost*. Its measurement unit is **1/second**. It defines how much it "costs" to shift a spike in time.
- **Spatial VPD:** The parameter Q represents a *space shifting cost*. Its measurement unit becomes **1/millimeter** (or 1/meter). It defines how much it costs to shift a spike by one millimeter along the spatial topography of the object.

Imagine exploring an object that is exactly 10 centimeters long:

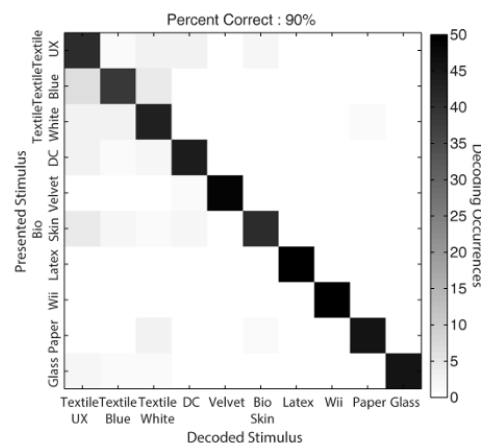
- **At a velocity of 10 mm/s:** The overall exploration takes 10 seconds. You record a specific pattern of spikes spread across a 10-second window.
- **At a velocity of 20 mm/s (double speed):** The overall exploration takes only 5 seconds.

If you look at the data in the time domain, doubling the velocity compresses the time scale by half. While the core pattern of spikes is essentially the same, it is squeezed into a 5-second window instead of a 10-second window. A time-based algorithm would struggle to match these two significantly different temporal patterns. If you represent both of these trials in the space domain, the x-axis maps to the physical 10-centimeter length of the object, not the duration of the touch. Assuming there are no major spurious spikes, the core pattern of spikes for both the 10 mm/s and 20 mm/s trials will perfectly align. The representation becomes invariant to velocity.

By utilizing the Spatial VPD (where Q is measured in 1/millimeter), researchers applied the same classification methodology used previously, but achieved far superior generalization.

- **High Accuracy:** The classifier achieved an accuracy of approximately **90%**.
- **Robust Generalization:** It successfully discriminated among 10 different tactile stimuli, and critically, it successfully identified the stimuli *across a wide range of different explorative velocities*.

To implement spatial representation biologically or artificially, the system requires a specific type of memory. The system must be able to memorize spikes and localize/associate them not with the *temporary event* of when they occurred (a timestamp), but with the *spatial location* of where they were produced on the stimulus.



ICT and BioRobotics applications

The neural representations discussed previously can be directly applied to Information and Communication Technology (ICT) applications. This includes fields like bionics, neural engineering, and bio-robotics, where applications range from restoring sensory feedback in amputees to processing data in robotic systems.

A critical advantage of utilizing spiking neural networks in these applications is their ability to drastically compress the dimensionality of datasets.

By passing raw sensor data through a spiking model (like the Izhikevich model), you can convert a continuous signal into a minimal pattern of spikes. This shift from a continuous time representation to an "event-driven" representation results in massive data compression. To understand why this massive scaling happens, we have to look at how data is stored in a table for each method:

1. Constant Time Sampling (Traditional Approach)

In traditional data processing, a sensor is sampled continuously at a fixed rate.

- **Data Storage:** Because the time steps are perfectly constant and predictable, you do not need to store the time itself. You only need to store the amplitude value (the Y-axis value) at each step.
- If a value is stored as a standard 32-bit floating-point number, it requires **4 bytes** of memory. If you sample data at 1 kHz (1,000 times per second), you must store 1,000 values.
- **Total Size:** 1,000 values $\times$ 4 bytes = **4 Kilobytes (KB) per second**.

2. Neuromorphic / Event-Driven Sampling (Spiking Approach)

In a neuromorphic architecture, data is only recorded when an "event" (a spike) occurs.

- **Data Storage:** Instead of recording constant amplitude values, you only record the **timestamp** of when a spike happens.
- If a timestamp requires the same 32-bit (4 bytes) resolution, but the signal only generates 4 spikes in a given second, you only need to store those 4 timestamps.
- **Total Size:** 4 spikes $\times$ 4 bytes = **16 Bytes per second**.

The dimensionality of the data is no longer tied to time; it is strictly proportional to the number of spikes.

These two representations are not perfectly equivalent. Converting a full continuous sequence into sporadic spikes reduces the overall "informativeness" of the raw signal. However, it preserves the crucial information needed to successfully carry out downstream tasks like classification and regression.

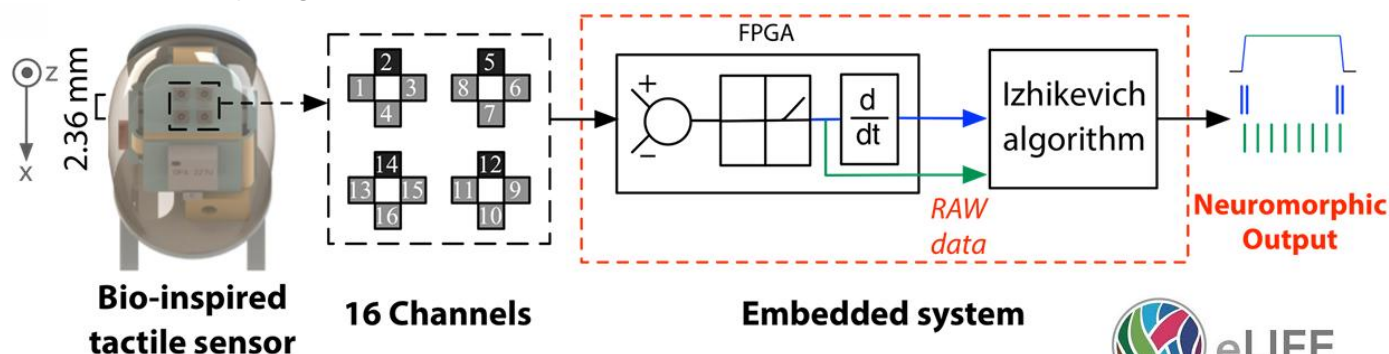
The power of neuromorphic implementation becomes obvious when scaling up to massive sensor networks.

Imagine a system with 1 million sensors (a "mega sensor"):

- **Using traditional 1 kHz sampling:** 1 million sensors $\times$ 4 KB/s = **4 Gigabytes of data generated every second**. This creates massive bandwidth and storage bottlenecks (similar to the challenges of processing high-density video).
- **Using neuromorphic spiking:** By drastically scaling down the data dimensionality to mere bytes per second per sensor, the network can handle a vastly increased number of sensors and channels without overwhelming the system's architecture.

To apply these spiking representations practically, researchers use a **"Soft" Neuromorphic Implementation**.

Unlike a "hard" solid-state neuromorphic device that intrinsically spikes due to its physical hardware, a "soft" implementation uses a standard, non-spiking sensor. The standard analog output of the sensor is passed through the differential equations of a spiking neuron model (like the Izhikevich model). The variations in the simulated membrane potential and recovery variables mathematically transform the non-spiking continuous data into discrete spiking events.



These generated spikes can be used to trigger neural stimulators. Beyond direct bionics, this architecture is used in:

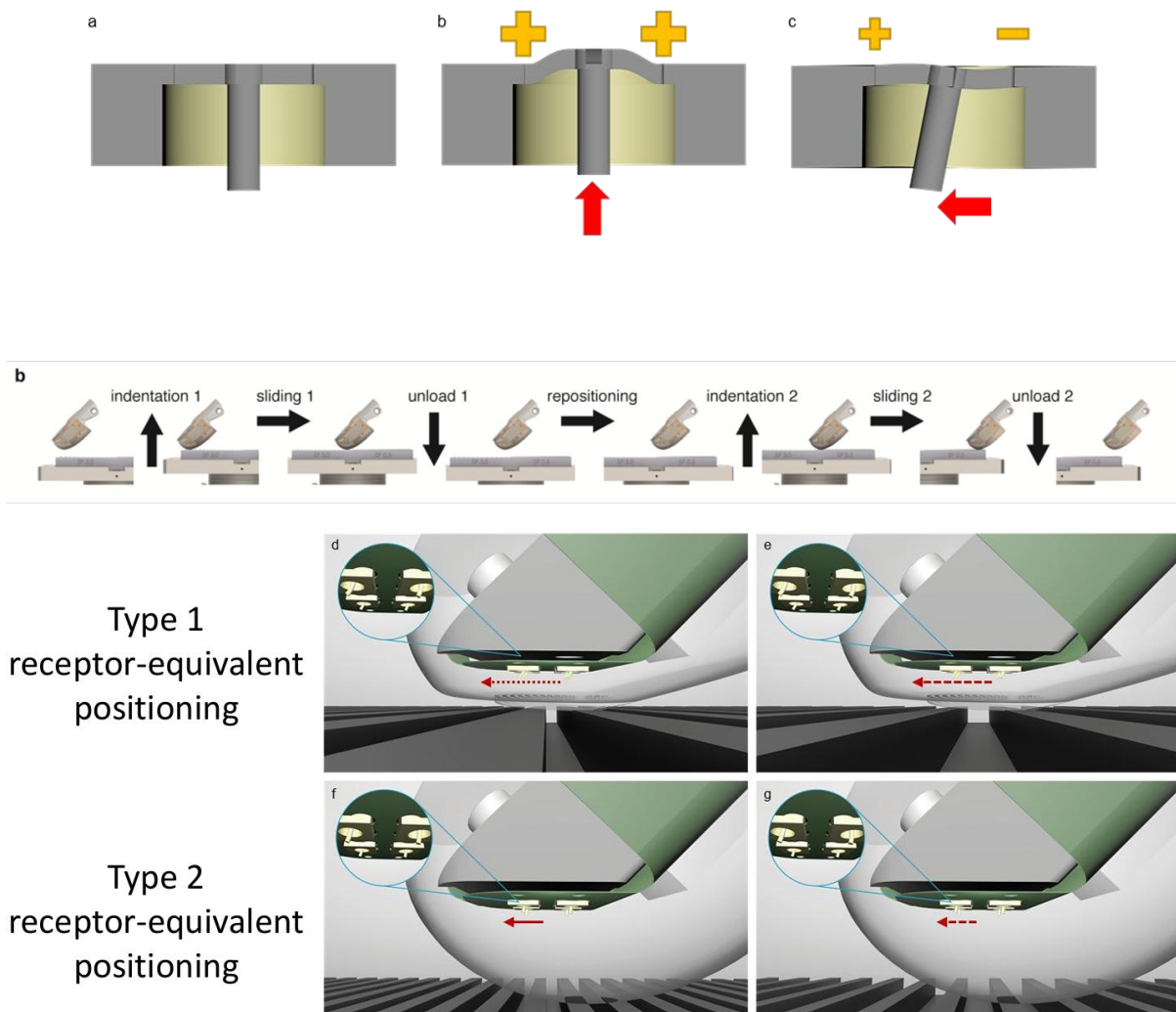
- **Telepresence / Bio-robotics:** A gesture-controlled robotic arm with sensorized fingers touches an object. The sensors' output is converted into spikes, which then trigger haptic feedback elements on a glove worn by the human operator.
- **Lower-Limb Prosthetics:** Sensorized insoles placed in a prosthetic foot generate data that is converted into spiking activity. These spikes are then conveyed to the processor units and motors of the prosthetic leg to trigger actions based on foot-ground contact.

Paper importante di Oddo del 2016

For the following experiments, the focus is placed heavily on tactile data gathered from sensors integrated into artificial fingertips.

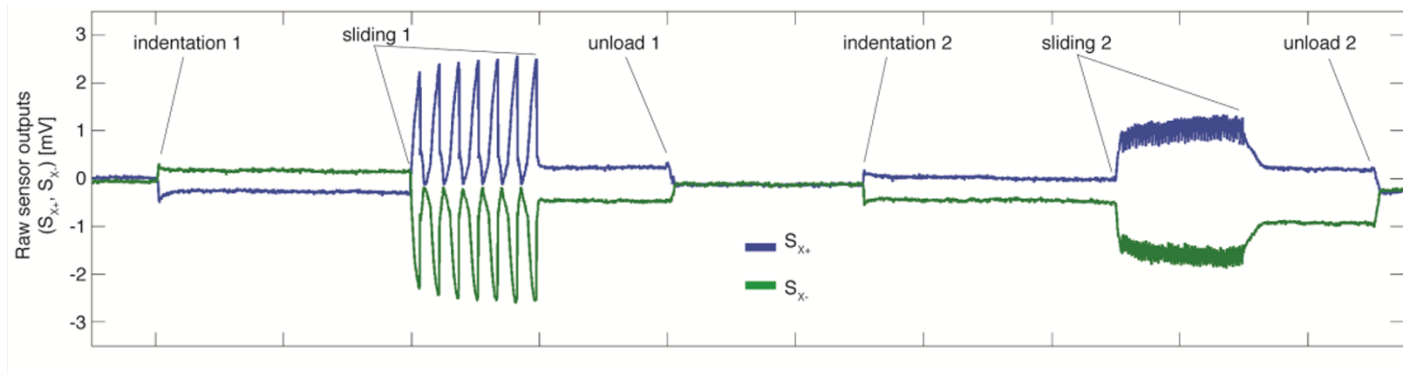
Sensors can be embedded at different depths within the artificial skin (polymer). Some are located very close to the surface (thin polymer layer above them), while others are implanted quite deep (thicker polymer layer). The sensors rely on piezoresistors implanted at the roots of a cross-shaped structure. How force is applied changes how these resistors deform:

- **Vertical Force (Normal Application):** Pushing straight down on the sensor causes a **Common Mode** deformation. The piezoresistors deform together in the same direction.
- **Lateral Force (Sliding Motion):** Applying lateral force (like sliding a finger across a surface) causes a **Differential Mode** of deformation. The tethers of the sensor are pushed in opposite directions—one goes positive (up), while the opposite one goes negative (down).

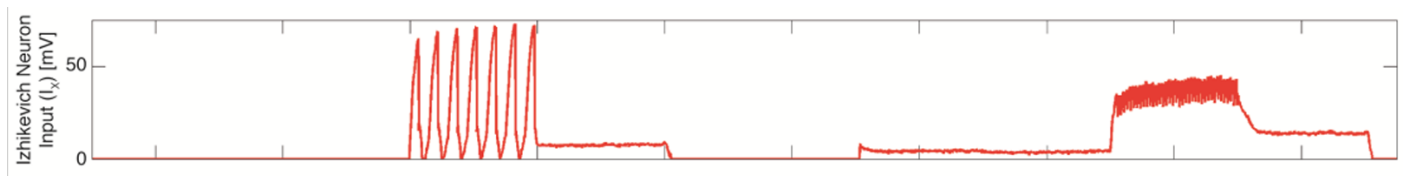


To prove that this system works, researchers tested the sensorized fingertip by sliding it across a specific topography: a grating (a surface with alternating ridges and grooves).

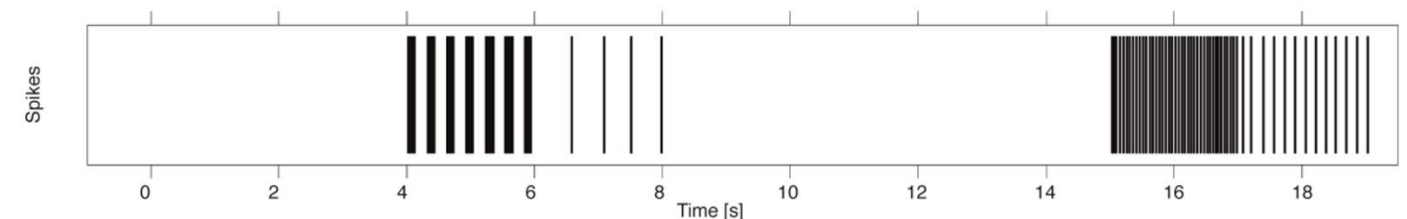
1. The Raw Sensor Output: As the sensor slides laterally across the ridges and grooves, the opposite tethers of the sensor move in counterphase (Differential Mode). If you look at the graph, one sensor output (e.g., the green line) goes up exactly when the opposite output (the blue line) goes down.



2. Calculating the Difference: To amplify the spatial pattern of the object, the system calculates the mathematical difference between these two counterphase signals. This produces the **red signal**, which perfectly traces the alternating physical bumps of the grating.



3. Generating the Spikes: This red difference signal is then fed as the input to the neuron model. The resulting output is a temporally modulated train of spikes. The distinct bursts of spikes physically align with the topography of the touched object.



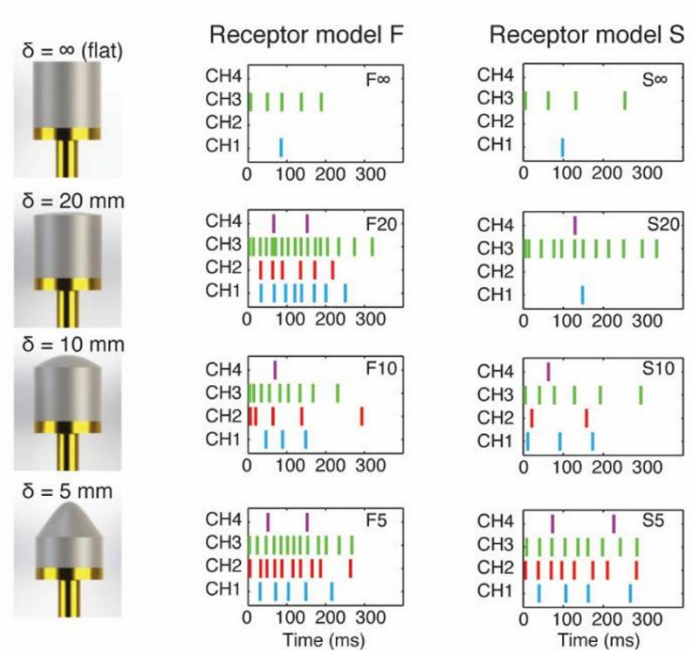
Because the pattern of the generated spikes so accurately reflects the physical space and topography of the grating, it can be used for direct human sensory restoration. By delivering a neural stimulation with this exact temporal structure to an amputee, the subject can accurately figure out in their mind what object the bionic finger is touching. This specific methodology was published in 2016 and is highly recognized in the field as the first demonstration in the world of the capability to provide **rich tactile sensory feedback** to a patient.

Focus (Video mostrato in classe): The professor manually applies pressure to the artificial fingertip. As he presses harder or softer, the continuous analog output from the sensors is fed into the **Izhikevich model**, which instantly converts the changing pressure into a modulated train of spikes in real-time. During the demonstration, the generated spikes are played out loud as audio. It is a very common practice among neurophysiologists to listen to spike trains rather than just looking at visual raster plots. The acoustic rhythm and frequency changes provide an immediate, highly intuitive understanding of the underlying pattern of spiking activity. Unlike the previous example which used a textured grating to demonstrate spatial topography, this specific demonstration focuses on a different spiking pattern (referred to as "5FH") to showcase real-time dynamic modulation based purely on varying degrees of applied pressure.

“Investigation of touch encoding in brain somatosensory cortex” paper del 2017

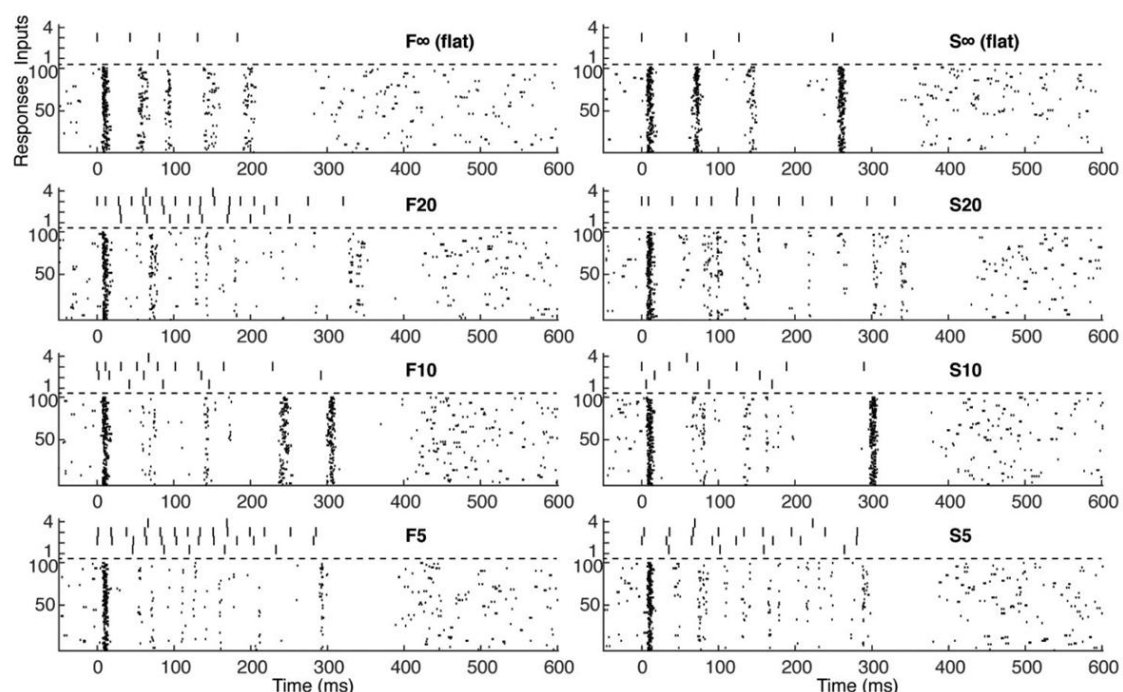
This section details the first in a series of animal studies designed to test the application of the spike neuron model (Izhikevich method) using tactile sensor output, before eventually translating these methods to human sensory neuroprosthetics. The experimental setup was composed of:

- The Stimulus:** An artificial finger is touched by an automated mechanism that moves up and down, applying pressure against the fingertip.
- Varying Curvatures:** The stimulus pushing against the fingertip is physically changed to test different curvatures. The curvatures tested are designated as 5, 10, 20, and "infinite" (which represents a completely flat surface).
- Peripheral Stimulation:** The artificial sensor's output is converted into a spatial-temporal pattern of spikes across different channels. These generated patterns are then used to physically trigger electrical neural stimulation on the fingertip of a rodent.
- Cortical Recording:** While the peripheral stimulation is applied, the rodent's brain activity is recorded in the somatosensory cortex using **patch clamp** methodology. This technique is highly powerful as it allows for extremely precise recording of intracellular or extracellular activity, specifically capturing temporal spiking with high fidelity.



On top of each major plot section, there is a representation of the specific pattern of spikes that was delivered peripherally to the rodent. These are labeled based on two variables:

- Receptor Model (F or S):**
 - F (Fast Adapting):** The stimulus was encoded using a derivative.
 - S (Slowly Adapting):** The stimulus was not derivative.
- Curvature (5, 10, 20, Infinite):** As mentioned, progressing from a sharp curve (5) towards a perfectly flat surface (Infinite).



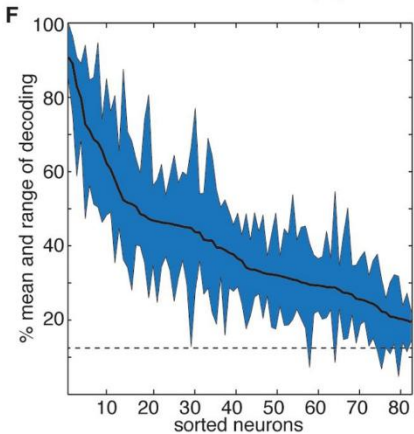
Below each specific input pattern is a raster block showing the recorded response of a single neuron in the cortex. The y-axis represents the number of times the experiment was repeated. In this study, there were **100 replicas** for each stimulus condition. When looking at the 100 replicas for a single stimulus (e.g., looking only at the F-Infinite block), you will notice a high degree of self-similarity. The spikes line up vertically across the 100 trials, creating thick, dark vertical lines. This indicates the neuron reliably fires in the same pattern when given the same input.

The critical finding is what happens when the stimulus is changed (e.g., comparing the F10 block to the F20 block). While some initial spikes may remain aligned across different stimuli, the overall pattern quickly diverges. Spikes that appear in the F20 response might be entirely missing in the F10 response, and vice versa. We can conclude by saying that each distinct stimulus (with its specific curvature and receptor model) produces a highly specific, characteristic activation pattern in the somatosensory cortex neuron.

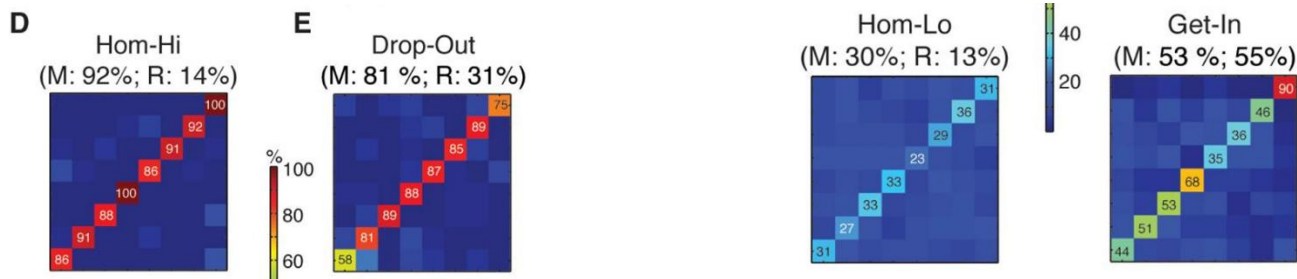
To translate cortical spike patterns into the identity of a tactile stimulus, researchers used a specific machine learning pipeline:

- 1. **Convolution:** Each spike is convolved with a kernel.
- 2. **Z-scoring:** A statistical normalization is applied.
- 3. **KNN (K-Nearest Neighbors):** A classification algorithm is used as the decoding strategy to identify the stimulus.

When neurons are sorted by their average decoding performance, they follow a **log-normal distribution**. This shape—where a small number of neurons are "experts" (high performers) while the majority are average or poor—aligns with theories proposed by scientists like György Buzsáki regarding general brain processes and neural activity distributions.



Neuron Type	Decoding Capability	Confusion Matrix Pattern
Hom-Hi (Homogeneous High)	Successfully decodes all stimuli with high accuracy and repeatability	Highly diagonal
Drop-Out	Recognizes a specific set of stimuli well but completely misses others	Diagonal with distinct missing elements
Hom-Lo (Homogeneous Low)	Performs poorly across all stimuli with very low recognition rates	Low values across the entire diagonal
Get-In	Correctly identifies only one single specific stimulus	Single high value corresponding to one stimulus



A major discovery of this study is how the brain functions as a collective rather than relying on single "expert" neurons. When applying a decoding rule to a group of cooperating neurons, perfect accuracy (100% decoding

for 8 stimuli) is reached with as few as **five neurons**. To simulate the complexity of real-life (where there are many more than 8 stimuli), researchers tested neurons that individually perform poorly (decoding rate < 50%, near chance level).

Even if individual neurons are "weak," if they are allowed to cooperate and information is integrated over a longer temporal window, the collective decoding performance increases significantly. This means that information is present in the neural population; even if individual units aren't "expert" decoders, the brain is capable of extracting collective information to identify sensory experiences.

To confirm the importance of temporal vs. spatial information, researchers utilized **shuffling** techniques to challenge their outcomes.

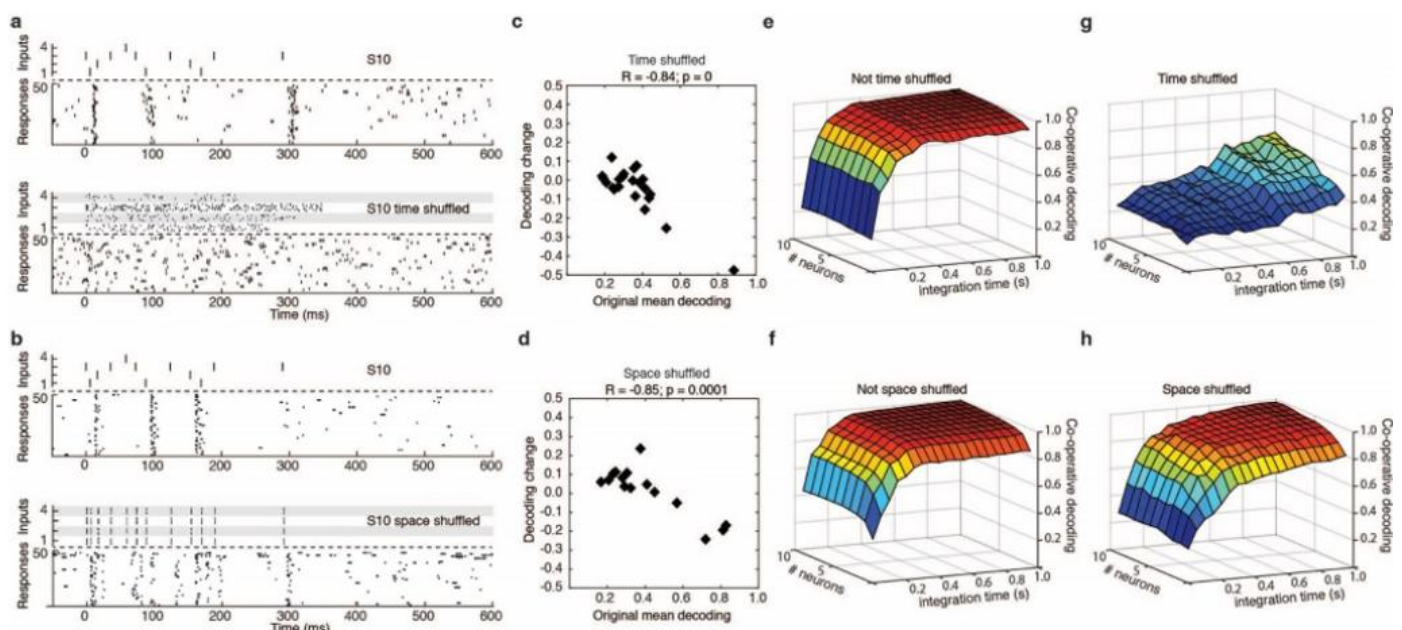
Temporal Shuffling

- **Process:** The timing of spikes within a channel is randomized, but the total number of spikes (spike count/density) is kept constant.
- **Effect:** This completely destroys the temporal structure of the signal.
- **Outcome:** Decoding performance is **disrupted drastically**. This proves that the precise timing of spikes is essential for the brain to identify stimuli.

Spatial Shuffling (Channel Randomization)

- **Process:** The timing of the spikes remains identical, but the channels (electrodes) to which they are delivered are randomized (e.g., swapping the order of a four-electrode array from 1-2-3-4 to 3-2-1-4).
- **Effect:** This is analogous to **rotating** the physical stimulus on the finger.
- **Outcome:** Spatial shuffling is **much less disruptive** than temporal shuffling. Performance remains quite high.

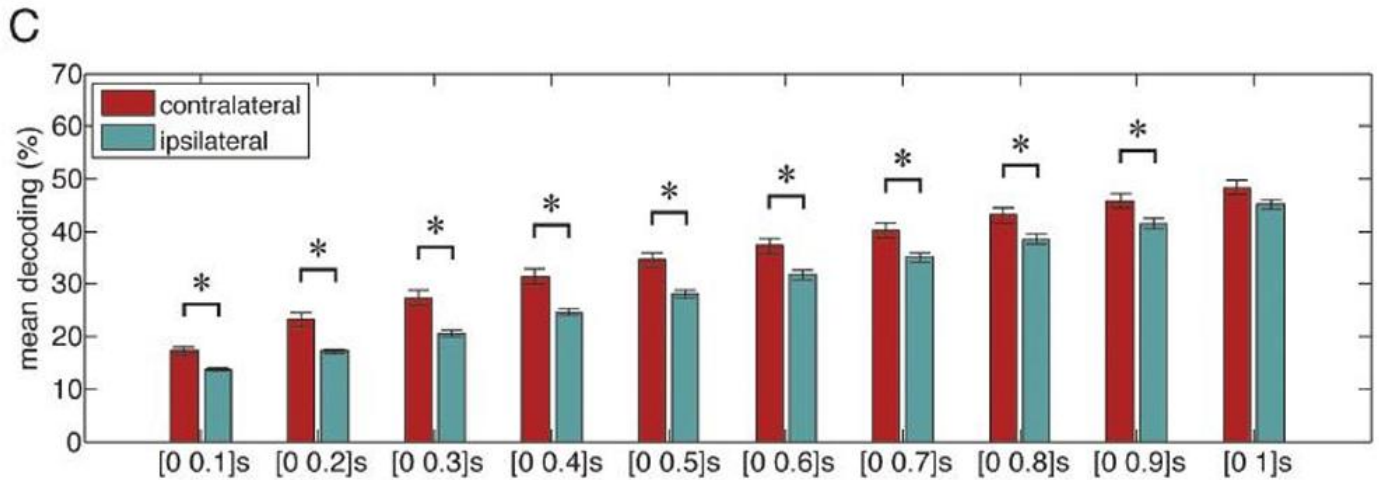
The fact that spatial shuffling has a limited impact suggests a degree of **rotational invariance** in the cortical representation. Much like how a person can recognize an object regardless of how it is rotated against their skin, the somatosensory cortex can decode a pattern even when the spatial mapping of the input channels is shifted or rotated, provided the temporal structure remains intact.



Paper n°2: ipsilateral vs contralateral decoding

A key aspect of somatosensory processing is how tactile information is distributed across the brain's hemispheres. Traditionally, sensory input projects **contralaterally** (e.g., touching with the right hand triggers activity in the left cortex), primarily passing through the thalamic region. Researchers investigated whether tactile stimuli could be decoded from both the contralateral and **ipsilateral** (same side) hemispheres by measuring decoding accuracy as a function of the **integration window** (the amount of time the brain has to process the signal).

- In the first 100 to 200 ms of stimulation, the contralateral hemisphere systematically contains a higher degree of information. The histograms show that the red bars (representing contralateral data) are significantly higher with statistical significance during these early stages.
- As the integration window grows longer, the amount of information available in the ipsilateral hemisphere increases until it becomes comparable to the contralateral side.



Understanding how information flows between hemispheres is crucial for studying brain connectivity and "connectomics." This has major implications for clinical scenarios:

- **Stroke Rehabilitation:** In stroke patients, one hemisphere may be damaged, leading to paralysis or loss of sensation on one side of the body. Studying inter-hemispheric data flow helps in developing better representation protocols, pharmacological solutions, and drug administration strategies.
- **Personalized Medicine:** Current protocols are investigating how different projections on both hemispheres can be leveraged to rehabilitate sensory perception in subjects affected by strokes or amputations.

3. Architecture – Based Decoding (boh mai fatte o forse non ho capito)

Second Order Neurons

Second-order neurons are fundamental components in sensory architecture and information processing.

Second Order Neurons in audition

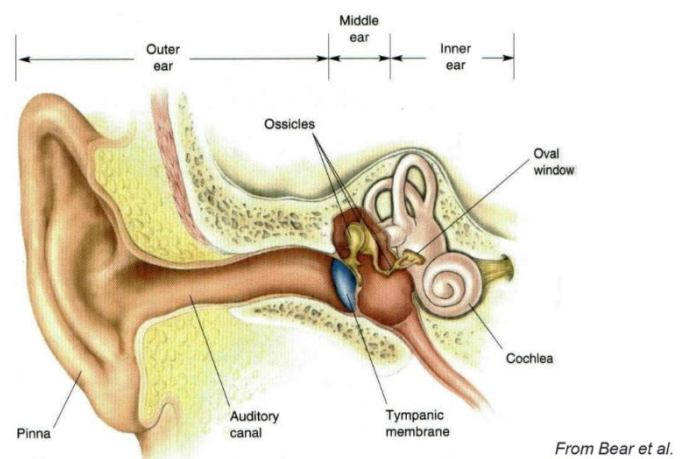
This part explores their function specifically within the **auditory system** to illustrate:

- Computational architectures utilized by second-order neurons.
- How technical implementations and engineering models can be grounded in biological physiology (relevant for both audition and touch).

The human ear is divided into three functional parts that work together to concentrate pressure waves and convert them into neural signals.

1. The External (Outer) Ear

- **Pinna:** The morphology of the pinna is designed to collect mechanical waves.
- **Auditory Canal:** Mechanical waves resonate here and are directed toward the **tympanic membrane** (eardrum).



- **Impedance Adaptation:** The tympanic membrane adapts the impedance of the external world to the internal world.

2. The Middle Ear

This section acts as a mechanical transformer using three small bones (**ossicles**):

1. **Malleus**
2. **Incus**
3. **Stapes** The motion of these ossicles elegantly transforms air pressure into pressure within the inner ear.

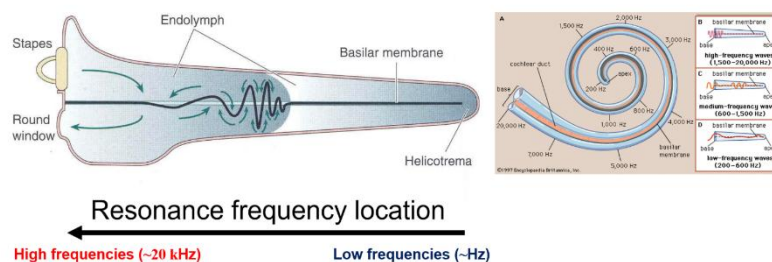
3. The Inner Ear

- **Cochlea:** A complex, snail-shaped structure containing the **oval window** and **round window**.
- **Stapes Function:** The stapes pushes pressure into the cochlea synchronously with the air pressure acting on the eardrum.
- **Basilar Membrane:** Located inside the cochlea, it has a **triangular shape** (larger at the base, smaller at the apex) ending at the **helicotrema**.

The auditory system performs a mechanical **Fourier transform**, decomposing sounds into their different frequency components. In particular, the ciliate cells are receptors distributed along the basilar membrane and that transform mechanical vibrations into neural activations. This allow a tonotopic mapping which is a spatial arrangement where different sound frequencies (tones) are processed in specific locations:

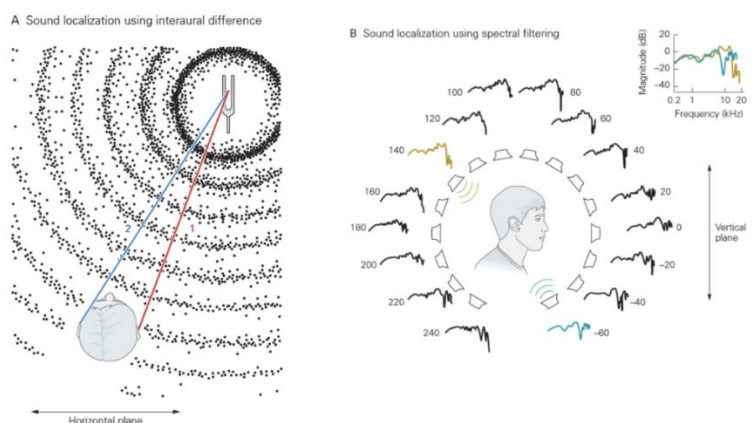
- **Base (Initial part):** Resonates at high frequencies (20 kHz), the maximum frequency humans can hear.
- **Apex (Helicotrema):** Resonates at low frequencies (200 Hz).

Humans progressively lose the ability to hear high frequencies as they age.



There are two primary mechanisms for localizing sound in space:

1. **Interaural Level Differences (ILDs):** Based on the difference in sound **amplitude** between the two ears.
2. **Interaural Time Differences (ITDs):** Based on the difference in the **time of arrival** of the sound at each ear.



Localization is defined using polar coordinates relative to the head:

- **Azimuth:** The angle with respect to the **vertical axis** (orientation in the horizontal plane). This relies primarily on ITDs.
- **Elevation:** The angle with respect to the **horizontal axis** (vertical plane). Localization here is achieved through **timbre transformation**. The asymmetry of the pinna changes the frequency content (timbre) of the sound depending on the elevation angle.

Jeffress Model

The Jeffress model is a robust, simplified model used to explain how we localize objects in the azimuthal plane using **Interaural Time Differences (τ)**.

If we assume the sound source is far away, the sound waves can be modeled as **rectilinear planar waves**.

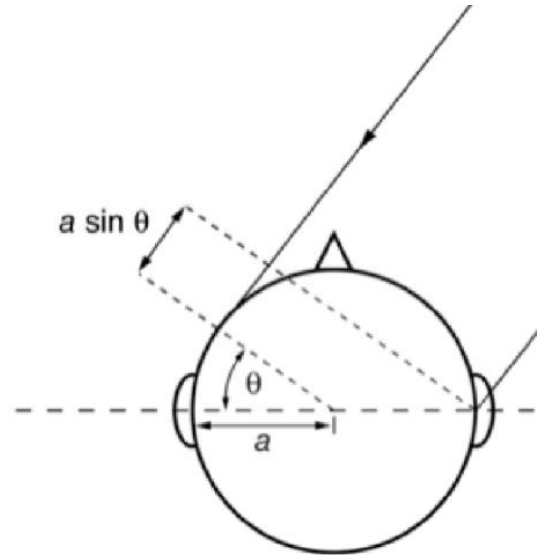
- If the angle θ (sound is directly in front), both ears are stimulated simultaneously (zero dephasing).
- If $\theta \neq 0$, the sound arrives at one ear before the other.

The time delay τ can be calculated based on the radius of the head and the speed of sound:

$$\tau = \frac{2a * \sin(\theta)}{c}$$

Where:

- a : The radius of the head.
 - c : The speed of sound in air.
 - θ : The azimuthal angle of the sound source.
- $2a * \sin(\theta)$: Estimates the full path length difference (diameter component) that the sound must travel to reach the second ear.



Phase – Locking Hypothesis

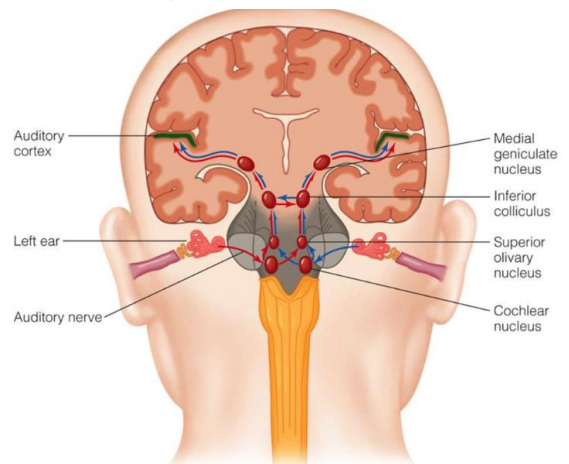
A fundamental hypothesis in auditory processing is **phase locking**, which is essential for temporal synchronization.

Definition: Neurons fire at a specific, consistent phase relative to the period of a sound wave.

While a neuron might not fire during every single cycle of a waveform, when it *does* fire, it occurs at the same phase without significant "jitter." This mechanism is primarily valid for **low-frequency sounds** (hundreds of Hz), which are the most common in daily life (e.g., the human voice). It does not apply to the maximum audible frequencies of 10–20 kHz. Because of phase locking, the intervals between spikes are approximately **integer multiples** of the waveform's period. First-order neurons associated with the ciliate cells in the cochlea produce this synchronized output, providing a regular firing rate that mirrors the sound's period.

The synchronization achieved through phase locking allows the brain to exploit a specific neuroanatomical structure for sound localization.

Firstly, the signal enters the **external ear**. Then, it reaches the **cochlear nucleus (CN)**. From the CN, signals project both **ipsilaterally** (same side) and **contralaterally** (opposite side) to the **superior olivary nucleus (SON)**. Every superior olivary nucleus receives input from both the left and right ears. For example, the left SON receives an ipsilateral projection from the left CN and a contralateral projection from the right CN.

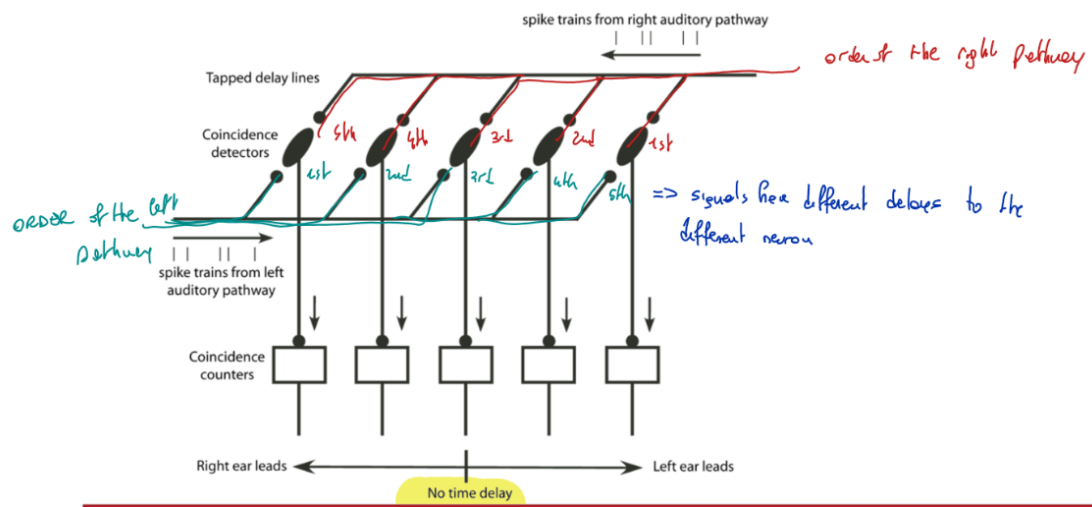


Coincidence Detection in the Jeffress Model

The **superior olivary nucleus** acts as an equivalent circuit for real-time localization. This architecture-based computing converts a time difference into a spatial position.

The system uses **neural delay lines**. As a signal travels from the cochlear nucleus to the SON, it must pass through different lengths of neural pathways to reach different neurons (e.g., neurons 1 through 5 in the didactic model).

If we look at the neuron 3 in the image, the delay from the left and right pathways is approximately the same. On the other side, peripheral neurons (like 1 or 5) receive a shorter delay from one side and a longer delay from the other.



Neurons in the SON function as **coincidence detectors**. Due to the firing threshold, a single spike from only one ear is usually insufficient to trigger the neuron. The neuron will only fire and propagate a spike if inputs from *both* the left and right auditory pathways arrive **simultaneously**. If a sound source is to the left, it reaches the left ear first (creating an acoustic delay τ). This is compensated for by the brain's internal architecture: the signal from the left ear must travel through a **longer neural pathway** than the signal from the right ear. When the neural delay exactly offsets the acoustic delay, the spikes arrive at a specific neuron at the same time, triggering it.

The localization process is essentially a conversion of a **time difference (Δt)** into a **neuronal position**. The specific neuron that activates serves as a "measurement indicator" or graduation scale for the sound's angle (θ). This architecture allows humans to localize sounds with extreme precision, distinguishing angles down to **10 degrees** or fewer. This requires the system to resynchronize elements in the order of **hundreds of microseconds**. This "architecture-based computing" is significantly more powerful for this task than standard digital signal processing methods like wavelet analysis, as it relies on the physical delays introduced by the neural pathways to compensate for sensory delays.

Second Order Neurons in Touch

→first part already explained in page 1-2 (population code in touch and coincidence)

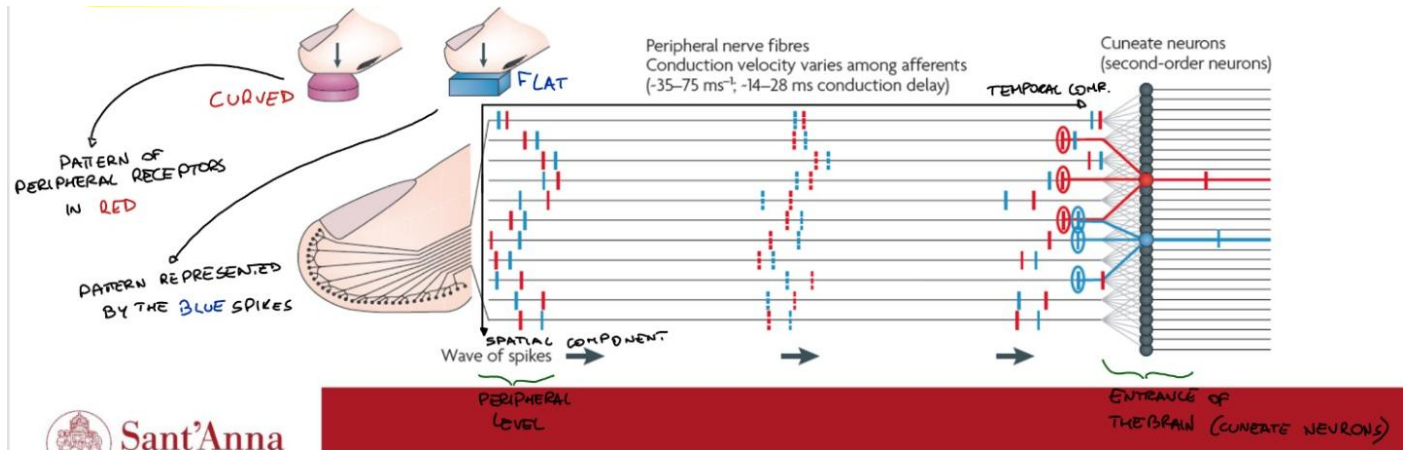
Positional Code in Second Order Neurons

Just as in the auditory system, second-order neurons in the somatosensory system (specifically in the **cuneate nucleus**) utilize a **positional code**.

The pathways between the first-order receptors and the second-order neurons introduce specific delays. A second-order neuron will only fire (spark) if the delays introduced by these neural pathways exactly compensate for the original delays between signals at the receptor level. When the original asynchronous signals are resynchronized by the pathway, they arrive at the second-order neuron simultaneously. This coincidence triggers the neuron to fire.

The architecture involves massive connectivity and a "randomization" of possible anticipation and delay values. This creates a vast array of combinations that can filter for specific spatiotemporal patterns.

In the image, synchronous spikes (marked by red lines) indicate compensated delays. If the signals are not compensated, they remain unsynchronized and do not trigger the same neuronal response. This allows different cuneate neurons to selectively filter spikes associated with different objects.

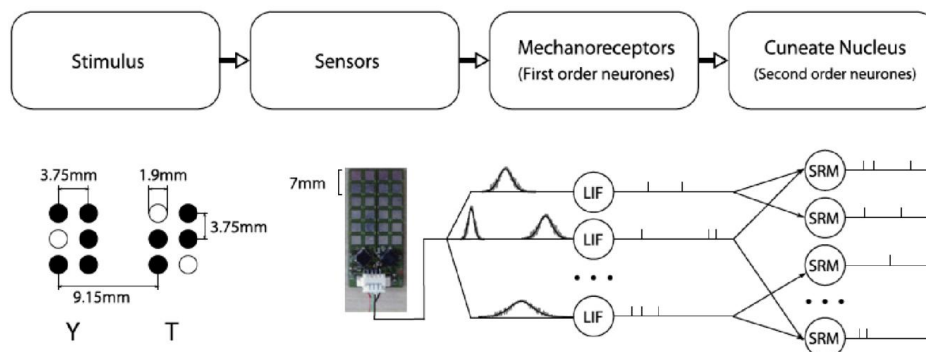


Case Study: Braille Recognition (prof. Arleo and Broni)

This physiological principle was exploited by Professor Arleo (in collaboration with the German Aerospace and using a **KUKA robot**) to solve technical engineering problems using tactile sensors. The goal was to recognize **Braille characters** using a bio-inspired neuron model.

The Setup:

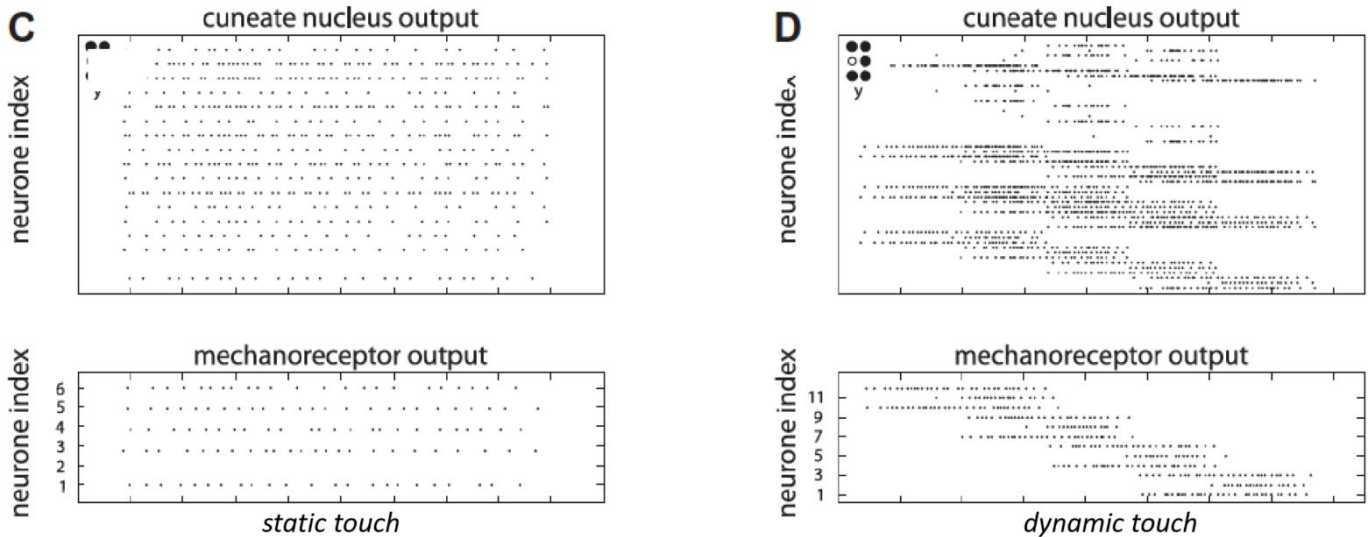
- **Sensors:** Tactile sensors mounted on a robotic arm (KUKA).
- **Neuron Model:** Instead of the more complex Izhikevich model, they used a **Leaky Integrate and Fire (LIF)** model.
- **Process:** The tactile sensor produces a temporal waveform based on the Braille character. The LIF neurons convert these waveforms into spikes, which are then directed to multiple second-order neurons through a complex connectivity strategy.



A crucial finding of this research was the distinction between different types of touch:

- **Static Touch:** This involves touching an object and keeping the finger pressed without movement. The researchers found they were **not** able to recognize different outputs using static touch alone.
- **Dynamic Touch:** This involves **tangential motion** (sliding). The spatiotemporal patterns generated by the first-order mechanoreceptors during dynamic stimulation result in specific spike patterns.

These patterns are separated and recognized by the second-order neurons in the cuneate nucleus. Essentially, the identity of the activated second-order neuron acts as a "look-up table" to identify the stimulus (e.g., which Braille letter was touched).



The success of this study was significant enough to be published in both **Engineering** and **Physiology** journals, highlighting a bidirectional relationship between the two fields:

1. **Neuroscience → Engineering:** The "neuro-sensitive hypothesis" tells engineers how the human brain processes touch. Engineers then use this hypothesis to build technology that solves technical problems (like robotic tactile recognition).
2. **Engineering → Neuroscience:** By successfully building a machine that produces the expected outcome based on a physiological model, engineers provide a **scientific proof of evidence**. Demonstrating that the architecture works in a machine helps us better understand how nature and human physiology actually function.

“Cuneate-based model for Neuromorphic artificial touch categ. Of stimuli” (Oddo paper)

The principles of the **Jeffress Model** (originally for auditory localization) can be applied to the sense of touch to recognize the **angle** of an object being touched. This acts as a "kernel" by which the brain decomposes complex geometries, textures, and textiles into simpler spatial-temporal combinations of lines, angles, and periodicities. To recognize the orientation of a stimulus (like a ridge), the system relies on the spatial distribution of receptors in the skin. Consider two sensors, **S1** and **S2**, distributed on the skin.

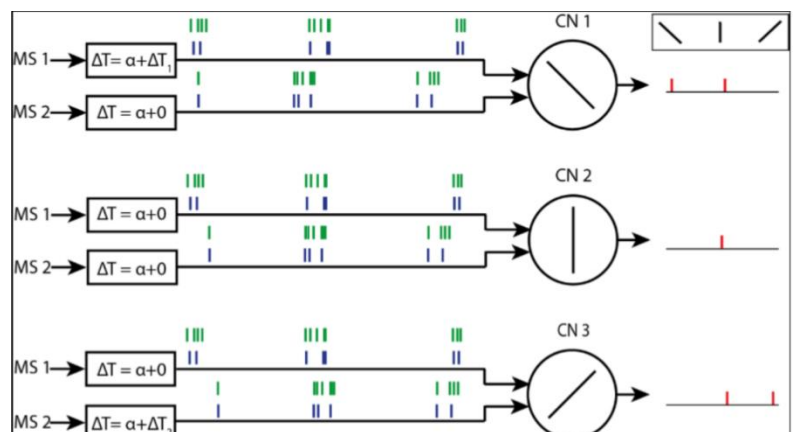
We can have two possible types of stimuli on the skin:

- **Vertical/Frontal Stimulus:** If the stimulus arrives at an angle of 0° , both receptors are stimulated **simultaneously**.
- **Angled Stimulus:** If the stimulus is tilted, one receptor is touched first and the second is touched later. The orientation of the angle determines which receptor is activated first and the duration of the delay between them.

By looking at the delay of activation from one receptor to the other, the system recognizes the angle of a static stimulus.

The processing architecture converts these temporal delays into a **positional code** in the second-order neurons.

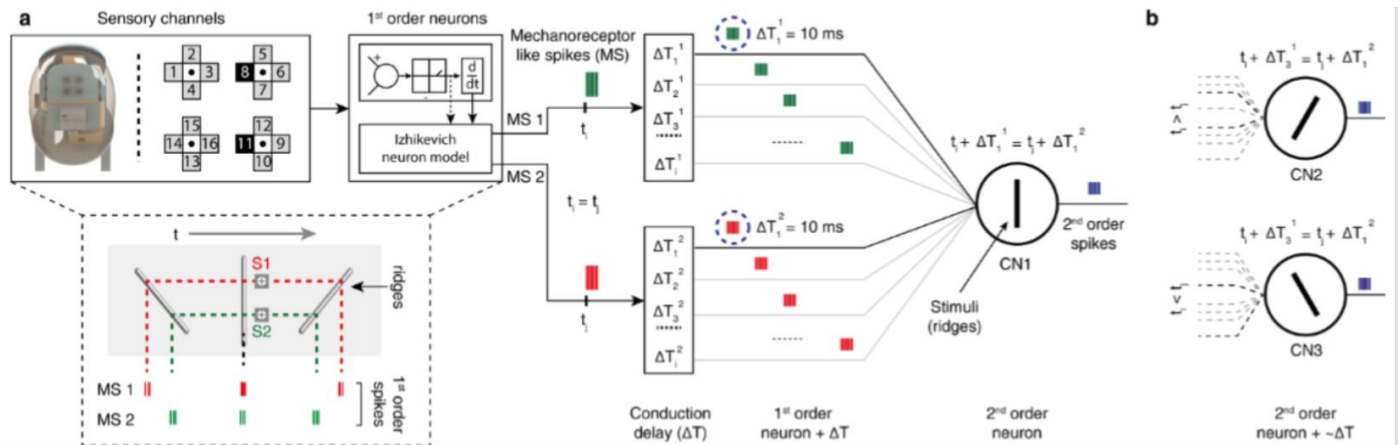
The system introduces secondary delays into the pathways in a "randomic" manner to create a wide array of possible combinations. Each sensor channel (Channel 1, Channel 2) is branched into multiple pathways, each with a progressively increasing delay (ΔT_1 , ΔT_2)



$T_2, \Delta T_3, \dots$). The architecture features massive **many-to-many connectivity**. While diagrams show a few discrete delays for simplicity, the biological reality involves a vast number of "micro-delays" covering all possible combinations of anticipation and delay.

The second-order neurons act as **coincidence detectors**, firing only when the internal neural delay compensates for the external stimulus delay:

- **Vertical Selection:** For a vertical stimulus (no delay between S1 and S2), the second-order neuron that activates is the one where the internal delays added to both channels are **equal**.
- **Angle Selection:** For a tilted stimulus, a neuron is selected that adds a higher degree of delay to the channel that was stimulated first and a lower delay to the channel stimulated second. This resynchronizes the spikes at the second-order level.



This architecture was tested using a **robotic system** in an experimental paradigm to recognize the angle of a physical ridge. The study demonstrated that adding differential delay lines in the pathway from the periphery to the second-order neurons successfully compensates for the delays produced at the cutaneous receptors. This provides evidence that differential delay lines can identify the "identity" (angle) of a tactile stimulus, mirroring how the Jeffress model identifies the source angle of a sound.

The mapping between the stimulus angle and the specific population of second-order neurons is consolidated through a learning process, likely starting in childhood.

- **Cortical Plasticity:** There is high confidence that plasticity occurs at the **cortical level**, where neurons consolidate the understanding of the signals arriving from the second-order neurons.
- **Second-Order Plasticity:** It is currently **unknown** if plasticity exists at the second-order neuron level. It remains a scientific mystery whether these connectivities are pre-programmed or if they undergo "pruning" and consolidation based on experience.

Implementing these complex, bio-inspired architectures in engineering presents significant technical hurdles due to the requirements for many-to-many connectivity and precise micro-delay lines. Current implementation strategies include:

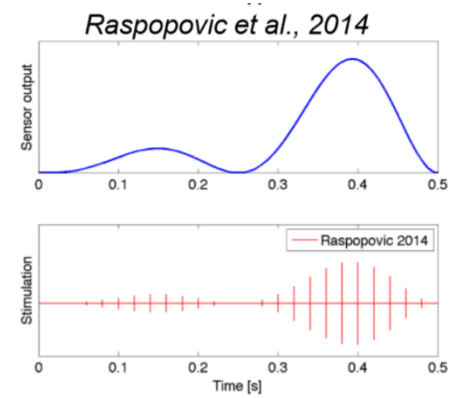
- **Software Models:** Utilizing **Izhikevich neuron** implementations to solve differential equations.
- **Soft-Hardware:** Moving away from standard environments like LabView toward specialized hardware.

Case Study: Neuromorphic Engineering in Bionic Limb Prosthesis

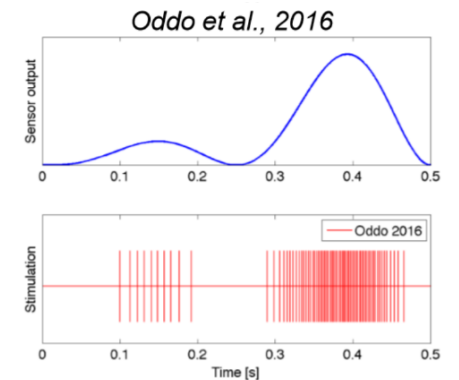
This section explores how neuromorphic principles are applied to restore sensory feedback in bionic limbs, moving from historical non-physiologic methods to modern, bio-inspired approaches.

In the early stages of sensory restoration (circa 2014), the feasibility of restoring feedback was demonstrated using **amplitude modulation**. However, this approach does not align with how the human nervous system naturally operates.

This non-physiologic involves modulating the spike current or using **Pulse Width Modulation (PWM)**. In PWM, there is a fixed frequency carrier, but the duration of the pulse varies to transfer a higher or lower amount of charge. However in biology, individual spikes do not carry information through their amplitude. Instead, information is conveyed via the **rate** (number of spikes per second) or the specific **population** of neurons that activates. As seen in the plot of the image, the inter-spike interval remains fixed, but the height (amplitude) of the spikes changes with intensity.



On the other side, the physiologic approach based on frequency modulation keeps the spike amplitude constant but varies the **inter-spike interval (ISI)**. A higher output from the sensors leads to a higher firing rate, meaning the ISI becomes smaller (the spikes are closer together). Shown on the right image, lower inputs result in a lower firing rate, while high-intensity inputs produce a very high frequency of firing.



2016 Paper by Oddo & Mazzoni “ Intra-neural stimulation elicits discrimination of textural features by artificial fingertip in intact and amputee humans”

A study published in 2016 tested these methodologies using microstimulation on both healthy subjects and an amputee.

The researchers explored two different points of access for the **median nerve**:

- **Elbow Level:** Accessible, but contains a high mix of both sensory and motor fibers. Sensory fibers bring info to the brain; motor fibers control muscles.
- **Wrist Level:** This is a more powerful stimulation site for sensory feedback. At the wrist, the median nerve is **mostly sensory**. Most motor components have already branched off to the intrinsic muscles of the hand. Stimulating here increases the probability of engaging sensory fibers rather than motor ones.

The experiment involved a **psycho-physical protocol** where subjects received electrical stimulation in the nerve while their brain activity was recorded via EEG.

A tactile stimulus (like a finger being touched and then sliding) triggers a specific pattern of spikes delivered to the nerve. The subject is presented with a pair of stimuli (one after the other). The subject must declare if the stimuli were:

- **First** (if the first was coarser).
- **Second** (if the second was coarser).
- **Same** (if both were identical).

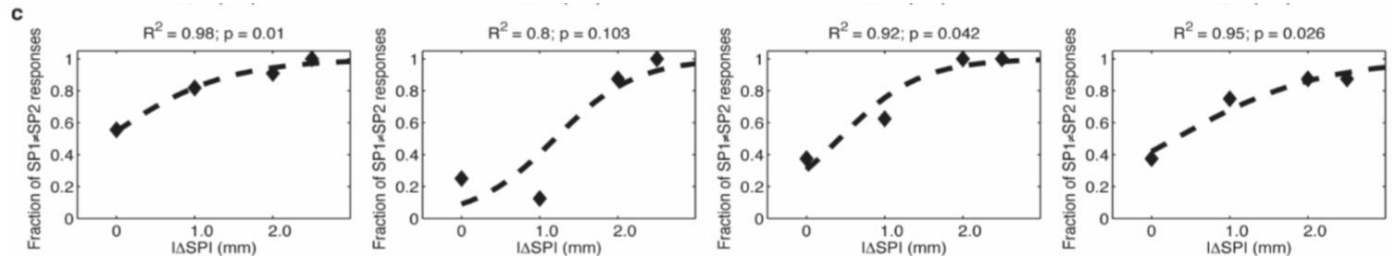
The performance results highlight a significant difference between acute and chronic neural interfaces. The amputees patients (chronic implantation) showed an accuracy of 96%, while healthy subjects (stimulated through a acute implantation only the 77%.

In the researcher's opinion, the amputee's superior performance (96%) was due to having a **chronic implantation**. This long-term interface allows for a much better mapping of the sensory-motor scheme. In contrast, healthy subjects with acute implantations have a lower ability to map neural stimuli into the perception of virtual objects.

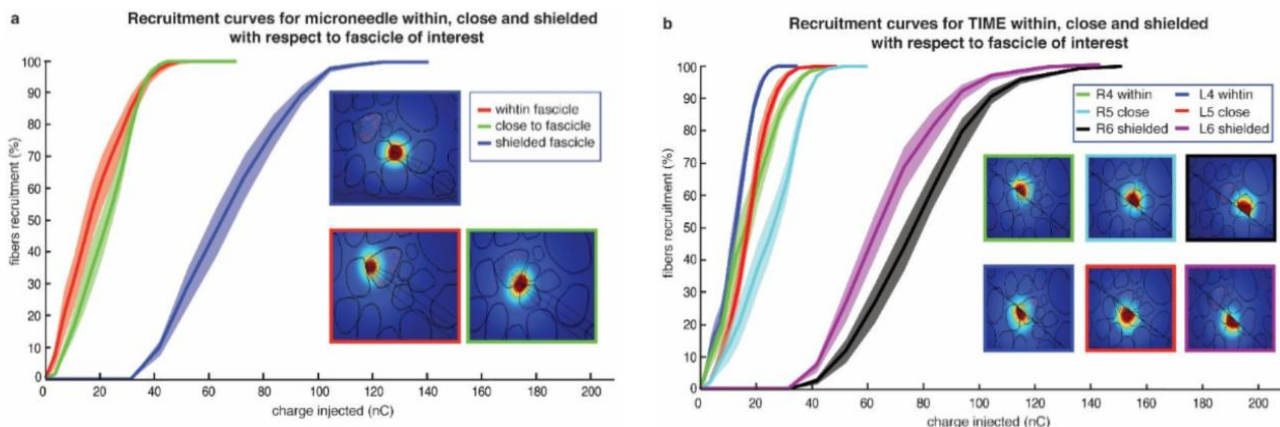
To evaluate the effectiveness of intraneural microstimulation, researchers conducted psycho-physical protocols with four subjects. The psychometric curves represent the relationship between the difference in stimuli and the rate of correct responses.

When stimuli are significantly different, subjects respond correctly at a higher rate. As the stimuli become more similar (smaller difference), the response rate drops toward the **chance level**.

In chronic implantation, decoding was highly accurate (96% overall), particularly for stimuli that were clearly distinct.



EEG was used as a control to test if artificial electrical stimulation mimics natural touch. Spike patterns (delivering specific currents) were administered to the **median nerve**. The cortical modulation observed in the **contralateral hemisphere** during electrical stimulation was similar to the modulation experienced when the same subject touched the same objects naturally. This confirms the **biomimetism** of the stimulation; the artificial spikes evoke a perception in the brain that the cortex recognizes as "natural."



A key engineering question is whether results from acute needle microstimulation (percutaneous insertion in healthy subjects) are translatable to chronic electrode implantation (surgery in amputees).

Researchers plotted the **percentage of neurons recruited** as a function of the **charge injected** (measured in nanocoulombs per spike). These are the so-called recruitment curves. The recruitment curves for acute and chronic interfaces showed the same shape. Because the recruitment behavior is similar, the neuromorphic patterns and charge levels tested in acute healthy subjects can be effectively translated to chronic clinical applications for amputees.

The study investigated which aspect of the neural spike pattern is responsible for the subject's ability to discriminate between different textures (spatial periods).

1. The Spiking Pattern (Bursting Activity)

Tactile information is often encoded in bursts of spikes rather than steady streams.

- **Coarse Stimuli:** May produce bursts of 4 spikes with longer intervals between them.
- **Fine Stimuli:** May produce bursts of 2 spikes with shorter intervals.
- **Very Fine Stimuli:** May produce single spikes with very small inter-spike intervals.

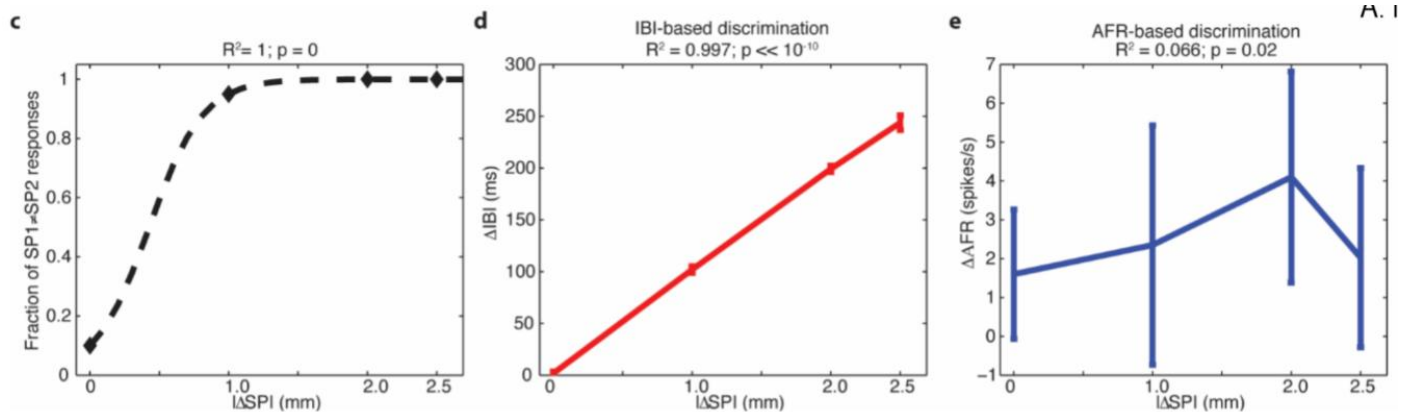
2. Inter-Burst Interval (IBI) as the Determinant

- IBI is the length of time between different bursts of spikes.
- There is a **monotonic, linear scaling** between the variation of the stimulus spatial period and the variation of the inter-burst interval (ΔIBI).

- The IBI was found to be the primary determinant of the psychometric response, allowing subjects to identify the geometry of the stimulus.

3. The Absolute Firing Rate (ΔP)

- Firing rate was **not** a good descriptor of the subject's performance.
- The firing rate modulates in a **non-monotonic** manner. On average, a pattern with 4 spikes at long intervals can have the same firing rate as a pattern with 2 spikes at short intervals. Since the firing rate does not justify the psychometric behavior, it is the temporal structure (IBI) that matters most for discrimination.



This approach serves two primary purposes:

1. **Rehabilitation:** It allows subjects to recognize tactile stimuli through neuromorphic processes that vary the spiking pattern based on the stimulus geometry.
2. **Neurophysiological Study:** It helps identify the specific determinants of tactile information. In this case, it was discovered that the brain uses the **Inter-Burst Interval** rather than the absolute firing rate to discriminate between tactile stimuli.

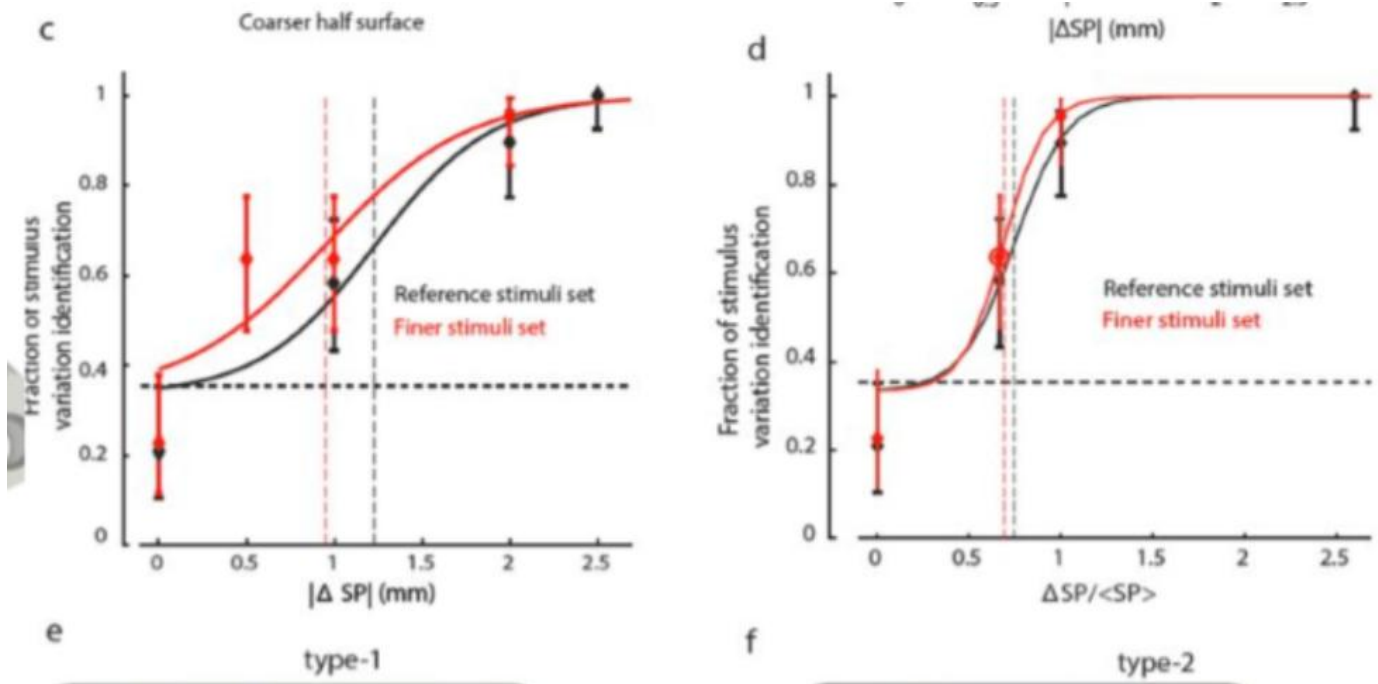
Paper by Oddo, Mazzoni and Micera on discrimination of naturalistic textures

A follow-up study examined how the specific type of receptor model affects the recognition of gratings and textures. The primary variable was the depth and spacing of the receptors within the tissue.

The type 1 receptors are modeled as being **close to the surface** of the stimulus. They are essential for recognizing **fine textures**. On the other hand, type 2 receptors represent deeply located receptors (similar to Ruffinian receptors) with a **higher spacing** between the fingertip surface and the receptor. They are separated from the stimulus by a "very, very thick material."

The study compared the performance of these two models using logistic curves that plot the probability of correct response against the stimulus difference (ΔP).

- Type 1 produces a smaller perceptual threshold. Subjects stimulated with a Type 1-inspired model were capable of recognizing stimuli even when differences were small (in the order of 10%, 20%, or 30%).
- In the type 2 the perceptual threshold increases significantly. The logistic curve has a **very small slope**, meaning there is little variation in response with small changes in the stimulus. Type 2 is **not capable** of recognizing similar stimuli; it can only discriminate between **gross textures** where the differences are very large.



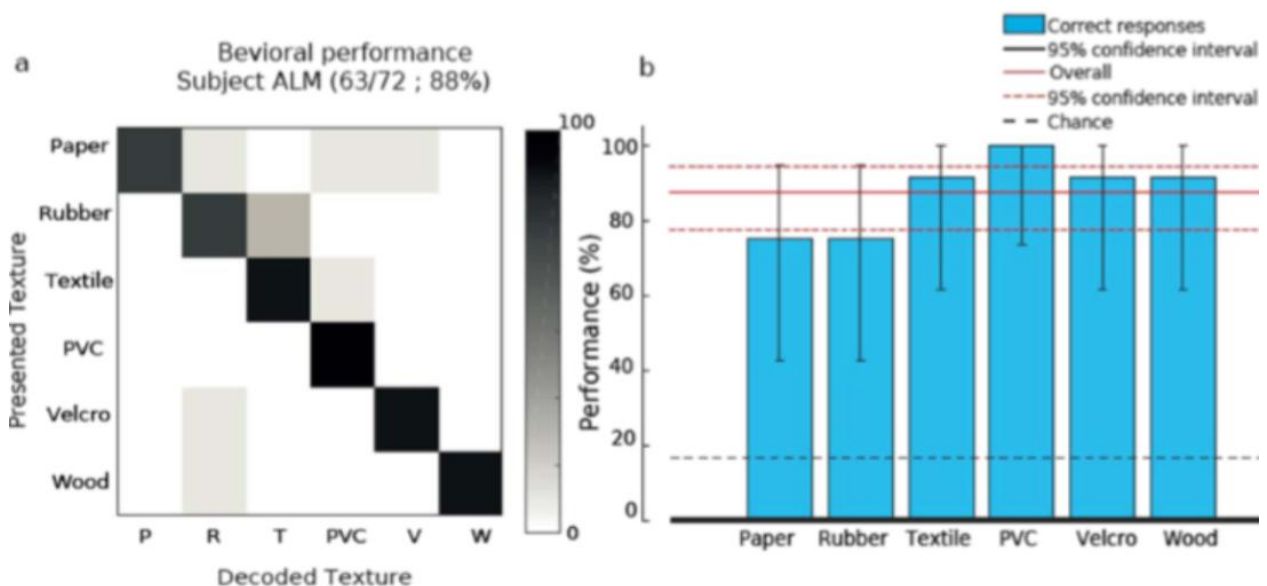
Based on the finding that Type 1 receptors are necessary for high-resolution information, researchers moved beyond simple gratings to **complex tactile stimuli**, specifically materialistic textiles (e.g., denim, silk).

Complex textiles were encoded into **neuromorphic patterns of spikes** using the Type 1 receptor model. These spike patterns were then administered via a chronic neural implant to an amputee.

The subject was successfully able to recognize the identity of the different materials.

Because the Type 1 model was used, the necessary information was preserved in the stimulation pattern. This allowed the subject to "mentally extract features" to identify the textiles. If a Type 2 model had been used, the information would not have been present in the signal, and the subject would have been unable to recognize the complex materials.

The study demonstrates that the choice of receptor model in neuromorphic engineering is critical for bionic applications. To restore the ability to perceive fine textures and complex materials in an amputee, the system must utilize a **Type 1 receptor model** to ensure the high-frequency temporal and spatial information is delivered to the nervous system.



Autonomic Regulation in Neuromorphic Engineering

This section introduces how the principles of neuromorphic engineering and neural computing extend beyond somatosensation (tactile sensing) to **autonomic regulation processes**.

While tactile sensing primarily regulates voluntary actions like grasping, it is also deeply linked to the regulation of autonomic systems, including cardiac activity. The regulation system relies on synchronized physiological behaviors:

- **ECG (Electrocardiogram):** Tracks the electrical activity of the heart.
- **Arterial Pressure:** Synchronized with the ECG to show the systolic and diastolic cycles of contraction and expansion that sustain blood flow.
- **Pulmonary Activity:** Manages the exchange of oxygen between the breath and the blood.
- **Mechanical Activity:** Monitored via synchronized mechanical sensors to observe the direct physical effects on the heart.

The raw signals captured from these mechanical and cardiac sensors are initially **amplitude-modulated** (moving continuously up and down), meaning they are not inherently neuromorphic. However, by applying the **Izhikevich neural model**, these continuous amplitude-modulated waveforms can be converted into spiking signals. Once converted, they can be used as direct feedback within autonomic regulation loops.

A collaborative research project among several institutions in Tuscany demonstrated how combining somatosensory and cardiac modeling provides critical clinical insights.

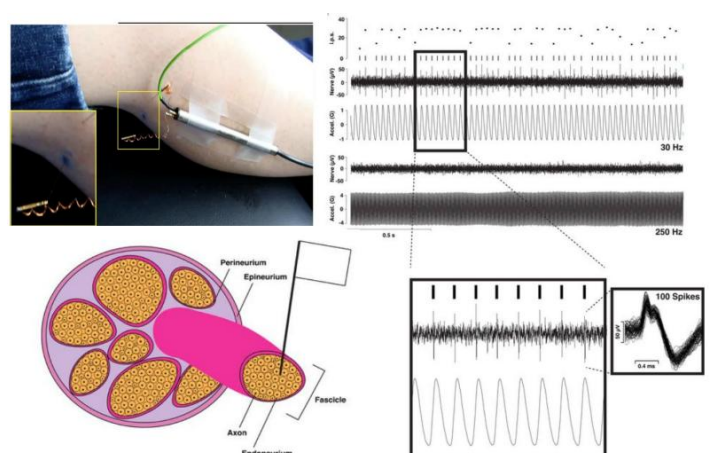
The architecture is used to evaluate patients with severe brain injuries who are in a **vegetative state**. Engineers administer peripheral neural stimulation to activate **afferent somatosensory pathways** while simultaneously recording brain activity using **EEG**. This setup serves as a novel prognostic technique to assess and track the objective progression of rehabilitation outcomes rather than acting as a treatment itself. Monitoring these pathways simultaneously yields vital data regarding autonomic functions, specifically blood pressure and cardiac activity regulation.

To gain a deeper understanding of blood pressure regulation and rhythmic cardiac neural activity, researchers transition from the upper limb to localized autonomic recording. Instead of recording from the median nerve in the upper limb (which is used to model tactile encoding from mechanoreceptors), microneurography is applied to cardiac-related neural pathways.

Muscle Sympathetic Nerve Activity (MSNA): This technique utilizes microelectrodes inserted directly into the nerve fascicles to record internal sympathetic activity. The microelectrodes specifically capture the activity of **efferent motor neurons** that innervate the smooth muscles surrounding blood vessels. Depending on how these muscles contract or relax around the vessels, blood pressure rhythmically increases or decreases. This continuous, rhythmic adjustment regulates systemic blood pressure according to the body's real-time needs.

The neural signals obtained during autonomic cardiovascular monitoring differ significantly in structure from those captured during somatosensory experiments.

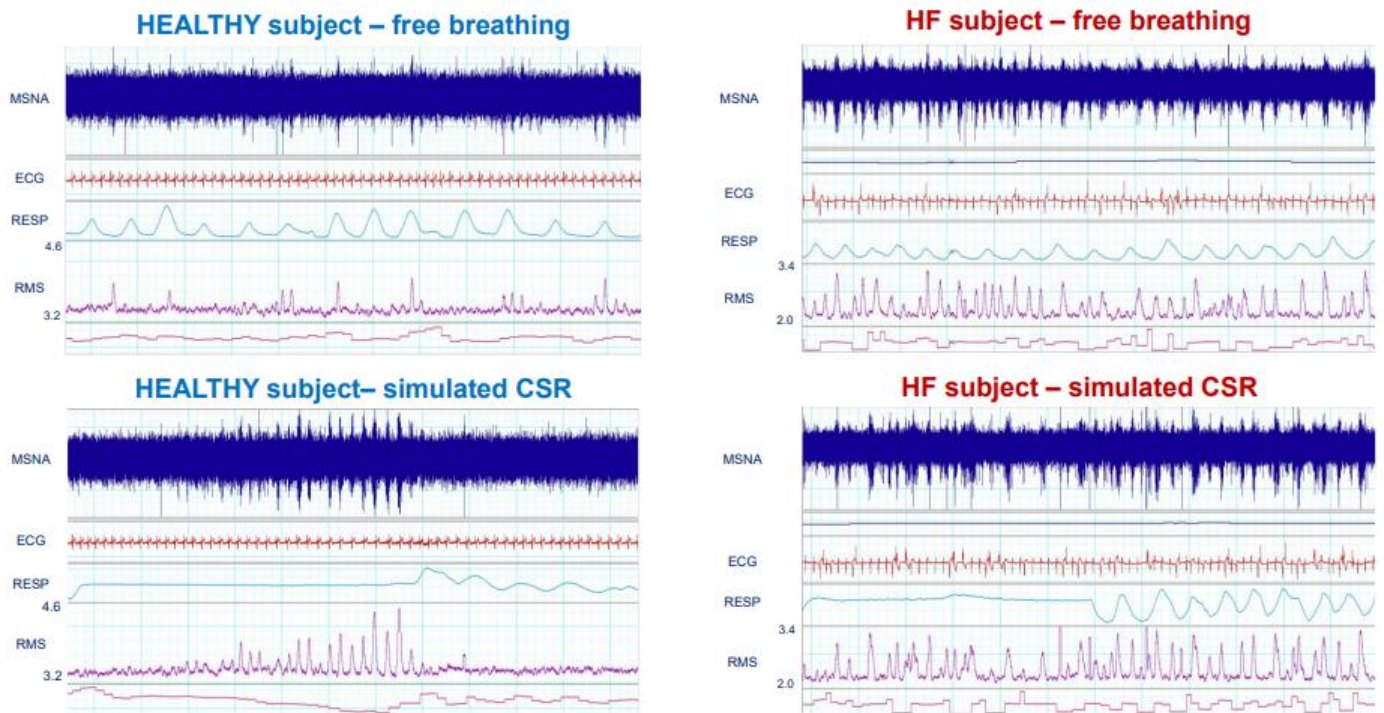
- **Somatosensory Signals (Single-Unit):** As seen in previous tactile experiments on the median nerve, it is possible to achieve single-unit recordings. This isolates the distinct spiking activity of individual mechanoreceptors.



- **Autonomic MSNA Signals (Multi-Unit):** Autonomic recordings capture **bursting activity** that represents electrical pulses delivered by the autonomic nervous system to trigger blood vessel contraction. These raw signals are **multi-unit**, meaning they consist of a dense superimposition of multiple concurrent axonal communication streams.

Because the sympathetic signal is a complex mixture of multiple neurons firing together, researchers utilize **spike sorting methods**. This technical processing approach allows them to:

1. Deconstruct the superimposed, multi-channel neural signal.
2. Separate the distinct electrical sources within the axonal pathway.
3. Identify the precise, individual contribution of each specific neuron to the overall vasoconstriction and blood pressure regulation behavior.



Spike sorting principles

The principles of neuromorphic engineering and neural computing can be utilized to model the precise relationships between autonomic variables. The network maps the interactions between the heartbeat (measured in terms of rate or frequency), blood pressure activation, and the corresponding neural signals recorded by directly accessing the **peroneal nerve**.

In clinical and bioelectronic medicine, these principles enable the development of implanted electrical circuits consisting of:

- A **sensor** that continually encodes the real-time status of the physiological system
- An **implanted stimulator** that targets a separate nerve using structured sequences of spikes.

Instead of delivering tactile sensory feedback (as seen in bionic limb prosthetics), this neuromorphic computing strategy is used to actively control autonomic processes involved in systemic blood pressure regulation and cardiac activity.

When dealing with multi-unit recordings from autonomic pathways, researchers use spike sorting to isolate individual axonal streams from the superimposed multi-channel data. Even within a dense multi-unit signal, the baseline activity of an individual neural unit can be extracted by defining a specific mathematical **template of a**



Multi-unit and single unit analysis – OO3 – TUNE-BEAM

spike. This feature is integrated into the analysis toolbox released by the researchers for microneurographic analysis.

The temporal precision and certainty of the source separation vary significantly depending on the density of the recorded units:

- **High-Quality Separation:** Achieved when the raw multi-unit signal is the result of only a few units (e.g., **3, 4, or 5 units** firing concurrently).
- **Low-Quality/Uncertain Separation:** Occurs when the microelectrode captures a highly complex signal resulting from a larger population (e.g., **20 units** or more). This drastic increase in overlapping streams complicates template matching.

The goal is isolating these individual units allows researchers to establish precise curves of correlation between distinct physiological quantities.

Application on colonoscopy

Moving beyond the field of limb prosthetics, tactile sensing technology can be expanded into other clinical medical domains, such as gastroenterology and colonoscopy.

Research has demonstrated the capability to successfully recognize and locate tissue nodules during colonoscopy procedures via **capsule endoscopy**.

The system utilizes a specialized **tactile**

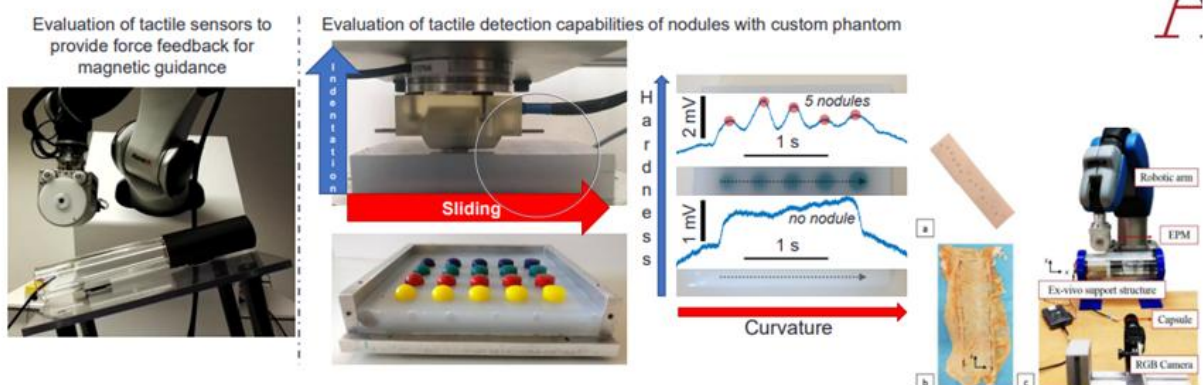
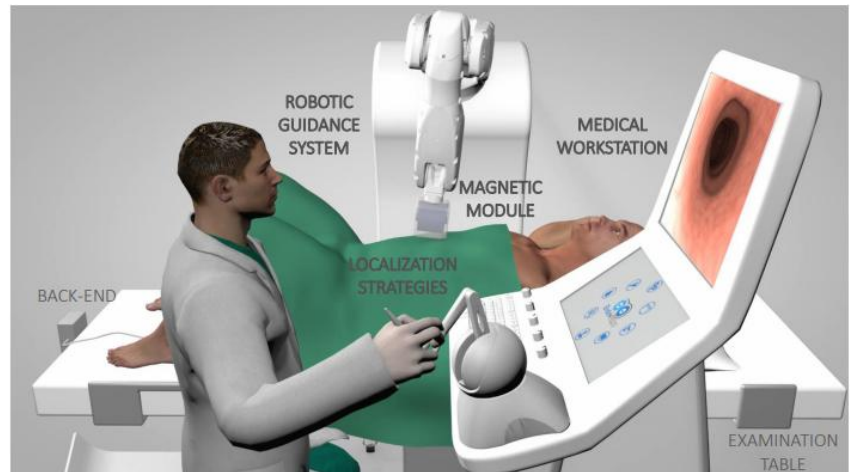
sensor array. The setup was validated experimentally by evaluating tactile responses across both synthetic tissue phantoms and real tissue samples using *ex vivo* models.

As represented in the experimental data, the baseline output of this tactile sensor array is inherently **amplitude-modulated**. The physical passage over different tissue nodules registers directly as distinct, separate peaks in the continuous amplitude signal.

The continuous, amplitude-modulated signal generated by the endoscopic tactile array raises an engineering design question: *Is it more efficient to process the raw amplitude signal directly, or transform it into a sequence of spikes?*

The research provides experimental evidence validating the transition to a neuromorphic encoding strategy due to two core operational benefits:

1. **Dataset Dimension Compression:** Transforming continuous waveforms into sparse spike patterns achieves a significant compression of the overall dataset size.
2. **Energetic Savings:** Relying on event-driven spike sequences instead of continuous sampling yields substantial energy savings across the entire processing hardware architecture.



Traditional Non – Neuromorphic Tactile Sensing Applications

Before diving fully into neuromorphic alternatives, it helps to look at how traditional tactile sensors operate across fields like **histology**, **wearable sensors**, and **collaborative robotics**.

Traditional sensor configurations rely heavily on continuous amplitude modulation. The signal intensity fluctuates continuously up and down to reflect changes in pressure, rather than altering a pulse or spike rate. In a typical multi-cell sensor array, the direct modulation of the output voltage or current correlates to the physical pressure applied across its surface.

The conventional method for capturing sensor data depends on **constant fixed-time sampling**:

- The sensor array is read at rigid intervals (for example, sampling exactly every **1 millisecond**).
- Each sampled data point is stored with high baseline precision (such as using **32 degrees of precision** as a standard example).
- This continuous processing occurs regardless of whether the physical environment is changing, leading to heavy data streams.

We can find the zinc oxide nanorods for tactile transducers that have different benefits: Self-powered ✓ Flexible ✓ High transducer density ✓ Large area sensor arrays ✓ Low-cost fabrication process.

Photonic and Optiical Sensing (Fiber Bragg Gratings)

Another major family of traditional—yet highly novel—tactile sensors includes **Fiber Bragg Grating (FBG) sensors**. The working principle is composed of:

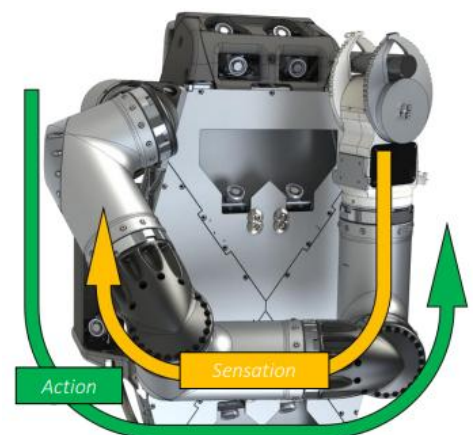
1. These optical fibers are embedded directly inside a **deformable polymer tissue**.
2. When external physical strain is exerted on the tissue, it stretches or compresses the embedded fiber.
3. This strain causes a physical variation in the **spatial periodicity of the grating** inside the fiber.
4. The structural shift results in an analog modulation of the reflected **wavelength** (a shift in peak frequency).

The instructor draws an important structural comparison between these sensors and biological systems:

- **Intra-neuron (Inside the cell):** Biological signaling and computing have an analog nature.
- **Inter-neuron (Between cells):** In biological networks and neuromorphic engineering models, communication between separate neurons happens via discrete **digital events** (on-off, 0 or 1, spike or no-spike).
- **Traditional Photonic Sensors:** Unlike the event-driven digital nature of neural networks, FBG sensors utilize a fully **analog encoding scheme** where the signal varies continuously along a spectrum.

This FBG sensor architecture was put to the test in a collaborative project with the **Jet Propulsion Laboratory (JPL)** of the American Space Agency:

- Researchers developed a robotic hand utilizing these optical fibers.
- Even though the system relied on traditionally amplitude-modulated signals, the sensory integration gave the hand the delicate feedback necessary to successfully manipulate **fragile** or **deformable objects** without breaking or distorting them.

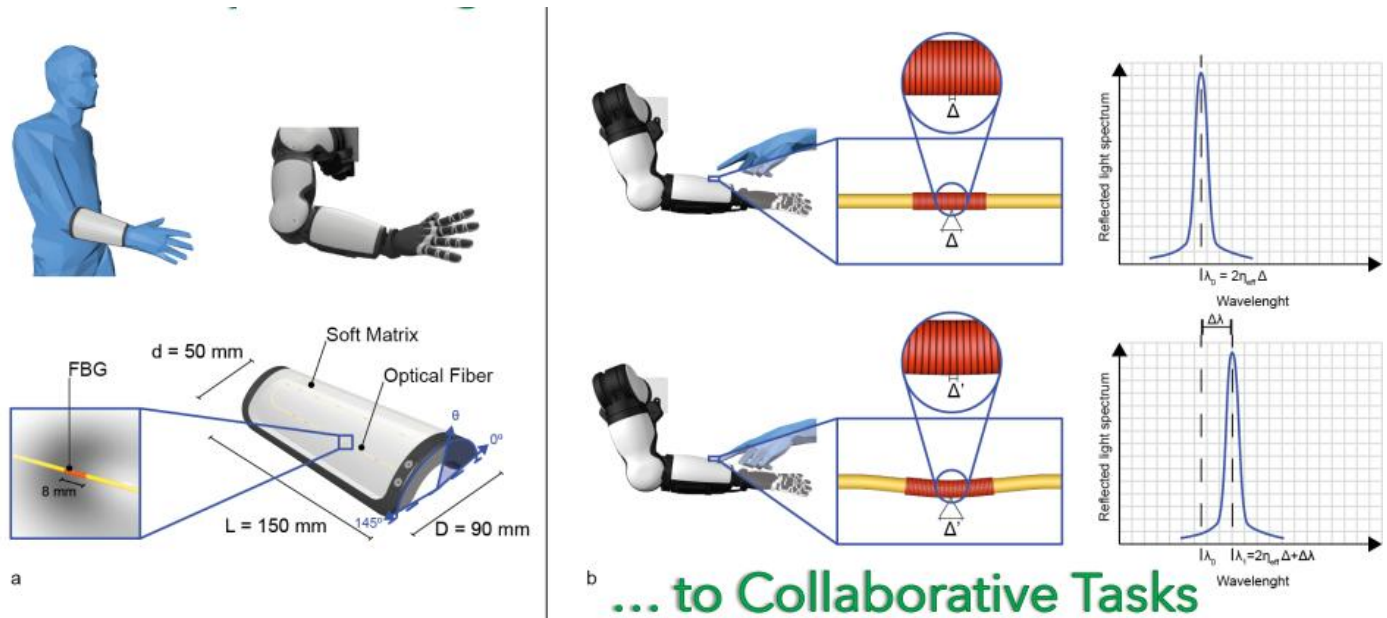


Tradional AI approaches

Transitioning toward neuromorphic designs requires first understanding how traditional AI handles FBG sensor data, and where it falls short.

When pressure is applied to a specific location on the distributed sensor array, the wavelength of individual transducers shifts. The magnitude of this shift maps directly to the amount of force applied. To turn this into a usable system, researchers construct large datasets using a specialized calibration platform:

- **The Indenter:** A automated probe controls the contact platform across known state variables: its precise **X, Y, and Z positions**.
- **Load Cell:** A calibrated load cell mounted on top of the probe measures the exact ground-truth forces, capturing both the **vertical Z-component** and any **lateral components of force**.
- **Data Correlation:** By matching the continuous wavelength shifts of the distributed transducers with the exact positional and force measurements from the load cell, a labeled dataset is established.



Once the dataset is collected, traditional AI methodologies are introduced to process the inputs:

- Models like **Convolutional Neural Networks (CNNs)** and **Multi-Layer Perceptrons** are trained on the dataset.
- **Inference Task:** The goal of the network is to take the distributed surface data and infer the **latent variable** (the exact contact force).

While traditional AI approaches excel from a pure measurement perspective, they present significant operational trade-offs:

Parameter	Performance Metrics
Metrological Accuracy	Highly effective. Positional errors are kept within tens of millimeters (or even down to a few millimeters), and force estimation errors are contained within a few millinewtons .
Computational Power Demand &	Extremely high. The resource overhead is heavily demanding during both the initial training phase and the real-time inference loops required to calculate the latent forces.

This stark contrast between high metrological precision and massive computational cost is precisely what motivates the shift toward **neuromorphic strategies**. Replacing continuous analog sampling with event-driven spike encoding compresses dataset dimensions and delivers substantial energy savings across the hardware architecture.

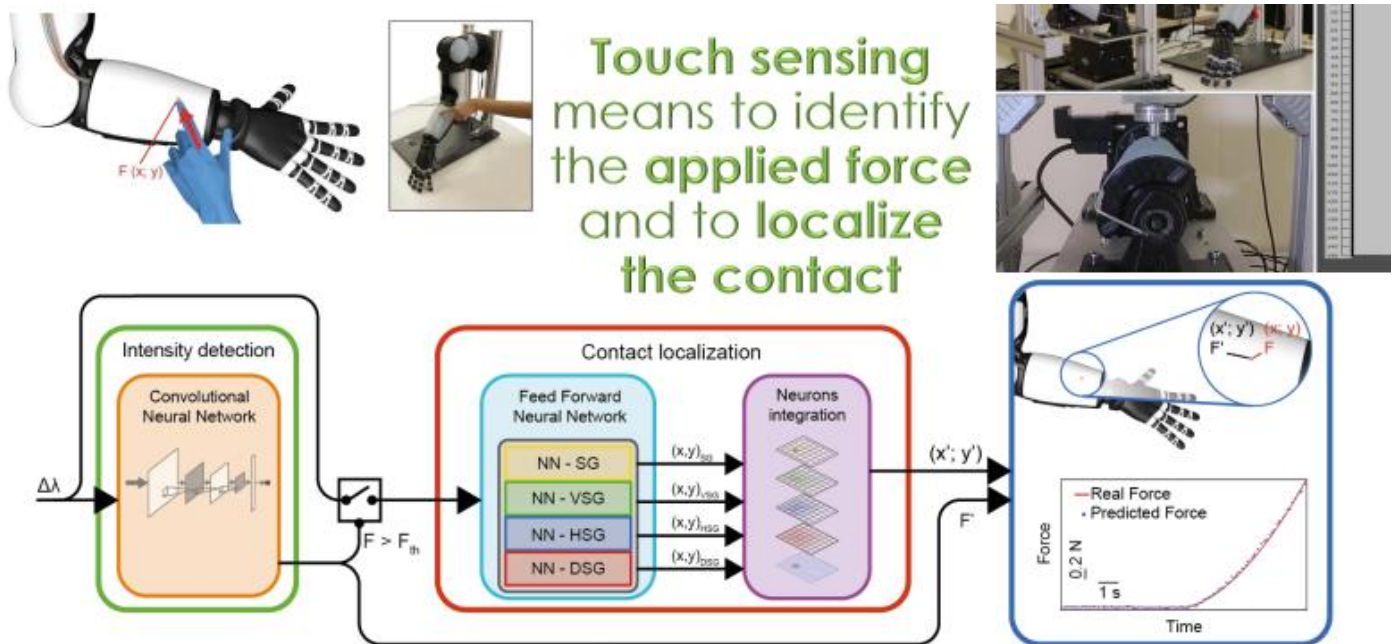
Shifting to Neuromorphic Coding

Traditional artificial intelligence methods excel at taking a continuous, single output variable—such as wavelength data variations sampled down to the milli-nanometer scale—and using convolutional neural networks to infer contact position, intensity, or contact force. However, neuromorphic engineering seeks to move beyond these continuous, rate-based models by implementing a **spatiotemporal time encoding strategy**.

Unlike traditional analog signals that vary continuously along a spectrum, communication between neurons relies on discrete, digital events:

- **Binary Activation:** A neuron is either active or not active (0 or 1); it cannot be partially active. If an action potential is successfully communicated, a spike occurs.
- **Intensity Modulation:** To represent a higher input intensity, the system modulates information in two ways:
 1. Producing **more spikes per unit of time** (increased firing rate).
 2. **Recruiting more individual neuron units** across a low-dimensional population.

Instead of performing mathematical inference through standard deep neural networks, neuromorphic systems convert raw sensor outputs directly into sequences of spikes using established computational neuron models, such as the **Izhikevich model** or **Integrate-and-Fire models**.



Once raw data is transformed into spikes, it enters a structured **connectome** composed of second-order neurons. These central neurons receive inputs from preceding layers through synapses. Rather than relying on traditional inference algorithms to minimize noise, the network accomplishes tasks through its specific structural connectivity and hardcoded rules:

- **Synaptic Weighting:** Synapses adapt and consolidate based on custom synaptic learning mechanisms.
- **Information Delivery:** Consolidated spikes are delivered sequentially to subsequent layers entirely through the physical network topology.

Network Topology

When designing the architectural layout of a spiking neural network, developers face a critical choice regarding how the topology of the connections is determined.

The Brute-Force Optimization Approach

In a totally assumption-free modality, the connectivity network can be defined using raw computational power. Developers can feed the design space to a supercomputer simulator, tasking it with determining the absolute best network topology to sustain a clean signal and extract latent variables (e.g., the position of contact with the skin) within a noisy environment. The supercomputer optimizes the network against a cost function, balancing power consumption, maximal resolution, and precision.

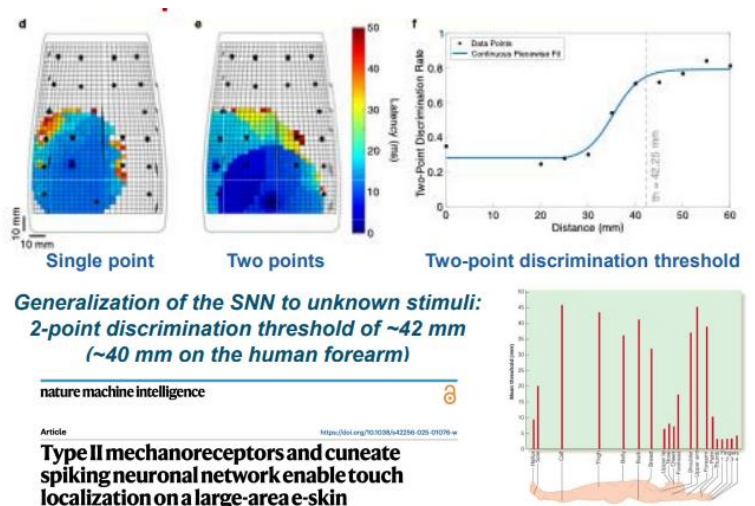
This approach is exceptionally expensive. Because it is free of architectural constraints, the supercomputer requires massive computational power and high energy consumption to thoroughly explore the massive design space.

The Bio-Inspired Assumption Approach

To bypass these computational bottlenecks, engineers introduce structural constraints derived directly from biological neuroanatomy. By observing how the human somatosensory connectome processes afferent information from the periphery to the central nervous system, developers can tightly constrain the exploratory space:

- **First-Order Layer:** Modeled precisely after the structural organization of neurons within the **dorsal root ganglia**.
- **Second-Order Layer:** Organized using a realistic combination of excitatory and inhibitory neurons, incorporating specific proportional weights regarding their overall numerosity.
- **Lateral Connectivity:** Constrained by incorporating interneurons—such as inhibitory interneurons placed structurally between the first and second layers, or between the second and third layers.

Research following up on the team's 2022 *Nature Machine Intelligence* paper (which originally used traditional non-neuromorphic AI networks to infer contact location and intensity) provides clear evidence that this bio-inspired approach is highly functional. Restricting the network configuration based on biological principles achieves metrological results (precision and resolution) comparable to unconstrained brute-force methods, but with a fraction of the computational and energy overhead. In this framework, training and evolution are focused entirely on the **evolution of synaptic weights** within a pre-defined, biologically accurate topology.



Experimental Validation

To test whether a connectome constrained by human somatosensory biology could effectively map tactile data, researchers simulated a classic psychophysical protocol: the **two-point discrimination threshold experiment**.

The Psychophysical Protocol

Used widely in experimental psychology to estimate the tactile resolution and acuity of the somatosensory system, this experiment involves applying either a single probe or two distinct probes randomly to the skin. The physical distance between the two probes is varied systematically from coarse distances to closer and closer intervals. Eventually, as the distance shrinks below a certain threshold, the subject can no longer differentiate them and perceives the two tips as a single stimulus.

Spatial Acuity and Receptor Density

The two-point threshold serves as an accurate proxy to estimate the underlying density of mechanoreceptors in different regions of the body:

- **Fingertips:** Characterized by a very high density of receptors, resulting in a low discrimination threshold and exceptional tactile acuity.
- **Forearm:** Possesses an intermediate receptor density; the tactile acuity is noticeably worse than the fingers, meaning the probes merge into a single perceived stimulus at a much larger distance.
- **Back of the Torso:** Contains a much lower density of receptors, yielding a significantly larger perception threshold.

Neuromorphic Simulation Results

When researchers tested the artificial skin using a connectome inspired by the human somatosensory network, they successfully generated **psychometric curves** for the artificial system. With the application of proper geometric rescaling, these curves modulated in a manner matching human psychometrics.

The alignment between artificial and biological psychometric curves highlights the dual, bi-directional value of pursuing neuromorphic methodologies:

1. **Engineering Efficiency (Biology to Technology):** Observing and copying physiological structures allows engineers to develop hardware systems, sensors, and robotics that perform better, exhibit higher precision, and operate with superior energy efficiency.
2. **Scientific Advancement (Technology to Biology):** Conversely, building these bio-inspired models allows scientists to look back at physiology and use technology to explain biology. Simulating these systems helps clarify the exact structural and neurophysiological determinants required to achieve high-acuity perception in human beings.

Energy Sustainability in Traditional AI

Previous advancements in tactile sensing and neurocomputing heavily prioritized performance metrics—such as contact force estimation and high spatial resolution—without optimizing for energetic efficiency. However, traditional AI solutions present severe energetic sustainability challenges.

However, training modern, unconstrained AI models requires immense power. Extreme theoretical alternatives include building dedicated nuclear plants around terrestrial data centers or moving data centers to the moon to exploit unlimited solar or nuclear power without Earth-bound environmental risks.

Current microelectronics companies are addressing this by integrating traditional AI inference capabilities directly onto edge devices, such as local microcontrollers.

While model training can technically occur on remote or lunar systems, real-time operation necessitates **inference on the edge** due to physical communication delays.

- Even if a data center on the moon were financially and structurally feasible, the physical distance introduces a severe **communication latency** between the Earth and the moon.
- **Latency-Critical Examples:**

- **Autonomous Navigation:** An artificial intelligence system navigating a car on Earth cannot afford to wait for data to travel to the moon and back to make driving decisions.
- **Robotic Manipulation:** A collaborative robot manipulating an object must perceive its surroundings and react to tactile feedback instantaneously.

Therefore, the optimal modern engineering framework splits tasks: models are trained on remote/local computing systems, but the actual **inference must run locally at the edge**.

To bypass the high power consumption of traditional AI at the edge, engineers are developing novel architectures. These systems combine **neuromorphic computing signaling** with network topologies and learning rules directly inspired by human biology.

The SCBO Architecture Study

Researchers evaluated specific connectivity configurations, such as the **SCBO topology**, to study how structural constraints alter power consumption. This architecture tests specific biological wiring features, including:

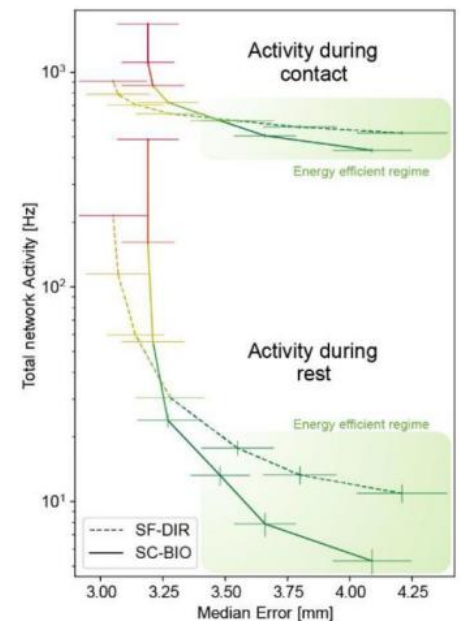
- The integration of inhibitory interneurons.
- The presence or absence of lateral connectivity within a layer versus purely feed-forward structures.

When compared against traditional AI models, these biologically-inspired spiking neural network architectures demonstrated a massive reduction in power:

- As shown in the experimental data, the neuromorphic architecture achieves the exact **same target error** in tactile localization as traditional methods.
- While the visual drop on the plot graph may appear modest at first glance, the graph uses a **logarithmic scale**. The actual transition represents a highly relevant reduction in energetic consumption—surpassing **an order of magnitude or more** in energy savings.

The core advantage of shifting from traditional deep learning to biologically-constrained spiking neuronal networks lies in achieving architectural optimality without sacrificing accuracy.

- **Preserved Accuracy:** By using biological topologies and custom learning rules to evolve synaptic weights, the system completely preserves its metrological performance (maintaining low millinewton and millimeter errors) while simultaneously operating in a low-power regime.
- **Shift in Paradigm:** Rather than fighting inference noise with massive, unconstrained neural networks, the neuromorphic system solves the processing task natively through the structural configuration of its connectivity.
- **Applications:** This energetic optimization is expected to drive significant breakthroughs in the fields of **collaborative robotics** and interacting **intelligent machines**, which remain the primary research focus of the laboratory. Past robots used to operate inside closed cages and kept separated from workers, namely within different areas. Future collaborative robots, instead, will share common spaces with workers, the robot will become a work-mate and will not be anymore just a potential replacement.



Take Home Message from the Course (skip)

Understanding the relationship between basic scientific discovery and applied engineering is essential for driving technological breakthroughs rather than simple, step-by-step developments.

To solve modern, complex challenges, the discovery engine of basic science and the innovation engine of applied engineering cannot remain isolated. Separating them results only in minor, incremental progress.

Instead, a continuous, bi-directional interplay must exist between:

- **The Discovery Engine (Basic Science):** Neuroscience, material science, mathematics, and general physics.
- **The Innovation Engine (Applied Engineering):** Application-oriented technologies and engineering systems.

In public speaking and scientific communication, a clear distinction must be made between these two concepts:

- **Discovery (Scientists):** The characterization and understanding of phenomena that already exist in nature.
- **Invention (Engineers):** The development and creation of novel systems or technologies that do not exist in nature.

The field of biorobotics is currently steering toward an upcoming global megatrend: **Physical AI**.

- Physical AI is heavily driven and enabled by touch science in engineering.
- Specifically, the core focus is centered on **neuromorphic touch science engineering**. This intersection—using neuromorphic methods to solve touch engineering challenges—defines the underlying mission of the laboratory and the specific case studies addressed throughout this course.

Technology and science share a reciprocal relationship, encapsulated by the phrases: "**Science for Robotics**" and "**Robotics for Science**." While scientific discoveries provide the primary inputs needed to fuel technological innovation, robotic systems simultaneously serve as advanced tools to help explore, model, and explain biological principles.

In both living organisms and intelligent machines, the overall computational workload is shared between the physical structure (**bodyware**) and the central processing unit (**brainware**).

- **Peripheral Pre-computation:** In humans, mechanoreceptors are located entirely within the body (the periphery) rather than the brain.
- **Workload Alleviation:** Despite their peripheral location, these receptors perform preliminary computations by directly extracting the specific spatiotemporal properties of physical stimuli. This localized processing drastically reduces the computational burden placed on the central processing structures.

While traditional metrology prioritizes metrics like strict linearity and low total harmonic distortion, forcing systems to be completely linear can block technological breakthroughs. Non-linearity is highly prevalent throughout human physiology, the nervous system, and the sensorimotor apparatus, offering significant operational benefits.

As an established engineering rule, always check the consistency of your measurement units and perform thorough dimensionality checks within your equations.

Methodological Note: While verifying your dimensions does not automatically guarantee that an entire equation is perfectly correct, it guarantees that severe mathematical errors will be caught and exposed immediately.

When sharing scientific findings and research outcomes with a wider public audience, engineers must do so responsibly. It is crucial to present data accurately without overstating, overclaiming, or exaggerating the actual capabilities and scope of the developed technology.

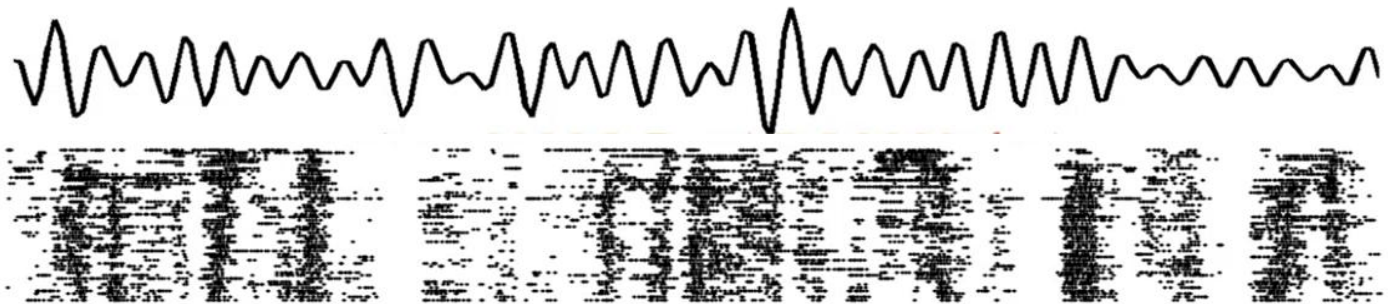
Computational Neuroengineering module

The computational Neuroengineering Laboratory studies information processing in the nervous system. In this field, there's a combination of methods from Computational Neuroscience (decoding and information analysis of biological and artificial neural data, spiking neuronal networks simulations) with the application-driven approach of Biomedical Engineering and Neuro Robotics. Understanding information processing is indeed a key feature for the development of neural interfaces, and such interfaces can in turn be used to validate neural models. Moreover, capturing neural coding dynamics is a basic step toward the development of biomimetic software/hardware for data processing.

Theoretical studies on the origin of **neural signals**, information transmission, and dynamics of neuronal networks are then complemented by a broad range of Biorobotic applications, spanning from invertebrates to humans, from sensory processing to decision making.

As examples of such applications, we are working on the analysis of **healthy and pathological neural dynamics** in the autonomic nervous system and in the basal ganglia to shed light on metabolic and neurodegenerative diseases, and we are contributing with the analysis of behavioral and neural responses to external stimuli to the development of novel upper and lower limb **neuroprostheses**. Recent modeling works include instead: sleep/wake transition in thalamic networks, synaptic and network factors determining the local field potential, and phase-of-firing code in single neurons.

Neural Coding



Understanding how the brain processes information requires tools to visualize neural activity and theoretical frameworks to decode it. The fundamental challenge in neuroscience is bridging the gap between physical neural activity (spikes) and subjective experience or behavior.

To analyze how neurons communicate, neuroscientists rely on different types of visual representations, depending on the scale of the recording.

1. **The Raster Plot (Discrete Data):** The lower image represents a raster plot, a highly common binary graph in neuroscience.
 - **X-axis:** Represents **time**.
 - **Y-axis:** Can represent either a **population of different neurons** or **repeated presentations of the same stimulus** to a single neuron.
 - **Data Points:** Each mark (or "black bean") represents the activation, or "spike," of a neuron. It treats the neuron's firing as a binary event (on/off).
2. **Continuous Extracellular Recordings (Global Features):** The upper image represents a continuous wave. Because the human brain contains over 10 billion neurons, recording the individual raster plot for every single neuron is technically unfeasible. In 99.9% of cases, researchers cannot access all neurons in a region of interest. Therefore, they must rely on **global features** recorded from extracellular brain activity, which are continuous rather than discrete (like Local Field Potentials or EEG).

If the brain contains all our thoughts, dreams, and decisions encoded in sequences of spikes, decoding it is like finding a specific book in a massive library. However, finding the book isn't enough; you need the right dictionary to understand it.

Neural coding is the science of building that dictionary. It aims to translate the subjective experiences of the brain's owner (e.g., the decision to speak, the memory of a dream) into the specific sequences of neural spikes that originate them, and vice versa.

Once the "dictionary" of a specific sensory input is understood, it can be used for advanced neuroengineering applications, such as restoring a sense of touch.

- **Imitating the Neural Dialect:** By perfectly understanding how neurons encode touch (e.g., how they translate a specific pattern of pressure), researchers can artificially recreate this exact "language" or "dialect."
- **Sensory Injection:** Using a neuroprosthetic device, these artificially generated, realistic spike patterns can be injected back into the nervous system.
- **Immediate Recognition:** Because the injected pattern is biomimetic (it mimics the natural neural voice perfectly), the sensory cortex immediately recognizes the information. The brain processes it as a natural stimulus without knowing it was artificially generated.

The brain is a highly **plastic and flexible** structure. It can learn to interpret completely foreign or arbitrary signals through trial and error.

- *Example:* A monkey can learn to modulate specific brain activity to move a cursor on a screen and get a juice reward, simply through training and neuroplasticity.
- *The Limitation:* Relying purely on plasticity takes a significant amount of time and training, and the control might never be perfect.

Conversely, if neuroengineers use **biomimetic coding**—injecting the exact natural language of the neurons—the process bypasses the need for long training periods. It becomes a **"plug-and-play"** system, allowing the subject to decode sensory stimuli on the very first day.

Extracellular recording Pipeline

To understand how the brain encodes and transmits sensory and motor information *in vivo* (in a living subject), neurophysiologists typically rely on **invasive, extracellular recordings**.

Unlike intracellular recordings (where you measure directly inside the soma of a single neuron in a petri dish), extracellular recording involves placing an electrode into the brain tissue.

- **The Challenge:** You are not recording the direct activity of one specific neuron. Instead, you are recording the **fluctuations in the extracellular medium's electrical potential**, which are modulated by the activity of nearby neurons.
- Because this is an **indirect measure of neuron activity**, the raw data must go through a specific signal processing pipeline to make sense of it.

Step 1: Filtering the Raw Signal

Since the recorded signal contains a mix of network-wide electrical changes and individual neuron firings, the first crucial step is to separate them using frequency filters.

- **Low-Pass Filter (typically < 250–500 Hz):** This extracts the **Local Field Potential (LFP)**. The LFP does not show individual spikes; rather, it reflects the slow-wave, average electrical activity of the surrounding neural network.

- **High-Pass Filter:** This removes the slow waves and isolates the high-frequency baseline noise along with discrete, fast electrical events. These discrete events are the **spikes** (action potentials) of individual neurons.
 - *Note on Multi-Unit Activity:* Sometimes the background noise is too high to confidently isolate individual spikes. In these rare cases, researchers use the whole time series to study the general area's activity, known as *multi-unit activity*.

Step 2: Spike Identification (Thresholding)

Once the signal is high-pass filtered, you are left with a baseline of noise and popping discrete events. How do you mathematically prove an event is a true spike and not just a random fluctuation in noise?

- **Gaussian Noise Assumption:** Researchers assume the background noise follows a standard Gaussian (normal) distribution.
- **Setting a Threshold:** They place a strict amplitude threshold, usually **3 to 4 standard deviations** away from the mean noise level.
- **Identification:** Because the statistical probability of noise reaching 4 standard deviations is a tiny fraction of a percent, any signal that crosses this threshold is confidently identified as a true neuronal spike.

Step 3: Spike Sorting (Clustering)

Identifying the spikes isn't the final step. Because the recording is extracellular, a single electrode tip usually picks up the activity of **1 to 5 different neurons** in its immediate vicinity. You know a spike happened at timestamp T₁, but did Neuron A, B, or C fire it?

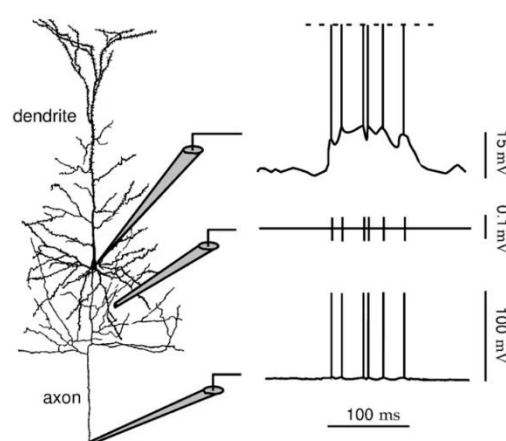
- **High Sampling Frequency:** Good laboratories record this data at a very high sampling rate (e.g., **20 kHz**, meaning 20 data points every millisecond). This high resolution allows researchers to see the exact geometric **shape (waveform)** of the 20 points closest to the spike's peak.
- **The "Fingerprint" Assumption:** It is a biologically grounded assumption that the specific waveform shape of an action potential acts as a unique fingerprint for a particular neuron.
- **Clustering:** Researchers group (cluster) the identified spikes according to their waveform shape. They associate every distinct shape with a different neuron (e.g., "All the tall, sharp spikes are from Neuron 1; the shorter, wider spikes are from Neuron 2"). This final operation is called **Spike Sorting**.

Neuronal Code

The neuronal code is a sequence of spikes. To understand the neuronal code, we must look at the anatomy of a neuron and how it processes electrical signals. The flow of information typically follows a specific path: Dendrites → Soma → Axon.

- **Dendrites** (Input): These branching structures receive inputs from other neurons. These inputs cause local changes in electrical potential.
- **Soma** (Cell Body): All the inputs traveling down the dendrites converge and are integrated at the soma.
- **Axon** (Output): This is the output channel that carries the signal away from the soma to the dendrites of other (postsynaptic) neurons.

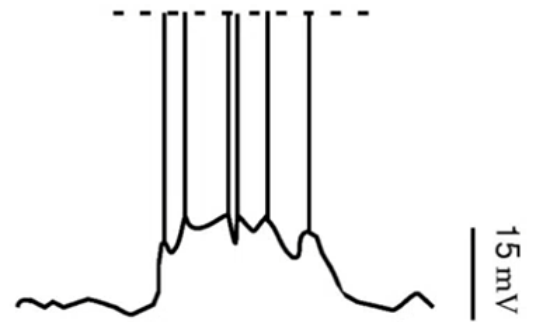
The way we perceive neuronal activity drastically changes depending on where and how we place our recording electrodes.



A. Somatic Electrode (Intracellular Recording)

If we place an electrode directly inside the soma, we can observe the full picture of the neuron's electrical state.

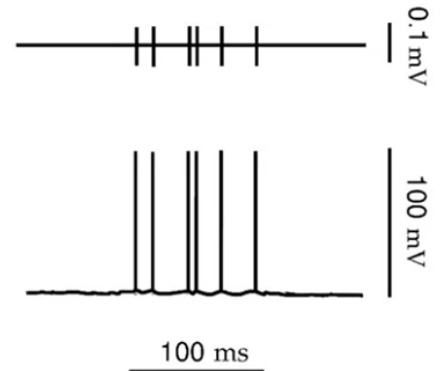
- **Subthreshold Activity:** We can see the rich fluctuations of the membrane potential caused by the continuous "battle" between excitatory and inhibitory inputs arriving from the dendrites.
- **Action Potentials (Spikes):** If these potential fluctuations integrate and reach a specific critical threshold, a discrete event occurs: a spike (Action Potential) is fired.



B. Extracellular Electrode & Axonal Electrode (The Output View)

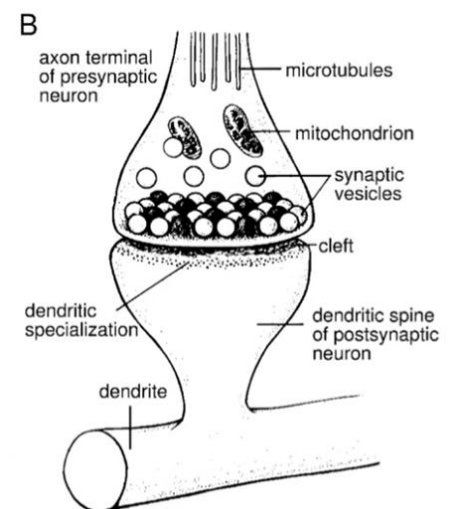
Things look very different when we record extracellularly or if we record from the axon.

- **Attenuation of Subthreshold Activity:** As the signal moves down the axon, or when viewed from outside the cell, all the complex subthreshold membrane potential fluctuations are attenuated and disappear.
- **Only Spikes Propagate:** The only information that survives and propagates over long distances (down the axon to the next neurons) are the spikes themselves.



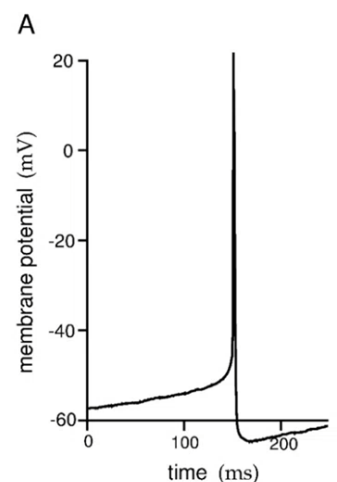
Synaptic Transmission

While information *within* a single neuron is processed electrically (membrane potentials and spikes), information transfer *between* neurons is primarily **chemical**. This occurs at the synapse, the microscopic gap (synaptic cleft) where the axon terminal of the presynaptic neuron almost touches the dendrite of the postsynaptic neuron.



The Mechanism of Communication

1. **Depolarization:** When an action potential (spike) travels down the axon and reaches the presynaptic terminal, it causes a drastic increase in the membrane potential, known as depolarization.
2. **Neurotransmitter Release:** This depolarization triggers the synaptic vesicles to release chemical messengers called **neurotransmitters** into the synaptic cleft. *Crucially, these vesicles only release neurotransmitters when an action potential arrives, proving that neurons communicate only through spikes, not through subthreshold fluctuations.*
3. **Receptor Binding & Ion Channels:** The neurotransmitters travel across the cleft and lock into specific receptors on the postsynaptic dendrite.
4. **Potential Change:** This binding acts like a key, opening ion channels in the postsynaptic membrane. Ions flow in or out, changing the local membrane potential. If the integration of these changes reaches the threshold, a new spike is fired.



Thus, neurons communicate information only through action potentials, not through subthreshold membrane fluctuations. Thus, **neuronal code** consists of a time series of stereotyped action potentials.

We can distinguish between **inhibitory** and **excitatory** synapses:

- **Excitatory Synapses:** The opened ion channels allow a net inward current that *increases* the membrane potential (depolarization), making it easier for the postsynaptic neuron to fire a spike.
- **Inhibitory Synapses:** The opened channels cause a *decrease* in the membrane potential (hyperpolarization), pushing the neuron further from the threshold and making a spike less likely.

Drugs mimics neurotransmitters' behaviour. External chemicals like nicotine or THC can alter brain activity because their molecular structures are "crazily close" to natural neurotransmitters. For example, nicotine binds to *nicotinic receptors* on the dendrites, opening ion channels and driving neural activity artificially, acting as an external chemical influence without any true presynaptic spike.

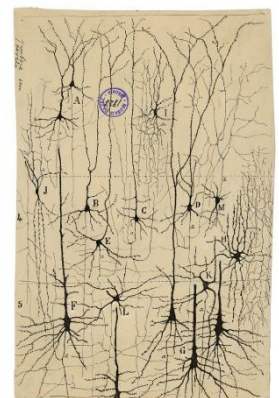
From a biological standpoint, the true alphabet of the brain is a sequence of spikes. However, from a **neuroengineering perspective**, the neuronal code is "*whatever works*" to extract useful information for devices, prostheses, or diagnostics. Engineers have access to different "windows" into brain activity, ranging from direct measurements to highly indirect proxies:

- **Spikes** (Single/Multi-Unit Activity): The only way to see exactly what the neuron is "seeing" and transmitting. It is the direct alphabet of the nervous system.
- **LFP** (Local Field Potential): Recorded via invasive extracellular electrodes, it reflects the average electrical activity of a local network of neurons rather than individual spikes.
- **ECoG** (Electrocorticography): Electrodes are placed directly on the surface of the cortex, under the skull. It provides a strong, relatively localized electrical signal.
- **EEG** (Electroencephalography): Electrodes are placed externally on the scalp (like an EEG cap). It records the coarse, generalized electrical activity of the underlying cortex. It is non-invasive but highly filtered by the skull.
- **fMRI** (Functional Magnetic Resonance Imaging): A completely indirect measure because it relies on neurovascular coupling. It does not measure electrical activity at all; instead, it measures energy consumption and blood flow to infer which brain regions (and thus, which groups of neurons) are currently active.

Neurons Networks

While understanding single neurons is fundamental, the brain rarely relies on a single neuron to perform a specific function.

Neurons act in **networks**. The nervous system is engineered to be highly redundant and robust. Functions are almost always performed by a *pool* or population of neurons rather than isolated cells. This ensures that if a few neurons fail or die, the overall function is not lost. As neuroengineers, looking at detailed, biological drawings of neurons (like the historical sketches by Ramón y Cajal showing thick or thin dendrites) is visually beautiful but not perfectly useful for building devices.



- The goal is to **translate the biological mass of neurons into abstract, functional mathematical forms**. We don't need to replicate the exact physical branching of a dendrite in a silicon chip; we need to extract and replicate the *network dynamic* (the computation it performs). This is the foundation of **Neuromorphic Engineering**.

When we analyze these network dynamics, we find **Network Motifs**: recurring patterns of connectivity that the brain reuses in different areas to perform specific computations. One of the most important and classic motifs, found across almost all sensory pathways (like vision and touch), is **Lateral Inhibition**.

We can imagine this as a competition mechanism related to sensory pathways.

Imagine you are looking at a visual scene. You have thousands of parallel excitatory pathways (fibers) carrying pixel-by-pixel information from your retina to your brain.

The Problem: Sometimes the most important ("salient") information is just slightly more active than the background noise. For example, one pathway might be only 5% more active than its neighbors. How does the brain prevent this tiny signal from being lost in the noise?

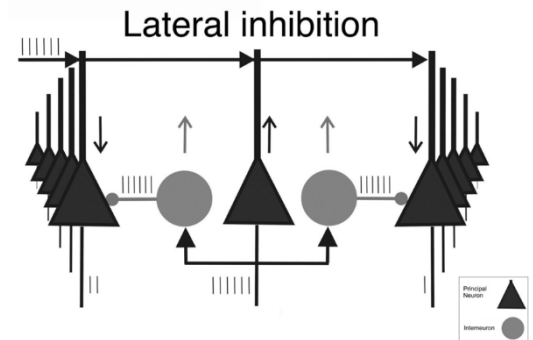
The mechanism can be described in three steps:

1. **Parallel Excitation:** Parallel fibers send excitatory signals to the next layer of the nervous system.
2. **The Inhibitory Counterpart:** Crucially, as an excitatory fiber sends its signal forward, it *also* branches off to excite an **inhibitory interneuron**.
3. **Cross-Inhibition:** This inhibitory interneuron then sends a suppressing (inhibitory) signal laterally to the *neighboring* parallel pathways.

Because of this cross-inhibition, a competition starts between neighboring pathways based on their activation levels:

- If Pathway A is slightly more active (e.g., 5% stronger) than Pathway B, Pathway A will stimulate its inhibitory interneuron *more strongly*.
- Therefore, Pathway A will inhibit Pathway B more than Pathway B can inhibit Pathway A.
- Very quickly, the slightly weaker signal (B) is completely suppressed, while the slightly stronger signal (A) is maintained or amplified.

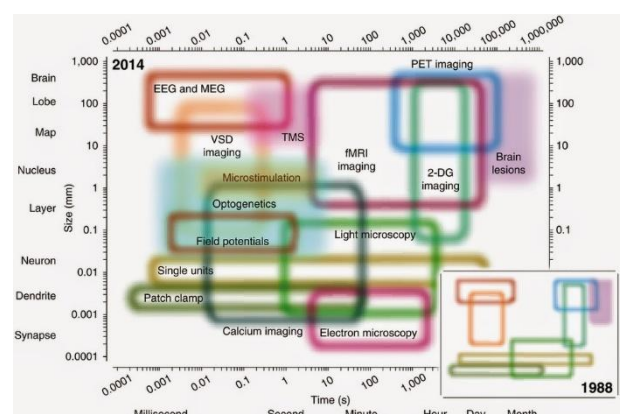
Lateral inhibition is a brilliant, highly efficient computation that **amplifies tiny differences** and acts as a contrast enhancer. It allows the brain to immediately extract silent/important information from a noisy background. This is exactly the kind of dynamic engineers want to implement in artificial neuromorphic circuits!



This final graph plots the various "windows" we have into the nervous system across logarithmic scales of **Time** (Temporal Resolution) and **Space** (Spatial/Anatomical Resolution).

The core takeaway is that **no single technique is perfect; every method involves a trade-off:**

- **Single-Unit Recordings:** Excellent temporal resolution (captures millisecond-level spikes), but very limited spatial scope.
 - *Modern Update:* While the fundamental methods haven't changed much, their scope has expanded. New high-density electrodes like **Neuropixels** now allow researchers to record from hundreds of individual neurons simultaneously, pushing this window slightly higher on the spatial y-axis.
- **EEG:** Excellent temporal resolution and fully non-invasive, but highly filtered by the skull, resulting in a very coarse spatial resolution.
- **fMRI:** Excellent spatial coverage (you can see the whole brain's energy consumption), but terrible temporal resolution. It only captures a slow, "low-pass filtered" version of neural activity.



Because we cannot currently place Neuropixels across the entire human cortex, we cannot see the whole brain at the single-spike level.

This is where **Computational Neuroscience** comes in. By designing mathematical and computational models, engineers can "connect the dots" between these different windows. Models allow us to link the microscopic, fast dynamics of single neurons to the macroscopic, slow signals seen in an fMRI, creating a complete, multi-scale understanding of the brain.

Neuronal Models

Before diving into specific mathematical models, we need to understand *why* we model neurons and *how* we choose the right level of abstraction. Just like Picasso's progressive abstraction of a bull, modeling in neuroscience is about simplifying the description of a phenomenon by finding the most relevant relationships within the data, stripping away unnecessary biological details to capture the core computation.

There are different reasons to study the modelling of neurons:

1. The "Greedy" Reason: Large-Scale Simulation (The Virtual Brain). Massive funding (like the Human Brain Project) has been poured into creating large-scale simulations like **The Virtual Brain**. This platform does not model every single biological neuron. Instead, it uses differential equations to model entire *regions* of the cortex (mesoscale representation). The crucial component of this model is the **connectivity matrix**. It is not just a binary anatomical map (who connects to whom); it includes **weights**. It describes whether an area connects to another strongly or weakly, and whether the connection is excitatory or inhibitory.
2. Winning a Nobel Prize in Medicine, Alan Hodgkin and Andrew Huxley derived the first set of differential equations describing the temporal evolution of a neuron's membrane potential. It all starts with the differential equation describing how voltage changes over time as a function of internal dynamics and external inputs. While biologically accurate, the HH model uses many complex differential and parametric equations for just *one* neuron. If you want to simulate a network of millions of neurons, your system will explode with parameters. This complexity forced neuroengineers to move toward simpler models.
3. The Abstract/Computational Approach (John Hopfield). John Hopfield (who recently won the Nobel Prize in Physics) took a completely different, highly simplified approach that laid the foundation for modern Artificial Neural Networks (ANNs). He completely ignored dendrites and membrane potentials. His "neurons" were simple bistable elements: either active (1) or inactive (-1). Hopfield emphasized that while the *anatomical* connections in a brain are fixed early in life, the *weights* of these synapses change constantly based on experience. **This change in synaptic weight is the physical basis of memory and learning** (Synaptic Plasticity). If your anatomical brain scans at 11:59 and 12:00 are identical, but you learned something in that minute, the difference is entirely in the synaptic weights.
4. Today, we are collecting an unprecedented amount of data about the brain. Without models, this data is just noise. We need computational models to extract the relevant functional mechanisms from this massive sea of data to build neuroprosthetics and cure disorders.

When moving forward to study specific neuronal models, we must apply a fundamental philosophical principle known as **Occam's Razor**. The Principle states "Among all models that can successfully explain the phenomenon under investigation, always choose the simplest one". If you have 70 data points and you build a complex model with 70 differential equations, you will perfectly reproduce the data. However, you will have understood absolutely nothing about the underlying mechanism because you are just mathematically overfitting the data. The method is to start with a very simple model. If it fails to explain the phenomenon, *then* you add a degree of realism and complexity, but never more than strictly necessary.

Cellular Automata Model

The **Cellular Automata** is the simplest possible network model. While scientifically obsolete for modern research ("in 2026 it is scientifically useless" for advanced simulation), it remains a powerful deductive tool to understand macro-level brain dynamics.

The main model's architecture features and rules are the following:

- **Bistable Elements:** Neurons are represented as simple cells on a grid. They can only be in one of two states: Inactive (White) or Active (Black).
- **Connectivity:** Each neuron is connected to its 8 immediate neighbors on the grid.
- **The Rule of Propagation (The Single Parameter):** The entire system is governed by a single parameter, N . N represents the average number of neighbors that a neuron activates when it turns from white to black (active).

By varying this single parameter N , we can observe drastically different global behaviors of the network.

We can divide the behavior of this network into three distinct mathematical regimes. Only one of these can represent a healthy, functional brain.

Case 1: $N < 1$ (Inhibition-Dominated System)

- **What it means:** On average, the activation of one neuron triggers *less* than one neighboring neuron. The inertia of the system not to fire is stronger than the excitation.
- **The Dynamics:** If you trigger a stimulus, it might activate a couple of neighboring cells, but the signal will quickly fade. Mathematically, as time (T) goes to infinity, N^T approaches 0.
- **Consequences → No Communication:** A neuron on one side of the network will never be able to communicate with a neuron on the other side. The signal dies out before it can propagate.
 - **Event Size:** Events are always of very limited size and short duration.
- **Conclusion:** This cannot represent a healthy brain, as functional communication across areas is impossible.

Case 2: $N > 1$ (Excitation-Dominated System)

- **What it means:** On average, the activation of one neuron triggers *more* than one neighboring neuron. The network is highly prone to excitation.
- **The Dynamics:** A small initial stimulus quickly multiplies. In a few steps, the activation spreads uncontrollably until the entire network is black (active).
- **Clinical Equivalent (Epilepsy):** This runaway excitation directly models an **epileptic seizure**, where an electrical fluctuation overrides the system and spreads massively due to an excess of net excitation.
- **Consequences:**
 - **Loss of Information:** While the system is highly *sensitive* (it reacts strongly), it is not *informative*. Because every stimulus, no matter how small or specific, results in the entire network turning black, the final state contains absolutely zero information about the initial stimulus.
 - **Bimodal Distribution:** Events in this regime are either very small (if they fail to take off) or massive (involving the whole network). There are no middle-sized, controlled events.
- **Conclusion:** This represents a pathological state (like epilepsy or migraine), not a healthy computing brain.

Case 3: $N = 1$ (The Balanced System)

- **What it means:** On average, the activation of one neuron triggers exactly one other neuron. The system sits on the critical boundary between fading out and exploding.
- **The Dynamics:** The signal propagates without dying out (like $N < 1$) and without taking over the whole network (like $N > 1$).
- **Consequences:**
 - **Optimal Communication:** Signals can travel across the network, allowing distinct areas to communicate without losing the specificity of the information.

- **Power-Law Behavior:** If the network is infinite, the system displays a power-law distribution. This means there is no single "characteristic" size for an event; the brain can generate events of all possible sizes and durations, allowing for complex, flexible computation.

The cellular automata model mathematically proves a fundamental biological rule that you must remember: “In a healthy nervous system, Excitation (E) and Inhibition (I) must always be balanced.”

This does not mean that every single neuron receives exactly equal excitatory and inhibitory inputs at every millisecond (if that were true, the brain would be completely silent and frozen). Instead, it means that over a **sufficiently long period of time** and across a **sufficiently large volume of tissue**, the network architecture guarantees that excitation and inhibition cancel each other out, maintaining the system near the critical state of $N=1$.

This result has also an important **clinical relevance**. Many neurological disorders can be theoretically traced back to a failure of this E/I balance:

- **Epilepsy:** The most obvious example of runaway excitation ($N>1$).
- **Migraines, Alzheimer's, Sclerosis:** Research suggests these also involve localized or systemic disruptions in how the brain balances its excitatory and inhibitory drives.

The Population Firing Rate Model (The Base Model)

The core philosophy of this model is simplification based on your research question: **Sometimes, you don't care about exact spike times**. For example, when studying slow-adapting mechanoreceptors in the fingertip, the physical pressure is encoded simply by the *average firing rate*. If your question only requires knowing whether a neuron is firing "more" or "less" over a window of time (e.g., 100ms), you model the **Rate (R)**.

Here is the foundational differential equation (which has a decaying exponential as solution) presented of the model:

$$\tau \frac{dR}{dt} = -R + \Phi(JR + I_{ext})$$

- τ is the relaxation time (time constant of the firing rate dynamics). This is derived from basic biological reality (referencing von Bertalanffy's *General System Theory*). No variable in a living, healthy system can go to infinity; it must always relax to a stable equilibrium. Therefore, a differential equation in biology *must* have a negative relaxation term ($-R$). In the absence of stimuli, the firing rate R relaxes back to a stable baseline (often assumed to be 0). If this term is missing, you are likely modelling a pathological state out of equilibrium (like a toxin injection).
- JR is the **network interaction** and J is the **connectivity matrix**. This is proportional to the strength of the synaptic connections within the population (if $J < 0$ the population is said to be inhibitory, while if $J > 0$ it is said to be excitatory). This describes how the population's own firing rate influences itself. If R is treated as a scalar (a global average), J is a scalar representing average connectivity. If R is a vector of individual rates, J becomes the full connectivity matrix.
- I_{ext} is the **external input**. This represents everything driving the neurons from *outside* the specific network you are modeling. This includes deliberate experimental inputs (like Deep Brain Stimulation in Parkinson's disease) but critically also includes the natural inputs arriving from the billions of other neurons in the brain that you simply aren't including in your specific model.
- Φ is the **transfer function** that gives the steady – state activity as a function of the inputs. Since the activity is a non negative quantity, Φ must be positive or zero for any value of the inputs, and is typically a monotonically increasing function. It introduces the biological threshold. If the combined input ($JR + I_{ext}$) is too weak, it remains sub-threshold, and Φ outputs 0 (no firing). If it crosses the threshold, Φ

translates that input into a positive firing rate (typically acting as a threshold-linear or monotonically increasing function).

$$\Phi(I) = \begin{cases} I - T & I > T \\ 0 & I < T \end{cases}$$

Where T is the threshold, and continuous sigmoidal functions such as the logistic function.

For a one – population network and constant inputs, the equation always converges to a fixed point starting from any initial condition. In an inhibitory network, only a single fixed point exists. In an excitatory network, and with a suitable non linear Φ , the network can exhibit bistability, between a low and a high activity state.

The Major Limitation: If you analyze this base dynamical system mathematically, it will only ever reach a stable fixed point R^* . **This base model never oscillates.**

To capture the complex rhythms of the brain (especially in the cortex), we must add a tiny, vital piece of biological realism: **Time Delay** (Δ). This is due to the finite conduction time from the site of action potential to the synaptic site.

When a presynaptic neuron fires, the postsynaptic neuron does not feel it instantly. Even if the electrical signal down the axon is fast (approximately 1 ms), the chemical process takes time: the neurotransmitter must be released, diffuse through fluid dynamics across the synaptic cleft, bind to receptors, and trigger the protein machinery to open ion channels.

This whole process takes a few milliseconds (e.g., < 5 ms). Therefore, the firing rate right now, $R(t)$, is influenced by the network's firing rate from a few milliseconds ago, $R(t - \Delta)$.

This updates our equation to:

$$\tau \frac{dR(t)}{dt} = -R(t) + \Phi(JR(t - \Delta) + I_{ext})$$

This delay leads to a qualitative change of the dynamics when interactions are purely inhibitory. In this case, the delay can lead to firing rate oscillations, which are not present in the absence of delays (Brunel and Hakim 2009). Let's look at a specific scenario: a purely inhibitory network ($J < 0$) receiving a constant excitatory external input ($I_{ext} > 0$).

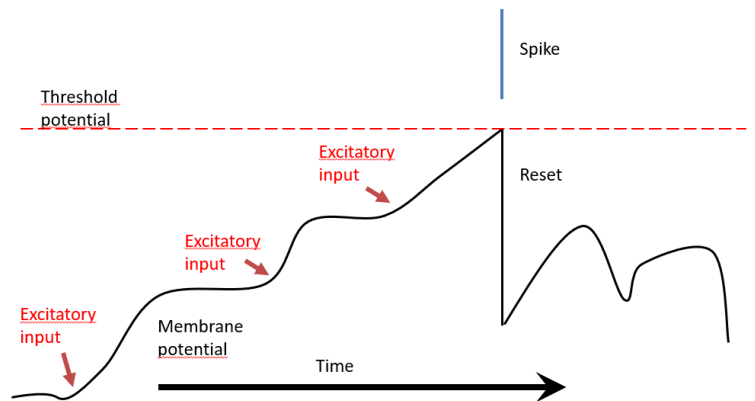
1. **Initial Firing:** The external excitation (I_{ext}) is applied. The previously silent neurons cross the threshold and start firing heavily.
2. **The Delay Window:** Because of the biological delay (Δ), the network's own strong inhibitory signals (JR) haven't reached the somas yet.
3. **Massive Inhibition Arrives:** After time Δ , the massive wave of self-inhibition finally hits. Because J is strongly negative, it completely overrides the constant excitatory input I_{ext} .
4. **Silence:** The derivative $\frac{dR(t)}{dt}$ becomes heavily negative. The firing rate crashes to 0. The neurons stop firing.
5. **Inhibition Fades:** Because the neurons are no longer firing ($R = 0$), the delayed inhibitory term $JR(t - \Delta)$ eventually drops to 0 as well.
6. **The Cycle Restarts:** With the internal inhibition gone, the constant external excitation I_{ext} is once again the dominant force. The neurons start firing again.

Conclusion: Even with a perfectly flat, constant external input, the addition of a biological delay forces the system to continuously alternate between firing and silence. The network generates its own **internal resonance**, producing the characteristic brain oscillations we see in cortical EEG data!

Integrate and Fire Neuronal Model

Overview and Motivation

The Integrate-and-Fire (I&F) model is widely celebrated in computational neuroscience for its elegance and simplicity. While heavily detailed morphological models exist, the I&F model abstracts the neuron's structure (ignoring complex dendrites and axons) to focus purely on the **membrane potential of the soma**.



- **From Rates to Precise Timing:** Unlike basic rate models that only look at the average firing frequency, the I&F model allows us to determine precise **spike times**.
- **Subthreshold Activity:** A major advantage of modeling the membrane potential is that it captures the *below-threshold* (subthreshold) activity—the internal dynamics of the neuron that occur even when a spike isn't fired.

Starting from the **basic integrate – and – fire Model**, this model captures the fundamental nonlinearity of a biological neuron using a very simple mechanism:

- **Integration:** As **excitatory inputs** arrive at the neuron, they increase the membrane potential step-by-step.
- **Threshold & Fire:** If the membrane potential accumulates enough charge to hit a specific **threshold potential**, the neuron fires an action potential (a spike).
- **Instantaneous Event & Reset:** The spike is treated as a sudden, instantaneous event. Immediately after firing, the membrane potential is artificially **reset** to a lower baseline level, and the integration process restarts.
- **Network Role:** Recording this spike as a discrete event is crucial, as it is then "injected" as an input to other downstream neurons in a network simulation.

The I&F model is drastically simpler than the Hodgkin-Huxley model. It does **not** explicitly simulate the complex biophysics of active sodium (Na⁺) and potassium (K⁺) ion channels. Instead, it relies on a simple mathematical rule (hit threshold → emit spike → reset) to replicate the outcome of those channels.

However, a neuron's potential doesn't only go up. Real neurons receive both types of signals:

- **Excitatory inputs:** Push the membrane potential *up* toward the threshold.
- **Inhibitory inputs:** Push the membrane potential *down* away from the threshold, making the neuron less likely to fire.

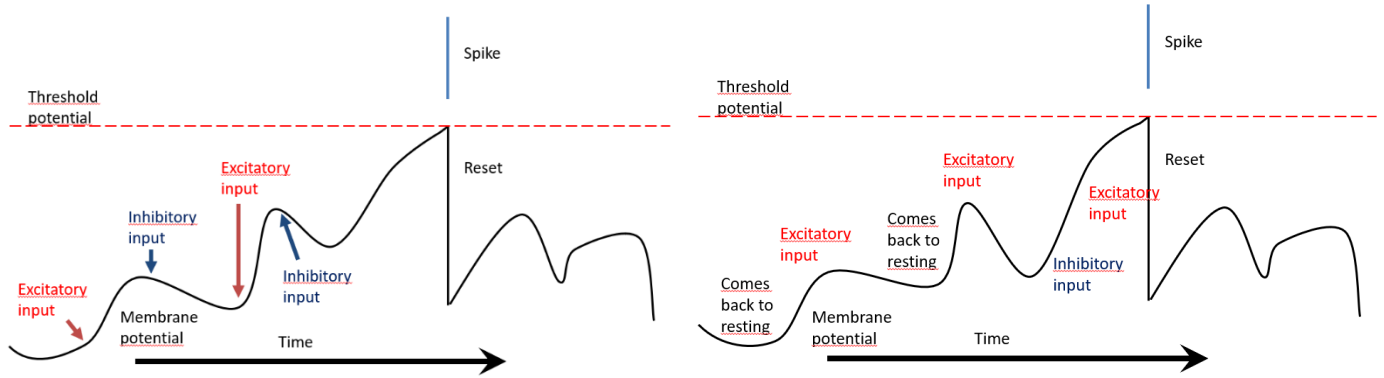
This leads to the **Leaky integrate – and – fire Model**.

In the basic I&F model, the membrane potential would stay elevated forever until another input arrived. Biological neurons do not behave this way due to passive ion channels that "leak" charge.

To make the model more realistic, we introduce the **Resting Potential**:

- **The Leak Mechanism:** If the membrane potential is left alone after an excitatory or inhibitory input, it will naturally decay and stabilize back to a baseline equilibrium known as the **resting potential**.
- **Three Core Factors:** The state of a LIF neuron is constantly determined by the interplay of three distinct factors:
 1. Excitatory inputs (driving it up)
 2. Inhibitory inputs (driving it down)
 3. The Leak / Resting mechanism (pulling it back to equilibrium)

Because of its balance of simplicity and biological realism, the standard non-leaky model is rarely used in practice. The **Leaky Integrate-and-Fire (LIF)** neuron is the standard foundational model used to build and simulate complex spiking neural networks.



Let's now analyze the equations related to this model. Before looking at the equations, it is crucial to establish a baseline. In this model, we care about the *difference* in potential, not absolute biological values.

- For simplicity, the **resting potential** (equilibrium) is defined as **0 mV**.
- All changes in membrane voltage are measured relative to this 0 mV baseline.

When the neuron is *not* firing a spike (subthreshold), its membrane potential is governed by a single differential equation that sums up the competing forces acting on it:

$$\tau_m \dot{V}_k(t) = -V_k(t) + V_{Ak}(t) - V_{Gk}(t)$$

- $\dot{V}_k(t)$: The derivative of the membrane potential of neuron k with respect to time. This represents the **rate of change** of the voltage.
- τ_m : The **relaxation time** (or membrane time constant). This determines how fast the neuron returns to its resting state after a perturbation. It is usually around **20 ms**, depending on the size of the soma.
- $-V_k(t)$: The **Leak term**. Notice the minus sign. If no inputs are arriving, this term forces the potential to decay back to the resting potential (0 mV).
- $V_{Ak}(t)$: The **Excitatory** post-synaptic potential (PSP). This is driven by **AMPA** synapses (main excitatory neurotransmitter). It represents the cumulative depolarization (increase in voltage) from external inputs.
- $-V_{Gk}(t)$: The **Inhibitory** post-synaptic potential (PSP). This is driven by **GABA** synapses (main inhibitory neurotransmitter). The minus sign in the main equation ensures this term hyperpolarizes the cell (decreases the voltage).

While other synapses like NMDA exist, this specific model is simplified to only include AMPA and GABA.

When a neuron receives an input spike, its voltage doesn't change instantly. The opening and closing of ion channels create a specific curve (a rapid rise followed by a slower decay). To model this realistic shape, the postsynaptic potential is described mathematically as a **difference of exponentials**.

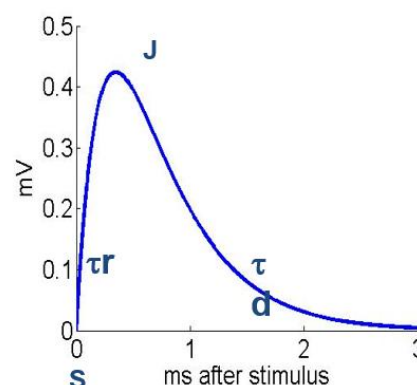
To keep the math simpler, this 2nd-order dynamic is split into two coupled 1st-order differential equations using an auxiliary variable X:

- **The Rise (Channel Opening):** $\tau_{rA/G} \dot{X}_{A/Gk}(t) = -X_{A/Gk}(t) + \tau_m \sum_{stim} J_k^{stim} S_k^{stim}(t)$
- **The Decay (Channel Closing):** $\tau_{dA/G} \dot{V}_{A/Gk}(t) = -V_{A/Gk}(t) + X_{A/Gk}(t)$

Let's analyze the terms:

- τ_r : The **rise time** of the PSP due to channel opening (usually **0.5 - 2 ms**).
- τ_d : The **decay time** of the PSP due to channel closing (usually **3 to 4 times longer than τ_r**).

- $S_k^{stim}(t)$: A binary function representing the incoming spike. It is **0** at all times, except when the neuron is actively receiving an input, at which point it becomes **1**. It is basically a long sequence of **0s**, which becomes a **1** only when a connected presynaptic neuron fires. This represents the fixed, underlying **anatomical connectivity** (the physical axons and dendrites).
- J_k^{stim} : The **strength** (or amplitude) of the incoming synapse. While the anatomical structure (S) remains mostly fixed after development, the functional weights (J) change constantly. **This continuous change in J is the mathematical basis of brain plasticity, learning, and memory.**



How it works in practice: When a spike arrives ($S = 1$), it triggers the auxiliary variable X to spike up quickly (governed by τ_r). X then feeds into the second equation, causing the actual synaptic voltage V to rise and then slowly decay (governed by τ_d), creating the characteristic curve seen in the graph.

Because τ_r is so fast (~ 1 ms), running a computer simulation requires a very fine sampling rate (using methods like Euler-Runge-Kutta). You must sample much faster than your smallest time scale—for example, updating the simulation every **0.05 ms** (1/20th of a millisecond) to ensure mathematical accuracy without wasting computational power.

We have a Super-threshold Dynamics. The differential equations above only handle the subthreshold buildup.

Summary of Key Variables

Symbol	Definition	Typical Value / Unit
V_k	Membrane potential of neuron k	mV
V_{thr}	Threshold potential required to fire a spike	mV
V_{res}	Reset potential after firing	0 mV
τ_m	Relaxation time of the membrane potential	~ 20 ms
τ_r	Rise time of the post-synaptic potential	1-5 ms
τ_d	Decay time of the post-synaptic potential	3-10 ms
τ_{ref}	Refractory period (time neuron cannot fire)	ms
J	Strength / Amplitude of the synaptic input	-

The model relies on an explicit, non-linear rule to handle the actual firing:

If $V_k(t^*) > V_{thr}$, the **threshold is crossed**

1. **A spike is emitted:** This event is recorded and transmitted to downstream neurons.
2. **Network Propagation:** This spike is injected into the input matrix (S) of all postsynaptic neurons connected to neuron K . This arrival is delayed by a short latency period (e.g., 0.5 ms) to account for the physical travel time down the axon.
3. **Voltage drops:** V_k is instantly reset to V_{res} (the reset potential, which is typically set at **0 mV** or slightly below).
4. **Refractory period:** The neuron cannot fire again for a set amount of time, denoted as τ_{ref} .

The Izhikevich Neuron Model

Limitations of the LIF Model: Adaptation

The Leaky Integrate-and-Fire (LIF) model has a major limitation: if you apply a constant external current, the neuron will fire at a constant, fixed rate forever.

Biological neurons, however, exhibit **adaptation**.

- **Definition of Adaptation:** A decreasing response (lower firing frequency) to a constant stimulus.
- **Why does the brain do this?** The brain prioritizes **novelty** and **discontinuity**. If a stimulus is constant and non-threatening (like the hum of a fan or the constant pressure of a shirt), peripheral neurons adapt and stop sending information to the brain, freeing up cognitive resources. We can have different levels of adaptation in the receptors (slowly vs rapidly adapting).
- **Conclusion:** If you are modeling peripheral sensory systems (like mechanoreceptors in the fingertip), you *must* use a model that captures adaptation. The Izhikevich model is designed specifically for this.

The model

Developed by Eugene Izhikevich, this model bridges the gap between the biologically realistic (but computationally heavy) Hodgkin-Huxley model and the computationally efficient (but biologically limited) LIF model.

It reduces the complex Hodgkin-Huxley dynamics into a **2-dimensional system of coupled ordinary differential equations (ODEs)**.

The two main variables are:

1. v : The membrane potential of the neuron.
2. u : A membrane recovery variable (which provides negative feedback to v). This is the **adaptation variable**.

The continuous evolution of the system is governed by these two coupled equations:

$$\begin{cases} \dot{v} = 0.04v^2 + 5v + 140 - u + I \\ \dot{u} = a(bv - u) \end{cases}$$

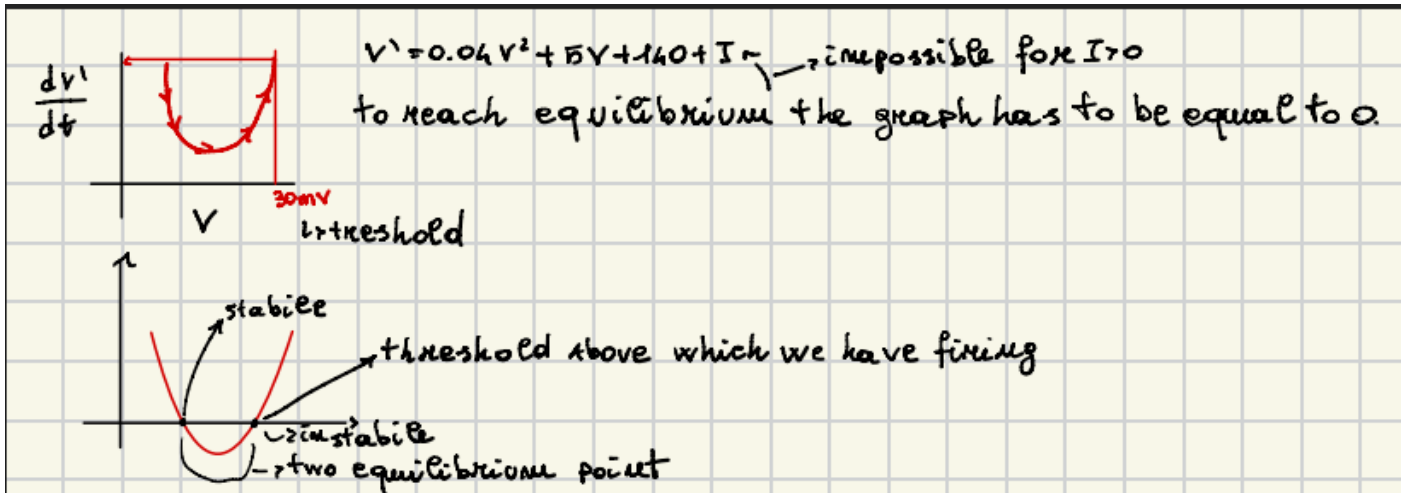
When the membrane potential crosses the spike apex (set at +30 mV), the variables are instantly reset:

$$\text{if } v \geq 30 \text{ mV} \rightarrow \begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases}$$

The **first equation** monitors the membrane potential v . Unlike the LIF model (which has a linear decay $-v$), the Izhikevich model has a **quadratic nonlinearity**. This mathematical shape is crucial because it allows the model to simulate the *actual generation* of the action potential (the upward spike) rather than just artificially drawing a line when a threshold is crossed.

If you plot $\dot{v}$ against v , you get a parabola. Depending on the parameters, this parabola intersects the x-axis ($\dot{v} = 0$) at two points. These are equilibrium points:

- **Stable Equilibrium (Resting State):** If the voltage is perturbed slightly from this point, the derivative forces it back. This is the normal resting potential.
- **Unstable Equilibrium (Spike Initiation Threshold):** If the external current (I) pushes the voltage past this point, the system enters a region of **positive feedback** ($\dot{v} > 0$). The voltage rapidly explodes upwards, naturally simulating the rising phase of an action potential.



The positive feedback would make the voltage go to infinity, which isn't biologically realistic. Therefore, an artificial limit is imposed: when v hits 30 mV, the spike is considered complete, and the reset rule is triggered. The second equation governs the recovery variable u , which provides the "friction" or negative feedback that causes adaptation.

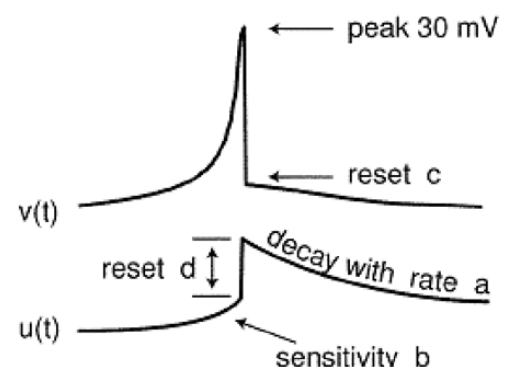
- **The Relaxation Term ($-au$):** Like all biological variables, u naturally decays back to its equilibrium over time. The parameter a dictates the time scale of this relaxation.
- **Coupling with v ($+abv$):** As the membrane voltage v increases (due to excitatory input), it forces u to increase as well.
- **The Negative Feedback ($-u$ in the v equation):** Look back at the first equation. Notice the $-u$ term. As u grows, it exerts a stronger *negative* force on the voltage v , making it harder and harder for the neuron to depolarize and fire a spike.

The real magic of the adaptation happens when a spike occurs.

$$\text{if } v \geq 30 \text{ mV} \rightarrow \begin{cases} v \leftarrow c & (\text{The voltage resets to a low baseline value, } c) \\ u \leftarrow u + d & (\text{The adaption variable umps up by a set amount, } d) \end{cases}$$

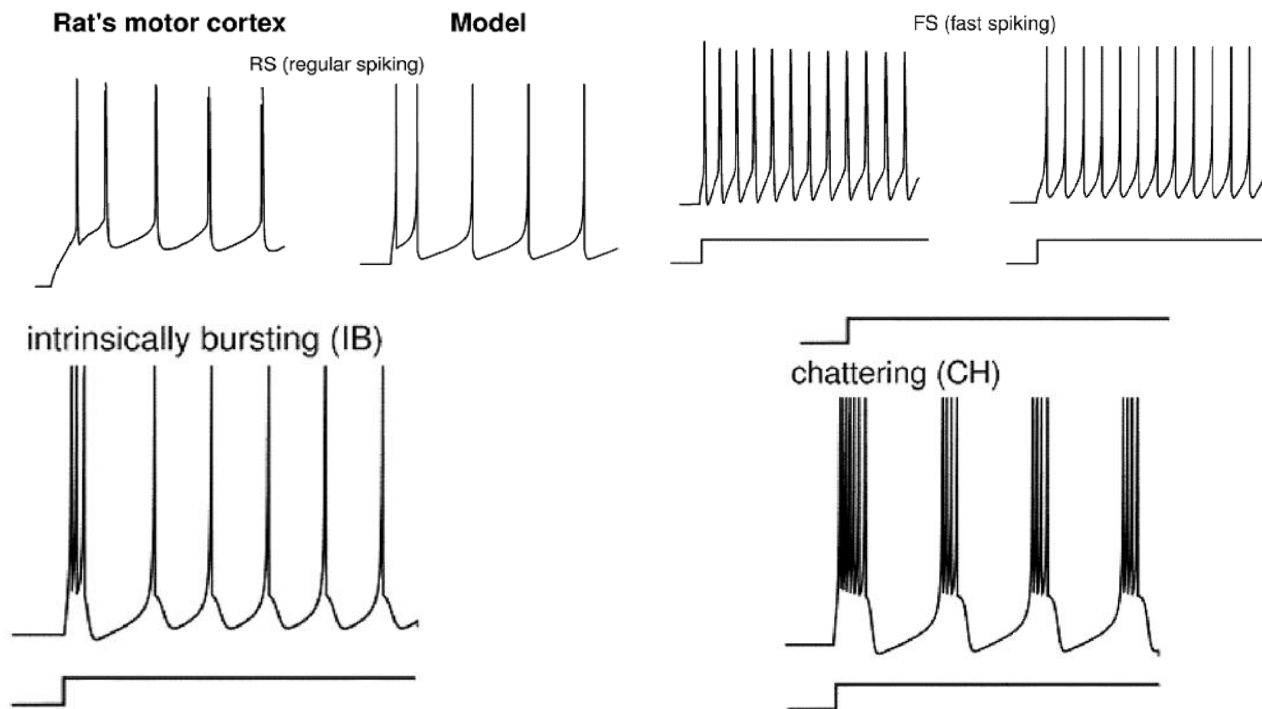
So we can summarize the adaptation mechanism into different steps:

1. A constant depolarizing current I is applied.
2. The voltage v rises toward the unstable equilibrium. Because v is rising, u slowly starts to rise as well.
3. The threshold is crossed, positive feedback kicks in, and the neuron fires ($v = 30$).
4. The reset rule triggers. Voltage drops to c . But crucially, u jumps up by the value d .
5. Now, the neuron tries to fire again because the constant current I is still there. However, u is now much larger than it was before the first spike.
6. Because u is larger, the $-u$ term in the voltage equation exerts a massive negative drag. It takes much longer for the external current I to overcome this drag and push the voltage back up to the threshold.
7. **Result:** The time between spikes (Inter-Spike Interval) gets longer and longer. The firing frequency decreases. The neuron has adapted.



8. If the input stops, u will slowly relax back to baseline governed by the parameter τ , and the neuron is ready to respond to novel stimuli again.

By tweaking the four dimensionless parameters (a, b, c, d), the Izhikevich model can replicate the firing patterns of almost every known type of cortical neuron (Regular Spiking, Intrinsically Bursting, Chattering, Fast Spiking, etc.).



- **Regular Spiking:** when presented with a prolonged stimulus, the neurons fire a few spikes with short interspike period and then the period increases (Frequency adaptation). Increasing the strength of the input, increases the interspike frequency, though it never becomes too fast because of large spike after hyper polarizations.
- **Fast Spiking:** neurons can fire periodic trains of action potentials with extremely high frequency practically without any adaptation (slowing down).
- **Intrinsically Bursting:** neurons fire a stereotypical burst of spikes followed by repetitive single spikes.
- **Chattering:** neurons can fire stereotypical bursts of closely spaced spikes.

The Reverse Potential

Up to this point, the basic Leaky Integrate-and-Fire (LIF) model has treated synapses as completely **memoryless**.

- In the standard equations, every time a presynaptic neuron fires, the synapse simply adds a fixed "weight" or voltage amount (J) to the postsynaptic neuron.
- If a presynaptic neuron fires once, it adds J .
- **The High-Frequency Problem:** if a presynaptic neuron fires at a very high frequency (e.g., one spike every 5 milliseconds), the standard LIF model just keeps adding J over and over again. The membrane potential unrealistically shoots up toward infinity.

Real biology does not allow voltages to rise infinitely. To understand why, we have to look at what actually happens when a synapse opens:

1. **The Channel Opens:** A neurotransmitter binds to a receptor and opens an ion channel (it does not physically "throw" ions into the cell; it just opens a door).
2. **The Driving Force:** Ions flow through this door *only* because there is an imbalance (a concentration and electrical gradient) between the inside and outside of the cell.

3. **Reaching Equilibrium:** If the presynaptic neuron fires constantly, the "door" stays open. Ions rush in, but eventually, so many ions have crossed the membrane that the internal and external environments balance out.
4. **The Thermic Analogy:** if you open a window between a hot room and a cold room, heat flows. But once both rooms reach the same temperature, keeping the window open doesn't cause any more heat to flow.

This biological point of equilibrium is called the **Reversal Potential** (E_{rev})

- **Definition:** The reversal potential of an ion is the specific membrane voltage at which there is **no net overall flow** of that ion, even if its channels are completely open.
- **The Consequence:** As the neuron's membrane voltage gets closer to the reversal potential, the "driving force" pushing the ions inside gets weaker and weaker, until it eventually drops to zero.

The current version of our LIF model uses current-based synapses, which ignore this driving force. It assumes the driving force is always exactly the same, no matter what the current membrane voltage is. To achieve this next level of realism—and to stop our simulated neurons from shooting up to infinite voltages during high-frequency inputs—we must introduce a mechanism where the synaptic input naturally decreases as the neuron approaches the reversal potential.

Study of Mazzoni and his PHD: Current-Based (CUBA) vs. Conductance-Based (COBA) Synapses

We are looking at two different ways to model the input arriving at a synapse. They look mathematically similar, but the physical implications are very different.

Current – based model (CUBA)

This is the standard Leaky Integrate-and-Fire model.

$$\tau_m \dot{V}(t) = -V(t) + J_{syn} S_{syn}(t)$$

Every time a spike arrives ($S_{syn} = 1$), a fixed amplitude of current (J_{syn}) is injected into the neuron.

Problem: The post-synaptic potential is fixed. It completely ignores the biological reality of the Reversal Potential. If you bombard the neuron with inhibitory inputs, the voltage $V(t)$ will drop infinitely into the negative, which is biophysically impossible.

Conductance – Based Model (COBA)

This model introduces the Reversal Potential (V_{syn}) to simulate the "thermic balance" we discussed earlier.

$$\tau_m \dot{V}(t) = -V(t) + g_{syn} S_{syn}(t) (V_{syn} - V(t))$$

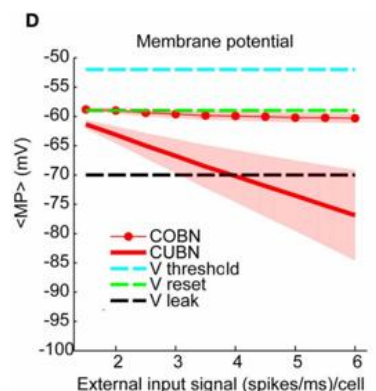
Instead of a fixed current, the synapse provides a fixed *conductance* (g_{syn} }, representing how wide the channel opens). The actual current that flows depends on the **driving force**: $V_{syn} - V(t)$.

The result of this change is that If the membrane potential $V(t)$ equals the reversal potential V_{syn} , the driving force is 0. No current flows.

Before looking at the statistical outputs (like firing rates), the actual traces of the membrane potential in a recurrent network subjected to **strong inhibition** were analyzed.

To analyze the graphs accurately, the voltage scale was refined into four distinct reference levels:

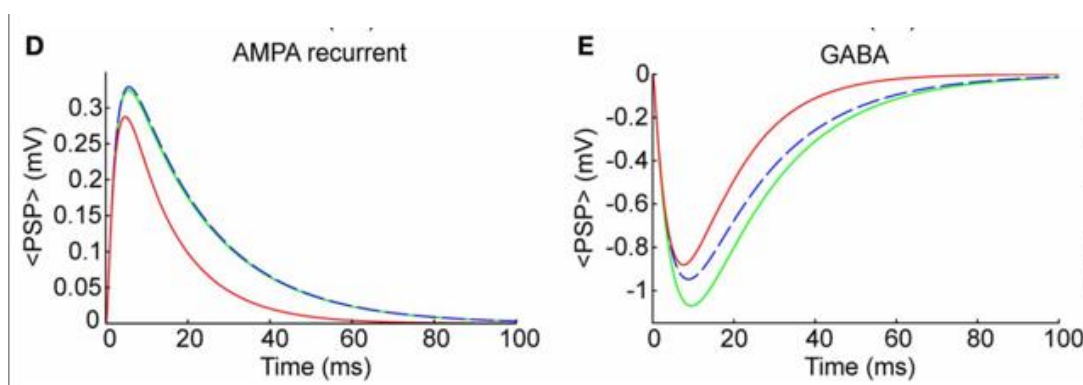
1. **Above V_{thr} :** This area is dynamically unreachable because the moment a neuron hits the threshold, it fires and resets.
2. V_{thr} : The firing threshold.
3. V_{eq} : The natural resting potential of the neuron.



4. V_{res} : The potential the neuron drops to immediately after a spike. (Note: For a more elegant and realistic model here, V_{res} is treated as slightly different from V_{eq} .)

To understand the COBA model, you must know where the reversal potentials sit on the voltage scale relative to the resting potential (V_{eq}) and the firing threshold (V_{thr}).

- **Excitatory Synapses (AMPA):** $V_E > V_{thr}$: The reversal potential is situated above the firing threshold. Because the neuron fires and resets as soon as it hits V_{thr} , the membrane potential never actually reaches V_E . However, as $V(t)$ gets closer to threshold, the excitatory drive gets slightly weaker.
- **Inhibitory Synapses (GABA):** $V_I < V_{eq}$: The reversal potential is situated below the resting equilibrium. The Self-Stabilizing Effect: As a neuron receives more and more inhibitory input, its voltage drops. But as it drops closer to V_I , the inhibitory effect becomes weaker and weaker. At the same time, because the voltage is now very low, it is extremely far away from the excitatory V_E , meaning any excitatory input that arrives will have a massively magnified, strong effect. This mechanism strongly stabilizes the network!



Fascinating Biological Tangent (Cherubini's Discovery): During fetal development, the baseline membrane potential of neurons is exceptionally low—so low that it sits *below* the GABA reversal potential. Because of this, when GABA channels open in a fetus, the ions flow the opposite way to reach equilibrium, making GABA an *excitatory* synapse during development, even though it is purely inhibitory in adults!

Why does the COBA model stabilize so strongly under heavy inhibition? The graphs reveal a continuous "push-pull" biophysical mechanism based on the distance to the reversal potentials:

1. **Inhibition becomes Weaker (The Push):** You hyperpolarize once, you hyperpolarize twice... but by the third time, the membrane voltage is getting very close to the inhibitory reversal potential (V_I). Because the driving force $V_I - V(t)$ is shrinking, every subsequent inhibitory spike has less and less effect.
2. **Excitation becomes Stronger (The Pull):** At the exact same time, because your membrane potential has dropped so low, you are now much *further away* from the excitatory reversal potential (V_E , which is situated way up above V_{thr}). Because this distance is now huge, the driving force for excitation is massive. If even a single excitatory spike arrives, its depolarizing effect will be significantly stronger than normal.

Since the COBA model inserts $V(t)$ into the synaptic term, the differential equations become much harder to solve (often requiring numerical integration). Textbooks traditionally argued that the COBA complexity was unnecessary (Occam's Razor) because "the results are basically the same."

Metric 1: Average Firing Rate (The 1st Statistical Moment, μ)

- **Result:** The textbooks were right. Despite the mechanical differences at the membrane level, if you look purely at the average number of spikes a neuron produces over time, the CUBA and COBA models yield almost identical results.

Metric 2: Variability / Noise (The 2nd Statistical Moment)

- **Result:** The textbooks were wrong. The models behave entirely differently when looking at the *variance* of the spike times.

To measure how "noisy" or variable a neuron's firing is, we use the Coefficient of Variation:

$$CV = \frac{\sigma}{\mu}$$

(Where σ is the standard deviation and μ is the mean).

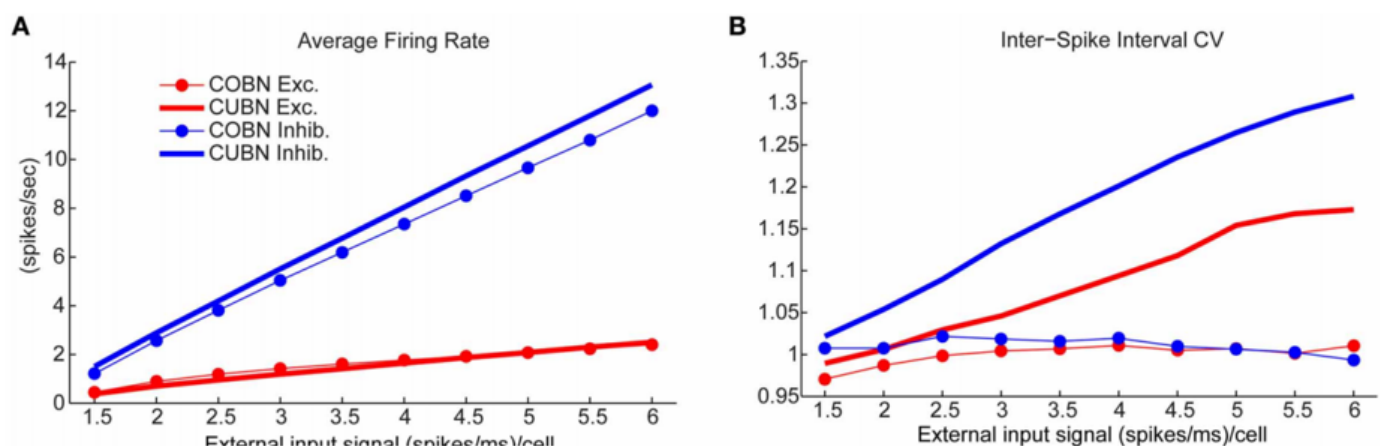
- **Biological Reality:** If you measure an isolated neuron in the real cortex, its CV is almost exactly **1**.
- **The Poisson Distribution:** In statistics, a probability distribution with a CV of 1 (where the mean equals the variance) is the Poisson distribution. It describes the probability of discrete events (spikes) occurring in a fixed interval of time (λ). Real cortical neurons act like Poisson processes.
- **Model Performance:**
 - **COBA (Conductance):** Successfully replicates biology. Regardless of the input level, the network stabilizes, and the neurons maintain a Poisson-like distribution ($CV \sim 1$).
 - **CUBA (Current):** Fails to replicate biology. The CV fluctuates wildly depending on the input. It is far too noisy.

Results of the study

When choosing which model to implement in a simulation or neuromorphic circuit, the answer depends entirely on your research goal:

- **If you only care about the Average Firing Rate:** Use the **Current-Based (CUBA)** model. It is mathematically simpler, saves computational power, and gets the average rate perfectly correct. The size (amplitude) of the voltage jumps caused by synapses is rigidly fixed.
- **If you care about Spike Timing Variability (Fluctuations & Noise):** You must use the **Conductance-Based (COBA)** model. CUBA fails to capture the realistic Poisson-like statistical variance ($CV = 1$) of real cortical neurons. In this model, the amplitude of the voltage jumps dynamically changes.

While individual neurons will still show local fluctuations in their membrane potential, this dual mechanism ensures that the **average membrane potential of the entire network is strongly stabilized**. The network actively resists being pushed too far in any one direction.



Exponential Integrate Fire Neuron

In biology, an action potential is not instantaneous; it lasts approximately **1 ms**. The classic Leaky Integrate-and-Fire (LIF) model ignores this completely, treating the spike as an instant event.

If a typical cortical neuron fires at **10 spikes per second** (10 Hz), the Inter-Spike Interval (ISI) is **100 ms**. In this case, skipping **1 ms** of spike duration doesn't change the overall dynamics much.

However, if you are studying systems with extremely high firing rates—such as the cerebellum, epileptic seizures, or clinical applications like Deep Brain Stimulation (DBS) for Parkinson's disease (which operates at **130 pulses per second**, meaning an ISI of just **8 ms**)—that single millisecond of spike generation becomes critical and cannot be neglected.

Instead of abandoning the LIF model for computationally heavy, multi-dimensional models (like Hodgkin-Huxley), researchers (like Nicolas Brunel) found a highly elegant solution: **add just one exponential term to the standard LIF equation**.

The continuous dynamics of the Exp-IF model are governed by this equation:

$$C \frac{dV}{dt} = -g_L(V - E_L) + \psi(V) + I, \quad \psi(V) = g_L \Delta_T \exp\left(\frac{V - V_T}{\Delta_T}\right)$$

- **The Linear Term** ($-g_L(V - E_L)$): This is the classic "leaky" part of the LIF model. It pulls the voltage back toward the resting equilibrium.
- **The Exponential Term** ($\psi(V)$): This is the new addition. It simulates the rapid opening of sodium channels during an action potential.
- **V_T (Spike Initiation Threshold)**: This is the critical voltage where the exponential term "wakes up" and takes over.
- **Δ_T (Slope Factor)**: This parameter controls how sharply the action potential rises once the threshold is crossed.

Unlike the standard LIF model, which draws an artificial line to count a spike, the Exp-IF model uses two distinct thresholds to naturally simulate the rising phase:

1. V_T (The Unstable Equilibrium): when the voltage V reaches V_T , the derivative crosses zero and enters a region of positive feedback. The exponential term dominates, and the voltage shoots up rapidly. You are actively simulating the rising phase of the spike.
2. V_{peak} (The Hard Cutoff): Because it's an exponential growth, the voltage would technically go to infinity. To stop this, a second threshold is imposed (e.g., 30 mV).

When the spike finishes rising and hits the cutoff, the standard reset applies:

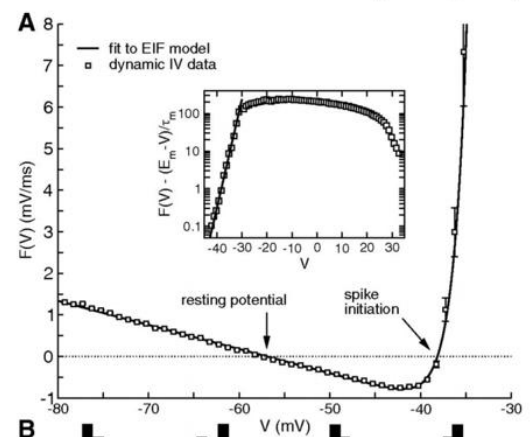
$$\text{if } V \geq V_{peak}, \text{ then } V \leftarrow V_{reset}$$

But the difference is that the, instead of having the whole equation designed as implementing as a second order equation polynomial and defining both resting potential and spike initiation, you have one term of the equation giving you the resting potential, and a second term of the equation giving you the spike initiation threshold.

There is a study which tested various models (LIF, Quadratic IF, and Exp-IF) against real biological data (intracellular recordings of sodium channels).

- **Quadratic IF (QIF)**: The parabola rises too slowly compared to biology.
- **Leaky IF (LIF)**: Completely misses the rising curve.
- **Exponential IF (EIF)**: Fits the real sodium channel activation almost perfectly!

The Exp-IF model is incredibly powerful. It captures the complex biophysics of sodium channels and high-frequency firing perfectly, but because it only adds a single exponential term, it remains computationally lightweight. If the firing rate is low, it behaves exactly like a standard LIF model; if the firing rate is high, the exponential term kicks in to provide the correct dynamics.



Adaptive exponential Integrate Fire Neuron

As we saw earlier, Eugene Izhikevich developed a brilliant, computationally light model that successfully captured neuronal adaptation (using the quadratic model and the recovery variable u).

Meanwhile, the "French community" (researchers like Brette and Gerstner) had developed the highly elegant **Exponential Integrate-and-Fire (Exp-IF)** model, which perfectly simulated the fast opening of sodium channels. However, their model lacked adaptation. Feeling "left behind" by Izhikevich's publication, they decided to update their model.

Their solution? They literally took Izhikevich's adaptation mechanism and plugged it directly into their Exp-IF model. To mask this direct copy-paste, they simply changed the name of the adaptation variable from Izhikevich's u to w .

The model

The beauty of the Integrate-and-Fire framework is its modularity. The AdEx model is simply the sum of parts we already know: **LIF + Exponential Term + Izhikevich Adaptation = AdEx Model**

Here are the two coupled differential equations that define the continuous dynamics :

1. Membrane Potential Equation: $C \frac{dV}{dt} = -g_L(V - E_L) + \psi(V) + I, \quad -w$
2. Adaption Variable Equation: $\tau_w \frac{dw}{dt} = a(V - E_L) - w$

As we see, the first one contains the classic Leaky term, the Exponential term (simulating the fast sodium channel spike) and the new Adaptation term (providing the negative feedback/drag). The second one is the exact same dynamic as Izhikevich's second equation. The variable w relaxes toward an equilibrium over a time scale dictated by τ_w , and it grows in response to the membrane potential (V). Subthreshold depolarization causes adaptation to slowly build up.

Just like the Izhikevich model, the real power of adaptation kicks in after a spike occurs.

$$\text{if } V \geq V_{peak} \rightarrow \begin{cases} V \rightarrow V_{reset} & (\text{The voltage drops back down}) \\ w \rightarrow w + b & (\text{Spike - triggered adaptation}) \end{cases}$$

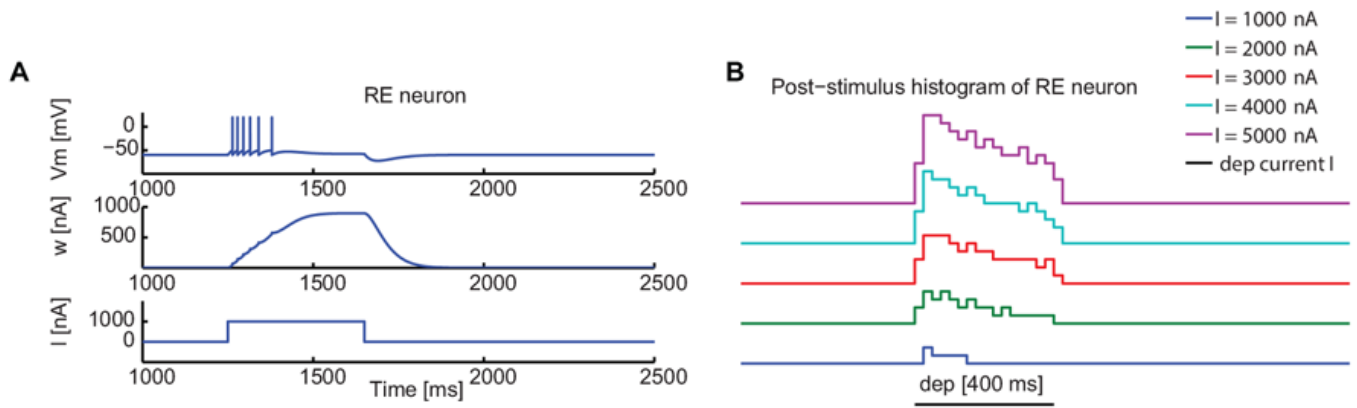
Again, this is exactly the same as Izhikevich's reset rule ($u \leftarrow u + d$), just using the letter b instead of d . By tweaking the parameters (a for subthreshold adaptation and b for spike-triggered adaptation), this model can reproduce a massive variety of complex biological firing patterns (bursting, chattering, adapting, etc.).

Important Note: The neurophysiological comparison (done by a swiss neurophysiologist Gerstner) only proved that the **exponential term** is the best mathematical approximation for the massive, fast voltage change caused by the opening of sodium channels. It did not test the adaptation (w) part. The adaptation mechanism is purely theoretical mathematics borrowed from Izhikevich because "it works."

Firing for rebound from inhibition

To understand this phenomenon, we are looking at a specific type of cell: the **Reticular Neurons of the Thalamus (RE cells)**. These are heavily inhibitory neurons known for having extremely **strong adaptation**. We are still using the AdEx model (or the Izhikevich model, the logic is identical). In these specific neurons, the parameter a (subthreshold adaptation) is very large. This means the adaptation variable w is highly sensitive to the resting membrane potential ($V - E_L$).

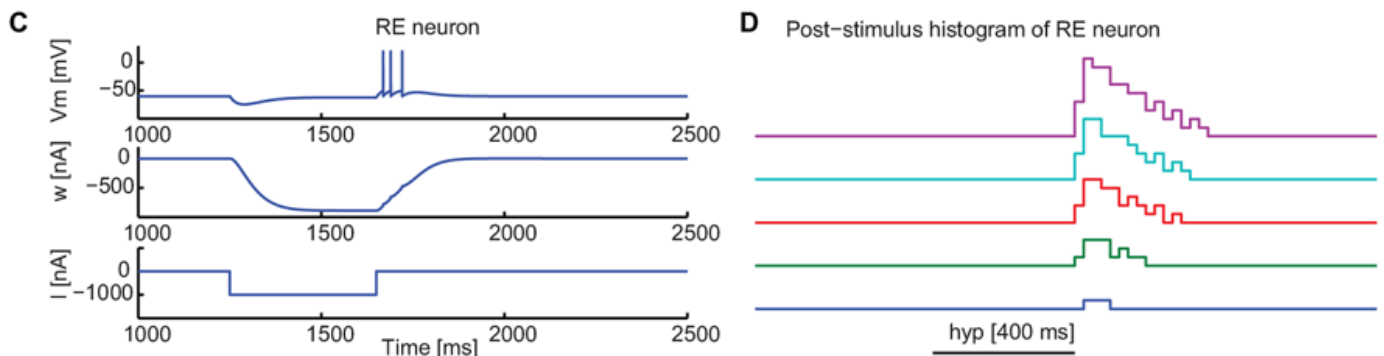
Strong Excitatory Current



As shown in the image, if you inject a strong, constant *excitatory* current ($I > 0$) into this neuron:

1. The voltage V rises above the leak/equilibrium potential (E_L).
2. Because $(V - E_L)$ is positive, the derivative of w becomes positive. The adaptation variable w starts to grow.
3. Because w is subtracted in the main voltage equation ($-w$), it acts as a strong inhibitory drag.
4. The neuron fires a few spikes initially, but because w grows so large, the $-w$ term eventually overwhelms the $+I$ term. **The neuron stops firing completely**, even though the excitatory current is still being applied.

Strong Inhibitory Current



In this case, we have two phases: the build – up (during the inhibition) and the release (which leads to the rebound burst).

In the build – up phase:

1. You inject a negative current (I_{ext} is very negative). The neuron hyperpolarizes; V drops far below E_L .
2. Now, look at the adaptation equation: $(V - E_L)$ is heavily **negative**.
3. Because the derivative $\frac{dw}{dt}$ is negative, the value of w itself drops and eventually becomes a **large negative number**.
4. Now, look back at the main voltage equation: you are subtracting a negative number ($-(-w) = +w$). The adaptation term has paradoxically become a **strong excitatory drive!**
5. *Why doesn't the neuron fire right now?* Because the massive negative external current (I_{ext}) is still holding the neuron down, masking the internal excitatory drive from w .

Then we have the release phase:

1. Suddenly, you turn off the inhibitory current. The negative external current (I_{ext}) instantaneously drops to 0.

2. The membrane voltage is now free to move, but the adaptation variable w does not change instantaneously. It takes time to decay back to zero (dictated by the time constant τ_w).
3. For a brief window of time (τ_w long), the external inhibition is gone, but the strong internal excitatory drive ($+w$) is still present!
4. This massive internal push rapidly depolarizes the neuron, causing it to fire a rapid burst of spikes.

This entire process is called **Rebound Bursting** or **Firing for Rebound from Inhibition**.

It is a beautiful example of how biological variables working on different time scales interact. The neuron doesn't fire *because* of the inhibition (it stays silent while inhibited); it fires because the sudden *removal* of the inhibition leaves the slow-moving adaptation variable (w) "caught" in an extremely excitatory state.

Summary and limitations of all the models

To wrap up the single-neuron dynamics, we can define a practical guide on how a computational neuroengineer chooses the right model for the job. You never start with the most complex model; you start simple and add realism only when the phenomenon demands it.

- **Cellular Automata & Rate Models:** Too simple. We completely discard them for neuroengineering purposes because they lack the necessary biological dynamics.
- **Leaky Integrate-and-Fire (LIF):** The starting point. This is the baseline model. If you just need a network of spiking neurons and don't care about specific biological quirks, you stop here.
- **AdEx / Izhikevich Models:** The next step up. You use these when your specific research requires **adaptation** or a **realistic action potential duration**.
 - *Example:* Modeling mechanoreceptors in the fingertip, which are strongly adaptive. If you use a LIF model here, your simulation will completely miss the core biological reality.
- **Hodgkin-Huxley Model:** The most complex point-model. You only use this when you don't just care about the *effects* of an ion channel, but you need to study the **dynamics of the ion channels themselves**.
 - *Example:* Studying the effect of a specific drug, a neurotoxin, or even just a massive intake of potassium (the "banana" example) on the specific gating variables of the cell membrane.

Every single model we have studied so far (from LIF to Hodgkin-Huxley) shares one massive limitation: they are all Point Neurons.

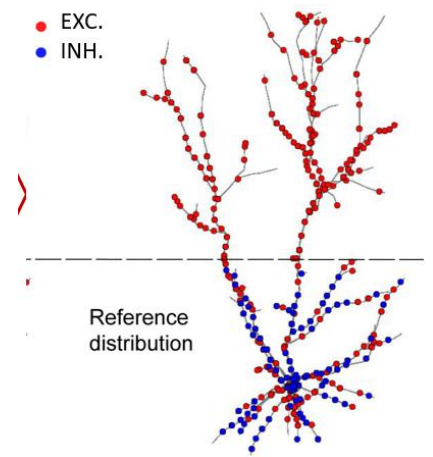
They describe the mathematical evolution of the membrane potential (V) at a single, abstract point in space. They assume the neuron has no physical body, no shape, and no spatial dimensions. While abstract math is fine for studying general network behaviors, real neurons are made of physical fat and proteins. While the soma (body) is tiny (micrometers), the **dendritic arborizations** can be huge, and **axons** can stretch for centimeters or even meters (e.g., down the spinal cord).

This physical geometry becomes absolutely critical in **Bioelectronic Medicine** (e.g., Deep Brain Stimulation or neuroprosthetics):

- **The Drug vs. Electrode Difference:** If you take a drug, it washes over the whole brain chemically. But if you implant an electrode, the stimulation is highly localized and selective.
- **The Volume of Tissue Activated (VTA):** When an electrode fires, it creates an electrical field. Often, the size of this electrical field (the VTA) is roughly the same size as the neuron itself.

Look at the morphological model of the Pyramidal cell in the image. It shows the sprawling dendrites (top), the soma (middle), and the axon (bottom). If you use a real physical electrode (like the laminar electrodes, which have multiple contact sites fractions of a millimeter apart), the exact physical placement changes everything:

- **Stimulating the top:** You are hitting the dendritic tree (likely excitatory synapses). The signal must physically travel down to the soma.
- **Stimulating the middle:** You are hitting the soma directly.
- **Stimulating the bottom:** You are hitting the axon. Furthermore, thicker axonal fibers are harder to entrain (depolarize) than thinner ones due to their physical properties.

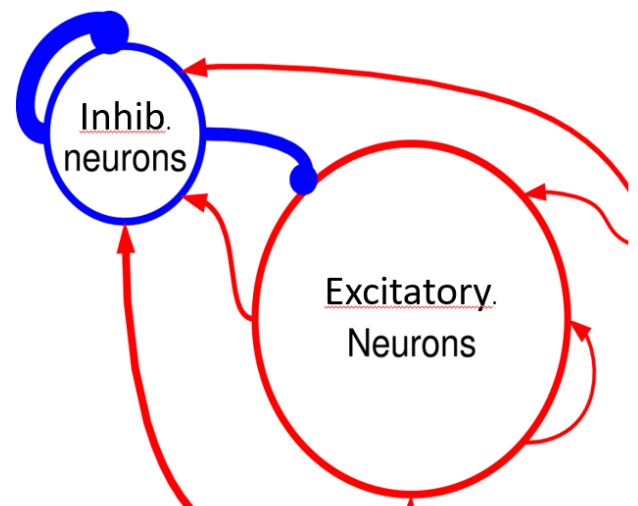


If you want to map exact neurophysiological recordings to specific neuron parts, or if you want to predict exactly how a localized electrical pulse will activate a neuron, you must abandon the "Point Neuron" simplification. You have to trade computational simplicity for spatial realism and move to **Morphological Models** (multi-compartment models).

Cortical Recurring Networks

Just as the Leaky Integrate-and-Fire (LIF) model is the standard baseline for studying single neurons, the **Local Cortical Recurrent Network** is the standard baseline for studying network dynamics.

- **Why "Recurrent"?** Unlike feedforward networks (where layer A simply passes information to layer B without loops), a recurrent network means the neurons are highly interconnected and continuously interact with each other. This is how the real cortex actually works.
- **Why "Local"?** The network represents a physically and functionally defined, small chunk of the cortex.



To understand the network, you must understand the physical space it occupies. The cortex is organized into functional units called **Cortical Columns**.

- **Horizontal Position (The Surface):** Moving across the surface of the brain changes the **Function** (e.g., moving from a column that processes visual data to one that processes auditory data).
- **Vertical Position (The Depth):** Moving down into a specific column reveals **6 distinct Layers**. The depth dictates the **Morphology and Connectivity** of the neurons.
- **The Specific Model:** The reference network shown in the image specifically simulates **Layer 4 of a visual cortex column**.

How do these columns actually process the world? The visual and tactile systems work in surprisingly similar ways because they both have to solve a **2D representation problem** (the retina is a 2D surface, and the fingertip is a 2D surface). In particular, the sensory field (what you see or touch) is divided into tiny "pixels". A specific group of neurons in the cortex is assigned to monitor *only* one specific pixel of that 2D space. That pixel is called the population's **Receptive Field**.

Some tests on monkeys were carried out in order to study the specificity of the visual cortex. When you insert an extracellular electrode into the visual cortex of a monkey, you know you are in the visual area, but you have *no idea* which specific "pixel" of the visual field those specific recorded neurons are looking at. (Just like probing a prosthetic nerve doesn't immediately tell you if it's the thumb or the pinky finger). To find out, the team flashed random lights (flickers) very quickly across every single pixel of a screen. A pixel-wise correlation was performed between the exact timing of the flashes and the firing rate of the recorded neuron. The correlation was exactly 0 for flashes all over the screen, except for one tiny specific spot where the correlation was 0.95. Thanks to this test, they were able to map that neuron's receptive field.

Why do we find *populations* of neurons dedicated to a single receptive field instead of just one single neuron? To answer this question, we have to think about two related concepts: Darwinian robustness and topography. The first tell us that visual and tactile information is critical for survival. If only one neuron processed a specific pixel and that neuron died, you would develop a blind spot. By having a whole population of interconnected neurons doing the same job, the system is incredibly robust. Furthermore, calcium imaging shows that neurons processing the exact same pixel are physically located close to each other in the cortex. This reflects the topography of our visual and tactile cortex.

Building of the Reference Network

Now that we know we are modeling a Local Cortical Recurrent Network (Layer 4, Visual Cortex), how do we actually build it in a simulation? Let's consider an example of a network with **N = 5,000 neurons**.

Note: 5,000 is a good number because it's computationally light for a modern laptop using LIF neurons, but large enough to show complex, emergent, nonlinear dynamics that smaller networks of 50 neurons cannot show

First of all, we have to consider the anatomical factor. Before connecting them, we must decide what kind of neurons make up these 5,000. It was proved that, in the cortex, the biological ratio between excitatory and inhibitory neuron is roughly 4 to 1. This means that in our network we'll have $N_E = 4000$ and $N_I = 1000$.

Then, we have to create the **connectivity matrix**. We need to create a $5000 * 5000$ binary matrix (1s and 0s) to represent the physical anatomy. We have to avoid the "mean – field" approximation which basically states: every neuron is connected to every other neuron. Connecting every neuron to every other neuron makes the math perfectly solvable, but it produces weird, biologically inaccurate results.

We can instead use the **Random connectivity** based on biological probabilities. For neurons sitting in the same column, processing the same receptive field, the probability of a directed connection existing is 20%. To build the matrix, we have to generate a random number between 1 and 0 for every cell. If it's < 0.2 , write a 1 (connected). Otherwise, write a 0 (not connected). The obtained matrix will follow the following rules:

- **The Diagonal is 0:** Neurons do not synapse onto themselves (no "autopsies").
- **It is Asymmetric:** Just because Neuron A connects to Neuron B, it does not mean B connects to A.

Because $p = 0.2$, each neuron will, on average, receive inputs from 1,000 other neurons (20% of 5,000). Of those 1,000 incoming connections:

- ~800 will be Excitatory (80%)
- ~200 will be Inhibitory (20%)
- **Crucial Note:** Because this is a random process, some neurons will have 805, some 792. This introduces **Static Architectural Noise (Heterogeneity)** into the network, making it biologically realistic.

Now we have the anatomy (the 1s and 0s), but we need the actual weights (the J matrix) to define how much voltage a spike transmits. Having 25 million different, unique J parameters is impossible to manage. Instead, we simplify the matrix into **4 quadrants**, assigning a single parameter to each type of connection:

1. J_{EE} : Excitatory $\rightarrow$ Excitatory
2. J_{EI} : Excitatory $\rightarrow$ Inhibitory
3. J_{IE} : Inhibitory $\rightarrow$ Excitatory
4. J_{II} : Inhibitory $\rightarrow$ Inhibitory

How do we determine the values of these J weights without having to measure them in a lab? We use a fundamental computational principle of the healthy nervous system (also seen in the cellular automata model): Over a sufficiently long time and space, a healthy nervous system must maintain a strict **Balance between Excitation and Inhibition (E/I Balance)**. Too much excitation leads to epilepsy, while too much inhibition freezes the system that cannot transmit information.

If we know there are **4 times as many Excitatory neurons** as Inhibitory neurons (a 4:1 ratio in the anatomy), how do we balance overall energy? The only way to achieve E/I balance is to make the **Inhibitory synapses roughly 4 times stronger** than the Excitatory synapses. This strength is represented by thicker arrows in the image above and mathematically $J_{inhibitory}$ (weights of the inhibitory connections) will be much greater than $J_{excitatory}$ (weights of the excitatory connections).

The network design relies on two complementary forces:

- From an anatomical perspective, we have an excitatory dominance with a ratio of 4:1;
- Looking at the synaptic weights, we have an inhibitory dominance with a ratio of 4:1;

This allows us to reach the perfect E/I balance.

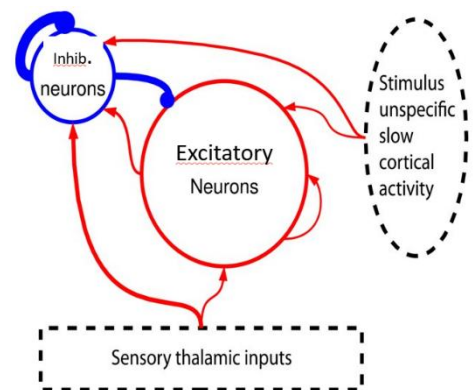
Introduction of external stimuli

In Neuromorphic Engineering, when we build cortical models like recurring networks, our primary objective is to capture how natural neuronal networks encode and transfer information. Specifically, this model focuses on how a fragment of the cortex processes **external sensory information**.

However, to accurately simulate biological information transfer, we must accept a fundamental rule: **we must simulate noise**. The brain is not a deterministic machine; it is highly noisy at every single stage of processing.

It is crucial to understand that the cortex *never* receives raw data directly from the external world. Sensory information undergoes multiple processing steps:

1. **The Transducer:** This is the peripheral receptor (e.g., photoreceptors for vision, mechanoreceptors for touch, chemical receptors for smell).
2. **Intermediary Steps:** The signal travels through various neural pathways.
3. **The Sensory Thalamus:** All sensory inputs (except smell) must pass through the thalamus before reaching the cortex. For the visual pathway, this specific thalamic region is called the **Lateral Geniculate Nucleus (LGN)**.



The Rule of Long-Range Connections: A useful rule of thumb in neurobiology is that long-range inter-area connections are **strictly excitatory**. Therefore, in our model, the input from the thalamus to the cortex is purely excitatory (>0); there is no excitatory/inhibitory balance at this specific transmission step.

What kind of message is the thalamus sending? It usually conveys simple information, such as the level of pressure (tactile) or spatial contrast (vision). The average firing rate of the thalamic neurons is a **monotonic function** of the input stimulus intensity (e.g., contrast level c). We define this average firing rate as $v_0(c)$.

If the thalamus has an average firing rate of 3 spikes per second, it does not act like a perfect metronome (spiking exactly every 333 milliseconds). Furthermore, not every neuron in our cortical network will receive the exact same identical signal. We must introduce variability. To mathematically approximate the variability of a single neuron's firing, we use a **Poisson process**. A Poisson process perfectly describes discrete, independent events (like spikes) occurring over time with a fixed average inter-event interval. Unlike a Gaussian distribution (which requires two parameters: mean and standard deviation), a Poisson distribution is defined by just **one parameter**. Because you cannot have negative events (you can't have -1 spikes), as the average firing rate increases, the distribution naturally broadens. In our model, each cortical neuron receives a specific number of inputs drawn from a Poisson distribution based on the average thalamic rate $v_0(c)$. This creates **input heterogeneity**.

The input from the thalamus is not the only thing driving the cortex. In fact, we must simulate the strongest source of noise in the brain: **the rest of the brain itself**.

Whether you are sleeping, resting, or playing chess, the baseline level of activity in your brain is roughly the same. Our model represents a tiny fragment of the cortex (equivalent to a single **receptive field**, like one specific square degree of your visual field). However, this fragment is constantly bombarded by signals from other neurons processing completely different receptive fields. To model this background chattering, we use the so called **Brown Noise**. If you record resting-state brain activity, the spectrum looks like Brownian noise ($1/f^2$). In this spectrum, low frequencies dominate (large, slow fluctuations) with smaller, faster fluctuations riding on top. However, true Brownian motion is biologically impossible because the fluctuations can mathematically grow to infinity. Biology is always bounded. To solve this, we model the spontaneous ongoing activity (μ_c) using an Ornstein-Uhlenbeck process. This is essentially bounded Brownian noise. It allows for the $1/f^2$ spectrum but uses a "leak" term to constantly pull the value back toward a baseline, preventing it from going to infinity.

Mathematical Inputs of the Model

To build a functional model, we must convert the physiological realities of the brain into mathematical equations. In our simulated cortical network, a specific neuron k receives three distinct types of input at any given moment:

1. Recurrent input from local cortex

This is the internal communication within our local network.

Neuron k receives spikes from its neighboring neurons.

From a mathematical point of view, The input is defined as the sum over all other neurons (n) except k itself.

- $C_{k,n}$: the structural connectivity weight (the "synapse" strength) from neuron n to neuron k . If this is 0, they are not connected.
- $\delta(t - t_n - \tau_L)$: The Dirac delta function. This mathematically represents the discrete arrival of a spike, where t is the current time, t_n is the exact time neuron n fired and τ_L is the **latency** (usually around 1 ms). This is the physical time it takes for the electrical spike to travel down the axon and cross the synaptic cleft to reach neuron k .

$$S_k^{pyr/int}(t) = \sum_{n \in pyr/int} C_{k,n} \delta(t - t_n^* - \tau_L)$$

"Ongoing activity" cortical inputs

$$S_k^C(t) = Poiss_k(v^C(t))$$

$$\tau_c \dot{v}^C(t) = \mu_c - v^C(t) + \sigma_c \sqrt{2\tau_c \eta(t)}$$

Ornstein Uhlenbeck process

"Sensory" thalamic inputs

$$S_k^{thal} = Poiss_k(v^{thal}(t))$$

2. Sensory input from the thalamus

This is the external stimulus arriving from the outside world, via the peripheral transducer and the sensory thalamus (e.g., the Lateral Geniculate Nucleus for vision). As discussed in the previous lesson, the thalamus has an average firing rate based on the stimulus intensity (e.g., watching a dynamic movie). Neuron k does not simply receive a smooth, continuous "average rate." Instead, it receives a discrete number of spikes drawn from a **Poisson process** parameterized by that average rate. If the average is 3 spikes/sec, most neurons receive 3, but due to stochasticity, some might receive 2, 4, or even 10. This perfectly models the biological input heterogeneity.

3. Background noise from the rest of the brain

This represents the spontaneous, task-unrelated activity bombarding our local network from the rest of the brain. As established earlier, this is modeled as **brown noise** ($1/f^2$), meaning it is dominated by slow, high-amplitude fluctuations. Because pure brown noise can drift to infinity, we use the **Ornstein-Uhlenbeck (OU) process**. The OU process generates white noise at very low frequencies, but once it crosses a critical time scale (τ_c), it behaves like bounded $1/f^2$ brown noise. Just like the sensory input, the OU process determines a baseline firing rate for the cortex, and the actual spikes delivered to neuron k are again drawn via a Poisson process to maintain biological randomness.

Focus (noise types):

- **White Noise:** A flat spectrum. All frequencies have equal power (generated by independent random events, like a Gaussian distribution).
- **Pink Noise:** A $1/f$ spectrum. More power at low frequencies. (Common in technology, rare in biology).
- **Brown Noise:** A $1/f^2$ spectrum. Heavily dominated by low frequencies. (Characteristic of natural systems and random walks).

When scientists measure the brain using extracellular electrodes (like an EEG or Local Field Potential, which record the average activity of thousands of neurons), the resting state always shows a $1/f^2$ brown noise spectrum. A French team believed the brain actually generates flat **white noise**. However, they argued that the physical extracellular medium (the fluid and tissue between the neurons and the electrode) acts as a low-pass filter. The medium physically absorbs and filters out the high frequencies, leaving only the low frequencies for the electrode to read.

On the other hand, a German team argued that the medium's conductance is flat (it doesn't filter anything). The $1/f^2$ spectrum is the true nature of brain oscillation over large scales. Think of the brain as a massive collection of oscillators. Brain activity has natural timing errors called **jitter** (e.g., a ± 2 ms timing variance).

- **High Frequencies (e.g., 100 Hz):** The period of a 100 Hz wave is 10 ms. A 2 ms jitter is massive relative to a 10 ms period. Therefore, high-frequency oscillations across thousands of neurons quickly fall **out of phase**. When an extracellular electrode averages thousands of out-of-phase high-frequency signals, they cancel each other out, tending toward zero.
- **Low Frequencies (e.g., 1 Hz):** The period of a 1 Hz wave is 1000 ms (1 second). A 2 ms jitter is practically nothing. Therefore, slow, low-frequency oscillations remain **in phase** across large neural populations. The electrode easily records these large, synchronized waves.

The German hypothesis won the debate. The $1/f^2$ spectrum is not a biophysical artifact of the tissue; it is a fundamental property of how neural oscillations summate and cancel out over space and time.

Canonical Circuits

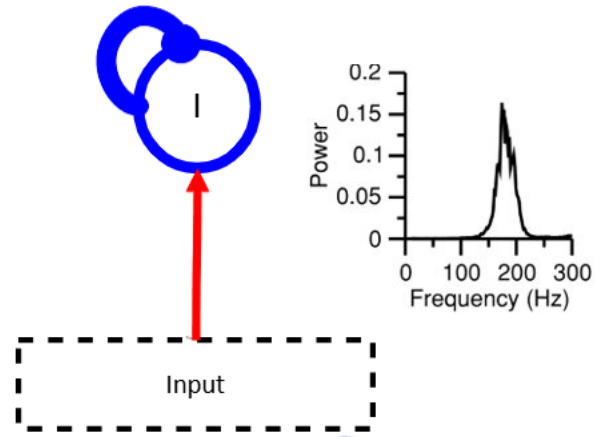
While we have largely focused on the temporal domain of single neurons, the brain is fundamentally a series of oscillators. The frequency domain is crucial for understanding brain function.

It is very rare for a single neuron to generate an oscillation on its own (so-called "pacemaker" neurons exist, but they are a tiny minority). Instead, the vast majority of brain rhythms emerge from network dynamics. Fascinatingly, different brain areas performing entirely different functions often rely on the same fundamental wiring diagrams to generate these rhythms. We call these recurring structural motifs **Canonical Circuits**.

Canonical Circuit 1: The Fast Inhibitory Network

The first canonical circuit consists of a pool of purely **Inhibitory (I)** neurons that receive a constant external excitatory input (K) and feature very strong self-inhibition. How does this generate a fast oscillation? Let's break down the cycle:

1. At time t_0 , all neurons are silent.
2. The steady external excitatory input activates the neurons, and they start firing.
3. As they fire, the recurrent self-inhibition builds up quickly. Because this self-inhibition is structurally very strong, it eventually overrides the external excitation.
4. The neurons completely inhibit themselves and fall silent.
5. Because they are silent, the self-inhibition stops. However, the external excitatory input is still there. We are back at Step 1, and the cycle repeats infinitely.



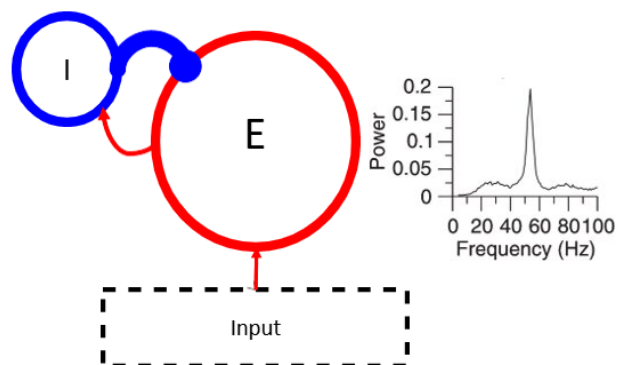
This cycle generates very **fast oscillations** (typically > 100 Hz). The speed is determined by the specific biological time constants (membrane time, synaptic latency, etc.), but the structural loop is very short, allowing for high frequencies.

Canonical circuit 2: The Excitatory-Inhibitory Loop

The second canonical circuit involves an **Excitatory (E)** pool and an **Inhibitory (I)** pool. The external input only targets the Excitatory neurons. (*Note: We deliberately ignore recurrent self-excitation in this isolated model, as E-neurons exciting themselves unchecked simply leads to epilepsy*).

Here is the step-by-step cycle:

1. At time t_0 , both E and I pools are silent.
2. The external input activates the E neurons.
3. The active E neurons send excitatory signals to the I neurons, waking them up.
4. The I neurons begin to fire and send a massive inhibitory feedback to the E neurons. This feedback is strong enough to completely silence the E neurons.
5. The "ungrateful" I neurons have now silenced their only source of input. Without the E neurons driving them (since they receive no external input), the I neurons also fall silent.
6. Both pools are silent again, but the external input to the E pool remains. The process restarts.



This generates **slow oscillations**. It is slower than the first circuit because it requires more synaptic steps (a disynaptic loop) to complete the cycle.

Optogenetics

A key skill in neuromorphic engineering is recognizing that a real piece of the cortex is a "network within a network." It contains both canonical circuits simultaneously: E-pools, I-pools, external inputs, self-inhibition, and self-excitation.

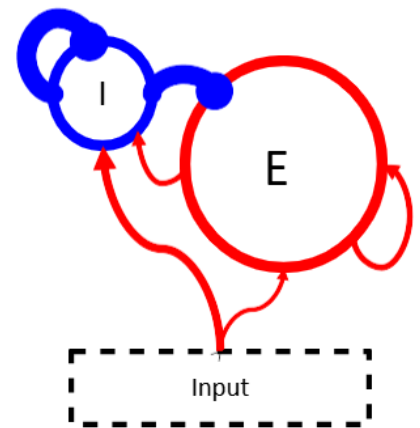
To prove how these sub-circuits interact, neuroscientists used a revolutionary technique called **Optogenetics**. Optogenetics involves genetically modifying specific neurons (e.g., only the E-neurons or only the I-neurons in a rodent) to become sensitive to specific frequencies of laser light. This gives scientists a "magic button" to instantly and reversibly turn specific neuron populations on or off.

Imagine a complete, oscillating cortical network. What happens if we use the laser to shut down specific pools?

- **Shutting down the Inhibitory (I) pool:** You are left with Excitatory neurons receiving external input and exciting each other without any brakes. As expected, this causes **epilepsy**.
- **Shutting down the Excitatory (E) pool:** Intuition suggests that removing excitation would kill the network's activity entirely. Surprisingly, the oscillation not only survived but became **faster**.
- Because by shutting down the E-neurons, you eliminate the "slow" canonical circuit. The only circuit left running is the purely Inhibitory network (Canonical Circuit 1), which naturally operates at much faster frequencies!

So in the complete network, both resonances are present and the input determines their interplay and hence the final frequency.

The **Gamma band (roughly 30 to 100 Hz)** is highly relevant and scientifically meaningful in this context. The complex interplay between the fast (inhibitory) and slow (excitatory-inhibitory) canonical circuits we just analyzed naturally generates structured oscillations exactly within this Gamma range. Therefore, when we see Gamma oscillations in the cortex, it is a strong fingerprint of these specific micro-circuits at work.



Testing of the Network Model

Until now, we have looked at how individual neurons respond to inputs (e.g., how firing rate increases linearly or super-linearly with input strength, depending on adaptation). However, we now know that when neurons are wired together into a complex network, they produce entirely new dynamics, such as the canonical oscillations discussed previously.

The next major step in Neuromorphic Engineering is to test the **information transfer properties** of our artificial network. To do this, we must compare our *in silico* model to a real biological system.

To validate the model, the research team collaborated with the Max Planck Institute, using data recorded from the visual cortex of anesthetized monkeys.

The experimental setup provided a complete map of the information pathway:

- **The Input:** A visual stimulus presented to the animal (e.g., a movie).
- **The Relay (Thalamus):** Recordings from the visual thalamus (the Lateral Geniculate Nucleus, or LGN), which receives the visual data and sends it to the cortex.
- **The Output (Visual Cortex):** Recordings of the actual network activity in the primary visual cortex (V1).

We know the exact input the biological visual cortex received from the thalamus, and we know the exact output it produced. If we feed that *same* thalamic input into our mathematical network model, does it produce the *same* output? If the transfer functions match, our model is biologically accurate.

Feeding a highly complex, chaotic signal (like a natural movie) directly into the model right away makes it too difficult to understand what the network is actually doing analytically. Therefore, the researchers broke the problem down into three manageable steps:

1. Response to a Constant Stimulus

While early neuroscientists used unnatural flashing lights to map receptive fields, the brain is evolved for structured natural scenes. The simplest biologically meaningful stimulus is not a blank screen (due to relative luminance processing), but a steady **spatial contrast**, such as a fixed pattern of stripes/gratings. In the graph, this constant stimulus produces a flat, steady thalamic firing rate (the thin green line). The overall input to the cortical neurons remains highly noisy due to the spontaneous brain activity discussed in previous lessons, but the *sensory signal* itself is constant.

2. Response to a Periodic Stimulus

Once the network's reaction to a constant signal is understood, the next step is to introduce periodic (oscillating) stimuli. This is mathematically strategic because any complex, real-world time series can be broken down into a sum of periodic components (via Fourier analysis).

3. Response to a Complex Stimulus

Finally, once the constant and periodic responses are mapped, the network is ready to process the arbitrary, complex thalamic activity generated by watching a natural movie.

Important: it was noticed an important difference between the mathematics model and the biology data recorded by the Institute

From a mathematical point of view, the equations we built (like the Integrate-and-Fire model) output the exact membrane potential of individual neurons and their discrete action potentials (spikes/firing rate). However, the data from the Max Planck Institute was gathered using extracellular electrodes. These electrodes do not penetrate single cells; instead, they sit in the extracellular tissue and record the Local Field Potential (LFP)—the blended, average electrical activity of thousands of surrounding neurons.

Therefore, we cannot directly compare the discrete spikes of our simulated neurons to the continuous LFP waves recorded from the monkeys. Before we can validate our network's information transfer, we must first figure out how to mathematically translate our model's individual neural spikes into a simulated Local Field Potential.

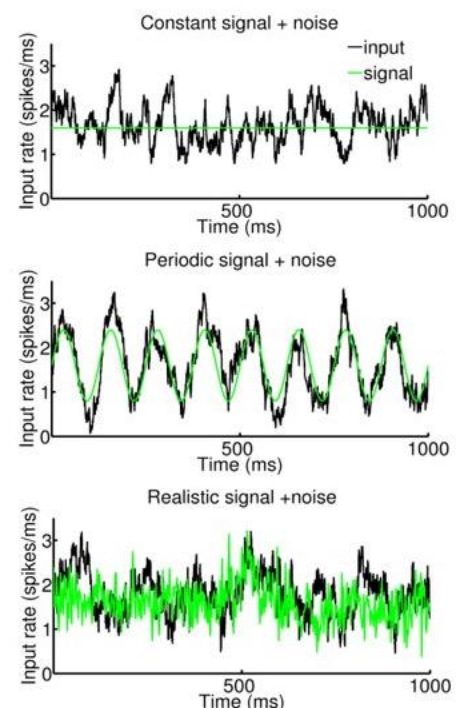
Ball and Stick Neuron Model

Up until now, the neurons in the model were "disincarnated"—they were treated as mathematical points in space whose entire existence was defined by their somatic membrane potential. To capture the extracellular field (the LFP), we must introduce spatial geometry using the principle of Occam's razor: adding only the simplest level of complexity necessary to make the model work.

In physical reality, the vast majority of a neuron's volume is not the soma, but the dendritic tree. Most synapses are planted on the dendrites, not the soma.

To capture the extracellular ion changes caused by these dendrites, we move to a **"Ball and Stick" model**:

- **The Ball:** Represents the soma.
- **The Stick (Cylinder):** Represents a simplified, single-cylinder approximation of the dendritic tree.
- **The Axon:** Ignored completely because it is too thin to significantly impact the extracellular field at this scale.

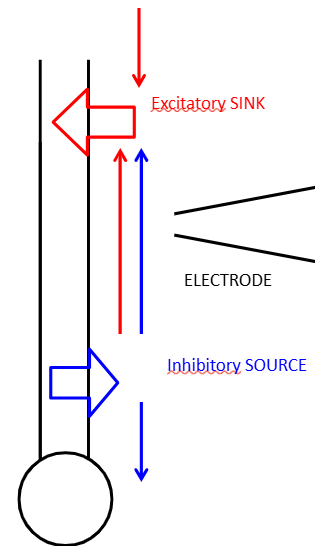


By reading extensive physiological literature, the researchers incorporated a crucial biological rule about where synapses are located on this "stick":

- **Inhibitory Synapses** are mostly **perisomatic** (located very close to the soma). When they open, they act as a current *source* (positive current).
- **Excitatory Synapses** are mostly **apical** (located far away on the dendrites). When they open, they act as a current *sink* (positive ions flowing in).

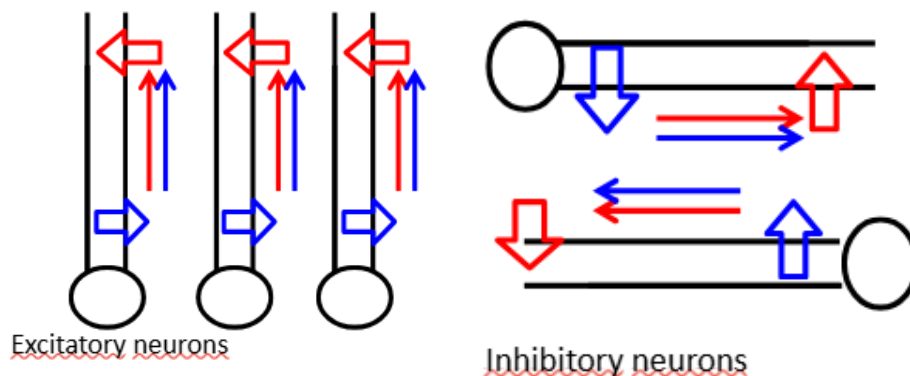
Inside the neuron, you simply subtract inhibitory inputs from excitatory inputs to get the net membrane potential. However, the LFP is recorded by an electrode sitting *outside* the neuron, usually somewhere along the dendrite.

Because the sink (excitatory) and the source (inhibitory) are physically separated by a finite distance on the neuron, they create a **dipole** (positive at one extremity, negative at the other). Because of this spatial separation, if you are measuring the extracellular space between them, the absolute values of these currents don't cancel each other out—they effectively sum together to generate the extracellular field.



However, understanding a single neuron's dipole is not enough. The LFP electrode records the average activity of thousands of neurons. We must account for how these neurons are physically arranged in a cortical column.

- **Excitatory (Pyramidal) Neurons = Parallel Dipoles:** Biological pyramidal neurons are arranged in highly structured, parallel columns. All their somas sit at the bottom, and all their apical dendrites point directly upward. Because they are perfectly aligned, their individual dipoles all point in the same direction. Consequently, they sum up linearly to create one giant, measurable macroscopic dipole.
- **Inhibitory (Interneurons) = Isotropic Dipoles:** Interneurons have a star-like, isotropic dendritic structure. For every dendrite pointing up, there is roughly one pointing down. Across a large volume of thousands of interneurons, these randomly oriented dipoles mathematically cancel each other out to zero.



By combining the geometry of the single neuron (the dipole) and the architecture of the network (parallel vs. isotropic), the researchers arrived at a brilliant and simple proxy for simulating the LFP.

The simulated LFP is calculated as the sum of the absolute values of the excitatory and inhibitory synaptic currents, but **calculated exclusively on the excitatory neurons**:

$$LFP \propto \sum (|I_{exc}| + |I_{inh}|)_{excitatory_neurons}$$

We use the absolute sum because the spatial separation on the dendrite creates a dipole where the magnitudes add up in the extracellular space. We calculate this *only* for the excitatory neurons because their parallel alignment allows their dipoles to survive the macroscopic averaging, whereas the inhibitory neurons' dipoles cancel out to zero.

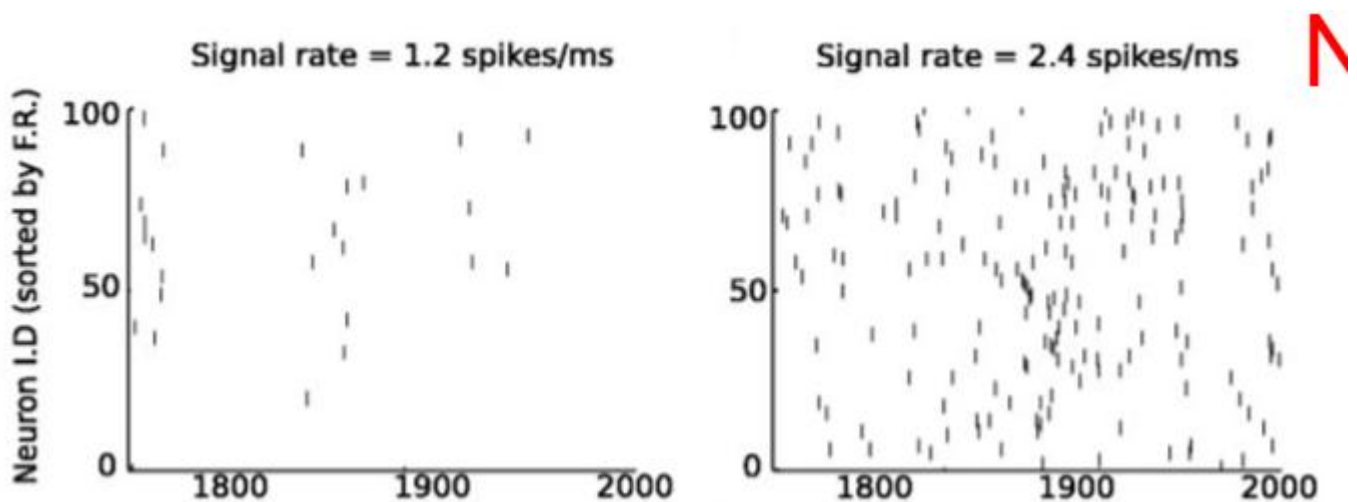
Emerging of the Oscillation Behaviour

Now that the complete model is built—incorporating constant thalamic input, spontaneous background noise, individual neuron dynamics, and a mathematical proxy for the Local Field Potential (LFP)—it is time to run the simulation and observe the network's behavior at different input intensities.

The network's response changes dramatically depending on the intensity of the constant sensory input it receives:

- **Low Input (e.g., 1.2 spikes/ms):** At this level, the network produces a baseline level of activity. (Note: 1.2 spikes/ms sounds impossibly high for a single neuron, but this number represents the *aggregate input* a neuron receives from its ~10,000 physiological neighbors, scaled down in the model to 1,000). At this stage, the activity is basically just noise. Nothing structurally interesting happens.
- **Intermediate Input (e.g., 2.4 spikes/ms):** When the input is increased, something remarkable occurs. If you look at the total firing rate of the 1000 inhibitory neurons, the 4000 excitatory neurons, or the simulated LFP, clear and regular **oscillatory peaks** emerge from the noise. The network has started to oscillate!

However, if you look at the raster plot (which visualizes the exact spike times of the 100 most active single neurons), **you will not see an oscillation**. The individual spikes look completely random and noisy. You can only visualize the oscillation if you sum up the activity of the entire population or look at the LFP. (This is precisely why physiologists rely on LFP recordings; recording just a handful of single neurons hides the global rhythm).



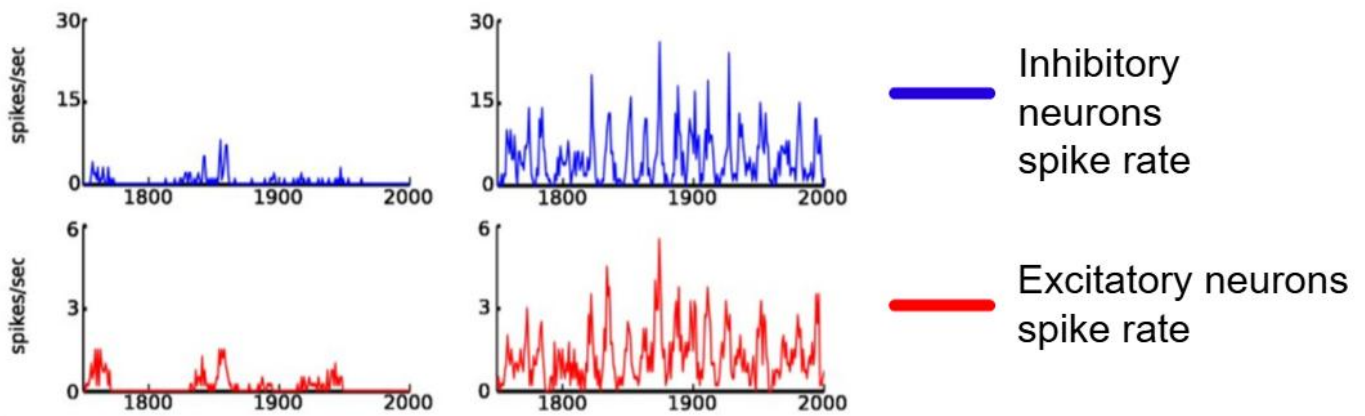
Weak Synchronization

This brings us to a fundamental concept in the physics of complex systems: an **emergent property**. The oscillation in this network is an emergent property because:

1. **The input is not oscillating** (it is a constant stimulus).
2. **The noise is not oscillating** (it is $1/f^2$ brown noise, which has no characteristic time scale).
3. **No single neuron is oscillating** (a Fourier analysis of any individual neuron's activity reveals zero rhythmic peaks).

Yet, through the interactions of these non-oscillating elements, the network as a whole oscillates. This specific regime is called **Weak Synchronization**. Therefore, we can identify three regimes of the network activity:

1. **Low Firing Rate:** Almost no organized activity (noise).
2. **Weakly Synchronized Regime:** (Crossed the first bifurcation threshold). The network oscillates, but individual neurons do not. **This is the normal operating state of the biological brain.**
3. **Strongly Synchronized Regime:** If the input is increased massively, the average input drives every neuron above threshold. Both the network *and* every single individual neuron oscillate together.

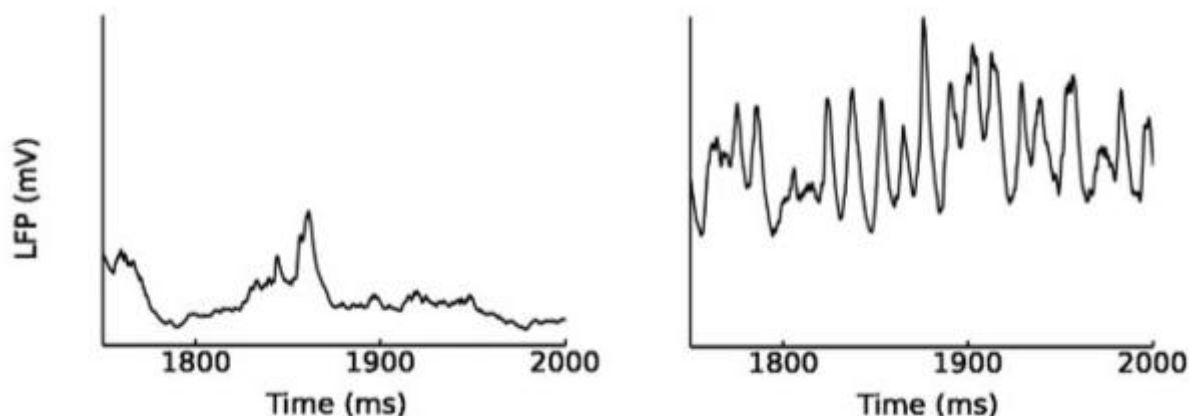


The weak synchronization oscillation is obtained by the combination of two factors: **Non-linearity (the firing threshold)** and **Noise**. Imagine a single neuron in a weakly synchronized network that is oscillating at 50 Hz.

- Because the network is oscillating, the *average* input this neuron receives fluctuates up and down periodically.
- However, this periodic input is **sub-threshold**. On its own, it is not strong enough to make the neuron fire. If there were no noise, the neuron would stay completely silent.
- In reality, the neuron is bombarded by random background noise. At the peak of the network's oscillatory cycle, this random noise gets added to the sub-threshold input.
- By pure chance, a small fraction of neurons (let's say 5%) will receive a "lucky" positive fluctuation of noise exactly at the peak of the cycle, pushing them over the threshold to fire a spike.

Crucially, the 5% of neurons that fire in cycle #1 will likely not be the same 5% that fire in cycle #2. Because the noise is random, a entirely different subset of neurons gets the "lucky" boost at each cycle. Therefore, a sufficiently large subpopulation fires at every single cycle to sustain the macroscopic 50 Hz oscillation in the LFP, but any individual neuron only participates sparsely and randomly. No single neuron is firing at 50 spikes per second; they fire much less frequently, only participating in a random subset of the cycles.

Now that the network is running and showing emergent oscillations, we can analyze the **power spectrum** of the simulated Local Field Potential (LFP) to see how it changes with different input intensities.



When we plot the LFP spectrum:

- **At Low Input (≤ 1.2 spikes/ms):** The network activity is dominated by spontaneous background noise. The spectrum perfectly mimics the bounded $1/f^2$ brown noise we modelled earlier. There is a very tiny flat "white noise" plateau at extremely low frequencies (because biological systems cannot mathematically go to infinity), but otherwise, it smoothly decays.
- **At Higher Input:** As the constant sensory input increases, the low-frequency noise remains largely unaffected. However, a distinct, powerful peak emerges right around **50 Hz**. The network has entered the weakly synchronized regime, generating true oscillations in the **Gamma band**.

Note: for a long time, researchers struggled to separate noise from true signals. Now, it is widely accepted that *any* neural power spectrum can be mathematically decomposed into two parts:

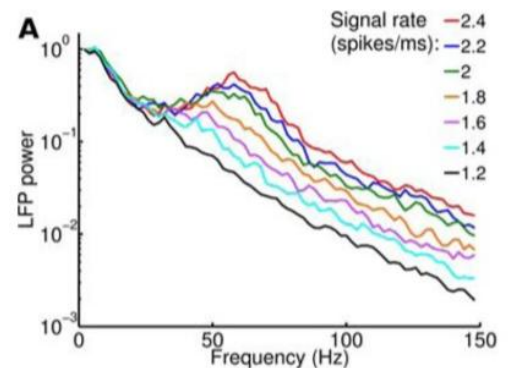
1. **An Aperiodic Component:** The background $1/f^\alpha$ noise (the overarching slope).
2. **True Oscillations:** The distinct periodic peaks (like our 50 Hz gamma peak) that rise above the aperiodic baseline.

To clearly see what the sensory stimulus is actually doing to the network, we need to mathematically strip away the $1/f^2$ background noise. We do this by computing the Relative Modulation. For every frequency, we calculate how much the input changes the LFP relative to the baseline (spontaneous) activity:

$$\text{Modulation} = \frac{\text{Input} - \text{Baseline}}{\text{Baseline}}$$

The results of the modulation show that:

- **Below 40 Hz:** The modulation is basically zero. The slow fluctuations are just background noise; the stimulus doesn't affect them.
- **Above 100 Hz:** There is some slight, broadband modulation (meaning the overall high-frequency "chatter" of the network increases).
- **Between 50 and 100 Hz:** There is a **massive peak**. This proves that the network's primary response to a constant sensory input is to generate a highly specific, structured oscillation in the Gamma band.



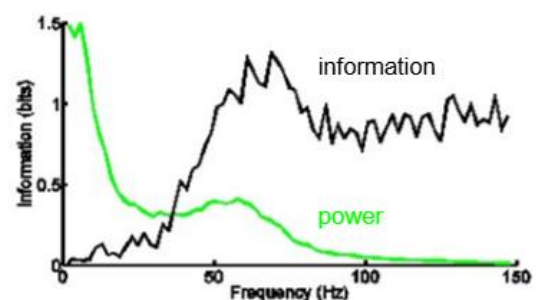
The model's simulation was confirmed by some biological data of a paper where another researcher had recorded the LFP from the **primary visual cortex (V1) of a monkey** while showing it constant spatial contrast patterns. It showed the exact same massive spike in modulation specifically in the **Gamma band (50–100 Hz)**. While the Y-axis magnitudes were slightly different (e.g., the biological signal increased by a factor of 5, while the model increased by a factor of 15), the *frequency bands* and the *network dynamics* were identical. The model successfully and accurately predicted how a real biological cortical network reacts to a constant sensory stimulus. We now have a network that biologically mimics the visual cortex. We know that when we feed it a stimulus, it responds by generating Gamma oscillations.

Integrate and fire models of local cortical excitatory – inhibitory networks

As shown in the left image, **the peak of neural activity (power) does not necessarily correspond to the peak of information.**

The graph displays two aligned graphs analyzing the Local Field Potential (LFP) across different frequencies (0 to 100 Hz):

1. **Top Graph (Power Spectrum):** Shows the absolute power of the LFP. Notice the massive spike in power at very low frequencies (below 20 Hz, particularly the green line).
2. **Bottom Graph (Information Spectrum):** Shows the actual mutual information (in bits) carried by those frequencies.



Despite the massive peak in power below 20 Hz, the bottom graph shows that the information carried at these frequencies is **exactly zero**. This high-amplitude activity is pure background noise (unspecific slow cortical activity).

The actual peak of information (1.25 bits) occurs in the Gamma band (roughly 74 Hz), even though the absolute power there is much lower. Information is driven by *stimulus-dependent variability*, not just raw signal

amplitude. Therefore, there is no *a priori* reason to assume that the frequency with the highest power is the most informative one.

In this specific experiment, the network was presented with **7 different signals** an equal number of times. The maximum possible information in this system is 2.8 bits. Looking at the information spectrum (bottom graph), you might notice multiple informative peaks within the Gamma band. For example, there might be a peak of 1.25 bits at 74 Hz, and another peak of 1.2 bits at 70 Hz. The redundancy within the Gamma band is extremely high. The different frequencies in this band are highly correlated; they are essentially carrying the exact same information about the stimulus. Computing multidimensional (joint) information across many frequencies is computationally very expensive. Because the frequencies within the Gamma band are highly redundant, adding more of them to your decoder will not yield any new information. Therefore, the best practical strategy is to **find the single most informative frequency peak (e.g., 74 Hz) and exclusively use that for decoding**.

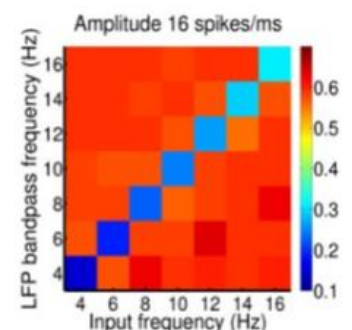
Network Response to Periodic Signals

When periodic oscillations are injected into a network, the network can translate the signal into an endogenous oscillation by activating its internal resonances. If the injected oscillation has a sufficiently low frequency and a sufficiently large amplitude, the system responds linearly (entrainment). For example, injecting a 2 Hz oscillation causes the network to oscillate at 2 Hz.

- **Low Frequency:** The input must be slower than the network's time constant (typically < 30 to 40 Hz). If it is too fast, it will start interfering with the locally originated gamma band rather than entraining the network.
- **Large Amplitude:** Most background noise within the network occurs at very low frequencies. The stimulus modulation must be large enough to override the effect of this low-frequency noise.

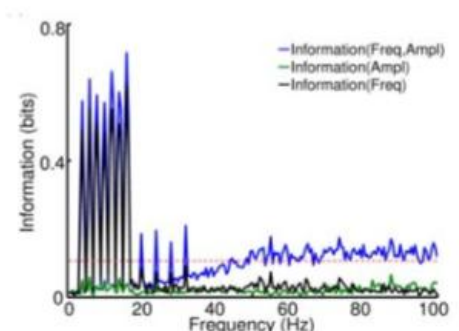
Because phase angles (0 and 2Pi) are continuous and connected, standard linear statistics cannot be used to measure phase variance. Instead, a specialized branch of circular statistics is applied.

- **Circular Variance Formula:** $CircVar = 1 - |\langle \exp(i\Delta\Phi(t)) \rangle_t|$
- **Phase Locking:** A low circular variance indicates that the phase difference between two oscillations is fixed, meaning they are successfully phase-locked.
- **Frequency Interference:** When plotting the correlation between input and output frequencies, circular variance is very low at low frequencies. However, as the input frequency increases, circular variance slowly increases because the external signal begins interfering with the internal gamma resonances, causing the system to lose strict phase-locking.



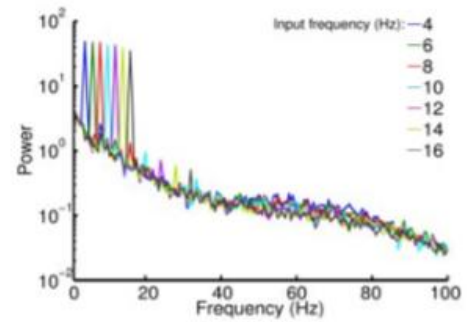
Injecting periodic signals (ranging from 4 to 16 Hz with varying amplitudes and noise) reveals how different frequencies carry different types of information.

- Periodic signals successfully entrain network Local Field Potential (LFP) oscillations, with the effect being strongest at lower frequencies.
- Injecting a 4 Hz signal results in a corresponding 4 Hz power peak superimposed directly on the background noise.
- Information about the lowest band of the signal spectrum is contained within that exact same band of the network LFP.



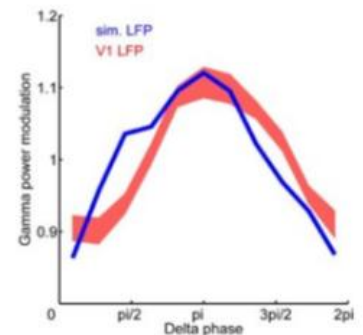
The network operates in an efficient regime where different frequency bands encode different features of the input simultaneously. This is known as **multiplexing**.

- **Low Frequencies:** Contain information strictly about the periodic (oscillatory) components of the input signal.
- **Gamma Band (High Frequencies):** Contains information about the average intensity of the stimulus.
- Because of this multiplexing, analyzing the spectrogram (time across frequencies) is heavily preferred over the standard time domain when studying continuous biological signals. By looking at different frequency bands from a single recording electrode, you can extract information about completely different aspects of the system.



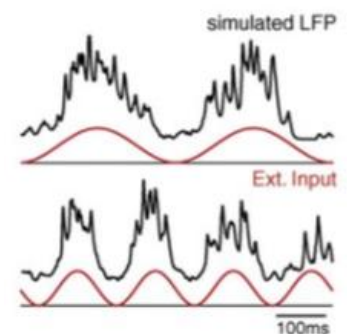
Biological systems frequently oscillate across multiple time scales simultaneously. Cross-frequency coupling occurs when the properties of a high-frequency oscillation depend directly on the phase of a low-frequency oscillation. When the network is stimulated with a sufficiently slow and large periodic input, two things happen simultaneously:

- The low-frequency component of the LFP tracks the input due to entrainment.
- High-frequency gamma oscillations cluster specifically at the peaks of the slow input wave (the "top of the mountain"), while the troughs (the "valleys") remain smooth.



Gamma band power is directly proportional to the strength of the input.

- Just as a constant strong signal (e.g., 3 spikes/ms) triggers gamma oscillations while a weak one (< 1.2 spikes/ms) does not, a slowly oscillating input causes dynamic changes in local intensity.
- At the peak of the slow signal, the local input intensity is high enough to trigger the gamma oscillations.
- At the trough, the input intensity drops, and gamma oscillations cease.

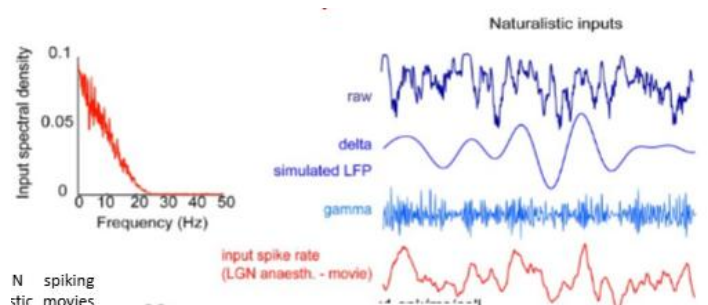


Notice that the input must be slow enough to allow the network time to generate the faster gamma oscillations during the peak phase. The amplitude difference between the peak and the trough of the signal must be large enough to cross the threshold for generating gamma waves. Ultimately, if you decompose the LFP into frequency bands, you will find that the power of the fast frequency (gamma) strictly depends on the phase of the slow frequency (delta). Experimental data often shows gamma power peaking precisely at a specific phase of the delta band.

Network Response to Realistic Stimulation

When moving from artificial inputs to realistic stimuli, it is important to test if the previously established rules for the network hold true.

For the network to properly segregate information into different frequency bands without interference, the spectral power of the input must be concentrated mostly at **low frequencies**. If the input contains high



frequencies, it interferes with the network's internal resonances.

The visual system naturally operates under this low-frequency condition:

- **Biological limits:** The refresh rate of the human retina is roughly 24 Hz.
- **Environmental physics:** In the real world, most objects move relatively slowly (well below 1 Hz).
- **Experimental setup:** The experiments used "naturalistic movies" based on LGN (Lateral Geniculate Nucleus) spiking activity. To capture truly natural spatial and temporal richness (rather than flashing or artificial lines), researchers used a "cat pen"—a light camera placed on a cat's head as it roamed a garden. Because the visual information from these naturalistic inputs is heavily weighted below 24 Hz, it perfectly matches the model's requirement for low-frequency input.

When the complex, naturalistic input is injected into the simulated network, the model successfully reproduces the information content found in experimental LFP (Local Field Potential) recordings. It achieves this by decomposing the complex input into two distinct pathways:

1. **The Low-Pass Version (Delta Band):** The slow, periodic components of the signal are transmitted linearly through the system. According to the model, the peaks of the input are associated with the troughs of the LFP delta band.
2. **The Average Signal (Gamma Band):** The average strength of the signal directly modulates the high-frequency internal resonances. Input peaks result in a larger amplitude of the LFP gamma band.

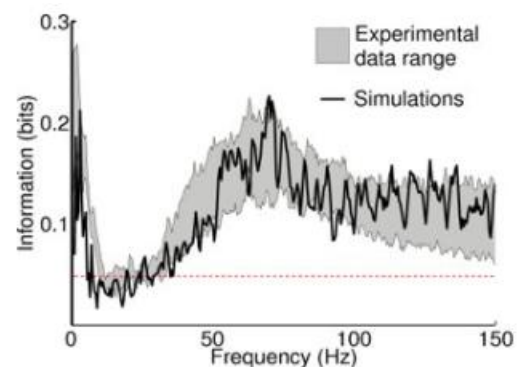
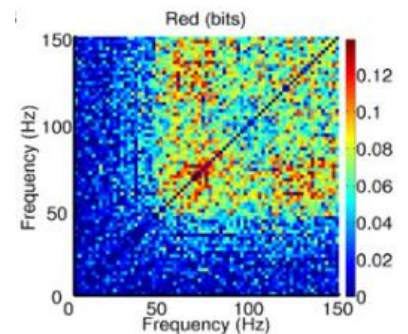
By plotting the information (bits) against frequency, the data reveals how the network efficiently handles data. There are two distinct channels of information, separated by a "silent channel" in the middle frequencies that carries no information:

- **Low Frequencies (< 10 Hz / Delta Band):** Encodes the slow, spectral components of the input signal.
- **Gamma Band (30 - 100 Hz):** Encodes the average signal rate (the overall strength or intensity of the input).

The model also successfully explains the mechanisms of redundancy and independence observed in the data:

- **Within the Gamma Band:** The frequencies within the gamma band are **redundant**; they essentially carry the same information about the signal's rate as a whole.
- **Within the Low Frequencies:** The different low frequencies carry information that is **independent** from one another.
- **Between the Bands (Multiplexing):** The gamma band and the low-frequency band are **completely independent** of each other. This is true multiplexing. Because the average value of a signal and its slow oscillatory component are distinct properties, a network can process a strong average signal alongside mild slow oscillations, or vice versa, simultaneously.

Ultimately, this model is considered successful not just because it replicates existing experimental data, but because it mechanistically explains *how* the brain segregates and multiplexes this information.



Analysis of Thalamus Oscillating behaviour

While the visual cortex actively processes information (e.g., translating signals into oscillations), the thalamus serves a fundamentally different purpose. It acts as a **gateway or filter** for sensory information, deciding which streams of information can pass through to the cortex and which cannot. The thalamus follows two gating rules:

1. **Attention:** The thalamus can selectively shut down specific channels; closed channels will not transmit information.
2. **Sleep:** When you sleep, your sensory receptors (like those in your ears) still function normally, but the thalamus blocks this information from reaching the cortex.

The thalamus is historically less studied than the cortex or the periphery, partly because it is presumed to have a lesser role in direct processing, and partly because it is impossible to record from non-invasively (unlike EEG for the cortex or EMG for the periphery). However, understanding it is critical for biomedical applications:

- **Neuroprosthetics:** Developing high-level sensory feedback for whole-arm amputated or tetraplegic subjects.
- **Deep Brain Stimulation (DBS):** Treating conditions like Tourette syndrome, which is caused by a dysfunction in the basal ganglia network. This network communicates with the cortex *through* the thalamus, making the thalamus a key gateway for both external and internal information.

It would be biologically dangerous if the thalamus simply dropped to zero activity during sleep to block inputs. Instead, during sleep, the thalamus generates strong, self-sustained endogenous activity.

- Sleep state activity is dominated by spindle (or theta) oscillations, typically in the **7–15 Hz** range (centered around 10 Hz).
- This oscillation is so strong that it acts as a barrier, stopping external stimuli below a given threshold from perturbing the system and reaching the cortex.
- The primary goal of the researchers' model was to explain the transition between these two distinct dynamical regimes: the awake state (driven by external stimuli) and the sleep state (dominated by these spindle oscillations).

When modeling the thalamus, standard non-adaptive Leaky Integrate-and-Fire neurons are insufficient. As a rule of thumb, the closer you move toward the periphery, the more relevant adaptation becomes. Therefore, the model requires **Adaptive Exponential Integrate-and-Fire neurons**.

Response to Depolarizing (Positive) Current

- If a constant excitatory input is injected, the neuron initially fires. However, because the adaptation is so strong, the firing gradually decreases and eventually stops entirely despite the constant input.
- This makes the neuron a **novelty detector**. It stops spending energy conveying information about constant, unchanging stimuli.

Response to Hyperpolarizing (Negative) Current

- If an inhibitory input is injected, it decreases the membrane potential. During this time, the internal adaptation variable decreases, effectively building up a net excitatory current (like pulling a rope in the opposite direction).
- When the inhibitory input is removed (releasing the rope), the adaptation variable takes time to return to zero. This imbalance causes the neuron to fire without any external excitatory input. This crucial phenomenon is called **firing for rebound from inhibition**.

Self – sustained Rhythmogenesis in basic oscillators

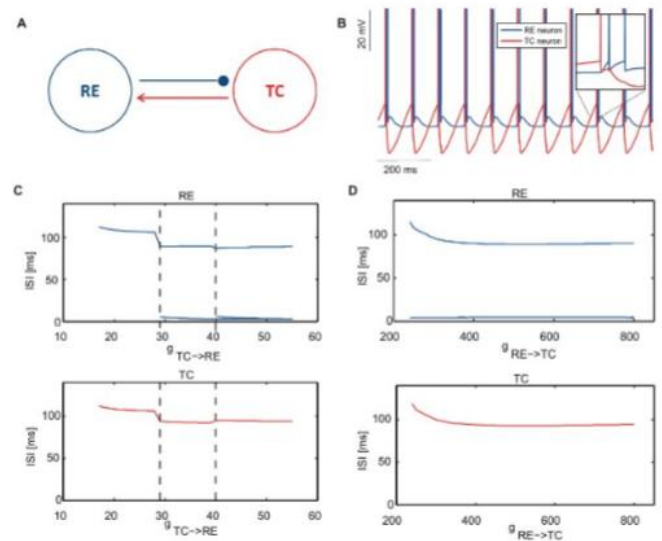
This mechanism of rebound firing allows the thalamic network to generate oscillations without needing a constant external drive. This is the third type of rhythmogenesis seen in the course: **self-sustained oscillation**.

The basic circuit involves two types of neurons:

1. **TC (Thalamocortical) Neurons:** Excitatory neurons that project to the cortex.
2. **RE (Reticular) Neurons:** Inhibitory neurons.

Cycle:

- **Step 0 (The Kick Start):** An external trigger temporarily activates the TC neurons.
- **Step 1:** Active TC neurons excite the RE neurons (both sets are now active).
- **Step 2:** The active RE neurons inhibit the TC neurons (only RE neurons are active).
- **Step 3:** Without excitatory input from the TC neurons, the RE neurons shut down (both sets are silent).
- **Step 4:** The TC neurons no longer receive inhibition. Due to their strongly adaptive nature, they fire via **rebound from inhibition**.
- **Step 5:** The cycle loops back to Step 1.



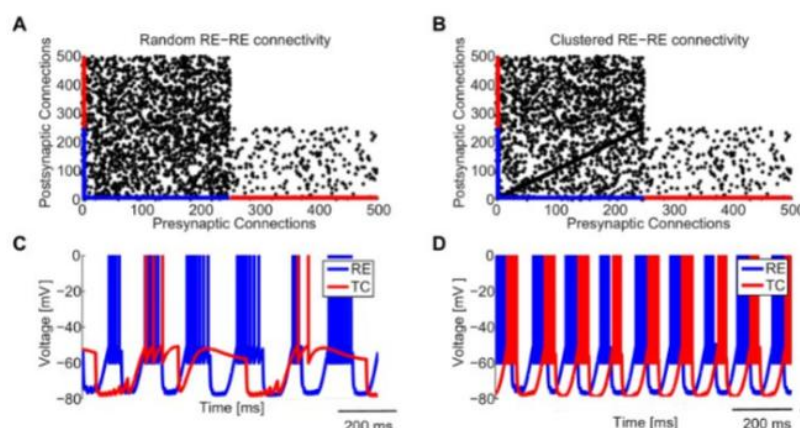
While this process continually consumes energy, it does not require a continuous external drive to keep oscillating. If the biological time scales (adaptation, synaptic opening/closing) are set correctly, the Inter-Spike Interval (ISI) settles at roughly **100 milliseconds**, which perfectly matches the **10 Hz** frequency of sleeping spindle oscillations. If the excitation in the network is altered slightly, the main ~10 Hz rhythm remains intact, but individual neurons may fire more than one spike per cycle. This results in two sets of Inter-Spike Intervals: one at ~100 ms (the rhythm of the bursts) and one at just a few milliseconds (the rapid succession of spikes within a single burst).

Thalamic connectivity

While a purely inhibitory network can also generate oscillations (relying on the "kickstart" and subsequent rebound from inhibition), modeling the full thalamic circuit requires a specific connectivity matrix.

The main differences between the connectivity in Thalamus and in the cortex:

1. **Cell Ratios:** Unlike the cortex, the thalamus has roughly an equal amount of excitatory (Thalamocortical - TC) and inhibitory (Reticular - RE) cells. In the model, this is approximated to 250 of each.
2. **No Excitatory-to-Excitatory Connections:** This is a unique and defining anatomical feature of the thalamus. TC neurons do not project to other TC neurons. In the connectivity matrix, the TC-to-TC quadrant is completely empty.



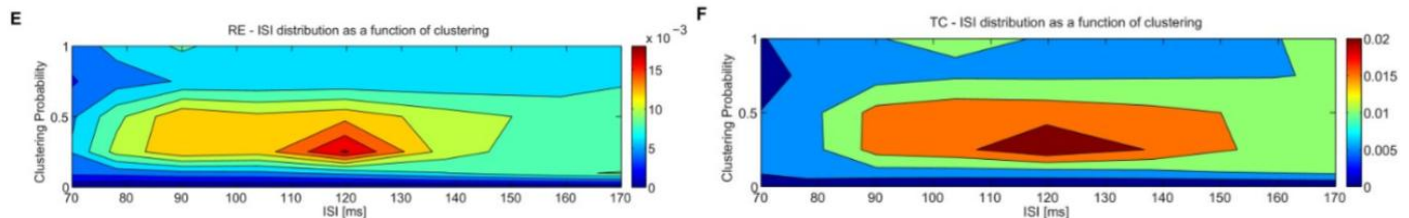
Initially, the model attempted to use random connection probabilities for the remaining three quadrants (as had worked previously for the cortex model). However, under random connectivity, the network completely failed to generate the characteristic sleep oscillations.

To solve the lack of oscillations, researchers applied a general neuroanatomical rule of thumb: long-range connections are primarily excitatory, while short-range (local) connections can be both excitatory and inhibitory. The hypothesis was that inhibitory (RE) neurons are more likely to connect strictly to their immediate neighbors. However, to define "neighbors," the network needed a spatial structure—the original unit numbers (0 to 500) were merely tags, not positions.

To introduce spatial distance without complicating the model with border effects (which occur in a simple line structure), the neurons were arranged in a **ring topology**.

- This is a 1D structure with well-defined distances and no borders.
- Neuron IDs became physical positions (e.g., inhibitory neuron 48 is now physically adjacent to 47 and 49).
- In the updated connectivity matrix, this local clustering appears as a dark diagonal band in the inhibitory-to-inhibitory (blue-blue) quadrant, representing neurons connecting only to their immediate neighbors.

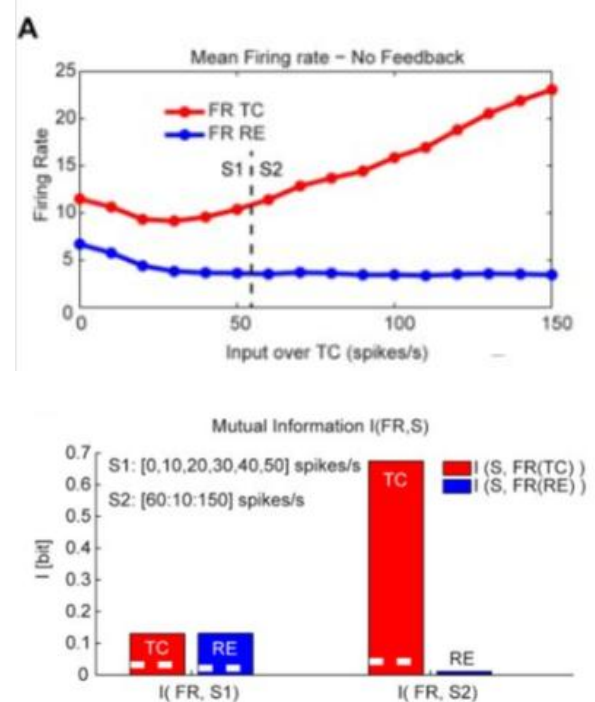
Once this local clustering was introduced, the network successfully generated stable, self-sustained oscillations. The clustering parameter must be carefully tuned: too much clustering prevents the oscillation from spreading across the network, while too little fails to generate it.



With the parameters tuned, the model successfully explains the functional transition between the sleep state and the awake state based on external input strength.

As you increase the external input index, the system eventually crosses a specific, unknown *a priori* threshold that triggers the state change:

- **Below Threshold (Sleep State):** The firing rates of both TC and RE cells change only mildly. They carry very little information about external input. Importantly, this state is **robust to small perturbations**, meaning weak external stimuli will not easily wake the system or break the spindle oscillations.
- **Above Threshold (Awake State):** Once the input is strong enough to cross the threshold, the excitatory (TC) neurons begin to **linearly encode** the input, actively translating the sensory information to the cortex. Conversely, the inhibitory (RE) neurons stop responding to the input variations. This mirrors real brain anatomy, as only the Thalamocortical neurons project to the cortex to deliver the awake sensory information.



This concludes the study of two distinct types of network resonances, driven primarily by the underlying individual neuron models:

- **Cortical Resonance:** Modeled using standard **Leaky Integrate-and-Fire (LIF) neurons**. In the cortex, adaptation is present but negligible. Rhythmogenesis is driven by the interplay of excitatory and inhibitory populations and external drives.
- **Thalamic Resonance:** Modeled using **Adaptive Exponential Integrate-and-Fire neurons**. In the thalamus, strong adaptation cannot be ignored. This adaptation leads to *firing for rebound from inhibition*, which allows the network to generate **self-sustained resonances** (like sleep spindles) without the need for a continuous external drive.

Introduction of Morphology in the LIF neuron model

While previous models (such as the thalamic ring network) introduced the concept of physical distance between neurons, they still treated the neurons as dimensionless points. However, there are scenarios where the physical body—the morphology—of the neuron must be considered.

- When doing simulations of neural stimulations or trying to accurately capture Local Field Potentials (LFPs), the physical size of the neuron becomes critical.
- Modern microelectrodes used for recording and stimulating are extremely small, often on the same order of magnitude in size as the neurons themselves.
- Because of this scale, the previous simplistic model of LFP generation (which treated the neuron as a basic dipole with apical excitatory sources and perisomatic inhibitory sinks) is no longer sufficient.

To accurately model morphology, a group of researchers in Norway utilize databases (like ModelDB) that contain actual 3D histological reconstructions of neurons, specifically capturing their complex dendritic structures (axons are typically too thin to be captured). First, a linearization process is required:

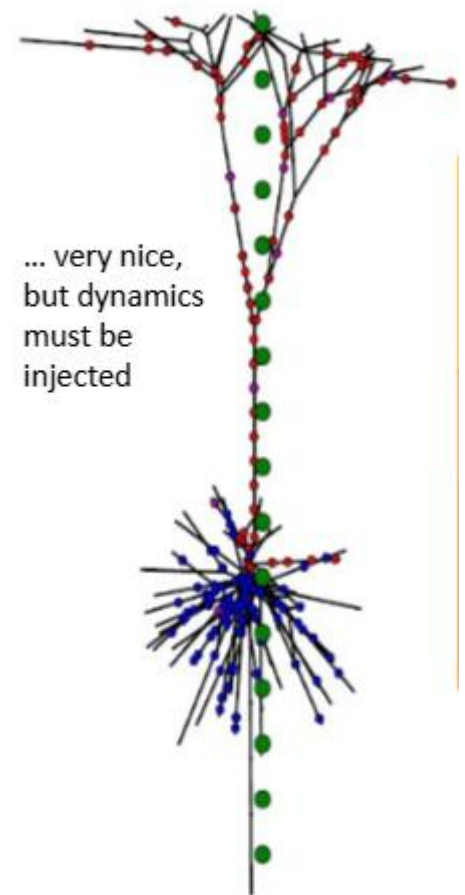
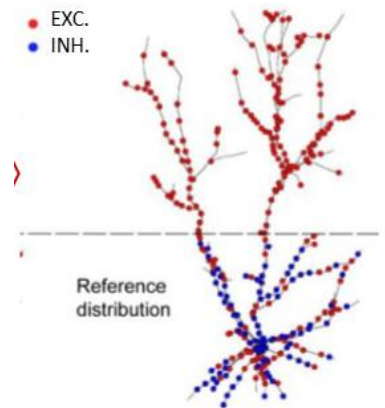
- The 3D reconstructions are simplified through a process called linearization. Based on a chosen resolution, the complex dendritic branches are converted into a series of straight lines or segments.
- A single neuron can be accurately represented by a few hundred lines (e.g., 194 segments).
- Each of these individual segments is treated as a distinct "compartment."

Secondly, for every single compartment, a specific set of Maxwell equations is formulated to compute electrical activity. For a neuron with 194 compartments, you have 194 sets of equations. These equations calculate:

- The current received from dendrites further away from the soma.
- The current sent down toward the soma.
- The exchange of information with the external world through synapses (including conditions for opening, speed, and ion transmission).

The morphological models also respect the biological distribution of synapses:

- **Excitatory synapses** (often visually represented in blue) are distributed widely across all dendritic bushes.
- **Inhibitory synapses** are primarily restricted to the lower dendritic bush closer to the soma.



Calculating the extracellular potential for every single point in the 3D space surrounding the neuron is incredibly computationally expensive, historically requiring the use of massive supercomputers (like those used for climate change simulations). However, the researchers thought that they do not actually need to compute the potential for the entire volume of space, but they only need to compute it where you are recording.

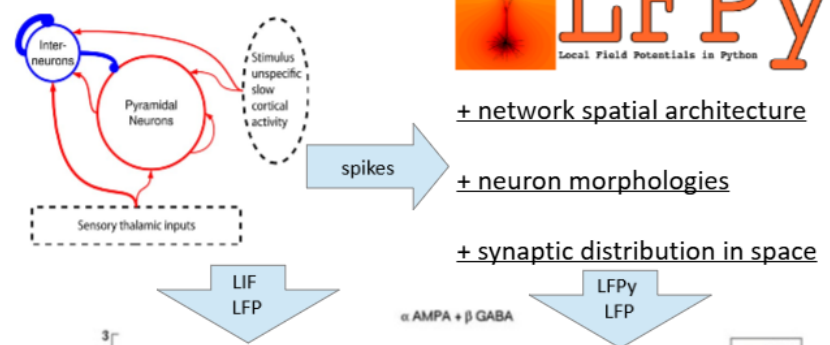
When doing morphological simulations, you must also simulate your neural interface. For example, you can superimpose a simulated laminar intracortical electrode with specific contact sites onto the neuron. By doing this, you only compute the Maxwell equations for those specific contact points (e.g., reducing the calculation from an infinite number of spatial points down to just 20 contact sites). This approach not only saves computational power but also allows researchers to test and optimize different electrode designs to maximize reading and stimulating efficacy.

Adding morphological realism comes at a heavy computational cost regarding network dynamics. If you have a network of 5,000 neurons, and each neuron consists of 194 individual compartments, it becomes nearly impossible to simulate dynamic network interactions (such as the continuous feedback loops of excitation and inhibition discussed in previous models). To solve this, Norway researchers using these highly detailed morphological models typically **inject assumed dynamics** directly into the system. Instead of simulating the interactive network from scratch, they apply a known dynamic pattern to study how the specific morphology shapes the resulting Local Field Potential.

Hybrid model: combination of morphology and dynamics

To overcome the limitations of individual models, researchers combined two different approaches:

- **The "Italian" Model** (by prof. Mazzoni): Generates network dynamics (spikes and basic LFP approximations) but lacks morphological realism.
- **The "Norwegian" Model:** Contains highly detailed morphological structures but requires external dynamics to be injected into it.

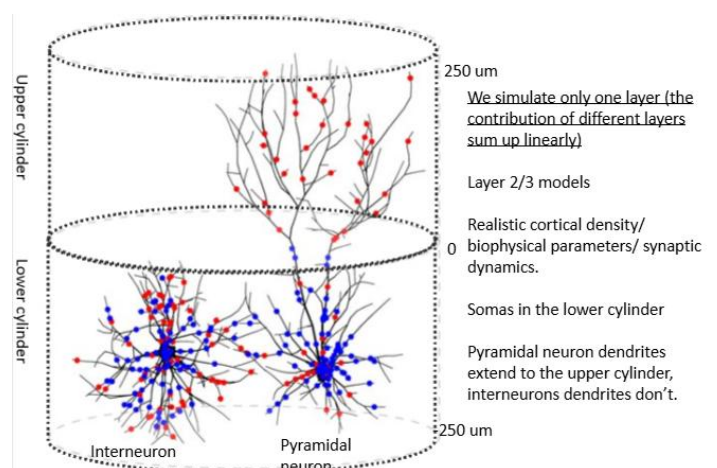


The dynamics generated by the simple network were injected into the detailed morphological model to produce a highly accurate, morphologically-detailed LFP.

- This detailed LFP served as the "ground truth."
- **Dimensionality Reduction:** The ultimate aim was to see if this complex LFP could be accurately approximated using the simple, basic variables of the dynamic model. Because a network contains many structural symmetries, many of the microscopic morphological details average out, making this dimensionality reduction possible.

To test this, researchers built a network of 5,000 morphologically detailed neurons arranged in a 3D space. The network was modeled as a cylinder, representing the size of a natural receptive field in the cortex:

- **Somas (Cell Bodies):** All somas (both excitatory and inhibitory) were placed in the lower part of the cylinder.



- **Inhibitory Interneurons:** Their dendrites are **isotropically distributed** (they branch equally in all directions). Their entire dendritic bush remains in the lower part of the cylinder.
- **Excitatory Pyramidal Cells:** They have classic apical dendrites that stretch upwards, occupying the upper part of the cylinder.

Using 5,000 identical replica neurons would cause artificial, unnatural neural synchronization. However, histologists only had about 30 actual 3D scans available. An algorithm was trained to take those ~30 base pictures and generate 5,000 unique, realistic neurons by applying spatial transformations (stretching, rotating, etc.). This ensured the necessary heterogeneity for a realistic simulation.

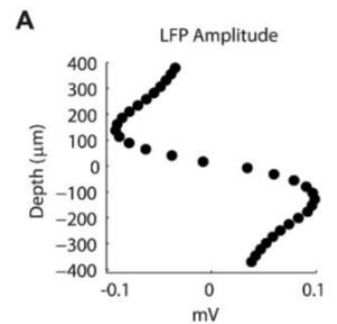
Results

Once the morphologically detailed LFP was generated, two major phenomena were observed and verified.

1. LFP Inversion (Phase Reversal)

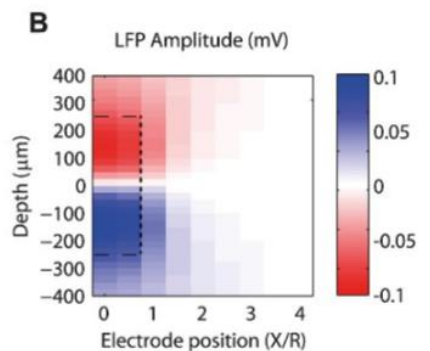
Because there is entering current in one part of the cell and return current in another, the LFP exhibits a strong spatial gradient across the depth of the cylinder. Since the LFP is a continuous function, there is a very thin spatial area right in the middle of the network (exactly between the upper and lower dendritic bushes) where the LFP deflection is **exactly zero**. This analysis gives the scaling factor of the LFP amplitude that cannot be predicted by the LIF network.

Note: You will also record zero LFP if your electrode is simply placed too far away from the network.

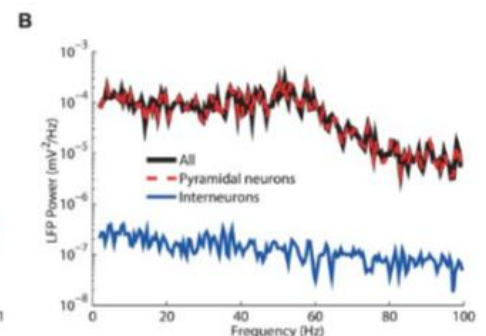
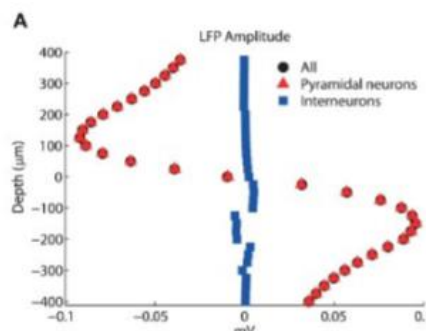


2. Validating the Interneuron Dipole Approximation

Previously, it was assumed that because inhibitory interneurons have isotropically distributed dendrites, they do not form a dipole and therefore do not contribute to the LFP. The detailed morphological simulation showed that interneurons *do* generate an LFP. However, the LFP generated by the interneurons is **three orders of magnitude smaller** than the LFP generated by the excitatory pyramidal neurons.



Therefore, the original assumption holds practically true: the contribution of inhibitory cells to the LFP can be safely neglected.



Approximation of the LFP with hybrid network model

The ultimate goal of combining the two models was to see if the highly detailed, complex Local Field Potential (LFP) generated by the morphological model could be accurately approximated using only the basic variables ("ingredients") available from the simpler Leaky Integrate-and-Fire (LIF) point-neuron network.

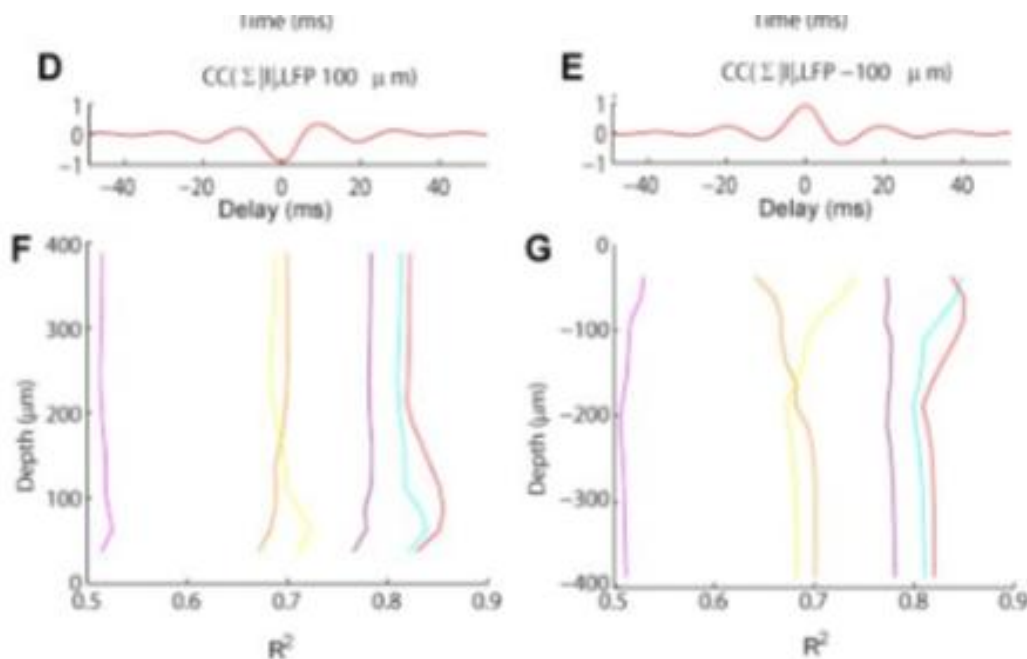
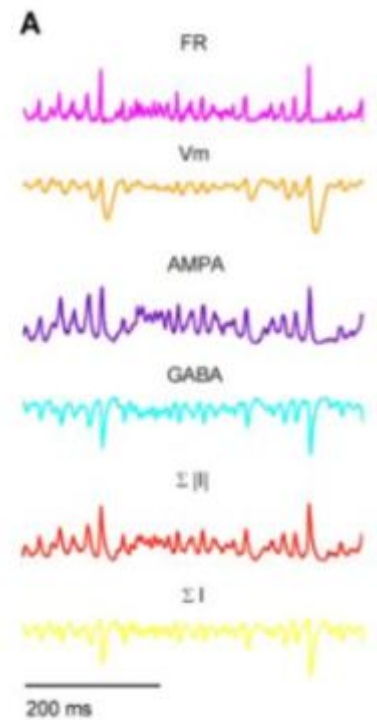
Researchers tested several simple variables to see which best predicted the detailed LFP:

1. **Firing Rate (FR):** The average spiking activity of the network.
2. **Membrane Potential (V_m):** The average membrane potential of the neurons.
3. **Synaptic Currents:** Specifically, a linear combination (Weighted Sum or WS) of excitatory AMPA currents and inhibitory GABA currents.

To evaluate the accuracy of each proxy, researchers used R^2 (the square of the linear Pearson correlation). This metric represents the fraction (or percentage) of the variance in the detailed LFP signal that can be explained by the simple proxy variable.

By plotting the R^2 values for each proxy against the morphological "ground truth" LFP, a clear hierarchy emerged. This finding challenged previous papers in the field, many of which relied on less accurate proxies.

- **Linear Combination of Synaptic Currents (AMPA + GABA):** This was a novel approach proposed by the researchers and proved to be the **best proxy**. It explains roughly **80%** of the LFP variance ($R^2 \approx 0.8$). It is a linear combination of the absolute values of the synaptic currents.
- **Average Membrane Potential (V_m):** This proxy is somewhat effective, explaining about **70%** of the variance.
- **Average Firing Rate (FR):** Despite being used in many previous studies, the firing rate is highly inaccurate as a proxy for LFP, explaining only about **50%** of the variance.



There is a fundamental physiological decoupling between the firing rate and the Local Field Potential. The two main reasons why firing rate fails to capture LFP variability:

1. **The Threshold Problem (Subthreshold Activity):** imagine a neuron that is currently receiving *only* inhibitory inputs. These inputs fluctuate over time, causing continuous changes in the neuron's synaptic currents and its membrane potential (which the LFP captures). However, because the input is strictly inhibitory, the membrane potential remains below the resting state and never crosses the threshold to fire. The firing rate remains exactly zero, completely failing to reflect the rich underlying synaptic activity.
2. **The "In Vivo" External Input Problem** When recording from a living animal (*in vivo*), a local neural network is not a closed system. A large fraction of the synaptic inputs the local neurons receive are long-range projections from other parts of the brain (exogenous activity).
 - If you only measure the local **firing rate**, you are only capturing the local, endogenous network activity.

- If you measure **synaptic currents** or **membrane potential**, you capture both the local activity *and* the massive fluctuations caused by the external inputs arriving at the network.

Note (GABAergic synapses role): a closer analysis of the synaptic currents revealed a counterintuitive finding regarding the origin of the LFP signal. While it was previously established that the physical bodies of inhibitory interneurons do not directly generate a significant LFP dipole, **GABAergic (inhibitory) synapses drive a massive portion of the LFP**. To maintain the balance between excitation and inhibition in a network where inhibitory neurons are vastly outnumbered, the inhibitory currents must be exceptionally strong or frequent. Therefore, the projection of GABA currents *into* the excitatory pyramidal neurons creates massive transmembrane current fluctuations in those target cells. Consequently, the activity of these inhibitory synapses is actually responsible for generating a large part of the LFP variance recorded in the surrounding tissue.

Conclusions on the model

Based strictly on the foundational findings of the study, the primary conclusions are:

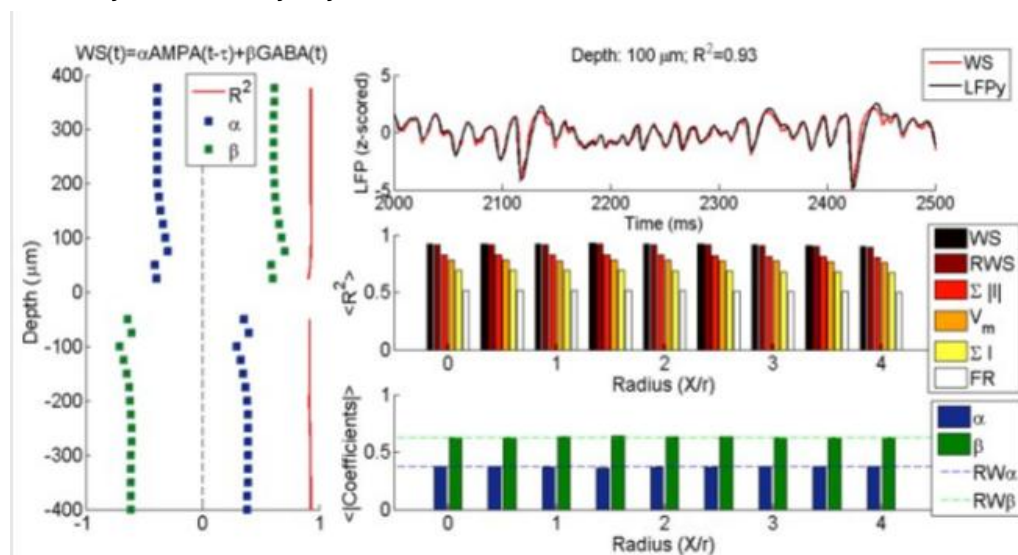
- **The Best Proxy:** The Local Field Potential (LFP) is best understood and approximated as the linear combination of the AMPA (excitatory) and GABA (inhibitory) synaptic currents.
- **The Flawed Proxy:** Firing Rate (FR) is a poor proxy for predicting LFP.
- **Optimization:** While basic absolute sums work well, you can optimize the approximation using a linear combination with specific coefficients, which yields the best results when excitatory and inhibitory currents take the same sign.

The Next Step is to check whether this model is robust. The researchers needed to test if this simple linear approximation holds true when the characteristics of the system change. They tested three main variables: Dynamical Regimes, Morphologies, and Synaptic Distributions.

Robustness across dynamical regimes

The network was tested across three different dynamical states based on external input:

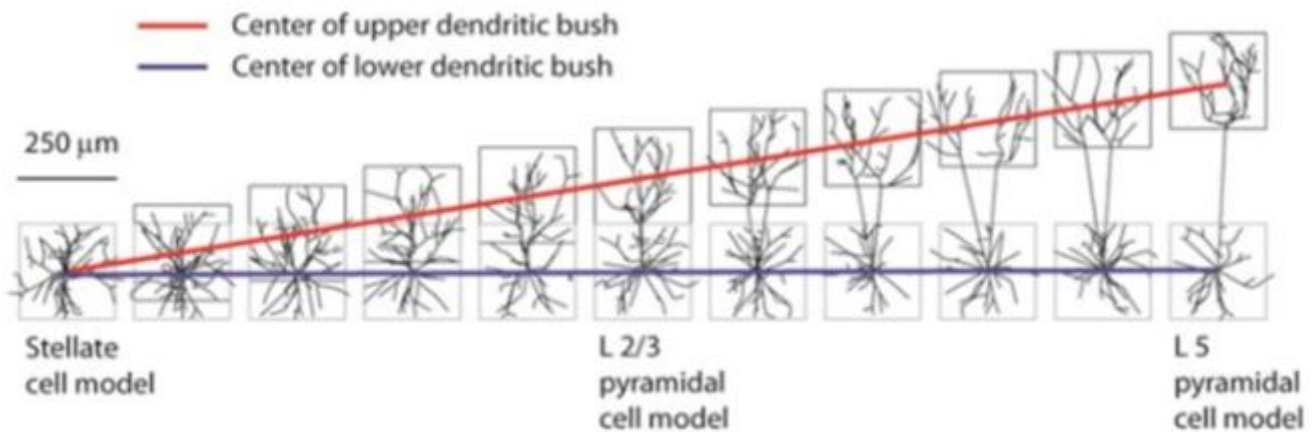
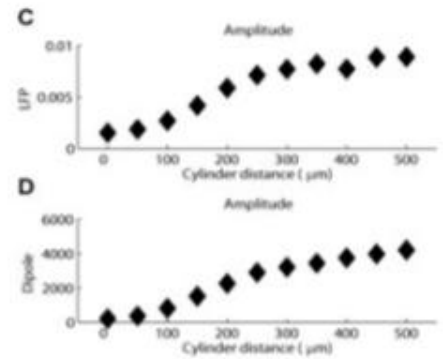
- **Strongly Synchronized Regime:** Every single neuron oscillates in phase. The linear approximation model works very well.
- **Weakly Synchronized Regime:** Individual neurons do not oscillate regularly, but the network *as a whole* oscillates. The model also works very well here.
- **Unsynchronized Regime (Low Input):** The model **fails** in this state. Because the input is extremely low, the network's activity is entirely noise-dominated. Without collective dynamics, the structural symmetries required to perform the dimensionality reduction disappear. However, since the overall network activity is so low anyway, the failure of the model here means "we don't miss much."



Robustness to Neuron Morphology

To verify the "dipole effect," researchers artificially manipulated the physical shapes of the simulated neurons.

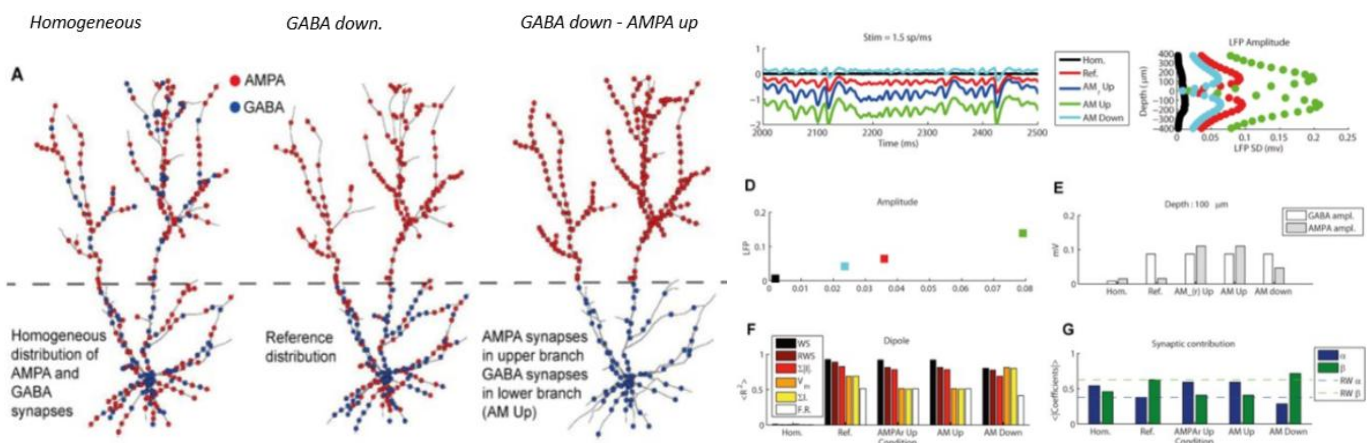
- **Pure Stellate Cells:** If a cell is perfectly symmetrical (stellate), the approximation model fails. The structural symmetry prevents the formation of a distinct electrical dipole, meaning any generated LFP is weak and completely noise-dominated.
- **Stretched/Pyramidal Cells:** As soon as the simulated neuron is stretched enough to create a spatial separation between its upper and lower sections, a true electrical dipole is generated.
- **Conclusion:** As long as there is an asymmetrical stretch to create a dipole, the model works perfectly. It does not matter how strong the dipole is or how far apart the poles are separated.



Robustness to Synaptic distribution

The reference model assumes a standard biological layout: inhibitory synapses cluster at the soma (bottom), while excitatory synapses are everywhere. What happens if we change this?

- **Homogeneous Distribution (Black Line):** If excitatory and inhibitory synapses are perfectly mixed and distributed evenly everywhere, their simultaneous activation creates local micro-dipoles pointing in opposite directions. These cancel each other out, resulting in essentially no net LFP.
- **Complete Partition (Green Line):** If excitatory and inhibitory synapses are strictly separated into two distinct physical zones, their synchronous activation (e.g., during gamma oscillations) generates a massive dipole pointing in the exact same direction. The currents compound together, producing a much stronger LFP. In this scenario, the model still works, but you must change the weight coefficients to account for the massive total extracellular currents.



Future prospects for the hybrid model

While computational power was a limiting factor 10 years ago, modern computational neuroscience can now apply these morphological models to much more complex, clinical scenarios.

Modeling Biophysical Stimulations Detailed morphology isn't strictly necessary for studying general states like sleep, but it is absolutely essential for understanding targeted biophysical stimulations, such as:

- Ultrasound stimulation
- Transcranial Magnetic Stimulation (TMS)
- Transcranial Direct Current Stimulation (tDCS)

The principles used to generate the LFP can be scaled up to simulate large-scale, non-invasive clinical readings:

- **EEG Generation:** This requires taking the extracellular currents from surface neurons and adding complex anatomical filters. Specifically, you must account for the folding of the cortex (sulci) and the fact that the skull acts as a strong low-pass filter.
- **fMRI Generation:** Because functional MRI measures blood oxygenation, not electrical activity, the network's electrical output must be coupled with a "**Balloon model**." This mathematical model links neural firing activity to local vascular blood consumption (the hemodynamic response).

EEG is clinically preferable because it is non-invasive. The ultimate ambition of this modeling is to solve the "inverse problem."

- **The Concept:** If we perfectly understand the transfer function from single-neuron morphology to the scalp, we could mathematically invert an EEG reading to pinpoint the exact single-neuron activity causing a patient's specific brainwave frequencies.
- **The Challenge:** This is an inherently ill-posed problem due to the "degeneracy of solutions." A weak electrical signal originating from many distant neurons looks mathematically identical to a strong signal originating from a few nearby neurons.
- **The Solution:** Using high-density EEG caps and advanced source-region reconstruction algorithms offers the best hope for solving this problem, representing a massive way computational neuroscience can directly support clinical diagnostics.

Aim of the hybrid model

Often in neurophysiological recordings, the only reliable data you have is the Local Field Potential due to noise, misplacement, or the physical nature of the electrodes. For example, EEG is essentially just the LFP recorded on the scalp, heavily low-pass filtered by the skull, bones, and cerebrospinal fluid.

The ultimate, highly ambitious goal of this theoretical model is to work backward: **If we only have the continuous LFP signal, can we reconstruct the underlying microscopic spiking activity, or even the external visual stimulus presented to the subject?** To test this, researchers used LFP recordings from the primary visual cortex of anesthetized macaques while they were shown video stimuli.

To make sense of the complex continuous LFP signal, it is more useful to analyze it in the **frequency (spectral) domain** rather than the temporal domain.

The spectrum of the Local Field Potential is driven by three major components. Because of the specific morphology of inhibitory neurons, the inputs they receive tend to average out to zero over a sufficiently large volume, so they are not the primary driver here

The fundamental equation breaks down the LFP spectrum into these components:

$$LFP(\omega) \approx J_{EE}S_{EE}(\omega)(\delta v_E(\omega) + \eta_E) + J_{EI}S_{EI}(\omega)(\delta v_I(\omega) + \eta_I) + \delta\mu_E(\omega)$$

- The first term represents the excitatory to excitatory input. δv_E is firing rate spectrum emitted by excitatory neurons. S_{EE} is the synaptic filter. This accounts for all the temporal filtering involved when a spike arrives (opening of the synapse, closing of the synapse, and the shape of the postsynaptic potential). J_{EE} is the synaptic strength. The stronger the synapse, the larger the postsynaptic potential, resulting in a larger deflection in the LFP. η_E is the noise added to ensure mathematical robustness.

- The same structure applies to inhibitory inputs, though they have different temporal filtering (S_{EI}), different strengths (J_{EI}), and rely on the inhibitory firing rate (v_I).
- μ_E is the spectral component of the external visual stimulus arriving from outside the local circuit.

Because the system of equations describing this network is far too complex to solve as a single giant equation, researchers use a mathematical trick: **separating the temporal scales**.

By assuming a clear boundary between fast and slow phenomena, you can split the complex system into two sets of simpler equations:

1. **Short Time Scale (Fast):** In this window, long/slow phenomena are considered almost stable (constant), so they can be ignored or treated as fixed values.
2. **Long Time Scale (Slow):** Over a longer period, fast oscillations average out to 0, meaning they can be safely ignored as a good approximation.

Note: Regarding the retina's refreshing rate, it's important to distinguish between two biological limits. For continuous, naturalistic motion, a refresh rate of around **24 Hz** is sufficient for the brain to perceive a continuous continuum. There is very little visual information processed above 30 Hz in naturalistic settings. If the visual stimulus is pushed with non-naturalistic "on and off" flashing lights, the brain can respond to much higher frequencies. For example, a novel therapy called **GENUS** uses 40 Hz visual stimulation to treat neurological reserves (like Alzheimer's). In these extreme cases, neurons can reach up to 40 or 50 Hz.

Therefore, placing the separation boundary at **50 Hz** is a safe assumption. Meanwhile, synaptic time scales (represented by S_{EE} and S_{EI}) are extremely fast—the slowest synapses close in just 5 milliseconds.

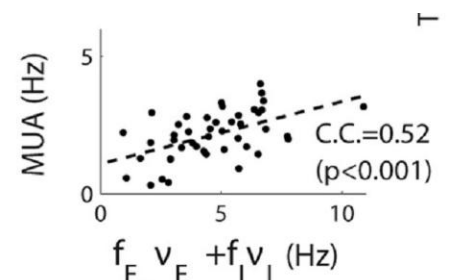
By separating the scales:

- **Low-Frequency LFP:** Because the synaptic parameters (J_{EE} , S_{EE} , J_{EI} , S_{EI}) act so fast, you can use the low-frequency component of the LFP to reconstruct the **slow input fluctuations**.
- **High-Frequency LFP:** Using neurophysiological assumptions about synaptic time scales and weights, you can use the high-frequency component to reconstruct the **total microscopic firing rate** (though disentangling the exact excitatory vs. inhibitory contributions remains difficult).

By applying this theoretical framework to the macaque recordings, the researchers tested what could actually be decoded from the mesoscopic LFP:

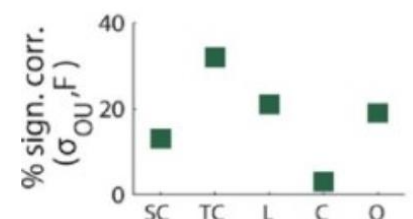
1. Reconstructing the Firing Rate (Microscopic Activity)

- The theoretical estimate of the firing rate was compared to the actual measured firing activity.
- They achieved a correlation coefficient of $r = 0.52$ (with peaks up to 0.7).
- When squared, this yields an **explained variance of roughly 50%**. This proves a significant correlation: we *can* reconstruct the general microscopic firing rate (MUA - Multi-Unit Activity) directly from the mesoscopic LFP.



2. Reconstructing the Visual Stimulus

- The reconstructed firing rate did *not* correlate well with complex external stimulus features like spatial contrast, orientation, or luminance.
- However, by using this simple equation on the LFP, they *were* able to reconstruct the **fluctuations of the temporal contrast**.
- While the correlation here is lower (explaining about **20% of the variance**), the ability to reconstruct the spectral evolution of the visual temporal contrast simply by reading the Local Field Potential is a highly significant step in understanding how information is encoded in the brain.



Applied Neuroprosthetics

This section transitions from the theoretical modeling of local field potentials to highly applied neuroprosthetics, specifically focusing on the **MBS Project**.

The foundational goal of this project is to create a closed-loop prosthetic system for amputees.

1. **Decoding (Motor Control):** The patient thinks about moving, and the EMG (Electromyography) from the muscles in their residual stump is decoded non-invasively to actuate different grasps in a robotic hand (e.g., the Prensilia hand).
2. **Encoding (Sensory Feedback):** When the biomimetic sensors in the robotic fingertips touch an object, that information must be sent back to the patient so they can physically "feel" it. This requires **invasive** writing (stimulation) directly into the nervous system.

To give back the sensation of all five fingers, surgeons must implant electrodes into the two main nerves that innervate the hand: the **Median nerve** and the **Ulnar nerve**.

Different types of electrodes were studied in order to understand which one was the most suitable and efficient:

- **Cuff Electrodes:** Initially, electrodes wrapped around the outside of the nerve. This was inaccurate because it only recorded/stimulated the border of the nerve.
- **Parallel Electrodes:** Electrodes were inserted to run parallel alongside the fibers. This allowed for multiple contact sites, but because they ran parallel along the same line of axons, the data was highly redundant.
- **Transverse Intrafascicular Multi-Channel Electrode (TIME):** The final, most advanced solution. The electrode is threaded inside and outside of the nerve fascicles transversely, almost like "knitting." This operation is extremely complex, requiring highly specialized neurology teams.

Sensory Algorithms for touch restoring

The critical missing link was the software: **How do we convert the continuous readout from the robotic sensors into a train of electrical spikes to inject into the patient's nervous system?** Over several years, the team iteratively improved this algorithm.

1. Truncated Linear Transduction (2014, Raspopovic)

The first, simplest approach mapped the force read by the sensor directly to the **amplitude** of fixed-frequency electrical pulses. The stimulation is carefully bound between two limits:

- **Lower Bound:** A minimum threshold to ensure the system ignores sensor noise or "glitches" (since sensory nerves should be completely silent if nothing is being touched).
- **Upper Bound:** A maximum safety limit. Because high electrical currents activate nociceptive (pain) pathways, the stimulation must never be unpleasant.

Every morning, the team manually calibrated the upper and lower bounds for each contact site, ensuring the stimulus was 10% higher than "I feel something" and 10% lower than "That is unpleasant." This method is completely non-neuromorphic. In biology, information is encoded in the *temporal patterns* of spikes, but biological spikes all have the same amplitude. Modulating amplitude is unnatural. Still, it allowed patients to distinguish hard vs. soft objects and basic shapes.

2. The Izhikevich Neuromorphic Model (Oddo)

The next major leap was using the **Izhikevich model** to generate actual spike trains based on the sensor data.

- **Neuromorphic Spiking:** Instead of changing the amplitude of a pulse, the algorithm feeds the temporal evolution of the sensor input (and in some cases, the derivative of the input to simulate fast-adapting neurons) into the Izhikevich equations.
- This allowed patients to discriminate fine tactile details, such as the coarseness or roughness of a texture, rather than just raw pressure.

3. The Bensmaia Skin Deformation Model

The most advanced level discussed involves adapting a model developed by Sliman Bensmaia. Bensmaia's model added a crucial middle step. Instead of mapping force directly to firing rates, the force is first used to mathematically simulate **skin deformation**. From that simulated skin deformation, the algorithm calculates the distinct responses of all four biological mechanoreceptor types: Merkel cells, Ruffini endings, Pacinian corpuscles, and Meissner's corpuscles.

Applying this highly detailed biomimetic model not only allowed for excellent discrimination of force and texture but crucially began to solve the problem of **Embodiment**.

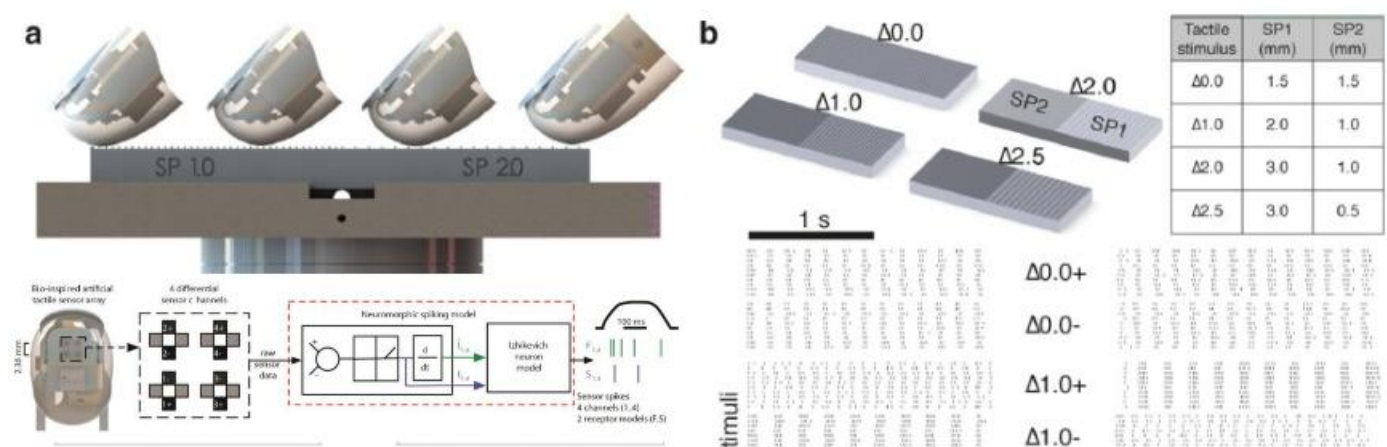
Problem of Embodiment

A major problem in neuroprosthetics is a high rate of rejection (where patients simply stop wearing the device). Because the implantation procedure is highly invasive (leaving the patient with a physical port protruding from their skin), the benefit must vastly outweigh the surgical cost. A patient may be able to control a robotic arm and even feel pressure from it, but they mentally perceive it as a tool—not as part of their own body. True embodiment occurs when the prosthesis is integrated into the patient's cognitive representation of their "self." If the goal was purely functional, one could use a non-mimetic feedback system—for example, playing a different pitch of sound in an earpiece depending on which robotic finger is pressed.

Non-mimetic feedback imposes a heavy cognitive load (the patient has to constantly "translate" the sound into a feeling) and never leads to embodiment. By pursuing highly neuromorphic algorithms that deliver natural-feeling electrical spikes mimicking actual skin deformation, researchers can trick the brain into accepting the robotic arm as its own flesh and blood.

Paper del 2016 fatto da Oddo e Mazzoni

The team conducted a classic experiment focusing on the discrimination of coarseness (differentiating between smooth and rough textures by sliding the robotic finger over them).



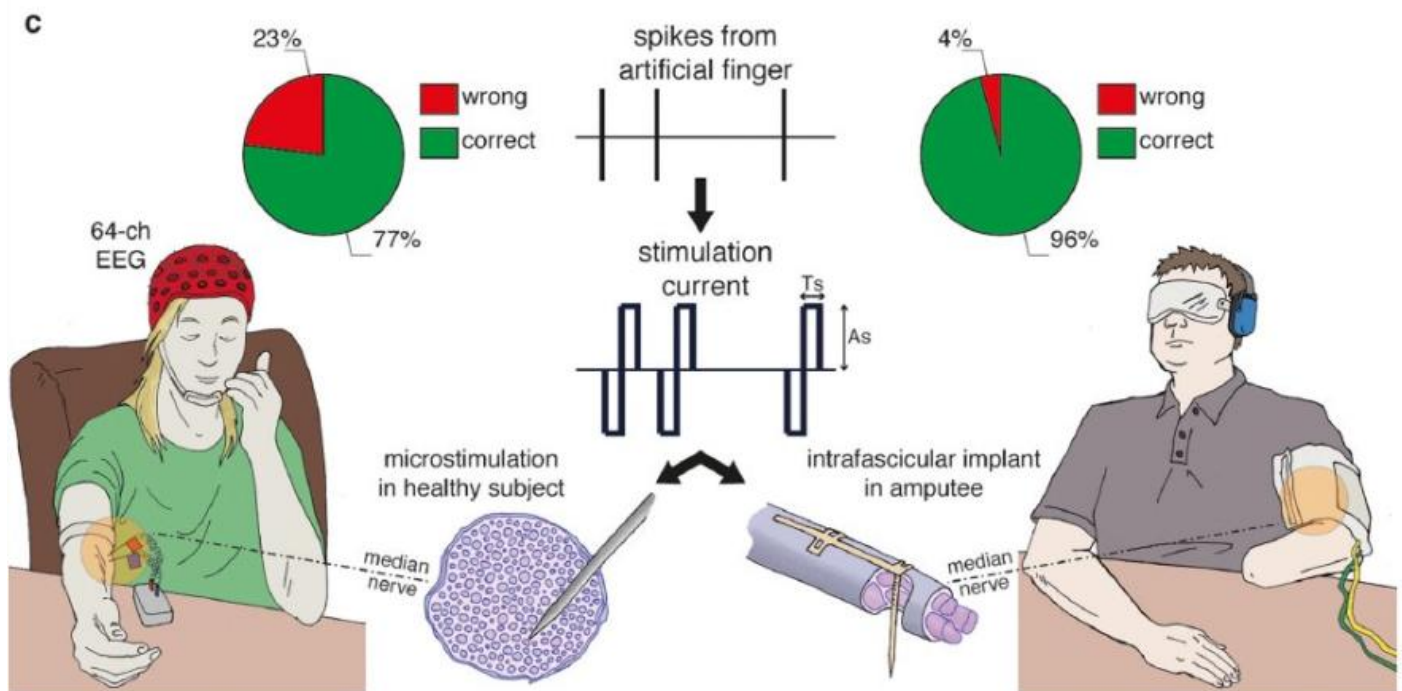
A remarkable finding emerged regarding the patient's learning curve. At the time, only one patient was enrolled in the study.

- The time the patient needed to learn the decoding part and achieve a **96% correct response rate** was essentially **zero minutes**. This was because the biomimetic adaptive Izhikevich neuron model provided highly natural feedback, the patient intuitively "grasped" the sensation immediately. This is the ultimate "plug-and-play" scenario.
- Referring back to the provocative question about using sound instead of neural stimulation (e.g., matching a specific musical note to a specific finger). If sound had been used, the patient would have needed significant cognitive effort and time to memorize the mapping. With biomimetic stimulation, no such translation was needed.

Because the study only had one amputee patient, a critical scientific question arose: **Is this a general principle, or just specific to this one individual?**

To test this, the team decided to replicate the procedure using healthy subjects.

- For healthy subjects, they used **intra-fascicular microstimulation**. This involves inserting a very thin microneedle electrode into the arm to target the fibers. It is mildly invasive, but much less complex than the amputee's implanted electrode.
- The amputee used a Transverse Intrafascicular Multi-channel Electrode (TIME)—which is "knitted" through the nerve with many small contact sites. The healthy subjects used a single-contact, larger microneedle. Because the physical elements were so vastly different, researchers had to prove these two studies were actually comparable.



Proving comparability

To prove comparability, the team turned to computational neuroscience, specifically relying on **morphological models**.

- **Functional Models (e.g., Izhikevich, Leaky Integrate-and-Fire):** Simulate the function (the firing/spiking) of the neuron. Useful for modeling mechano-sensors.
- **Morphological Models:** Simulate the physical structure and electrical spread. While LFP modeling looks at how electricity generated by a neuron spreads *outward*, this scenario flips the question: When we inject electricity, how does it physically spread to reach the neurons, which ones does it hit, and how strongly?

To build this morphological model, the following steps were taken:

1. **Imaging:** Start with a 2D transversal cut image of the nerve.
2. **Estimation:** Estimate the sizes of the fibers, the distance between bunches of fibers (fascicles), and the conductance properties of the intra-fiber material.
3. **3D Extrapolation:** Convert these 2D estimates into a 3D structural model.
4. **Finite Element Modeling (FEM):** Simulate the whole physical electrode (not just a single point source like in some LFP models, because the physical shape of the electrode matters here).
5. **Volume of Tissue Activated (VTA):** Simulate exactly how the electrical field spreads from the contact site through the tissue.

When surgeons open an arm, nerves do not come with color-coded tags; you cannot visually tell which fascicle is an afferent (sensory) or efferent (motor) nerve. If you stimulate the wrong fiber, you might just cause a muscle twitch in the stump instead of providing sensory feedback.

The goal is to target a specific fascicle. This is mapped using **Recruitment Curves**, which plot how many fibers of a specific fascicle are activated as a function of the injected electrical charge and the electrode's position.

Depending on where the electrode lands, you face three scenarios:

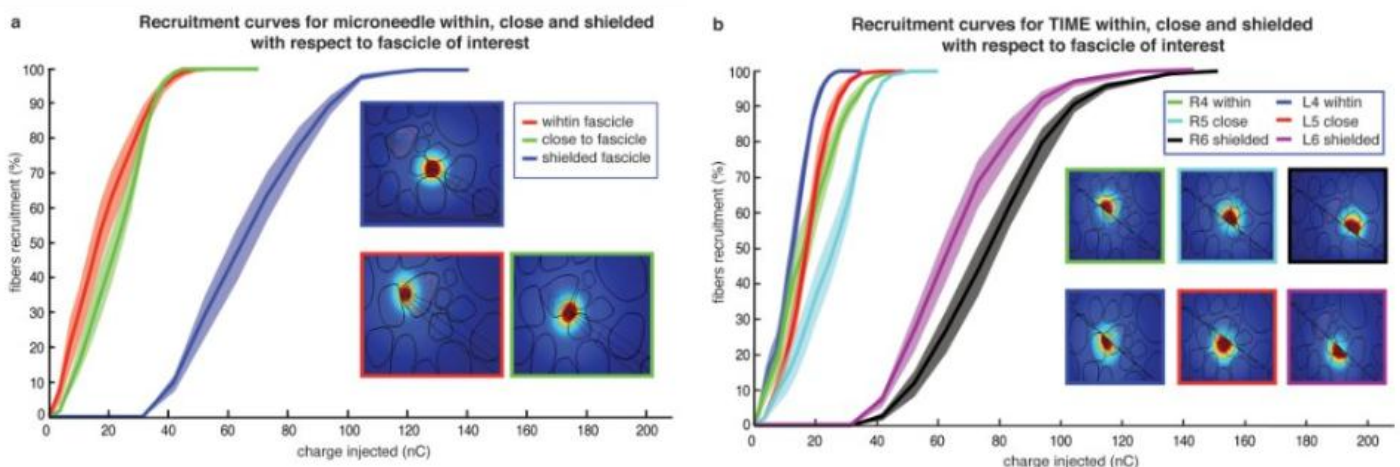
1. **Lucky:** The contact site is directly inside the target fascicle.
2. **Average:** The contact site is close to the target fascicle.
3. **Unlucky (Shielded):** A fascicle you do *not* care about is physically positioned between your electrode and your target fascicle, shielding it.

If you are shielded, you can still recruit your target fascicle by simply injecting a massive amount of current. However, if you do this, you recruit *everything* in the area, completely losing **selectivity**. The goal is always to find the minimal amount of charge that triggers *only* the target fascicle.

Both the microneedle (healthy subjects) and the TIME (amputee) suffer from bad performance if they are "shielded" by the wrong fascicle. However, they solve it differently:

- **Microneedle Solution:** Because it is a single needle, researchers must physically move it slightly and hope to get lucky with a better position.
- **TIME Solution:** The knitted electrode cannot be moved once surgically placed. Instead, it relies on redundancy. It features many contact sites (e.g., 16 per nerve). The researchers know many of these sites will be useless (shielded or poorly positioned), but out of 16, a few will perfectly target the right fascicles. You simply switch to the next contact site until you find one that works.

By running these morphological simulations, the team proved that despite the extreme differences in the physical shape of the microneedle and the TIME electrode, their fundamental recruitment curves were remarkably similar. This successfully validated that the findings from the microstimulation in healthy subjects were scientifically comparable to the findings from the amputee patient.



Paper del 2018 di Valle e Mazzoni

A major issue in the field of neuroprosthetics is a lack of standardization. Research groups frequently propose novel stimulation approaches, prove they work, publish them, and move on. To truly advance the science, the field must converge toward a standard.

To address this, the research team designed a "playoff" or competition between two distinct algorithms tested on the **same patient** in a double-blind setup.

The hardware stimulator was the TIME (Transverse Intrafascicular Multi-Channel Electrode) cable, the robotic hand, and the finger sensors were kept identical.

The two models compared were:

1. **Linear Amplitude Modulation (LAM):** Uses a fixed frequency, and only the amplitude of the electrical pulse is modulated. This is a completely **non-neuromorphic** approach (similar to the earliest 2014 studies).
2. **Frequency Modulation (FM):** Modulates the frequency of the pulses. While it doesn't generate complex spike trains using a mathematical model (like the Izhikevich model), it represents at least one step toward a neuromorphic approach.

The core scientific question was to rigorously determine the functional differences between purely modulating frequency versus modulating amplitude.

To accurately map these sensations, a student developed specialized software to record where feelings manifested on the patient's **phantom hand**. Having a perception of the phantom hand was actually a **crucial selection criterion** for patients in this study. This phenomenon typically occurs in patients who undergo amputation later in life. Because they are older, the portion of the sensory cortex originally dedicated to that hand does not fully reallocate to other body parts. Because the brain area is still waiting for input, random metabolic fluctuations or signals from neighboring brain areas can trigger it. This causes the patient to feel sensations, pain, or "itches they cannot scratch" in a hand that is no longer there.

Delivering even meaningless electrical stimulation can superimpose actual stimuli over these random metabolic fluctuations, which often makes the patient feel better. For the researchers, the existence of the phantom hand proved that the cortical area devoted to processing those nerve inputs was still active and available.

Experimental results

The testing revealed detailed differences in how the two algorithms performed regarding where the sensation was felt and what it actually felt like.

1. Localization and Perception Migration

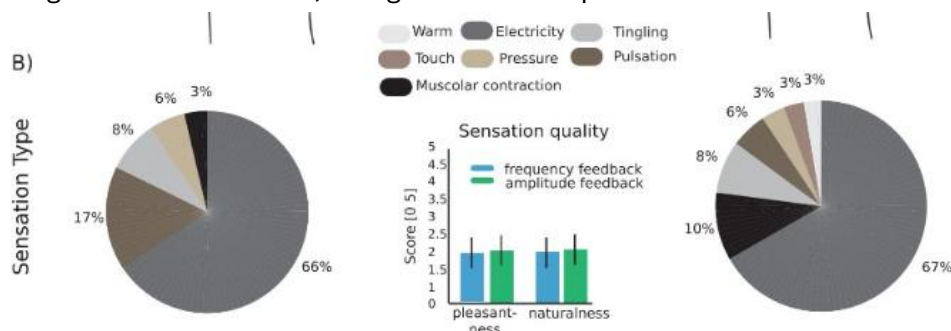
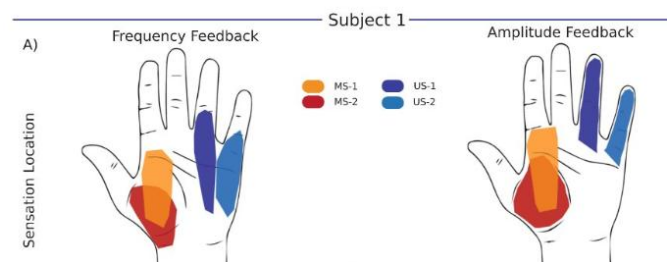
In the first days after the implant, the patient felt the electrical stimulation locally (e.g., near the stump).

However, over a timescale of **weeks**, the perception **migrated** to the correct anatomical locations on the phantom hand. The stimulation of the **Median nerve** correctly migrated to the corresponding fingers (thumb, index, middle). The stimulation of the **Ulnar nerve** migrated to the outer fingers.

Amplitude Modulation (LAM) proved to be better at localizing the feedback directly into the specific fingers.

2. Quality of Sensation: What did it feel like?

Patients were asked to verbally describe the feelings they experienced. The worst-case scenario ("Muscular Contraction") occurred about 10% of the time. If the patient feels a muscle contraction, it means the electrode has accidentally stimulated an efferent (motor) fiber instead of an afferent (sensory) fiber. Looking at the pie charts, the most common feeling reported was an unnatural "electricity," meaning the patient knew they were being shocked rather than touched. True feelings of "touch" were incredibly rare. For Amplitude Modulation, patients felt actual touch only 3% of the time. Frequency Modulation (FM) was slightly more effective and realistic at producing a "touch" sensation, though still far from perfect.



3. Naturalness and Pleasantness

Patients were asked to rate the stimulation on a scale (0 to 5) based on how "natural" and how "pleasant" it felt.

- **The Result:** Both Amplitude and Frequency modulation scored essentially a **zero** for both categories.
- **Conclusion:** Neither stimulation felt very natural, and neither felt very pleasant.

While these simpler algorithms (LAM and FM) allowed the patients to successfully decode information (like distinguishing shapes or textures based on discrimination), the patients were not genuinely happy using them because the sensations were distinctly unnatural and electronic. This justifies the deeper dive into the highly complex, fully neuromorphic models discussed earlier in the lecture.

Information Capacity

To effectively encode a signal, the system must provide an adequate number of possible, discriminable responses. If a patient can only feel "something" versus "nothing," that encodes just 1 bit of information, which is insufficient for complex tasks.

The researchers analyzed how many different levels of sensation the patients could discriminate between a minimum (I_{min}) and maximum (I_{max}) stimulation intensity.

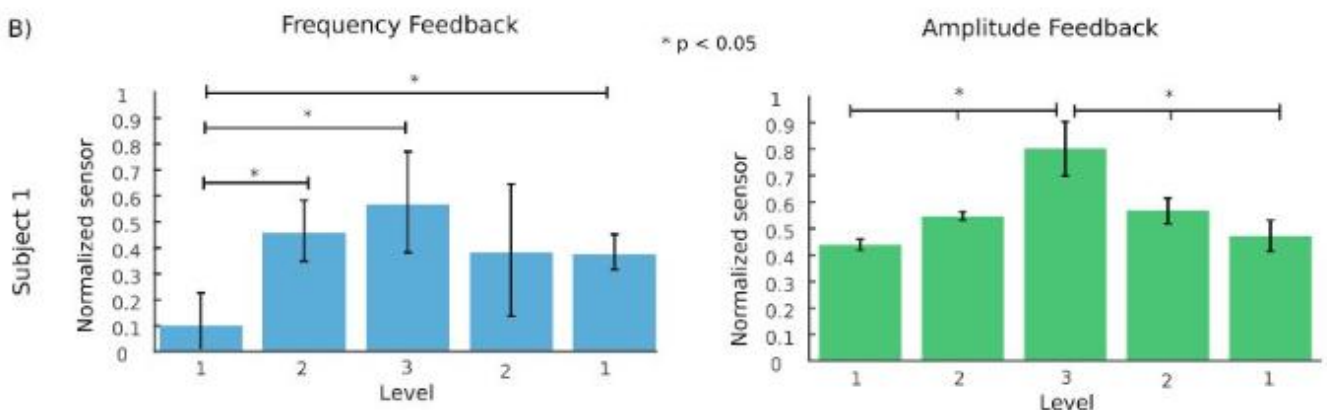
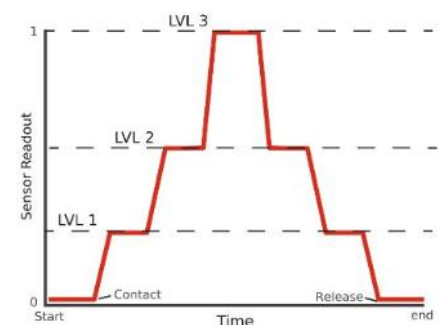
Amplitude Modulation (LAM) consistently provided a significantly higher number of discriminable feedback levels compared to frequency modulation.

The critical flaw with pure frequency feedback is **adaptation**. At the peripheral level, mechanoreceptors adapt (which can be simulated using adaptive neuron models). Further down the neural pathway, particularly in the thalamus, there is also adaptation. If the thalamus receives continuous, high-frequency stimuli, it eventually stops "taking notice" and ignores the input. For amplitude modulation, adaptation occurs over hundreds of milliseconds (a relatively long time). But for frequency modulation, once you increase the stimulation to higher frequencies, adaptation happens extremely quickly, completely muting the sensation.

To test this practically, patients were asked to exert a specific force on an object (ramp-up), maintain it, and then release it (ramp-down) using the robotic hand.

- Amplitude Modulation: Successfully allowed the patient to control both the ramp-up and ramp-down phases.
- Frequency Modulation: Worked during the ramp-up phase, but completely failed during the ramp-down phase. To communicate maximum strength, the FM system injected the maximum frequency. This immediately triggered rapid neural adaptation.

Because the patient's nervous system had "tuned out" the high-frequency stimulus, they felt absolutely nothing when trying to regulate the downward release phase.



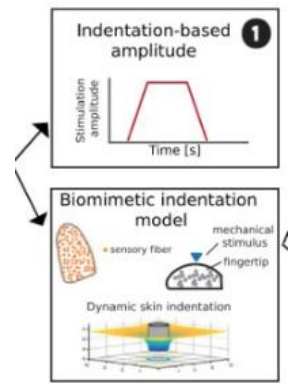
“Model battle” paper 2018 (Valle & Mazzoni)

Realizing that pure Frequency Modulation failed due to adaptation, and pure Amplitude Modulation failed in naturalness, the researchers set up a definitive "Model Tournament" to find the optimal hybrid solution.

They constructed a multi-layered decision tree (bifurcations) to test various encoding strategies in a head-to-head competition for naturalness.

1. Bifurcation 1: Simulating Indentation

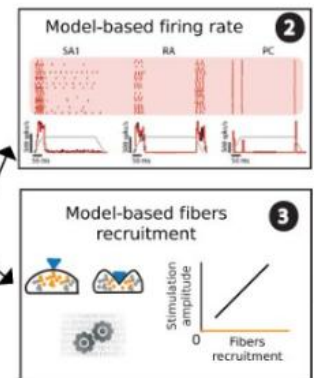
Do we simply map the raw readout of the sensor's force, or do we mathematically simulate the **physical deformation of the skin**?



2. Bifurcation 2: Selecting the Mechanoreceptors

Do we simulate just one type of neuron, two types, or all three relevant mechanoreceptors: Slowly Adapting (SA), Rapidly Adapting (RA), and Pacinian corpuscles (PC)?

The patient felt the sensation was significantly more natural when the model explicitly simulated the behavior of **all three types of neurons** rather than just a subset. The second bifurcation was thus fixed to use all three.

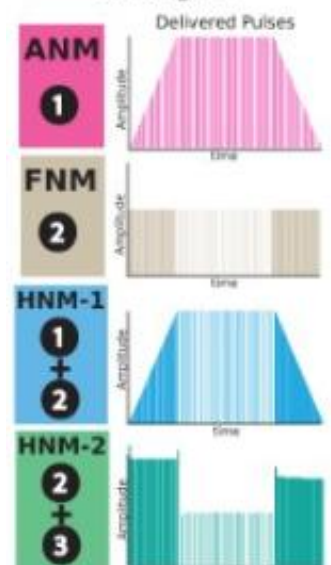


3. Bifurcation 3: The Encoding Strategy

The core of the tournament was comparing four distinct stimulation strategies:

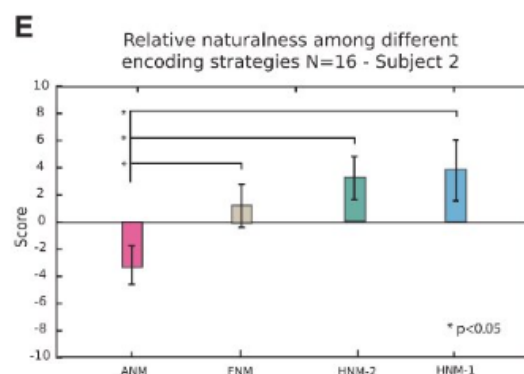
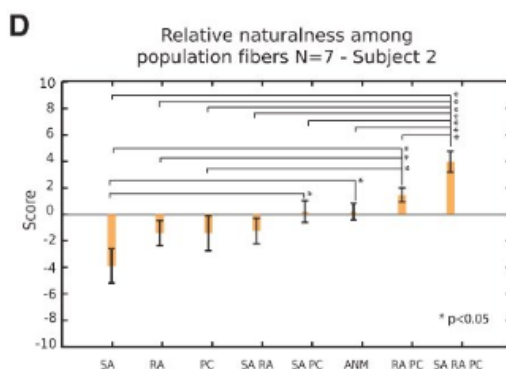
1. **Amplitude Neuromorphic Modulation (ANM / LAM):** Non-neuromorphic. Fixed frequency pulses where only the amplitude is modulated based on force.
2. **Frequency Neuromorphic Modulation (FNM / FM):** Temporal modulation. Generates a train of spikes using a model, and for each calculated spike, a pulse of *fixed amplitude* is injected.
3. **Hybrid Neuromorphic Modulation 1 (HNM-1):** Generates biomimetic spike *times* with a model, but the *amplitude* of those injected pulses is actively modulated by the raw readout of the force.
4. **Hybrid Neuromorphic Modulation 2 (HNM-2):** Generates biomimetic spike *times*, but the *amplitude* is actively modulated by the mathematical model of **skin deformation**.

Sensory encoding strategies



The patients rated the relative naturalness of these approaches.

- **ANM (Amplitude only)** lost all naturalness comparisons.
- **FNM (Frequency only)** was better than ANM but still felt highly unnatural.
- **The Winners (HNM-1 & HNM-2):** The hybrid models that combined spike-timing modulation (highly neuromorphic) with amplitude modulation (zero neuromorphic) were felt as the most natural by far.



Why did combining a highly neuromorphic timing model with a completely unnatural amplitude modulation feel the *most* natural to the patient? It solved the mystery of why the crude 2014 amplitude-only experiments worked at all.

In a natural biological finger, when you press harder on an object, two things happen:

1. **Rate Coding:** The mechanoreceptors already engaged start firing faster (spike timing).
2. **Population Coding (Recruitment):** The pressure physically spreads, triggering *new* mechanoreceptors that were previously silent. More fibers are recruited.

The researchers are not injecting chemical neurotransmitters into synapses; they are shooting raw electricity into a nerve bundle using a physical electrode. If you increase the amplitude of the electrical pulse, the physical electric field grows, creating a larger Volume of Tissue Activated (VTA). A larger VTA physically shocks more adjacent nerve fibers.

The hybrid approach (HNM) perfectly mimics biology by combining two unnatural things:

- The mathematical model provides the correct biomimetic spike timings (mimicking the rate coding).
- The artificial amplitude modulation expands the electric field, artificially recruiting more neighboring nerve fibers (perfectly mimicking biological population coding/recruitment).

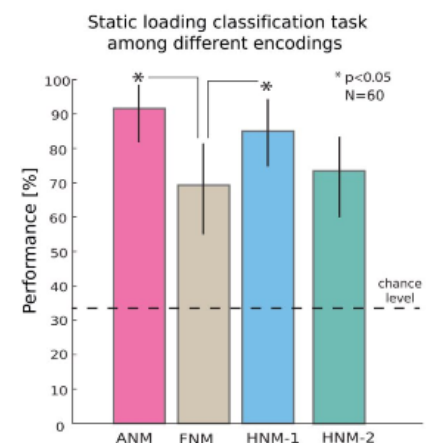
By artificially forcing both rate coding and population coding simultaneously, the brain is tricked into feeling a highly natural sensation.

Results on encoding static loadings

After the tournament for "naturalness," the models were tested functionally to see how effectively they encoded physical information. The first test measured **Static Loading**: How many different, distinct levels of pressure could the patient actually perceive and discriminate when force was applied?

In this case, the researchers didn't need the models to compete directly; they could objectively measure the functional performance of each strategy independently.

- The strategies that utilized amplitude modulation—specifically pure **Amplitude Neuromorphic Modulation (ANM)** and the hybrid **HNM-1**—were the optimal performers.
- Modulating the electrical amplitude perfectly mimics the biological recruitment of new nerve fibers proportional to the applied force. This artificial recruitment is actually the best mechanism to encode static loading/pressure.



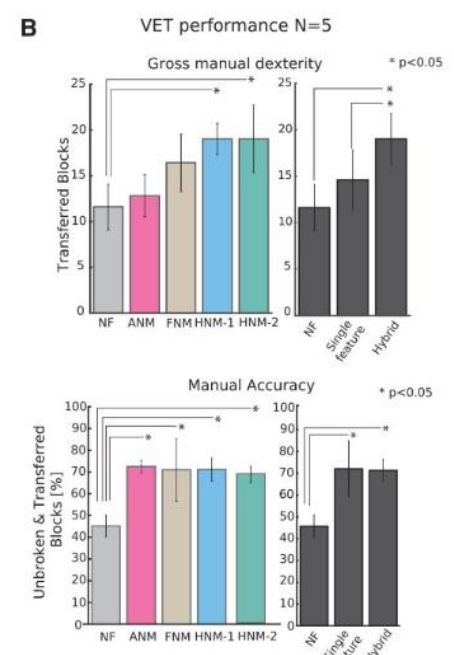
Results on grasping

To test practical skill and dexterity, the team designed a specific functional test using a custom object.

- The Object was a deformable cube embedded with springs. It acts as a fragile object; if the patient applies more than **1.2 Newtons** of force, the cube irreversibly changes shape (it "breaks").
- The patient must grasp the cube and move it from one side of a barrier to the other without breaking it.

Results of the Functional Test:

- When looking at the *fraction* of blocks transferred successfully without breaking them, all four sensory encoding strategies



performed very similarly (succeeding about two-thirds of the time, which is vastly superior to the control condition with zero feedback).

- The crucial difference appeared in the *total amount* of blocks transferred within a given time limit. The models with naturalistic/hybrid feedback (like HNM) allowed the patient to transfer significantly more blocks (**about 50% more**).

Because the hybrid feedback felt natural, it required drastically less **cognitive effort** to process. The patient didn't have to consciously think about translating an unnatural electrical buzz into a physical pressure; their brain instinctively understood the natural-feeling feedback, allowing them to perform the task much faster.

Embodiment of the model

While discussing the results with other scientists (specifically Olaf Blanke), a new question arose: If the feedback feels natural, does it actually create **embodiment**?

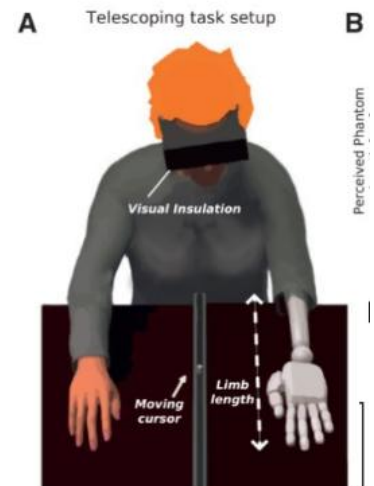
To test this, they looked at the patient's **phantom hand**.

- The sensation of a phantom hand is driven by the activity of the sensory cortex (the homunculus) that used to correspond to the missing limb. Normally, an amputee feels their phantom hand directly attached to their physical stump.
- After the patient performed the tasks using the naturalistic/hybrid feedback models (the models that utilized correct spike timings), researchers blindly asked the patient: "How long is your phantom arm?"

The patient reported that their phantom arm had physically elongated. The phantom hand was now perfectly superimposed over the physical robotic hand. This phenomenon is called **telescoping**.

This was an unintended "bonus" of the experiment. The primary goal was simply to elicit natural stimuli. However, proving that natural sensation (combined with functional amplitude modulation) automatically gives the patient a sense of **embodiment** was a massive breakthrough. It directly solves the problem of patients rejecting prosthetics because they feel like foreign tools rather than parts of their own body.

This specific modeling approach—combining highly detailed mechanoreceptor simulation with physical amplitude modulation—has become the standard in the field for the last eight years.



Paper by Romeni, Mazzoni and Valle on the peripheral neural interfaces

With the modeling of the mechanoreceptors established, the next open challenge was formalizing the **morphological part**. The goal was to build a comprehensive *in silico* (computer-simulated) testing environment. This environment allows researchers to simulate peripheral nerve stimulation, test various electrode designs, and define rules for optimal electrode shapes to achieve a specific stimulation without needing physical trials every time.

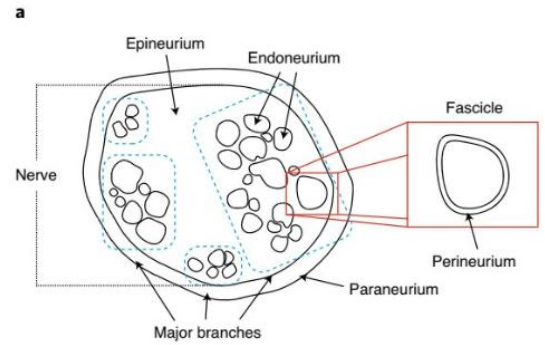
1. Geometry and Histology (Step Zero)

The foundational step to building the morphological model is **histology**—taking transversal cuts of the nerve to map the sections of all fibers. Using the exact, chaotic biological shapes of actual fascicles is computationally prohibitive. Instead, the model simplifies the geometry using circles or ellipses. Because actual histology is done *a posteriori* (after the fact), a predictive model needs to *generate* a set of possible shapes. Based on known statistical distributions of sizes and shapes, the system randomly generates non-overlapping circles/ellipses to represent the fascicles.

2. Defining Electrical Conductivities

To accurately model how electricity spreads, you must assign distinct electrical properties to the different biological materials within and around the nerve.

- **Paraneurium:** The tissue outside the nerve (highly relevant when testing cuff electrodes that wrap around the outside).
- **Epineurium:** The interfascicular material filling the space between fascicles.
- **Perineurium:** The sheath or "skin" surrounding each individual fascicle.
- **Endoneurium:** The material inside the fascicles.



The model must also differentiate between the conductivities of myelinated vs. unmyelinated fibers.

3. Spatial Segmentation

In some nerves (like the sciatic nerve), the internal architecture is well-segmented—fibers going to specific areas are neatly grouped on the left or right. In other nerves, it is highly mixed. If the segmentation is known, it is explicitly built into the model.

Interface Dynamics: Real-World Physical Factors

Once the biological nerve is modeled, the **interface geometry** (the shape, materials, and properties of the electrode) is introduced. There are two physical factors that heavily influence the model:

1. **Tissue Deformation:** Initially neglected by the team, the physical presence of the electrode significantly alters the spatial location of the fascicles. Even thin electrodes cause deformation:
 - **Extra-neural electrodes** compress the fascicles.
 - **Intra-neural electrodes** displace the fascicles.
2. **Cicatrization:** When a foreign body (the electrode) is introduced, the nerve reacts by scarring. This cicatrization decreases electrical conductivity over time. Developing materials that minimize this immune reaction to maintain stable conductivity is an entire sub-field of Brain-Machine Interface (BMI) research.

The 3D Simulation Pipeline

To bring this all together into a functioning 3D simulation, researchers follow a specific pipeline, making careful assumptions to keep the computation manageable:

1. **Level of Amputation:** Define exactly where on the limb the interface occurs.
2. **Probability Densities:** Use the known probability distributions for fiber diameters.
3. **Neglect Tortuosity:** Assume the nerve fibers run relatively straight. While actual nerves twist and turn (tortuosity), including this 3D twisting makes the model unnecessarily complex ("drives you crazy"), so it is usually relaxed or neglected.
4. **Generate Populations:** Create the fascicles based on varying densities, ensuring they do not superimpose, and separate the sub-populations (e.g., modeling efferent and afferent fibers slightly differently).
5. **Discretization:** The contact sites where stimulation occurs must be discretized (never modeled as a continuous field) to avoid cross-contamination in the simulation.
6. **Waveform Selection:** Because it is an *in silico* environment, researchers are not limited to standard square pulses and can test any stimulation waveform.

Practical application

This methodological framework yields no biological results on its own, but it is an incredibly powerful tool. For example, it allows researchers to compare completely different electrode designs.

- **Cuff vs. Intra-neural:** Cuff electrodes are much easier and safer to surgically implant than intra-neural ones. Using this simulation, researchers can determine the precise conditions under which a cuff electrode is "good enough" (i.e., when you don't lose too much precision compared to an internal electrode).
- **The Ultimate Output:** For any given electrode and injected current, the model calculates the **Recruitment** (how many fibers are activated) and the **Selectivity** (whether you are targeting the correct fibers).

Paper by Oddo, Mazzoni and Valle on discrimination of naturalistic textures

In the work, the research team shifted focus from discriminating basic levels of force or object shapes to discriminating complex **natural textures**. The goal was to prove that the neuromorphic system carried enough information to distinguish between various materials: glass, paper, leather, rubber, textile, PVC, Velcro, and wood.

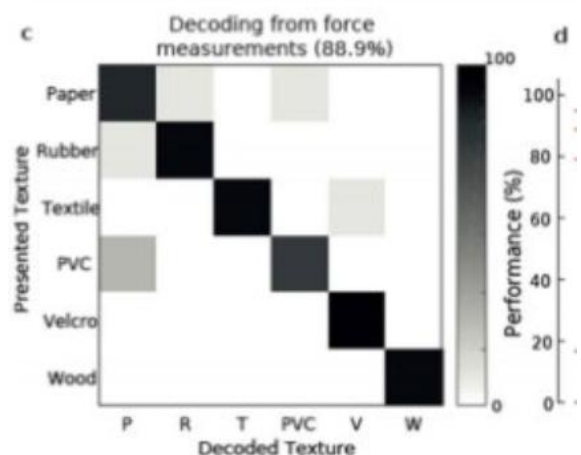
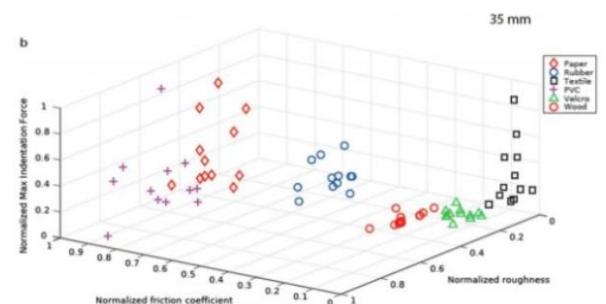
Biological skin is complex. Different mechanoreceptors not only have different adaptation levels but are also located at **different depths** within the skin (e.g., Type 1 closer to the surface, Type 2 deeper). The team wanted to reproduce this "morphological computation" by placing sensors at different depths in the robotic finger to simulate the respective receptive fields. The robotic finger did not have enough physical space/locations to accommodate sensors for both types of receptors simultaneously. They proved they could simulate depth differences in principle, but for the actual experiment, they simplified the system to use only **Type 1** receptors, as they provided more useful information for this specific task.

To understand if the textures were objectively distinguishable, the team clustered the materials in a 3D space based on their physical properties. They found that exactly **three dimensions** were sufficient to discriminate among them:

1. **Indentation Force:** Indicates how hard versus soft a material is.
2. **Roughness:** Indicates how physically textured or changeable the surface is.
3. **Friction Coefficient:** Indicates the resistance to sliding.
 - Friction and roughness are two distinct dimensions. For example, an absolutely smooth surface like glass has zero roughness, but if you try to slide a finger across it, it "sticks" (high friction).

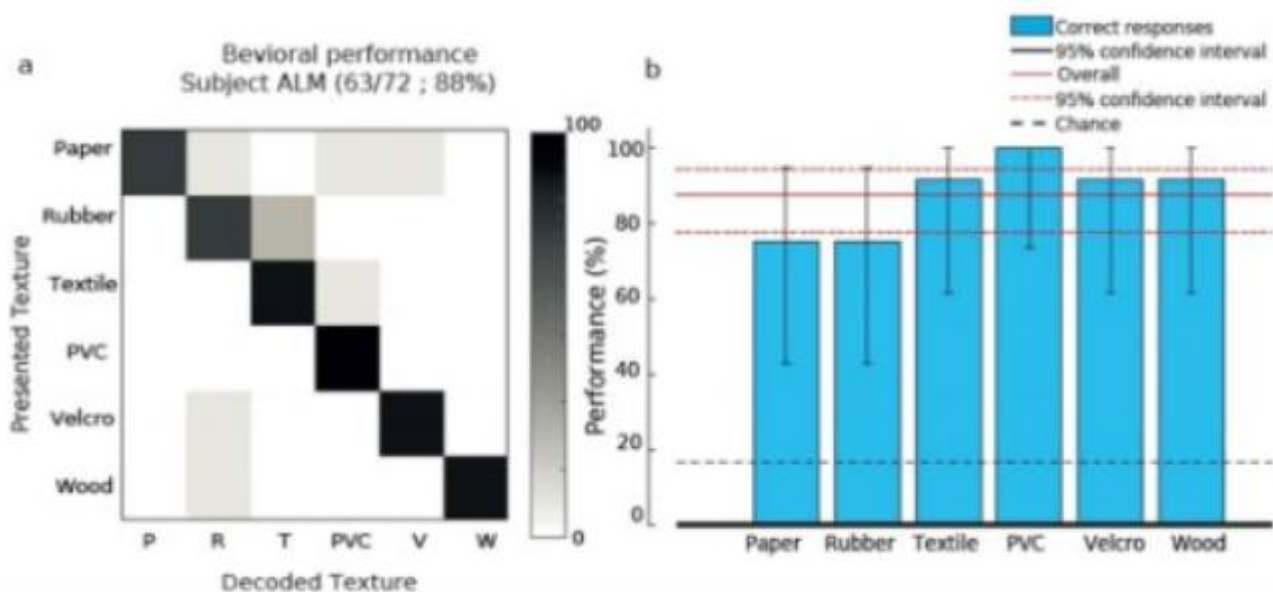
Before testing the patient, the researchers needed to know if the artificial sensors—prior to any neural conversion—were actually capable of differentiating these materials based on the force measurements.

- The sensors achieved a **90% accuracy** rate in discriminating the textures.
- Looking at the "Neuromorphic Sensor" confusion matrix, the high accuracy is visible on the diagonal. The off-diagonal elements show minor confusions, specifically mistaking PVC for paper, and rubber for paper.



Finally, the system was tested on the actual patient to see how well she could perform using the artificially injected neural feedback.

- The subject was able to successfully decode the textures with an **88% accuracy** rate. Her specific confusion matrix showed that she mostly confused paper and rubber, while successfully identifying the rest.
- The researchers wanted to understand *why* this relatively simple feedback model worked so well. They found that the patient's accuracy could be entirely reproduced by looking at just three specific properties of the injected spike trains:
 1. **Indentation Spike Count:** The number of spikes generated during the first reaction (the initial touch/press *before* sliding).
 2. **Overall Spike Count:** The total number of spikes generated *during* the sliding motion.
 3. **Inter-Spike Interval (ISI) Coefficient of Variation:** This measures how regular (or irregular) the timing is between the spikes, not just the total count.



Because the combination of these three macro-dimensions (Indentation Spike Count, Sliding Spike Count, and ISI Regularity) was completely sufficient to achieve 88% accuracy, it suggests an important limit to human perception in BMIs. While finer, microscopic details—like the exact mathematical pattern of every single spike—might technically carry more data, that level of detail is **not appreciated or utilized by the patient**. The nervous system effectively discriminates textures based on the broader regularity and volume of the spikes.

Information Theory

We have been studying cortical networks primarily to understand how they process information. To make this concept more immediate and easier to analyze, we focus specifically on **sensory pathways**.

In our model, the network represents a receptive field in the visual cortex. We analyze how this network handles progressively more complex inputs, keeping in mind that any biological input is always composed of two parts:

1. **A Signal Component** (the actual stimulus, like a visual pattern).
2. **A Noise Component** (the background spontaneous activity of the brain).

We discovered how the strength of a signal is encoded by neural activity. When we look at the power spectrum of the network's activity, we see the baseline $1/f^\alpha$ background noise. However, as we increase the intensity of the sensory input, the network develops distinct oscillations.

It is critical to understand that these are **Endogenous Oscillations**:

- A network can oscillate because the external input itself is oscillating (exogenous).
- a network can oscillate because of its own **internal resonance**, even if the external input is perfectly constant.

In our model, the input from the thalamus was constant, but due to the internal architecture of the network (the rhythmogenesis mechanisms we discussed previously), increasing the input intensity triggered an internal resonance. This generated a prominent oscillation in the **Gamma band (around 50 Hz)**. Because our mathematical prediction perfectly matched biological reality, we proved that the gamma band is **modulated** by the input.

We know the gamma band is modulated by the input (some might casually say "encoded"), but now we must ask a much more mathematically and statistically rigorous question: **Is this specific frequency band actually carrying information about the input?**

Saying that a signal is "modulated by the input" is *not* the same thing as saying it "carries information about the input." The fact that neural activity is modulated by an input is a **necessary but not sufficient condition** for it to carry information about that data.

To prove that the gamma band actually transfers meaningful data, we must move beyond simple spectral analysis and introduce the mathematical tools of Information Theory.

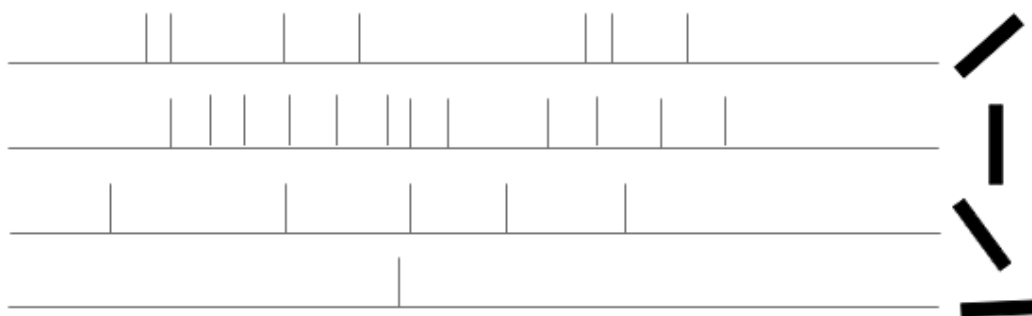
Definition of Information (1959)

Before we can mathematically define what it means for neural activity to "carry information" or be "redundant," we need to understand how neurons respond to stimuli in the physical world. To do this, we have to start from the 1959 paper by David Hubel and Torsten Wiesel. Hubel and Wiesel conducted experiments on anesthetized cats, inserting electrodes into their primary visual cortex to record single-neuron activity.

During these experiments, they defined the concept of the **Receptive Field**: the specific, restricted area of the visual field that a single neuron is sensitive to.

- To trigger a response, they found that these neurons act as basic "Lego bricks" for vision by detecting **contrast in movement**.
- For example, a neuron will fire spikes when a bright light bar moves against a black background, provided that the movement happens strictly *inside* that neuron's specific receptive field.

Hubel and Wiesel noticed that a neuron doesn't just care *where* the stimulus is; it also cares about the **direction and orientation** of the stimulus.



If you slide the light bar across the receptive field, the neuron's firing rate changes depending on the angle:

- **Preferred Direction:** The specific angle that generates the maximum number of spikes.
- **Slightly Off-Angle:** Generates fewer spikes.
- **Orthogonal (Perpendicular) Angle:** Generates very few spikes (not strictly zero due to background noise, but a minimal amount).

This behavior is called **Response Tuning**. This allows to build the **Tuning Curve**:

- **The X-axis (Function of):** The property of the stimulus being tested (in this case, the **angle/orientation** of the moving bar in degrees).
- **The Y-axis:** The neuron's response (the **average firing rate** or number of spikes). The curve usually looks like a bell shape, peaking at the neuron's preferred orientation. This curve represents **Encoding**: observing how the stimulus dictates the average neural response over many trials.

As computational neuroscientists, the goal is often to go backward. This is called **Decoding**: reading the spike output of a neuron and trying to guess what stimulus the animal is actually looking at in real-time. However, decoding is incredibly difficult because the tuning curve only represents an *average* over many repeated trials.

- If you present the exact same moving bar to the exact same neuron multiple times, you will get very different spike patterns and spike counts on each individual trial.
- Because of this high trial-to-trial variability, the **error bars** on the tuning curve are massive.
- If the noise (error bar size) is as large as the difference in the average firing rates between two different angles, you cannot accurately reconstruct or decode the stimulus based on a single trial.

We also have to consider that biological systems are inherently noisy and therefore it is almost impossible to decode a single trial.. The four main sources of noise in the brain are:

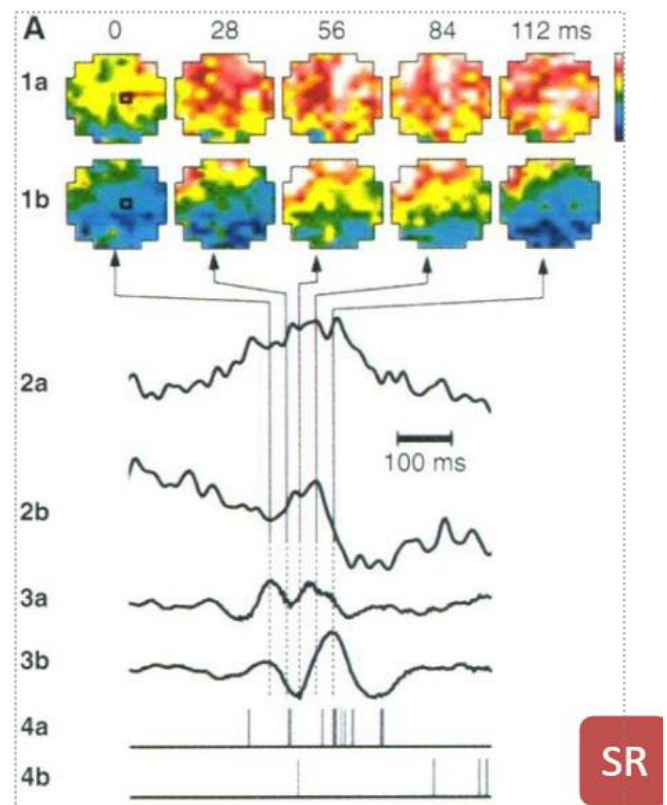
- **Sensory Transduction Noise:** The physical sensors (like the fluid dynamics and photoreceptors in the eye) are not 100% reliable every single time.
- **Cellular Noise:** The neuron's membrane is porous. Ion channels can leak or open spontaneously even when they are not supposed to, causing random fluctuations in the membrane potential.
- **Synaptic Noise:** This is the most significant source of noise. The synaptic cleft relies on fluid dynamics to transport neurotransmitters from the pre-synaptic button to the post-synaptic receptors. We can have false negatives in case the neurotransmitter is released but misses the receptors or fails to lock in properly, so the signal dies. On the other hand, false positives include random molecules that lock into the receptors by chance, triggering a signal when there shouldn't be one (this is also exactly how many drugs work).
- **Motor Noise:** Variability in the transmission and execution of physical movements.

Despite this massive amount of noise, we can still successfully grasp a moving object most of the time. How does an engineered system cope with such unreliable components? The answer is **Redundancy**. The brain does not rely on just one neuron to monitor a receptive field. Instead, the entire network surrounding that specific neuron, as well as several separated, distinct neural populations, are all assigned to process the exact same receptive field. By pooling the activity of many noisy neurons, the brain overcomes individual failures and lesions, creating a highly efficient macro-system out of noisy micro-components.

Extracting signal from ongoing brain activity

To understand how hard it is to decode a stimulus from neural activity, let's analyze a classic experiment using **Calcium Imaging** in the visual cortex of cats. Calcium imaging allows researchers to visually record the activity of a neural population. The neurons literally change color based on their membrane potential. When a visual stimulus is presented ($t=0$), the evolution of the network's activity over the next 100+ milliseconds looks completely different from one trial to the next, even though the cat recognizes the exact same stimulus every time.

A fundamental paper described the brain's massive amount of ongoing background activity with a brilliant analogy: **The effect of a stimulus is like the additional ripples caused when throwing a stone into a wavy sea.** The physics of the ripples (the stimulus) are there, expanding from the center, but good luck finding those tiny circles on top of massive, chaotic ocean waves (the ongoing brain noise).



To see the ripples, you must perform **Baseline Subtraction**.

1. Before the stimulus is presented ($t < 0$), you estimate the baseline activity and its variability.
2. You project that baseline forward and linearly subtract it from the post-stimulus activity.
3. Once the baseline "waves" are mathematically removed from the single-neuron data, two previously unrecognizable time series suddenly reveal the exact same underlying structure (e.g., two distinct peaks in activity caused by the stimulus).

To definitively prove that a specific neural feature carries information, we cannot rely on a single presentation of a stimulus. We must use multiple **trials (or repetitions)**.

Let's take as example the recordings from a monkey's visual cortex (Spikes and Local Field Potential) during four separate presentations of the same movie scene. For a neural code to carry information, it must be composed of features that are **different from scene to scene** (so the brain can discriminate between them) but **similar from trial to trial** (so the brain knows it is a reliable signal, not just random noise). By looking across all four trials, we might spot a highly reliable feature—such as a sudden, sharp decrease in the firing rate (inhibition) at the exact same moment in every trial. Because this feature is stable across multiple repetitions, it is highly unlikely to be noise; it is carrying information.

Correlation vs Connectivity

The brain is a massively interconnected network. The **Watts and Strogatz "Small World" network theory** shows that with just a few long-range connections, an entire system can be heavily linked.

However, we must strictly separate two concepts in neuroscience:

1. **Connectivity:** There is a physical link (or a series of links) between Node A and Node B.
2. **Correlation:** The functional state of Node A actually affects the state of Node B.

Just because two objects are connected does not mean they are correlated. For example, all neurons in the nervous system are connected. But in a waking state, most are functionally uncorrelated. The neurons processing the left side of your visual field are physically connected to the rest of the brain, but their activity must remain independent (uncorrelated) from the neurons processing the right side of your visual field at the first level of processing. They are connected, but separated by so many intermediate variables that they act independently.

Because neural variables are embedded in such a complex network, how do we define what is a "signal" and what is "noise"? The answer is that **it is always arbitrary**.

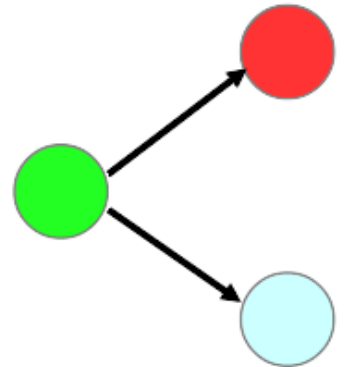
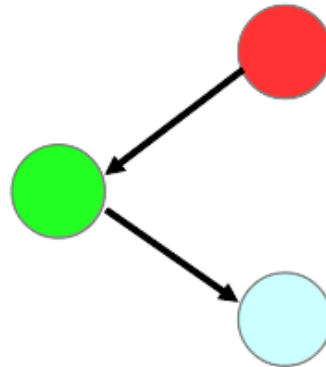
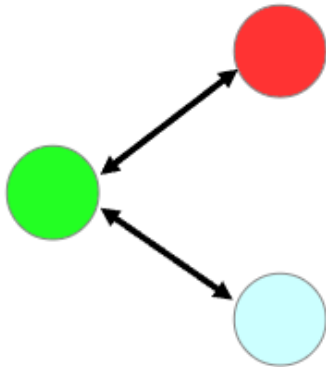
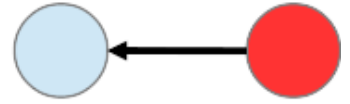
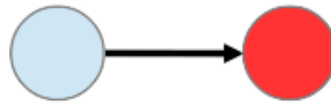
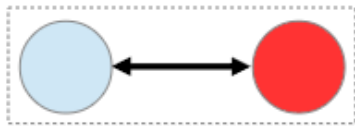
- **Signal:** The relationship between the specific pair of variables you have chosen to study.
- **Noise:** Literally everything else.
- *Example:* If you are studying the visual cortex, any electrical activity leaking over from the auditory cortex is considered "noise" for your specific experiment.

Finally, when studying the correlation between our defined "signal" variables, we must remember the golden rule of complex systems: **Correlation does not imply causation**.

If Node A and Node B are highly correlated, it could mean:

- **Unidirectional:** A drives B, or B drives A.
- **Bidirectional:** They mutually affect each other.
- **Mediated (Third-Party):** They have no direct influence on each other whatsoever, but they are both being driven by a hidden "Node C" that we aren't recording.

Therefore, trying to reconstruct the actual physical wiring of a neural network just by looking at functional correlations is an inherently **ill-posed problem**.



Correlation

Before moving into Information Theory, we must first understand the standard tool used to measure relationships: **Pearson Correlation**. If we have two variables (e.g., two time series, X and Y), the Pearson correlation measures how they co-vary relative to their respective means, weighted by their individual variability.

- **Positive Correlation (up to +1):** When X is above its mean, Y is also above its mean. If you plot them against each other, they form a line sloping upward.
- **Negative Correlation (down to -1):** When X is above its mean, Y is below its mean. They form a line sloping downward.
- **Zero Correlation (0):** When X is above its mean, Y does whatever it wants. The variables act independently, creating a scattered cloud of points.

$$\text{corr}(x, y) = \frac{\text{Covariance}}{\sqrt{\sum_t (x(t) - \bar{x})^2} \sqrt{\sum_t (y(t) - \bar{y})^2}}$$

Standard deviations

When you run a correlation, you actually get two distinct numbers,

and it is vital to understand the difference between them. The **correlation coefficient (R)** is the number between -1 and 1. It tells you the *strength* and *direction* of the linear relationship. On the other hand, the **significance (p)** tells you the probability of getting this exact correlation coefficient **purely by chance** (the null hypothesis).

- If $p < 0.05$, it means there is less than a 5% probability that random data would produce this strong of a correlation.
- **How to test the null hypothesis:** take the original time series and randomly scramble the temporal order of the data points. This preserves the mean, standard deviation, and overall distribution, but destroys any real relationship. If you correlate your data against 100 of these randomized series and fewer than 5% show a high correlation, your actual result is significant.

The relationship between coefficient and significance heavily depends on your dataset size:

- **Small Dataset (e.g., 4 points):** You might get a perfectly straight line and a massive correlation coefficient ($R = 0.99$). However, it is **not significant** because the chances of 4 random points aligning by pure luck are very high ($p > 0.05$).
- **Massive Dataset (e.g., 1000 points):** The points might look like a messy, scattered cloud, giving a very low coefficient ($R = 0.20$). However, because there are so many points slightly leaning in one direction, the probability of that happening by pure chance is practically zero. Therefore, it is **highly significant**.

Correlation across time and autocorrelation

Standard correlation checks if $x(t)$ and $y(t)$ affect each other synchronously. However, to see if the *past* of one variable affects the *future* of another, we introduce a **Delay (d)**.

We calculate the correlation between $x(t)$ and $y(t - d)$.

- If $d = 0$, it is standard synchronous correlation.
- If $X = Y$, you are correlating a variable with its own past ($x(t)$ vs $x(t - d)$). This measures how much the past state of a network affects its present state.

Spurious correlation

If you test enough random variables against each other, eventually, you will find two completely unrelated things that strongly correlate by pure statistical chance. An example of statistical absurd correlation is The divorce rate in Maine vs. per capita consumption of margarine.

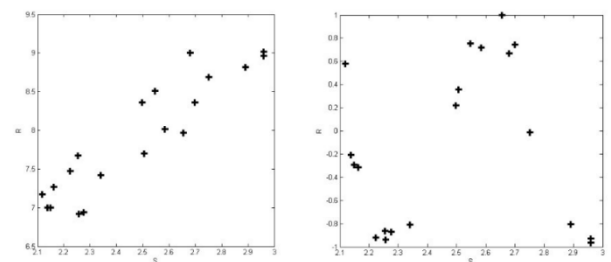
To avoid claiming that margarine causes divorces, you must use **Multiple Comparison Correction**. You cannot just blindly test thousands of variables; you must establish a functional hypothesis first.

To prove this point to the neuroscience community, researchers put a *dead salmon* in an fMRI scanner, showed it violent and peaceful images, and tested all 1,000,000 pixels (voxels) in the scan for correlations. Because they did not correct for multiple comparisons, they actually found two pixels in the dead fish's brain that "significantly correlated" with the violent images. This was a stark warning to scientists: if your resolution is high enough, you *will* find false positives if you don't correct your math.

Linear Correlation

Pearson correlation only cares about linear relationships.

If you have a dataset generated by the equation $R = \cos(S) + \text{noise}$, looking at the scatter plot will reveal a perfectly obvious, beautiful wave shape. The variable R is clearly completely dependent on S . However, if you run a Pearson correlation on this data, you will get a coefficient close to **0**, and a massive p -value ($p = 0.70$). The math will tell you these variables do not affect each other, which is demonstrably false.

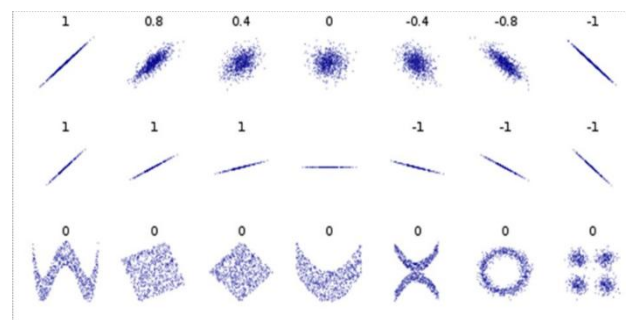


$p < 0.0000000001$ & $R = 0.902$
means related..

...but does $p <= 0.69$ & $R = -.075$
mean unrelated???

On the left image, we can visualize the math limits of the linear correlation:

- A perfect diagonal line = 1 or -1.
- A scattered cloud = 0.
- **A perfect horizontal line = Undefined.** Because there is zero variability on the Y-axis, the formula forces you to divide by zero. The math literally breaks down, even though the relationship (Y is constant regardless of X) is obvious.
- Complex shapes (circles, squares, U-shapes) = 0.



Because Pearson correlation produces **false negatives** whenever a relationship is non-linear—and because the basic dynamic unit of the brain (the neuron) is inherently non-linear—linear correlation is ultimately an insufficient tool for computational neuroscience. To find these hidden, non-linear relationships, we must abandon Pearson and move to a much more powerful tool: **Information Theory**.

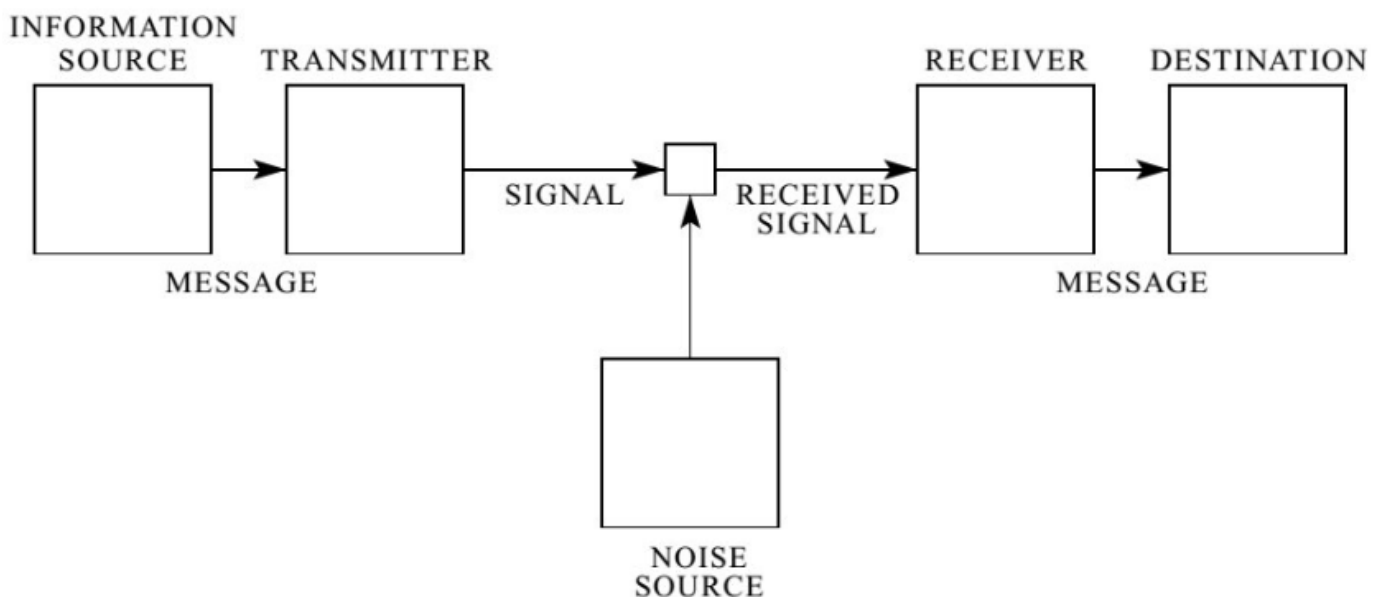
Shannon's Information Theory

In the 1940s, while working for a telephone company (during an era when industries heavily funded basic research), Shannon was tasked with creating a general theory of communication that could apply to *any* medium. His genius stroke was to take the concepts of **thermodynamics** (which studies the entropy and states of gases) and apply them to **symbolic representations**.

Shannon defined a universal pipeline for how information is transferred from a sender to a receiver.

The pipeline follows these steps:

1. **Information Source:** The idea or concept you want to communicate.
2. **Transmitter (Encoding):** Converting the idea into a specific code or signal (the "way" you say it).
3. **The Channel & Noise:** The signal travels through a medium, where background noise inevitably corrupts it.
4. **Received Signal:** The receiver gets a messy combination of the original signal plus noise.
5. **Receiver (Decoding):** The receiver must interpret the noisy signal back into the original code.
6. **Destination:** The information arrives in the receiver's mind.



For this communication pipeline to work, both the sender and receiver must agree on a limited set of symbols, known as the **Code**.

As an example, we can consider as code the language used in the lecture itself:

- **The Information:** Concepts about Neuromorphic Engineering.
- **The Code:** English. Even though the professor and students share Italian as a mother tongue, they agreed to use a limited set of English symbols to encode the information.
- **Failed Encoding:** If the professor chose to encode the lesson in Chinese, the flow of information would be near zero because the receiver does not have the tools to decode those symbols.

Other examples of codes could be with fingers or written language.

Entropy

How do we mathematically measure if a specific code (like English, Chinese, or Morse code) is an effective choice for carrying information? The first step is to measure the **Uncertainty** of the stimulus from the *receiver's perspective*.

When you agree on a code (like the English dictionary), you have a finite set of symbols. As a listener, you don't know exactly what the next word will be, but you have expectations. In a Neuromorphic Engineering class, the probability of the next word being "neuron" is much higher than the probability of it being "elephant". Your uncertainty about what the next symbol will be represents the system's capacity to communicate. If there are

many possibilities, a lot of information can be conveyed. To measure this uncertainty mathematically, Shannon used the exact same formula used for the entropy of gases in thermodynamics. This is called the **Initial Entropy of the Stimulus**.

The entropy is *not* just a raw count of the number of possible stimuli. Instead, it is calculated using the probability of each individual symbol occurring.

The mathematical formula for the entropy (H) of a set of symbols is:

$$H = - \sum p_i \log_2 p_i$$

Where p_i is the probability of a specific symbol occurring. H measures the amount of uncertainty inherent to a random variable, i.e. the amount of information necessary to describe it completely.

To measure information, we must first define our variables mathematically based on the set of symbols (or stimuli) we are using to communicate.

According to the equation, we define the following:

- **S** : The entire set of possible stimuli we have decided to use (e.g., the entire English dictionary, or a specific set of images).
- **s** : An individual element (stimulus) belonging to this set ($s \in S$).
- **$P(s)$** : The a priori probability of getting that specific stimulus s .

The starting uncertainty—the **Initial Entropy** of the stimulus set—is calculated using this formula:

$$H(S) = - \sum_{s \in S} P(s) \log_2 P(s)$$

Because $P(s)$ is a probability, it is always a number between 0 and 1. The logarithm of a fraction is a negative number. Therefore, the minus sign at the front of the equation ensures that Entropy (H) is always a positive value.)

Calculating the entropy of the entire English dictionary is computationally heavy. To understand how this works in practice, consider the classic board game **Guess Who?** as an example.

In this game, the goal is to communicate one specific signal (the chosen character) out of a limited, well-defined set of symbols.

- **Total Symbols**: There are 24 characters on the board.
- **Probability ($P(s)$)**: Assuming your opponent picks a character completely at random, the probability of any specific character being chosen is $1/24$.

When all stimuli have an equal probability ($1/N$), the entropy formula simplifies beautifully. The summation and the fraction cancel out, leaving just the base-2 logarithm of the total number of items:

$$H(s) = \log_2 24 \approx 4.585$$

The unit of measure for this base-2 logarithm is the **bit**. In Information Theory, entropy represents **the average number of binary (Yes/No) questions it takes to reduce your uncertainty to exactly zero, given knowledge of the underlying distribution and taking the optimal question-asking strategy**. Therefore, a starting entropy of 4.585 bits means that, mathematically, you can always win a game of Guess Who? with exactly 5 perfectly optimized questions, regardless of which character your opponent picked.

How does this translate back to the brain? When we present an animal with a standard set of visual stimuli in an experiment, we are not showing them every image in the universe. We have a defined set S . If we present them randomly, the starting entropy is simply $\log_2$ of the number of images.

The goal of the brain (or the computational neuroscientist) is to figure out **which response from the neurons is the most useful "question" to ask to reduce uncertainty about the stimulus**.

Once we ask a question (or observe a neural response), we need to measure how much uncertainty is left over. This is called the **Residual Entropy of the stimulus given the response ($H(S|R)$)**.

$$H(S|R) = - \sum_{r \in R} P(r) \sum_{s \in S} P(s|r) \log_2 P(s|r)$$

Let's say we use the question, "Are you a woman?" as our probe (our neural response feature). In the game, there are exactly 5 women and 19 men.

- **R (Set of Responses):** The possible answers are "Yes" or "No".
- **$P(r)$ (Probability of the response):** $P(\{\text{Yes}\}) = 5/24$. $P(\{\text{No}\}) = 19/24$.
- **$P(s|r)$ (Conditional probability of a stimulus given the response):**
 - *If the answer is "No":* The 5 women now have a probability of 0. The remaining 19 men each have an equal probability of $1/19$.
 - *If the answer is "Yes":* The 19 men now have a probability of 0. The remaining 5 women each have a probability of $1/5$.

If you plug all of these probabilities into the formula, you get a **Residual Entropy of 3.85**.

This question was good because it allows to successfully reduce your uncertainty from 4.585 bits down to 3.85 bits. However, it was not the *optimal* question. To win hard and fast, you want to eliminate as much entropy as possible. If the game allowed you to ask, "Is your character in the first two rows?" you would split the 24 characters exactly in half (12 and 12). Whatever the answer was, you would discard exactly half of the board. Because we are working in base-2 logarithms, discarding exactly half the options decreases your entropy by exactly **1 full bit**. This is mathematically the most effective question you can ask.

Conditional Probability in Information Theory

Calculating the starting entropy of your stimulus ($H(S)$) is relatively straightforward. The real challenge in neuroscience is estimating the **conditional probability**—the probability of a stimulus occurring *given* a specific neural response ($P(S|R)$).

The Information Theory is powerful than the Pearson correlation because it makes **zero assumptions** about linearity. You simply assume that the probability of S given R is not flat.

However, zero assumptions come at a heavy cost: **You need massive amounts of data**.

- If a neuron can fire 0, 1, 2, or 3 spikes, you have 4 possible responses.
- To get a reliable estimate of the probability of every response given every stimulus, you must present the same stimulus over and over again—at an absolute minimum, a number of times equal to the number of possible responses, but usually much more.

Once you have the starting entropy ($H(S)$) and the residual entropy ($H(S|R)$), calculating the actual information the response carries is simple subtraction.

$$I(S; R) = H(S) - H(S|R)$$

Information quantifies the average reduction of uncertainty after a single trial observation.

- The smaller your residual uncertainty, the more you have learned about the system, and the more information R is carrying about S .
- It is measured in **bits**. Gaining 1 bit of information means you have reduced your uncertainty by a factor of 2 (like getting the correct answer to a perfectly balanced Yes/No question).

As an example, If you try to guess whether the first person to enter the classroom was a man or a woman, how much information does that question carry? Because the classroom has 4 women and 1 man, the distribution is highly asymmetrical. Therefore, the question carries **less than 1 bit** of information.

- If the answer is "Man" (the unlikely event), your uncertainty drops to zero instantly.
- But if the answer is "Woman" (the highly likely event), you are still left guessing among 4 people.
- **Rule of Thumb:** The most informative question you can ask is one where the answers are perfectly equiprobable (50/50), as it guarantees getting rid of half the options regardless of the answer.

Properties of Mutual information

Mutual information has several strict mathematical properties that dictate how it is calculated and interpreted within computational neuroscience:

1. **Commutative:** The information the response carries about the stimulus is exactly equal to the information the stimulus carries about the response.
2. **Defined Positive:** Information cannot be negative. In the worst-case scenario, observing the neural response tells you absolutely nothing, leaving your residual entropy equal to your starting entropy. In this case, $I(S; R) = 0$. You cannot be *more* confused after observing the data.
3. **Maximum Value:** The information gained cannot exceed the starting entropy of the stimulus ($H(S)$), nor can it exceed the starting entropy of the response ($H(R)$). The maximum possible information is whichever of those two starting entropies is smaller.
 - If $H(S|R) = 0$ (meaning the neural response completely and perfectly identifies the stimulus), then $I(S; R) = H(S)$.

Through logarithmic properties, the formula $I(S; R) = H(S) - H(S|R)$ can be rearranged into the most useful formulation for computational neuroscientists:

$$I(S; R) = \sum_{s \in S} P(s) \sum_{r \in R} P(r|s) \log_2 \frac{P(r|s)}{P(r)}$$

It is more powerful because of experimental design. As the experimenter, you control the stimuli. You can decide to present every stimulus an equal number of times. If you do this, $P(s)$ simply becomes $1 / \{\text{Total number of stimuli}\}$. It becomes a constant that you can effectively pull out of the complex calculations, making the math much easier to manage.

The absolute core of this formula is the ratio at the end: $\log_2 \frac{P(r|s)}{P(r)}$

This ratio dictates whether a neuron cares about a stimulus at all.

If the probability of the neuron firing *when a specific stimulus is present* ($P(r|s)$) is exactly equal to the probability of the neuron firing *in general* ($P(r)$), then the ratio is 1, the logarithm is 0, and the **Information is 0**. If the neuron fires exactly the same way whether the stimulus is there or not, it does not carry any information about that stimulus.

Because this formulation relies strictly on probabilities ($P(r), P(r|s)$), **it makes absolutely no prior assumptions about the functional relationship** between the variables.

- **Prevention of False Negatives:** If you assume a specific hypothesis—such as a linear relationship—you risk encountering false negatives, which cause you to mischaracterize variables as completely independent just because they are not connected by a linear relationship. Mutual information successfully captures dependencies regardless of the underlying mathematical shape.
- **The Data-Consuming Penalty:** The trade-off for this flexibility is that the algorithm is highly data-consuming. It does not explicitly reveal the exact relationship, and it requires extensive experimental sampling to map the probabilities. This is especially problematic for the conditional probability $P(r|s)$, meaning you must present every single stimulus several times across numerous repetitive trials to get a reliable estimate of how likely each response is for each stimulus.

Information as an Upper Bound for Decoding

One of the most important properties of Information Theory in neuroscience is its relationship to decoding. Specifically, **the mutual information between a set of stimuli (S) and a set of responses (R) acts as an upper bound (a superior bound) to the performance of any ideal decoder.**

Information Theory itself will not tell you *how* to build the decoder (it makes no assumptions about the functional relationship), but it will definitively tell you if a decoder is even possible. Let's analyze the two extreme cases:

1. **Maximum Information ($I(S; R) = H(S)$):** If the mutual information perfectly equals the starting entropy, it means your residual uncertainty is zero. This guarantees mathematically that **a perfect decoder exists**. You can perfectly identify the stimulus just by looking at the neural response.
2. **Zero Information ($I(S; R) \approx 0$):** if the mutual information is tiny (e.g., 1% of the starting entropy), it is a waste of time to try and decode it. Even if you use a massive, 7-layer Convolutional Neural Network with Transformers, it will fail because the foundational information is simply not present in the neural responses.

When you have a rich set of neural recordings and want to decode a patient's condition or intent, you should follow these steps:

1. **Feature Selection:** First, use Information Theory to measure which neural features (e.g., specific neurons, EEG frequencies) carry the most information about the stimuli/condition.
2. **Build the Decoder:** Discard the useless features and build your decoder *only* using the features mathematically proven to carry high information.

Continuous vs Discrete Responses and Binning Strategies

Neural responses can present either as discrete values or continuous data streams, which drastically alters how probabilities are sampled.

Discrete Variables

Discrete variables, such as a simple **spike count** (the total number of spikes a neuron produces during a stimulus window), result in clear integer numbers. Computing absolute probabilities in this scenario is straightforward because you can easily count exact frequencies from your trial data (e.g., $P(4) = 1/8$ or $P(5) = 25\%$).

Continuous Variables and the Necessity of Binning

When recording continuous variables with floating-point numbers—such as measuring the continuous spectrum of an Electroencephalogram (EEG) or a Local Field Potential (LFP)—the data generated by a computer features a long string of decimal digits.

- **The Probability Flattening Problem:** Because of this high precision, every specific floating-point level of the spectrum will appear exactly once or zero times across your recording. Consequently, the absolute probability of encountering any exact number becomes zero or one, making it impossible to obtain a serious or useful statistical estimate of $P(r|s)$ or $P(r)$.
- **The Resolution (Binning):** To resolve this, continuous response variables must undergo discretization via binning. If a continuous value falls within a specified interval, it is assigned to a discrete state (e.g., labeling intervals as state "1", state "A", or separating a spectrum into "high" and "low" bins).
- **Target of the Quantization:** While both variables could theoretically be binned, the experimenter already directly controls and knows the presentation probability of the stimulus (S). The unknown factor that strictly requires binning is the neural response (R).

The practical selection of your binning properties is a critical choice that directly alters your analytical results based on specific core rules:

- **Positive Correlation with Entropy:** The finer your binning strategy (i.e., shortening the bin length to increase the total number of bins), the higher the number of available possible states becomes. This directly **increases the starting entropy** of your code.
- **Capacity Matching Rule:** If you are trying to decode a stimulus that possesses a starting entropy of **3 bits**, your response code must use **at least 8 distinct bins** ($2^3 = 8$), which provides a capacity of exactly 3 bits. This ensures your response code is mathematically capable of encoding the entire starting entropy (or the specific fraction of it you want to recover). If you attempt to decode a highly complex,

high-entropy stimulus using an overly restrictive binary code (2 bins = 1 bit), you can never recover the full information.

To give your code the highest capacity to extract all available information about a stimulus, you must maximize its starting entropy. The optimal way to achieve this is by implementing **equipopulated bins** (bins configured so that an equal number of data points fall into each interval).

Think of entropy as your uncertainty regarding an occurring event. If you divide your responses into 8 distinct bins, but 100 events land in the 8th bin while only 1 event lands in each of the remaining 7 bins, your uncertainty is low because you will naturally bet on the highly likely 8th bin.

Mathematically, uncertainty and entropy reach their absolute maximum when you have multiple boxes *and* every single box has the exact same likelihood of containing the object you are searching for. Therefore, adjusting bin boundaries to ensure they are equipopulated maximizes your analytical capacity.

Practical Case Studies in Information Scaling

2 Neurons Example

Let's look at a practical example where we record from a single neuron in the visual cortex. We present 2 different images (Stimulus 1 and Stimulus 2) over 4 separate trials and count the number of spikes. (Note: If there were no noise, we wouldn't need Information Theory. We use these complex tools precisely because neural systems are noisy).

Let's firstly analyze the case of perfect information. Because we have 2 stimuli presented an equal number of times, the starting entropy is simply $\log_2(2) = 1 \text{ bit}$. Essentially, the uncertainty is a single Yes/No question: "Is it Stimulus 1?". When we plug these values into our Information equation, the result is exactly **1 bit**. Therefore, the residual entropy is 0 bits. Because there is 0 uncertainty left, we know a perfect decoder exists. By looking at the data, the decoder rule is very easy to spot: If the neural response is > 6 spikes, it is Stimulus 1. If it is < 6 spikes, it is Stimulus 2. There is no overlap.

R	Stim 1	Stim 2
Trial 1	10	5
Trial 2	9	3
Trial 3	8	5
Trial 4	10	4

$$I(S;R) = 1 \text{ bit}$$

Let's now consider the case of zero information. In this case, the neural responses heavily overlap. A spike count of 5 occurs for both Stimulus 1 and Stimulus 2. A spike count of 4 also occurs for both. If we do the math, the information here is approximately 0 bits. Because the information is basically zero, there is absolutely no decoder you can build to accurately separate these stimuli based on this specific neuron's responses.

R	Stim 1	Stim 2
Trial 1	7	5
Trial 2	5	3
Trial 3	8	5
Trial 4	4	4

3 Neurons Example

Imagine we take the "perfect decoder" neuron (R_1) from the previous example, which could perfectly discriminate between Stimulus 1 and Stimulus 2. What happens if we introduce a third stimulus (S_3) to the system?

R_1 still gives a distinct response for S_1 . However, its response to S_3 largely overlaps with its response to S_2 . The starting entropy of the system has increased. Because we now have 3 stimuli presented equally, $H(S) = \log_2 3 \approx 1.585$ bits. However, If we calculate the mutual information $I(S; R_1)$, it is **still exactly 1 bit**. This is because R_1 is still only answering a single Yes/No question: "Is the presented stimulus S_1 , or is it

$$I(S;R) = 0.045 \text{ bit}$$

R2	Stim 1	Stim 2	Stim 3
Trial 1	3	1	9
Trial 2	1	0	8
Trial 3	0	3	8
Trial 4	2	2	9

not?" It perfectly identifies S_1 , but it is completely blind to the difference between S_2 and S_3 . The system is now more complex, and a single neuron is no longer sufficient to completely solve it.

To solve this more complex system, we move the electrode and find a second neuron (R_2). R_2 does the exact opposite of R_1 : it gives the same overlapping response for S_1 and S_2 , but gives a distinct, unique response for S_3 .

In the "flat," one-dimensional world of R_1 alone, S_2 and S_3 look identical. In the flat world of R_2 alone, S_1 and S_2 look identical. To optimally decode the stimuli, we must stop looking at single neurons and start looking at **combinations of responses** (a multi-dimensional vector). Instead of computing the probability of a single response given a stimulus $P(R|S)$, we compute the probability of a specific *combination* of responses given the stimulus: $P(R_1, R_2 | S)$.

If we plot R_1 on the X-axis and R_2 on the Y-axis, creating a two-dimensional representation of the data, the three stimuli separate perfectly. We can now discriminate all of them. The amount of information carried by this combination of neurons

is called **Joint Information**, written mathematically as $I(S; R_1, R_2)$. Because this 2D representation perfectly decodes all 3 stimuli, the Joint Information is equal to the starting entropy:

$$I(S | R_1, R_2) = \log_2 3 \approx 1.585 \text{ bits}$$

Now we run into a mathematical quirk that reveals how neural networks actually function.

- The information carried by R_1 alone: $I(S; R_1) = 1 \text{ bit}$.
- The information carried by R_2 alone: $I(S; R_2) = 1 \text{ bit}$.
- The linear sum of their individual information: $1 + 1 = 2 \text{ bits}$.

However, we just established that the Joint Information of the entire system is only **1.585 bits**. What happened to the missing 0.415 bits of information?

The "missing" information is known as **Redundancy**. Because both neurons answer partially overlapping questions about the environment, the information they carry is not entirely unique. When the joint information of a population is *less* than the sum of the individual information of each neuron, the system is redundant.

$$Red = I(S; R_1) + I(S; R_2) - I(S; R_1, R_2)$$

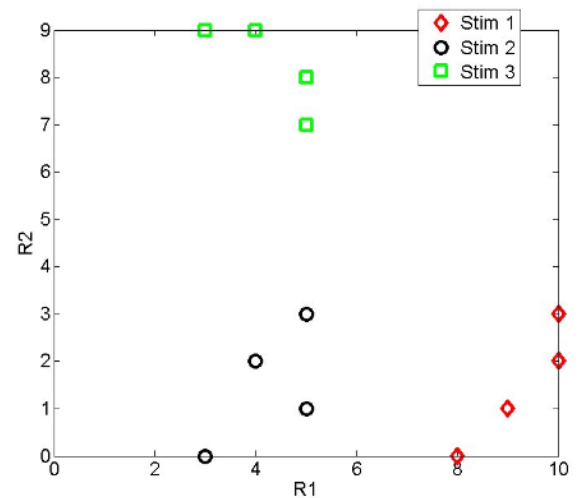
If R_1 and R_2 carried the exact same information (they both only fired for S_1 and ignored everything else), their Joint Information would be identical to their individual information (1 bit). In this case, looking at both neurons together tells you absolutely nothing more than looking at just one of them. The Redundancy would be equal to the Information itself.

While joint information can technically combine any set of variables—such as multiple separate neurons, distinct channels of an EEG, or different survey inputs—calculating it is highly complex. Estimating joint conditional probabilities requires tracking combinations of states for every stimulus condition, which demands an immense, often unattainable number of experimental trials.

Because of this data limitation, **joint information is practically never calculated for more than two channels at a time**, as gathering enough trials to compute triplets or higher-order combinations is unfeasible.

To analyze high-density neural recordings containing many channels without running out of trial data, researchers use a strict pairwise workflow:

1. **Individual Ranking:** Calculate and rank the independent mutual information carried by each individual channel across the entire recording array.
2. **Pairwise Joint Analysis:** Take the top-ranked informative channels and calculate all pairwise joint information values ($I(S; R_i, R_j)$.) two channels at a time.

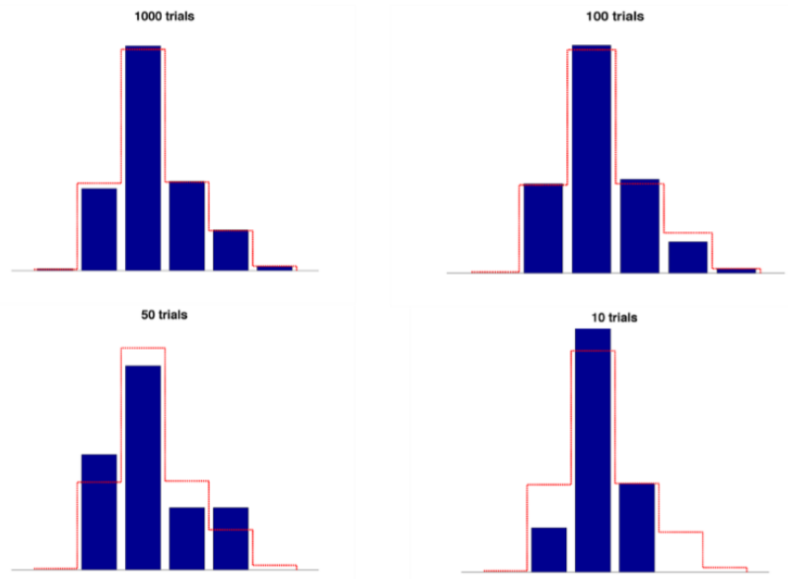


3. **Redundancy Mapping:** Compute all pairwise redundancies using the redundancy formula. This systemic map reveals if two highly informative channels fail to reinforce one another due to complete information overlap, allowing engineers to select the most efficient, non-redundant combination of features for their final decoder.

Bias Estimation

When transitioning from theoretical computer models (*in silico*) to experimental recordings of human patients (*in vivo*), gathering data is constrained by practical limitations. While models can simulate infinite data streams, human patients experience fatigue, lack of attention, and time constraints. This limits the number of trials an experimenter can collect, introducing statistical errors known as **bias**. In experiments, true underlying probability distributions are never directly observed. Instead, researchers record **histograms** of how often an event occurs and treat these frequencies as probabilities.

- Rare events are inherently difficult to estimate accurately with a small number of samples.
- **Definition of Bias:** In this context, **bias** is a reliable, systematic shift in a specific direction that prevents the calculated values from being purely objective.



The accuracy of a probability distribution estimate scales as a monotonic, saturating function of the number of trials.

Number of Trials	Quality of Estimation	Impact on Information Theory Analysis
10 Trials	Highly Insufficient	The estimation is poor. Less likely responses are frequently missed entirely and estimated at zero, distorting the true set of possible response states.
50 Trials	Reasonable / Transitioning	The empirical histogram begins to properly converge toward the real underlying distribution.
100 Trials	Optimal	For a limited set of responses (e.g., 4 possible responses), 100 trials provide a near-perfect estimation.
1,000 Trials	Diminishing Returns	Due to the saturating nature of the curve, adding more trials incurs significant computational costs, time requirements, and patient fatigue without offering meaningful improvements.

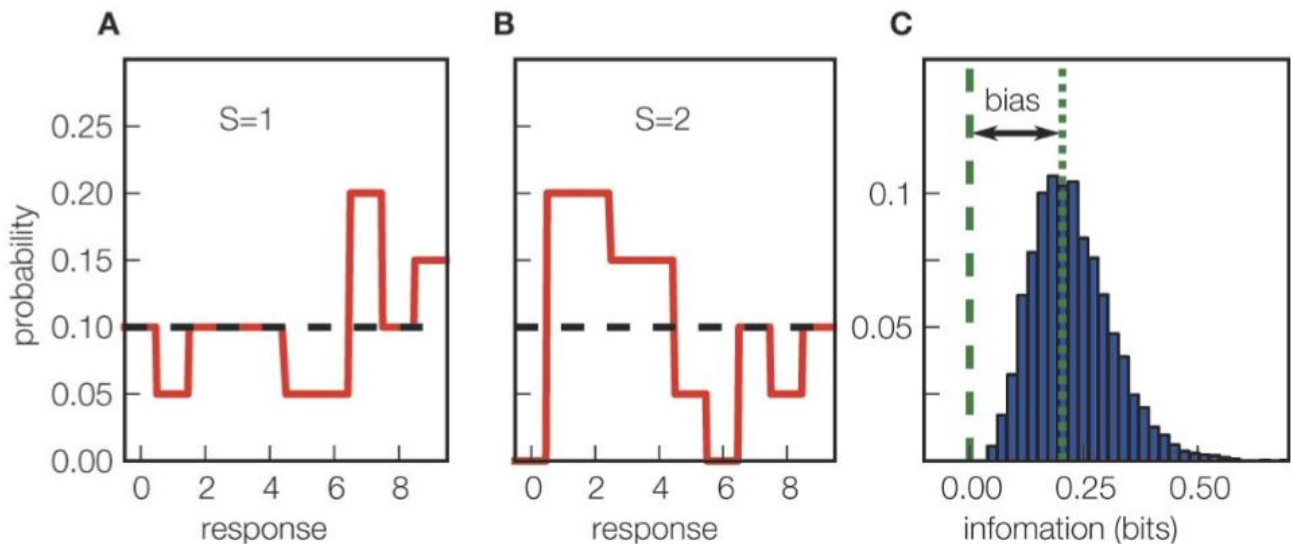
Curse of Dimensionality and Overestimation Problem

When evaluating joint variables or complex codes, the data required to accurately map responses scales exponentially. The total number of possible responses can be expressed through a geometric growth rule:

$$N_{responses} = (N_{InfoChannels})^{N_{BinsPerChannel}}$$

Because the number of channels (symbols) is elevated to the power of the number of bins, the total pool of possible response states grows exceptionally fast, drastically worsening the data requirements and the resulting estimation bias.

A critical property of estimation bias is that it does not create symmetric, zero-mean Gaussian noise. Instead, **bias always acts as a positive noise, leading to a systematic overestimation of mutual information.**



The bias of a given functional F of a probability distribution P is defined as the difference between the trial-averaged value of F when the probability distributions are computed from N trials only and the value of F computed with the true probability distributions (obtained from infinite number of trials).

Imagine recording from a single neuron while presenting two different stimuli (S_1 and S_2).

- The neuron is completely uninformative and does not care about the stimulation. Its true response can fall into any of 10 possible bins (0 to 9 spikes).
- The true probability distribution is completely flat; every response has an identical probability of exactly 10% for both S_1 and S_2 .
- **True Mutual Information:** Because the response probabilities are identical under both stimulus conditions, the true mutual information is exactly **0 bits**.

If you run this experiment with a limited sample size of 20 trials per stimulus, random sampling fluctuations prevent the tokens from distributing perfectly evenly across the bins.

- Purely by accident, the recorded histogram for S_1 might show that "Response 1" occurs 5% of the time, while for S_2 it occurs 20% of the time.
- When these asymmetric histograms are plugged into the mutual information equation, the mathematics interpret this random fluctuation as structural dependency.
- The equation concludes that observing "Response 1" makes S_2 far more likely than S_1 , treating a random sampling error as true discriminative information.

The properties of the resulting bias will be the following:

- **Strictly Positive Direction:** Because mutual information is mathematically defined as a positive quantity, accidental structural deviations cannot cancel out to create negative information. They always accumulate into a positive value.
- **Distribution of Errors:** Generating 100 independent random samples of 20 trials for this uninformative neuron yields a distribution of information values (visualized by the blue histograms on the slides). While

some random runs fall near the true value of zero, many show gross overestimations. Crucially, every single run overestimates the truth.

- **Non-Zero Baselines:** This inflation is not limited to uninformative neurons. If a neuron is genuinely informative, the bias adds directly to its true value:

$$\text{Observed Info} = \text{True Info} + \text{Bias}$$

This mathematical overestimation intersects with scientific confirmation bias. Researchers invest time into recordings expecting a neuron to be informative. Because limited data errors always inflate information values, a small dataset will naturally produce a positive result that appears to validate the researcher's hypothesis. Ethical scientific practice requires recognizing this mathematical artifact and collecting a sufficient number of trials or applying correction techniques to ensure the calculated information is not driven entirely by bias.

Analytical Bias Correction Methods

To recover the true mutual information without collecting an impossible number of real-world trials, researchers can use analytical correction frameworks based on Taylor series expansions.

Quadratic Extrapolation

Bias is defined as the systematic difference between the average observed information at a finite sample size N and the true information at infinite samples ($N \rightarrow \infty$):

$$\text{Bias} = I_{\text{observed}}(N) - I_{\text{true}}(N) = \sum_i \frac{C_i}{N^i} \approx \frac{C_1}{N} + \frac{C_2}{N^2} + \dots$$

As the sample size N approaches infinity, this positive, monotonically decreasing bias function approaches zero. Using a Taylor expansion, this function can be modeled as a sum of the inverse powers of the sample size. Where $C_1, C_2, \dots$ represent non-negative coefficients. In practice, estimating the first two dominant terms (C_1 and C_2) provides a highly accurate description of the bias behavior.

To estimate these unknown C coefficients from an existing dataset of size N^* without making strict assumptions about data normality, researchers use an artificial data-fractioning workflow:

1. **Full Dataset Evaluation:** Compute the observed mutual information using the complete available dataset at its maximum sample size (N^*).
2. **Half-Data Splitting ($\frac{N^*}{2}$):** Artificially divide the total dataset exactly in half. Recompute the mutual information for each half. To ensure stability, this operation is repeated across all possible non-redundant configurations of splitting the data in half, and the results are averaged. Because the sample size has decreased, the calculated information will worsen (increase) due to the higher bias inherent to smaller samples.
3. **Quarter-Data Splitting ($\frac{N^*}{4}$):** Subdivide the dataset further into quarters using the same exhaustive splitting approach to extract another data point.
4. This protocol yields multiple distinct data points relating sample size to observed information ($N^*, \frac{N^*}{2}, \frac{N^*}{4}$). Since the general mathematical shape of the inverse-power function is known, these points are sufficient to solve for the dominant coefficients C_1 and C_2 . Once these coefficients are mapped, the function can be mathematically extrapolated backward to find the asymptote where $N \rightarrow \infty$, revealing the unbiased estimate of the true information.

Bootstrap method

An alternative, data-driven approach that requires fewer mathematical assumptions than quadratic extrapolation is **bootstrapping** (or permutation testing). Rather than attempting to model the analytical behavior

of the bias curve mathematically, bootstrapping generates a statistical baseline directly from the collected experimental data to quantify how much information is appearing purely by chance.

To understand the core mechanics of bootstrapping, consider a simplified correlation analysis between two separate continuous variables over time.

Suppose you record two distinct time series and want to compute their **Pearson correlation coefficient**. Even if the two underlying processes are perfectly independent and uncorrelated in reality, analyzing a finite dataset (e.g., only 20 temporal observations) will inevitably contain accidental fluctuations that happen to align simultaneously.

Because of these limited observations, the absolute value of the calculated correlation will be systematically overestimated, manifesting as a false positive correlation or an artificially inflated false negative correlation. The magnitude of this artifact is strictly dependent on the overall size of your dataset (N).

To measure the exact strength of this sample-size artifact without creating entirely artificial data points, you can implement a shuffling procedure:

1. **Isolate the Vectors:** Retain the original time series vector A and time series vector B.
2. Take vector B and completely randomize (shuffle) the temporal positions of its data points while keeping vector A fixed.
3. Because you are using the exact same data points of vector B and simply rearranging their order, you preserve all of its fundamental non-temporal statistical properties:
 - **Mean**
 - **Standard Deviation**
 - **Skewness**
 - **Kurtosis**
4. This process completely breaks any actual time-dependent functional relationship between variable A and variable B.

Any residual correlation observed between the unaltered series A and the temporally shuffled series B is mathematically guaranteed to be an artifact of the finite sample size. By repeating this randomization process many times (e.g., generating 100 uniquely shuffled versions of vector B), you can construct a clear baseline of random artifacts:

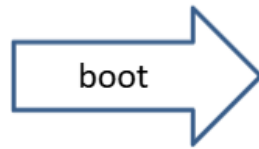
- Compare the 100 baseline artifact values against the true correlation value computed from the original, unshuffled data streams.
- Count how many times a shuffled, fake correlation matches or exceeds the magnitude of the true, original correlation. If this occurs fewer than 5 times out of 100, you have demonstrated that the probability of obtaining a correlation value that high purely by chance is less than 5% ($p < 0.05$). This confirms that the original correlation contains genuine structure.

Example to neural information matrices

The identical permutation logic can be applied directly to discrete neural data arrays to isolate information estimation bias.

Imagine a concrete experimental scenario where you record the spike rate of an individual neuron (Neuron A) across a series of 5 trials while presenting 3 distinct stimuli (S_1, S_2, S_3). This data forms an experimental response matrix.

	Stim 1	Stim 2	Stim 3
Trial 1	4	3	1
Trial 2	3	2	1
Trial 3	3	2	1
Trial 4	3	3	2
Trial 5	5	2	1
Tot	18	12	6



	Stim 1	Stim 2	Stim 3
Trial 1	1	3	4
Trial 2	3	1	2
Trial 3	1	3	2
Trial 4	3	3	2
Trial 5	1	5	2
Tot	9	15	12

Because you only have 5 trials per stimulus, calculating mutual information directly from the raw histograms of this matrix will yield an inflated value due to significant sampling bias. To quantify this specific bias:

1. **Matrix Scrambling:** Take all the recorded response values inside the matrix and randomly reassign them across the different stimulus categories, scrambling their positions.
2. **Artifact Calculation:** Recompute the mutual information equation using this scrambled matrix. Because the true functional link between the specific stimulus and the response has been broken, any mutual information still calculated by the equation represents the baseline bias caused by the small sample size.

Once the bootstrap baseline has been established via multiple permutations, it can be utilized in two different ways:

- **Bootstrap Subtraction:** The average information value obtained from the scrambled iterations is treated as a direct proxy for the bias term and subtracted from the raw uncorrected calculation:

$$I_{corrected} = I_{raw} - I_{scrambled}$$

- **Significance Testing:** A small number of permutations (e.g., 20 permutations) can be analyzed to evaluate how likely it is to obtain the true observed information value by chance, thereby confirming the statistical significance of the neural feature.

In practical computational neuroscience research, it is common to combine both toolsets. Researchers use **Quadratic Extrapolation** to calculate the stable, absolute value of mutual information, while simultaneously running the **Bootstrap Test** to confirm the statistical significance of that information.

Note: Failing to implement any mathematical corrections can cause raw histograms to generate up to a **50% overestimation** of the true mutual information. Conversely, utilizing appropriate bias correction techniques successfully counteracts these sample-size artifacts. This allows researchers to recover highly stable, objective mutual information values even when working with highly constrained datasets.

Quantifying and adjusting for this bias alters the data collection requirements for patient-facing experiments. Consider an evaluation where an experimenter must assess whether a localized brain region can successfully discriminate between a set of 5 distinct stimuli.

Without Bias Correction, you will require a number of trials much greater than the number of responses. With bias correction, you will require a highly feasible minimal number of trials.

If you compute mutual information directly from raw histograms without applying correction algorithms, the number of experimental presentations per stimulus must be **significantly higher** than the total number of possible response states (R) or the chosen dimensionality of the code. This is necessary to ensure the probability distributions are sampled perfectly.

When implementing an appropriate bias correction framework, this data requirement is relaxed. The number of trials per stimulus simply needs to be **equal to or greater than** the number of available response states (R). This lower limit ensures that every valid response state has the opportunity to be observed at least once, preventing the risk of missing a response state entirely, without requiring 5 to 10 times more repetitions from the patient.

The ultimate goal of using these analytical tools is to extract the optimal neural features for a decoding system. To isolate the most informative signals across experimental trials, the selected features must simultaneously satisfy two distinct core biological criteria:

- **Sensitivity:** The chosen neural feature must exhibit distinct, highly separable response profiles when moving from one stimulus to another. If the responses to different stimuli are too similar, the feature lacks the capacity to discriminate between conditions.
- **Reliability:** The chosen neural feature must maintain a stable, highly consistent response profile across separate, repetitive presentations of the exact same stimulus. If the intra-stimulus trial responses vary wildly due to internal noise, the feature cannot be decoded reliably.

Spike Count Hypothesis

The most fundamental coding mechanism investigated in the nervous system is the **spike count hypothesis** (also referred to as the **firing rate hypothesis**).

Historically, it was unknown whether neurons conveyed information through variations in the physical characteristics of an action potential, such as:

- Amplitude
- Current information
- Signal width
- Waveform shape

Experimental evidence has demonstrated that **all action potentials are identical for a given neuron**. Because their shape and size are uniform, action potentials must be treated as discrete, uniform units. Consequently, the primary mechanism available for information transmission is the quantity of these units fired within a given timeframe.

While closely related, a distinction exists between these two terms based on the observation window:

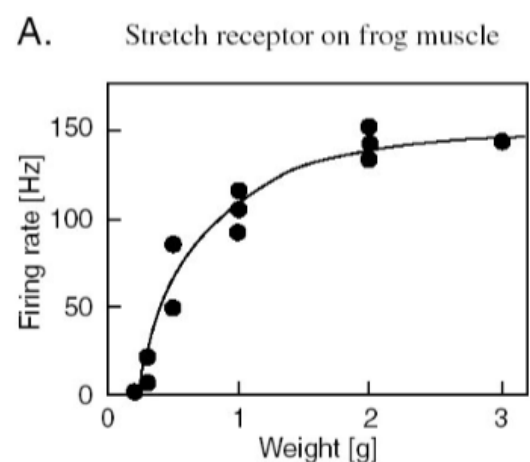
- **Spike Count Hypothesis:** This term is used when assuming a highly structured, well-defined observation time window (such as the exact duration of a stimulus presentation) where the total number of spikes is directly counted.
 - *Example:* If Stimulus A is presented for 1 second, it may yield 9 spikes. If Stimulus B is presented for the same 1 second window, it may yield 7 spikes.
- **Firing Rate Hypothesis:** This refers more broadly to the frequency of spike generation over time.

In neurobiology, Occam's razor dictates that before implementing highly complex, multi-dimensional coding schemes, researchers must first assume that neurons are executing the simplest possible operation. The simplest operational assumption is that a neuron encodes properties by generating more spikes.

Encoding Stimulus Intensity

The spike count code successfully accounts for **intensity measures** in **more than 80% of experimental observations**, particularly within peripheral sensory pathways.

In intensity coding, the identity of the stimulus space is already established by the active channel (e.g., a specific muscle, sound frequency, or receptive field). The decoding objective is exclusively to determine *how intense* the active stimulus is.



Biological systems do not behave linearly when evaluated over a sufficiently wide operating range. Firing rate functions consistently display non-linear, saturating curves.

At a stimulus level of "zero grams" or absolute silence, the system exhibits minor background firing noise, but no stimulus-driven signal. Firing rates elevate strictly as the input strength increases.

Some experimental examples of intensity coding could be:

- **Stretch Receptors:** Increasing the physical weight applied to a muscle stretch receptor drives a corresponding, non-linear increase in the total number of action potentials fired.
- **Auditory Frequency Receptors:** Individual sound receptors in the ear are tuned to a highly specific frequency channel. The intensity of that frequency (measured in decibels) is encoded directly by the output firing rate of the receptor neuron.
- **Primary Visual Cortex (V1):** When presenting a completely static visual stimulus to a localized receptive field, an increasing level of stimulus contrast results in a non-linear elevation of the neuron's firing rate.

Intensity vs Identity Coding

While the terms sound similar, they represent entirely distinct computational problems for the central nervous system.

Coding Paradigm	Analytical Focus	Neural Location	Coding Strategy
Intensity Coding	Determines the magnitude or strength of an established variable within a known receptive field, sound frequency, or muscle group.	Highly dominant in peripheral neurons and early sensory cortices.	The stimulus set is encoded monotonically by the number of spikes fired in a designated time window.
Identity Coding	Determines the specific nature, category, or profile of the stimulus itself.	Primarily observed within the higher-order associative cortex .	Requires selective modulation of spike counts across specialized neural architectures.

Encoding Identity in Face – selective Neurons

While peripheral neurons rely on spike counts exclusively for intensity, specific regions within the higher-order associative cortex utilize spike counts to encode **stimulus identity**. The primary example of this phenomenon is found in **face-selective neurons**.

From a purely geometric perspective, human faces are highly uniform, structurally resembling a simple emoticon. However, primates and humans are highly social animals. From a Darwinistic perspective, survival is dictated not merely by physical strength, but by an individual's capacity to navigate relationships with others. Subtle geometric variations between two individual faces, or shifting expressions on the same face, carry vital information regarding impending actions. Because of this evolutionary pressure, primates possess an entire specialized region of the brain dedicated exclusively to decoding the identity of other human beings.

The properties of Face-Selective Neurons are:

- **High Selectivity:** A face neuron responds strictly to faces. It does not respond to simple visual stimuli (such as bars or high-contrast lines), proving it is not a standard visual processing neuron.
- **Identity Tuning:** A given face neuron does not respond to all faces universally; it displays selective activation for some faces while ignoring others, thereby establishing a code for identity.

The true identity selectivity of these associative cortex neurons is demonstrated by tracking their firing rates across three distinct presentation profiles:

1. Showing a complete image of the individual yields a strong baseline firing rate, indicating the neuron recognizes the identity of the individual.
2. Isolating the image to present *only the face* causes the neuron's firing rate to **increase** beyond the whole-person baseline.
3. Presenting the entire body and context but completely obscuring the face—under conditions where the individual can still be recognized by their clothes or frame—causes the neuron's firing rate to be **halved**.

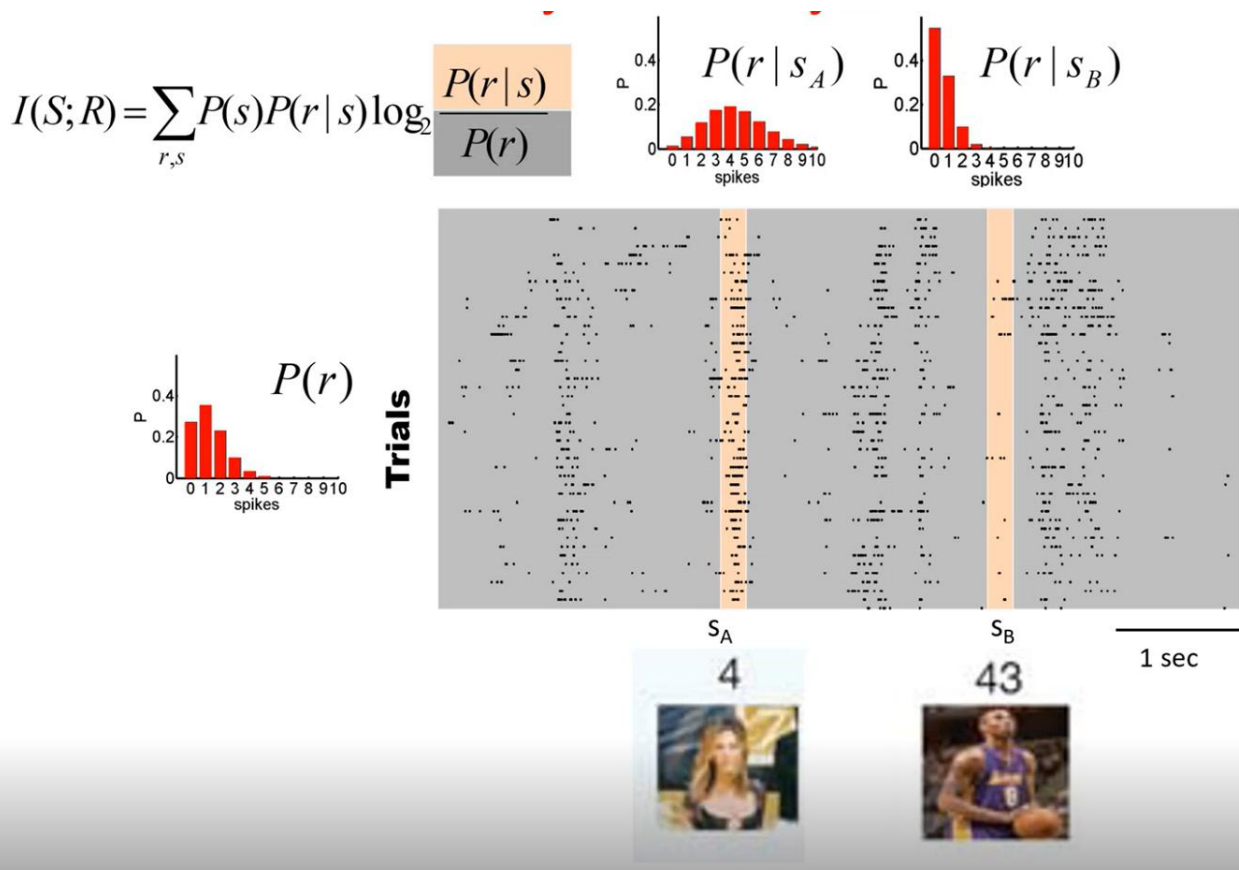
This severe drop in output when the face is removed mathematically proves that the spike count code of this neuron is highly selective for face identity.

Example of Case Study: Single-Neuron Face Identity Selectivity

A prominent area of study in human computational neuroscience (referred to in the lecture as the 90s Glenn Neuroscience paradigm) involves recording intracranial signals from human patients undergoing epilepsy surgery. These clinical settings provide a rare opportunity to study high-level visual representations directly from individual cells in the human brain.

During these surgical recordings, researchers tested neural responsiveness to high-level visual stimuli by presenting patients with pictures of famous individuals, particularly Hollywood celebrities.

- In one patient, a specific neuron was discovered to be exceptionally sensitive and uniquely tuned to the identity of **Jennifer Aniston**.
- The neuron's response was highly reliable and invariant to physical alterations in the image. It fired consistently regardless of whether her hair was cut in a specific style or if the photograph was captured in bright light or deep shade.
- The cell was not tracking low-level geometric or visual features; it was selectively firing for the concept/identity of the face. While other neurons in the population would form the "face code" for other famous individuals like Kobe Bryant or Julia Roberts, this particular unit was strictly tuned to this individual.



Mutual information from Neural Matrices

To quantify exactly how much information a single neuron carries about stimulus identity using a spike count code, researchers apply classical information theory equations.

To gather enough data to construct probability distributions, the experiment follows a strict protocol:

- **Stimulus Set:** A sequence of 200 unique pictures is compiled.
- **Trials:** The entire sequence of pictures is presented to the patient a total of 30 times.
- **Observation Window:** The spike count of the single neuron is recorded during a strictly uniform, well-defined time window corresponding to the presentation of each picture.

To compute mutual information, three probability terms are required:

1. **Stimulus Probability $P(s)$:** the probability of presenting any given stimulus is determined entirely by the experimental design. If 200 individual pictures are shown with equal frequency, the probability for each specific stimulus s is fixed to $1/200$.
2. **Absolute Response Probability $P(r)$:** the absolute probability of the neuron emitting a specific response (number of spikes) is computed by aggregating data across all collected trials and time windows. With 30 trials across 200 distinct images, the total data pool consists of $30 \times 200 = 6,000$ unique response windows. Because the spike count is a discrete integer value, it provides its own natural binning structure without requiring arbitrary categorization. For a typical selective neuron, compiling these 6,000 windows reveals an absolute distribution profile. For example, the neuron might be completely silent (0 spikes) for 25% of the total presentation windows, and fire exactly 1 spike for approximately one-third (or slightly more) of the windows. This comprehensive distribution represents the historical mean of the neuron.
3. **Conditional Response Probability $P(r|s)$:** The conditional probability represents the specific distribution of responses when one exact stimulus s is shown. While $P(r)$ is structurally sound because it is derived from 6,000 available windows, $P(r|s)$ is highly fragile. For any single stimulus (such as Stimulus A, Jennifer Aniston), there are only 30 trials available to estimate its specific distribution profile. The significant drop in available data points from 6,000 windows down to 30 windows highlights the extreme difficulty and fragility of estimating conditional probabilities accurately in real-world neural data.

Once $P(r|s)$ is derived for a specific stimulus, it is compared directly against the overall absolute response baseline $P(r)$. You can see it by looking at the image in the previous page. For the Jennifer Aniston stimulus, the conditional probability distribution $P(r|s_A)$ looks completely different from the average baseline $P(r)$. When these values are entered into the mutual information equation, the ratio $P(r|s)/P(r)$ deviates significantly from 1, yielding non-zero information values.

Information is not solely carried by high spike rates. Suppose Stimulus B represents Kobe Bryant. If the conditional distribution $P(r|s_B)$ reveals that the neuron responds with an unusually low spike count compared to its historical average, this deviation also carries substantial information.

Any response profile that systematically deviates from the global average is mathematically informative. It tells the decoder both what the stimulus *is* (e.g., an exceptionally high spike count implies Stimulus A) and what it *is not* (e.g., an exceptionally low spike count implies Stimulus B).

Information Theory and Bayesian Decoding

While information theory and neural decoding share the exact same underlying mathematical components ($P(s)$, $P(r)$, and $P(r|s)$), they serve entirely different analytical purposes.

Metric / Tool	Analytical Focus	Core Output	Operational Objective
Information Theory	Characterizes the full, descriptive relationship between the entire stimulus and response datasets.	Complete probability distribution maps.	System description and feature optimization.
Bayesian Decoding	Forces a definitive behavioral or categorical decision based on an individual observed response.	A singular, optimized prediction or "bet" on a single stimulus.	Action execution under suboptimal or real-time constraints.

In a Bayesian decoding framework, the algorithm circumvents dealing with an ambiguous probability distribution by selecting the singular stimulus state that maximizes the posterior probability given an observed response r :

$$\hat{s} = \max_s P(s|r)$$

In an ideal scientific environment—with ample dataset sizes, infinite processing time, and no external pressure—information theory and decoding should be implemented sequentially as complementary tools to isolate functional biomarkers (such as decoding motor intentions for a tetraplegic patient steering a wheelchair to the right, or detecting early indicators of Alzheimer's disease).

Confusion Matrix in Decoding

The most critical area of interest in a standard confusion matrix is the **diagonal**. The diagonal elements represent correct classifications, where the predicted stimulus matches the presented stimulus.

If all experimental stimuli are presented an equal number of times during the trial, the classification **accuracy** is calculated as the normalized average of this diagonal:

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If all experimental stimuli are presented an equal number of times during the trial, the classification **accuracy** is calculated as the normalized average of this diagonal:

$$Acc = \frac{\sum \text{Correct predictions}}{\text{Total Observations}}$$

While accuracy measures correct classifications along the diagonal, it is entirely blind to the distribution of errors occurring outside of it.

Consider a scenario where a decoder achieves an accuracy of 70%, leaving a remaining 30% rate of errors/mistakes. From an engineering standpoint, the spatial distribution of these errors matters significantly:

- Uniform Distribution:** The 30% of mistakes are scattered randomly and evenly throughout the upper and lower triangles of the matrix.
- Clustered Distribution:** The 30% of mistakes are tightly concentrated within specific "islands" or blocks in the upper and lower triangles.

Engineers heavily prefer **clustered mistakes**. When classification errors are clustered predictably, it indicates systematic misclassifications that can be leveraged or corrected using secondary algorithms. Accuracy alone cannot reveal this structural nuance because it only evaluates the diagonal elements.

To properly quantify the structural patterns across the entire matrix (including error distributions), you can compute the **mutual information between the predicted stimulus and the observed stimulus**, denoted as $I(S; \hat{S})$. If two decoding systems share the exact same accuracy (e.g., 70%), the system with clustered mistakes will yield a **higher mutual information value** than the system with uniformly distributed mistakes. Information theory is valuable not only *before* decoding to isolate optimal features, but also *after* decoding on the resulting confusion matrix to assess decoding efficacy at a much finer scale than raw accuracy allows. The structural flow of data from stimulus presentation to feature extraction and final decoding is strictly governed by the **Data Processing Inequality**.

When implementing a decoder, the mutual information between the true presented stimulus (S) and the final decoded prediction $\hat{S}$ is strictly upper-bounded by the mutual information contained between the stimulus (S) and the raw response features (R) extracted from the data:

$$I(S; \hat{S}) \leq I(S; R)$$

The amount of information present in the recorded features sets an absolute physical limit on how well any decoder can perform.

If a preliminary information analysis reveals a low value for the feature information $I(S; R)$, the downstream decoding process is mathematically guaranteed to fail, and the confusion matrix will yield poor results. This is why the ideal pipeline requires identifying features using information metrics before attempting to train a decoder.

To optimize a decoding system, you can choose to refine the decoding algorithm or improve the underlying dataset. However, if you record from a non-informative brain region—such as recording from the auditory cortex to decode visual information—the features themselves contain no relevant information. In this scenario, implementing advanced machine learning or convolutional methods is useless, because the algorithm cannot decode information that does not exist in the raw features.

Comparative Analysis between Information Theory and Decoding Techniques

Information theory and decoding algorithms offer complementary strengths, differing fundamentally in their mathematical assumptions, data requirements, and engineering applications.

Analytical Dimension	Information Theory	Decoding Techniques
Primary Advantage	Quantifies single-trial discriminability on an objective, meaningful scale.	Highly accessible to implement and computationally straightforward to calculate.
Data Requirements	Requires large datasets to accurately map complete probability distributions.	Robust and highly stable even when working with a limited amount of data.
Model Assumptions	Makes zero assumptions. Works perfectly in highly non-linear situations.	Requires structural assumptions. The algorithm must establish a classification framework.
Underlying Mechanism	Evaluates total knowledge and complete statistical relationships across the data.	Generates explicit categorical predictions by defining decision boundaries (e.g., hyperplanes in a Support Vector Machine).

Analytical Dimension	Information Theory	Decoding Techniques
Application Context	Ideal for system description, feature selection, and discovering optimal biomarkers.	Provides a direct implementation pathway for downstream engineering and clinical applications.

Spike Timing Coding Hypothesis

The **spike timing coding hypothesis** expands upon the traditional spike count framework by asserting that information is carried not just by the total number of action potentials, but by the precise chronological moments at which each individual spike is emitted.

Spike timing is a direct structural expansion of the spike count code. If you know the exact timestamp of every spike (e.g., Spike 1 at t_1 , Spike 2 at t_2 , Spike 3 at t_3), you automatically know the total count. Therefore, **spike timing information inherently contains all spike count information**.

When evaluating timing codes, researchers must ask: *Do we actually extract additional information by knowing the exact timing of these spikes?*

- For basic intensity measurements or straightforward identity decoding tasks, exact timing usually does not provide extra informational value beyond the raw count.
- In specific neural systems, however, substantial extra information is present, making a temporal analysis highly valuable.

To conceptualize the difference between count codes and timing codes, consider how human fingers can represent data:

- Traditional finger counting uses fingers as uniform units. Raising one finger or two fingers signifies a specific quantity. In this framework, the physical position of the fingers does not matter; only the total sum is read.
- This framework treats each finger as a distinct spatial slot carrying a binary value (0 or 1). By using just two fingers as independent bits, you can implement a positional binary code capable of representing four distinct numbers: 00, 01, 10, and 11. Here, the exact spatial configuration and pattern of the fingers are what carry the information.



Converting Spike Counts to Spike Timing

Transitioning an analytical framework from a count code to a temporal code requires changing how experimental time windows are processed.

This methodological transition is clearly illustrated by studies of rodent sensory processing. Because rodents live and navigate primarily in the dark, they extract spatial information about their environment by monitoring mechanical vibrations of their whiskers.

When recording from the somatosensory cortex during these movements, researchers observe distinct neural profiles corresponding to mechanical states:

- No Contact:** The whisker moves freely through the air without touching an object.

- **Contact:** The whisker physically impacts a surface, triggering immediate sensory spikes.

To extract temporal patterns from these cortical responses, researchers alter their time-window processing:

Traditional Count Window:

[===== Total Response Window =====] -> Count Spikes

Fractional Timing Bins:

[Bin 1] [Bin 2] [Bin 3] [Bin 4] [Bin 5] [Bin 6] -> Track Patterns

Instead of integrating the neural output across one long, continuous response window to compute a single total spike count, the total window is subdivided into a sequence of multiple smaller chronological bins.

Altering the window structure directly impacts the statistical dimensionality of the dataset:

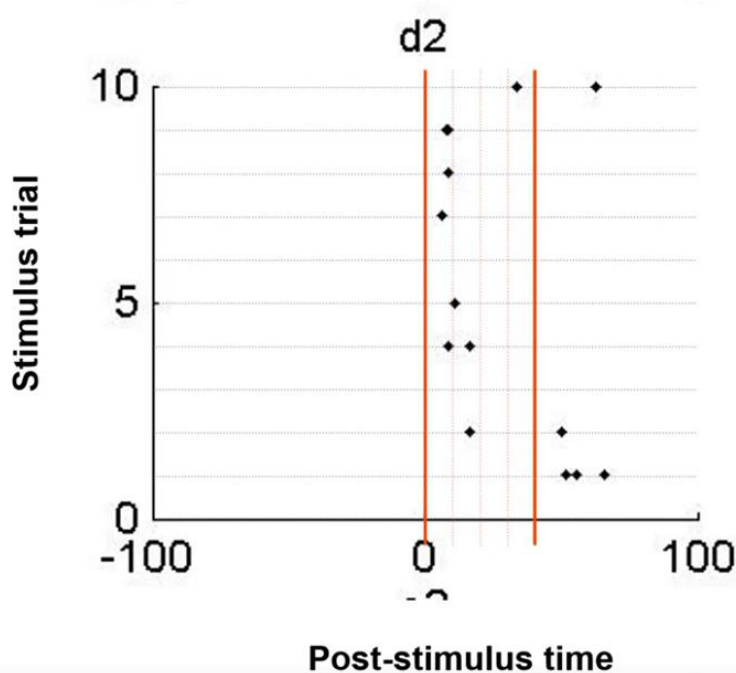
- In a typical rodent whisker trial, if the maximum number of recorded action potentials in a full window is 4 spikes, the count code yields exactly 5 possible response states: 0, 1, 2, 3, or 4.
- Suppose you subdivide that same window into 4 smaller fractional bins. If these bins are tiny enough that it is physically impossible for more than one action potential to occur inside a single bin, the response vector becomes purely binary. Each position in the vector contains either a 0 or a 1.

This structural modification causes the total number of possible response states to expand exponentially from 5 states to $2^4 = 16$ possible combinations.

This dimensional expansion means the timing framework evaluates a response space with much larger entropy. While this expanded state space can theoretically convey significantly more information, it becomes far more difficult to compute highly accurate probability distributions because the available experimental data is spread thin across 16 states instead of 5.

If the temporal bins are left wide enough to allow for multiple spikes (e.g., allowing 0, 1, or 2 spikes per bin), a 4-bin array expands to $3^4 = 81$ possible response states. Such high dimensionality makes accurate statistical analysis virtually impossible with standard experimental sample sizes, rendering the approach impractical.

The response window is divided into subwindows



Spike count	Spike timing
1	0001
1	1000
1	1000
1	1000
0	0000
1	0100
2	1100
0	0000
1	0100
0	0000
0	0000

Up to 5 numbers

$2^4 = 16$
numbers

The response becomes a binary vector

Classifying Neural Typologies

Neural codes can be divided into distinct categories based on how downstream reader neurons process spike sequences over time.

1. Rate Code / Count Code

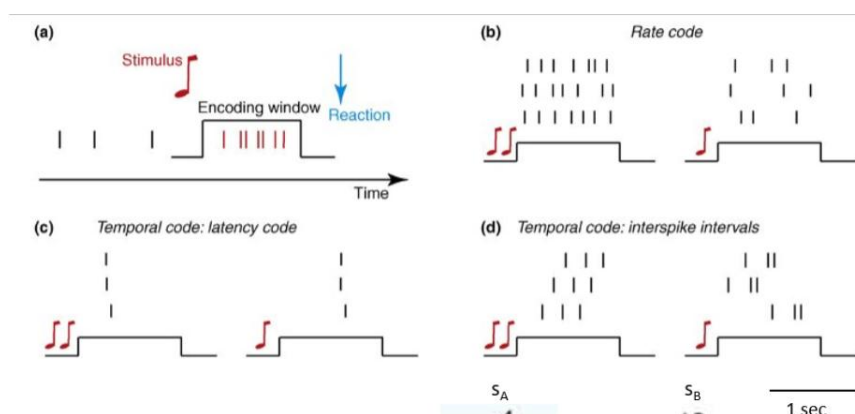
- **Mechanism:** Two different stimuli are discriminated entirely because one elicits a large volume of spikes while the other elicits a lower volume.
- **Temporal Factor:** The exact positions and intervals of the action potentials carry no meaning.
- **Downstream Decoding:** Highly straightforward for a single downstream target neuron to read and decode.
- **Classification Note:** This is **not** a temporal code because the signal is integrated and summed over the entirety of the response interval.

2. Temporal Code based on Latency

- **Mechanism:** Stimuli are discriminated strictly based on *when* an action potential occurs relative to the arrival of the stimulus (e.g., an early spike vs. a late spike).
- **Temporal Factor:** The decoder can differentiate inputs even if the total spike count is identical (e.g., exactly 1 spike occurs in both conditions), based entirely on its position in time.
- **The Downstream Decoding Blindspot:** It is mathematically **impossible** for an isolated downstream reader neuron to decode absolute latency on its own. While an external experimenter has access to a clear timeline showing the exact onset of the stimulus, a neuron deep within the brain cannot see this external starting point.
- **System Requirements:** A latency code can function only if it is explicitly linked to an **internal clock signal** or if it operates as a **relative latency code** measuring the difference in arrival times between two separate neurons (such as the computational framework mapped by the *Jeffress model* for sound localization). Without an internal or relative reference framework to mark the onset, absolute latency values are functionally useless to the brain.

3. Temporal Code based on Inter-Spike Intervals (ISI)

- **Mechanism:** Stimuli are discriminated by the precise structural arrangement of the time intervals separating consecutive action potentials.
- **Temporal Factor:** The absolute number of spikes remains completely identical across conditions, but the chronological pattern of the intervals changes.
- **Analytical Note:** This framework tracks the precise sequential pattern of single-trial intervals, not the broad statistical distribution of ISIs.
- **Downstream Decoding:** Highly straightforward for a single downstream target neuron to decode without requiring any external clock or absolute reference, because it only needs to monitor the incoming interval pattern.
- **Analytical Challenge:** This code remains exceptionally difficult for researchers to model and analyze due to the high dimensionality generated by the massive variety of potential temporal patterns.



Definition of Response Window and Binning Problems

When analyzing neural codes, establishing the precise boundaries of the temporal analysis window is a fundamental challenge. The relevant observation window is typically defined as the exact duration of the **presentation of the stimulus**. This was the implicit assumption utilized during the "90s glam" celebrity face-sequence experiments.

Restricting the window strictly to the stimulus presentation period introduces a clear biological approximation: it assumes there is no neural reverberation or lingering activity after the stimulus terminates.

It is methodologically impossible to extend the analysis window indefinitely—such as tracking the entire timeline from the presentation of Stimulus A until the patient executes an entirely different action or goes to sleep. While treating the window as matching the stimulus presentation is an approximation, it functions relatively well in practice. Nonetheless, defining the boundaries of a stimulus and its corresponding response window is significantly trickier than it appears.

To look for temporal patterns, a large response window must be discretized into smaller sub-bins. Choosing the size of these sub-bins determines the **coarseness of the temporal patterns** under investigation.

```
[===== Total Observation Window =====]

Sub-millisecond Bins (Too Small): → [|]  <- Action potential split across boundaries
Finer Resolution Binning:         → [ 10 ms ] [ 10 ms ] [ 10 ms ] [ 10 ms ]
Coarser Resolution Binning:       → [  20 ms  ] [  20 ms  ] [  20 ms  ]
```

When setting these sub-bin parameters, researchers face physical and mathematical constraints that create hard and soft boundaries.

The Hard Lower Boundary (Small Bins)

Analysts cannot arbitrarily shrink sub-bins down to sub-millisecond scales. If a bin is too narrow, a single action potential could easily be split directly down the middle, leaving half of the spike in one bin and half in the adjacent bin. Therefore, the physical duration of an action potential enforces a strict, **hard lower boundary** on bin size.

The Soft Upper Boundary (Large Bins)

To achieve a clean, mathematically simplified binary code, the optimal goal is to ensure that *at most one spike* can occur within any single sub-bin. This requirement establishes a **soft upper boundary** toward larger bins. While a binary code is sometimes sufficient to capture the underlying dynamics, in other neural systems, it is not.

Rule of Thumb Formula: A practical starting point for setting the sub-bin size is to select a value slightly below the inverse of the average inter-spike interval:

$$\text{Bin Size} < \frac{1}{\text{Average Inter} - \text{Spike Interval}}$$

However, this remains a highly flexible, soft recommendation.

Because the ideal time frame is rarely known a priori, spike timing code analysis must always be approached **parametrically** by systematically varying the temporal resolution scale to find which setting works best:

- **20 ms Intervals:** A wider window choice where it is rare, but still entirely possible, for two spikes to fall into the same bin.
- **10 ms Intervals:** A finer, more detailed wave of resolution that reduces multi-spike bin occurrences.

Unless a researcher possesses a highly specialized, advanced intuition regarding the exact operational time scale of a given circuit, they must test the data across a shifting spectrum of temporal resolutions.

A prominent, dedicated computational tool designed to extract and evaluate spike pattern information is the **Victor-Purpura distance**. This metric operates by calculating a distance score that reflects how close or how far apart two distinct temporal spike patterns are from one another. It has already been described in the Distance Based Methods.

Example Paper: Tactile Discrimination via Whisking (CD vs. Sandpaper)

Rodent whisking serves as an excellent model system for studying high-sensitivity tactile discrimination. The mapping and tactile sensitivity of whiskers in the rodent somatosensory cortex are highly prominent, drawing a direct functional parallel to how the human somatosensory cortex maps the sensitivity of different fingers.

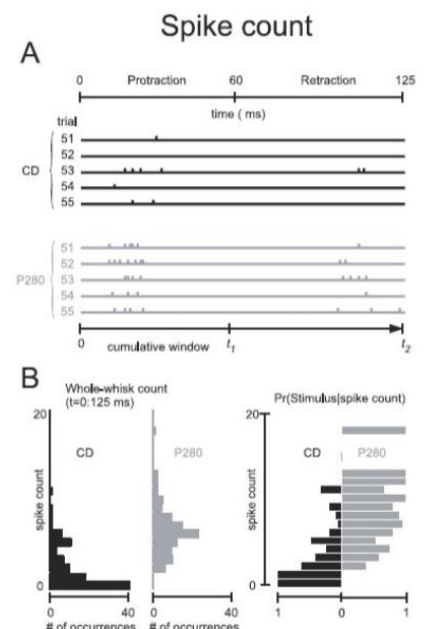
In a standard discrimination task, an animal is presented with two distinct surfaces to evaluate using its whiskers:

- **Smooth Surface:** A standard compact disc (CD).
- **Rough Surface:** Sandpaper of varying degrees of coarseness (ranging from smooth to highly coarse).

To determine if the animal can successfully perceive and utilize the tactile information, a verification setup is established: a juice reward is placed in a box located in the direction of the smoother texture. Driven by this incentive, rodents successfully perform the task, demonstrating an acute ability to distinguish surface textures through active whisking.

The sensory experience of whisking consists of discrete **protraction** and **retraction** phases. When tracking the neural responses over the entire duration of the whisker touching the surface, a clear visual difference emerges in the spike raster plots:

- **Spike Density:** There is a visible increase in spike density (represented as clustered dots in the data plots) when the whisker interacts with the sandpaper texture compared to the smooth CD, with these differences becoming particularly pronounced during the retraction phase.
- **Response Distributions:** Evaluating the overall distribution of the neural responses reveals distinct phase-locked profiles. The response distribution for the smooth CD stimulus characteristically peaks at 0, whereas the response distribution for the rough sandpaper stimulus (P280) peaks around 5.
- **Posterior Probabilities:** These distinct distributions allow for the calculation of the posterior probability of the stimulus given a specific spike count, $P(\text{stimulus} \mid \text{spike count})$:
 - A **high spike count** indicates a significantly higher probability that the animal is interacting with the sandpaper.
 - A **low spike count** indicates a higher probability that the animal is touching the smooth CD.



Fine Texture Discrimination

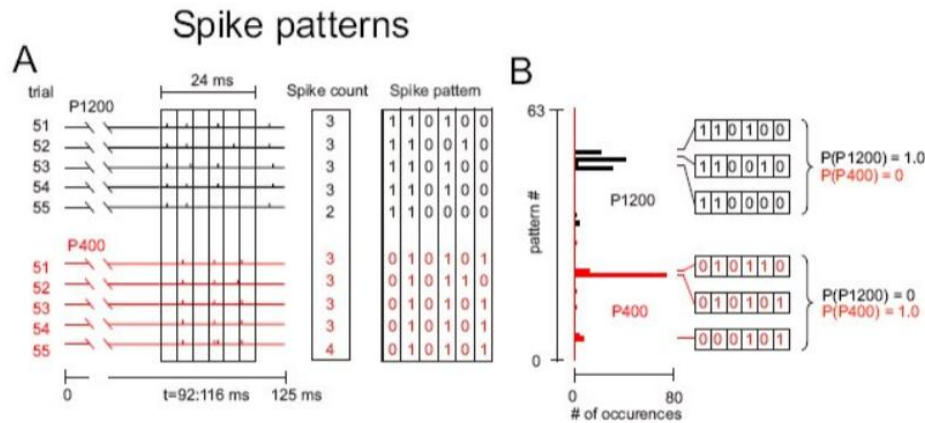
While a simple spike count code provides sufficient information for basic binary classifications (such as distinguishing between a smooth object and a rough object), it fails when the animal is required to process fine texture details.

Rodents are capable of recognizing minute variations between different grades of sandpaper coarseness. However, when recording from neurons during the presentation of two distinct sandpapers, the absolute number of action potentials fired is nearly identical. Because the spike count profiles are highly similar, a raw count code cannot resolve fine structural differences.

By applying fractional temporal binning to the response windows, researchers uncovered reliable, highly distinct temporal patterns hidden within the equal spike counts:

- **Sandpaper Coarseness Type 1 (P1200):** The neuron consistently exhibits an initial burst of intense firing at the onset, which then steadily decreases over time.

- **Sandpaper Coarseness Type 2:** The neuron fires in a highly regular, evenly distributed temporal pattern throughout the observation window.



Comparing the mutual information carried by the spike count versus the spike timing code across both the peripheral nervous system and the cortex reveals a clear functional divide:

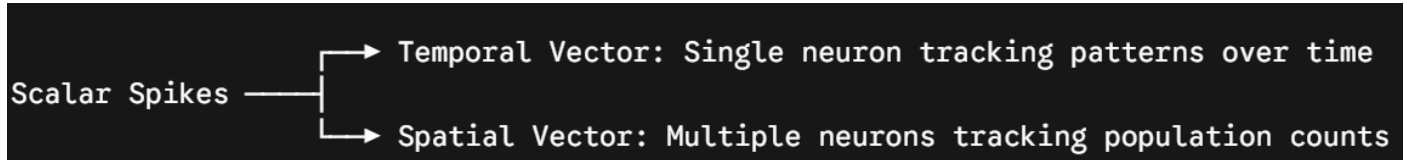
- **Global Discrimination:** Discriminating between free whisking (moving through the air without touching anything) versus actively touching an object, or identifying a CD versus sandpaper, can be achieved with near-perfect accuracy using a standard spike count code.
- **Fine-Scale Discrimination:** When attempting to differentiate between two similar levels of sandpaper roughness, the decoding performance of the spike count code drops drastically low in both the periphery and the cortex. However, substantial additional information is available within the **spike timing code**, demonstrating that fine texture details are captured by temporal spike patterns rather than cumulative counts.

Overcoming Single Neuron Limits

The highly selective "Jennifer Aniston neuron" represents a rare, exceptional case where the identity of a complex high-level stimulus is fully captured by the spike count of a single cell. In standard neural architectures, this mechanism is mathematically insufficient.

A single neuron operating purely on a spike count code faces severe informational constraints. If a cell's response range is limited to low integer values—such as firing 0, 1, 2, or 3 spikes—it can only convey a maximum of 4 states of information. Because no single neuron possesses a wide enough dynamic firing range to single-handedly map intricate inputs, it is mathematically impossible for complex visual or tactile stimuli to be encoded solely by single-neuron spike counts.

To overcome this informational bottleneck, the nervous system employs two primary strategies to transform a restrictive scalar value into a high-dimensional vector space:



- **The Temporal Approach (Single-Neuron Timing Vectors):** This strategy maintains single-neuron tracking (as seen in the whisker experiments where all whiskers except one are removed) but increases dimensionality by turning a single scalar spike count into an extended vector across time.
- **The Spatial Approach (Population Count Vectors):** This strategy abandons reliance on single units, instead tracking the spatial pattern of spike counts across an entire population of co-active neurons. The response shifts from a single scalar to a high-dimensional spatial vector, recording exactly how much each individual member of the ensemble fires (e.g., Neuron A fires 3 spikes, Neuron B fires 2 spikes, etc.).

Population vs Labelled Line code

A paper in computational neuroscience formalized how information is extracted from neural ensembles by drawing a clear distinction between a standard population code and a labeled line code.

1. Standard Population Code

- **Mechanism:** Information is extracted directly from the collective sum of the electrical activity across all recorded neurons within the population.
- **Correlations and Noise:** This framework explicitly accounts for the fact that neighboring neurons are structurally correlated but may possess different noise profiles.
- **Informational Efficacy:** Summing the collective pool of activity is fundamentally more informative than reading a single neuron because the aggregation helps compensate for independent noise variations across the individual cells.

2. Labeled Line Population Code (Labeled Code)

- **Mechanism:** Rather than discarding individual identities by summing the total activity, this framework preserves the exact origin of every spike. A distinct operational "tag" or "label" is placed on every single spike rate, explicitly tracking exactly *which* neuron produced *which* specific spike count.
- **Informational Efficacy:** By maintaining separate communication channels, a labeled line configuration can carry substantially more information than a simple pooled population sum.
- **The Computational Trade-off:** Preserving individual labels forces the data into a much higher dimensional space, making the underlying probability distributions significantly more difficult to calculate and analyze.

This high-dimensional, labeled line architecture is precisely what the cortex is forced to utilize when it must resolve fine, micro-scale differences that would otherwise be completely lost or obliterated within a simple population sum.

Frozen Noise vs True Noise Simulation

Spike timing precision is not a fixed architectural property of a neuron; rather, the temporal reliability of neural responses depends strictly on the **identity of the stimulus** being processed. To demonstrate this, researchers use specific types of sensory noise paradigms.

- **White Noise Baseline:** By definition, white noise refers to a signal possessing a completely flat power spectrum. In visual terms, each individual frame displayed on a screen is generated completely at random, meaning every frame is statistically independent from both the preceding and following frames.
- **True Noise Stimulation:** In a true noise experimental setup, the system generates a completely different, unique random noise sequence for every single trial.
- **Frozen Noise:** This occurs when an unstructured stimulus (like white noise) is generated at random exactly *once*, and then that exact same sequence is saved and presented identically trial after trial across a large number of runs.

Frozen noise is utilized when researchers want to present an unstructured stimulus but require it to remain perfectly reproducible across all experimental cases to assess response reliability.

Experimental Context: Cat Thalamic Responses

Researchers recorded neural activity from the **thalamus of a cat** subjected to visual stimulation using two distinct stimulus categories: spatially uniform frozen noise and natural movies. There was no conscious task associated with this experiment; the goal was to observe pure sensory encoding properties.

Response to Spatially Uniform Frozen Noise

When the cat's thalamus is stimulated with rapidly repeating frozen white noise, the resulting raster plots reveal exceptionally fine, narrow vertical columns.

- The system achieves an incredible **millisecond precision**, meaning that across dozens of repeated trials, the action potentials stack perfectly one right above the other with minimal temporal variation.
- **Information Density:** Even though it is an unstructured noise signal, repeating the exact same frozen noise sequence over and over turns it into a high information-density message. To cope with such rapid, independent frame transformations, the thalamus must engage an amazing degree of temporal precision.

Response to Natural Movies

When the exact same thalamic neurons are presented with natural movies, the structural properties of the raster plots alter visibly. While there are clear clusters of activity at specific points in the movie and long silences at others, the spikes exhibit a much larger **temporal jitter**.

- **Jitter Manifestation:** The columns in the raster plot become distinctly wider and blurred. The spikes are no longer precisely aligned one above the other, demonstrating several milliseconds of temporal jitter (described as a few milliseconds to loose alignments depending on the specific frame transition).

•

The relaxation of temporal precision during natural movies compared to frozen noise is driven by two primary underlying factors:

- **Temporal Constraints Beyond Refresh Rates:** the difference in precision is not merely a consequence of the screen's hardware refresh rate; it is driven by the structural statistics of the visual inputs themselves.
- **Long-Term Temporal Correlations:** natural movies are characterized by prominent, long-term temporal correlations. Because physical objects move continuously and relatively slowly across our visual field (whether watching a fast runner like Usain Bolt or any other natural scene), there is an inherent mathematical coherence from one frame to the next.

Because Frame M remains highly correlated with and similar to Frame $M - 1$ and Frame $M + 1$, the visual pathway has no functional need to extract entirely novel, independent pieces of information every single millisecond. The response to frozen noise proves that the thalamus is physically capable of generating responses accurate to a few milliseconds. However, when processing natural scenes where such extreme precision is functionally unnecessary due to input correlations, the nervous system relaxes its temporal constraints and allows the spike timing jitter to expand.

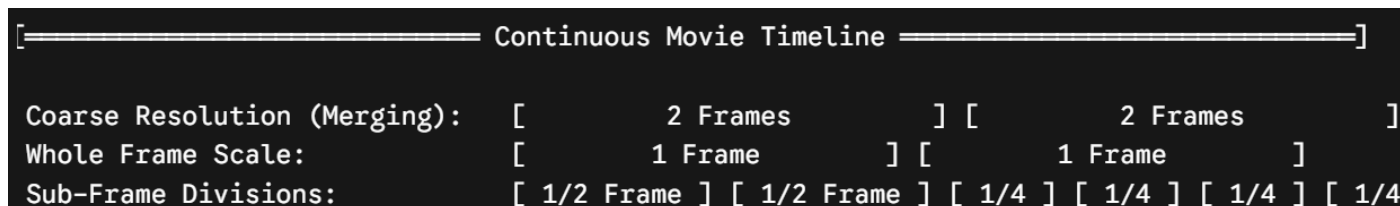
Paper: Frame By Frame Discrimination in Visual Cortex

When analyzing how complex, dynamic visual inputs are handled in cortical structures, researchers can apply the same information-theoretic frameworks used for isolated, uncorrelated images to continuous movies.

In experiments where movies are presented to monkeys, the analytical objective shifts from separating completely independent, uncorrelated static images to **discriminating between different chronological frames of a continuous movie**. The core investigative question becomes: *How much information does a single cortical neuron carry about the specific differences from one frame to the subsequent frame?*

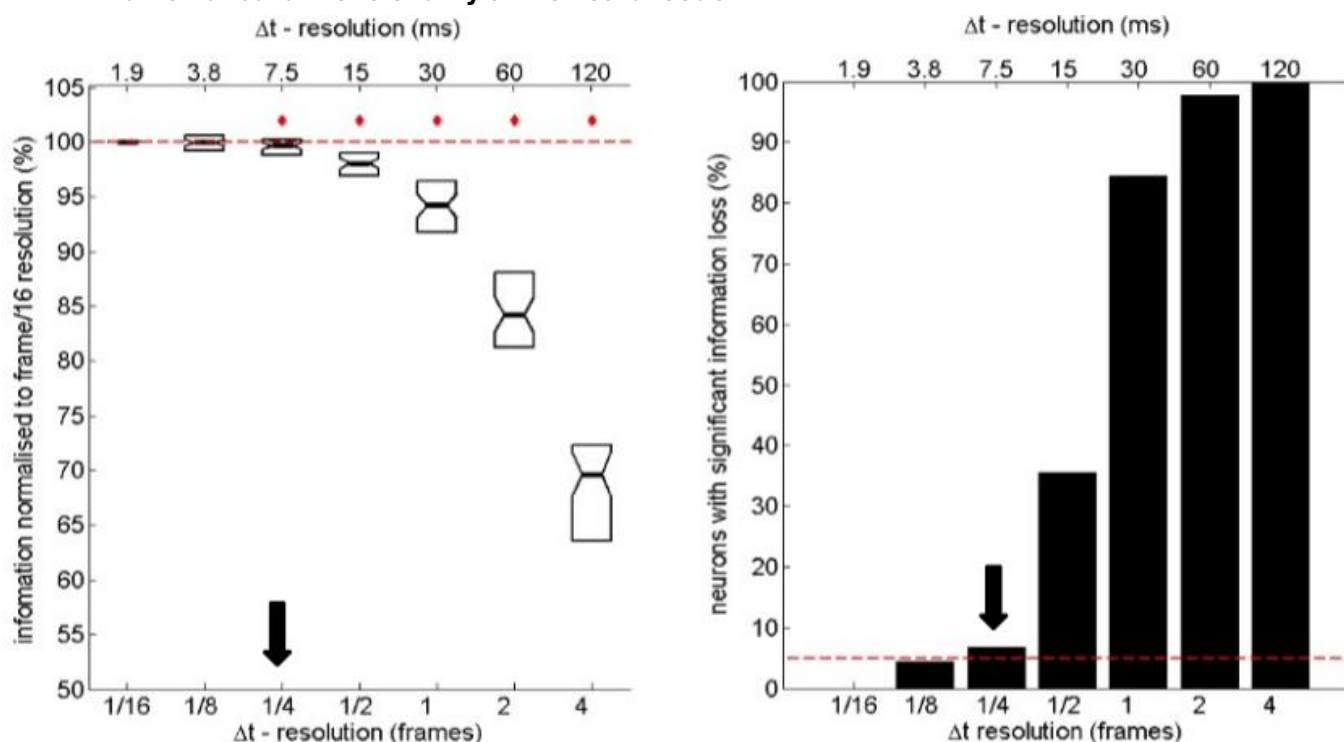
A standard movie frame spans a duration of several milliseconds. On the rapid operational timescale of an individual neuron, a single frame is actually a prolonged epoch. This raises the biological possibility that crucial sensory information is embedded **below the single-frame scale (sub-frame resolution)**. Specifically, does the exact chronological position of an action potential within the boundaries of a single frame convey additional identity information?

To rigorously test for the existence of sub-frame information channels, the temporal windows representing movie frames must be systematically subdivided and evaluated.



Varying the temporal sorting resolution alters the captured mutual information in a non-monotonic fashion:

- **Coarse Resolution (Merging Frames):** When the analysis resolution is widened to encompass two frames or four frames, the window merges independent frames together. This causes a massive drop in resolution, and a significant portion of the information enabling scene discrimination is completely lost.
- **Finer Resolution (Sub-Frame Splitting):** To probe sub-frame dynamics, researchers divide the frame window into halves (1/2 frame) and quarters (1/4 frame). Shrinking the bins down to these finer resolutions causes the measured information to increase.
- **The Dimensionality Trade-Off:** Even though finer temporal resolutions yield more information before reaching the absolute physical limit of a single action potential, shrinking the bins increases the mathematical **dimensionality of the neural code**.



Because increasing the code dimensionality risks introducing statistical noise and sampling errors, researchers must establish a rigorous baseline to separate true information gains from dimensional artifacts.

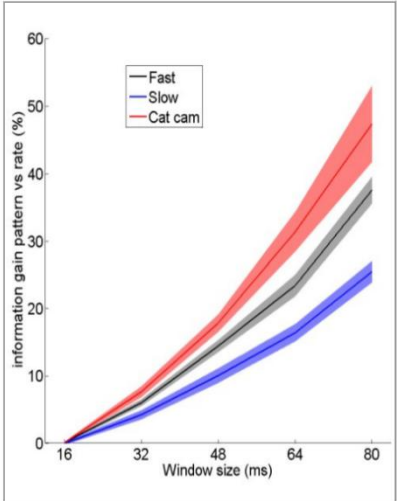
- **The Significance Line:** In information-vs-resolution plots, a strict threshold of significance (represented by the **red line** in the experimental data) is established using a **bootstrap test**.
- **The Saturation Knee:** Transitioning from an integrated whole-frame window down to a 1/2 frame or a 1/4 frame resolution provides a small but statistically significant increase in information. This proves that meaningful data is actively encoded by the precise temporal position of spikes down to **one-fourth of a frame**.
- **The Efficiency Cutoff:** Pushing the analytical resolution even further down to 1/8 or 1/16 of a frame fails to yield a significant information boost; the negligible gain is no longer worth the statistical effort. This parameter modulation behaves exactly like adjusting the Q parameter within the Victor-Purpura distance metric: it allows researchers to pinpoint the exact "knee" where the information saturation band flattens out.

The data collected from these sub-frame binning paradigms provides a formal, information-theoretic definition for a neuron's temporal fidelity.

The formal definition of temporal precision is the **coarsest temporal resolution that remains completely sufficient to extract all available information from the neural code**. In the visual cortex movie experiments, because scaling down to resolutions finer than 1/4 of a frame produces no statistically meaningful increase in mutual information, 1/4 of a frame is classified as the precise operational boundary of that code's temporal precision.

Information Flow speed in different visual stimuli

Building on prior findings from the cat thalamus—which proved that temporal precision tightens during rapid, independent white noise stimulation—researchers sought to determine if the visual cortex of primates adapts its temporal resolution dynamically based on the velocity of information flow. The primary Hollywood movie used in the initial primate trials contained a natural mix of slow and fast scenes. However, tracking the footage revealed that the actors themselves were not moving significantly faster during the "fast" sequences. Instead, the illusion of speed was generated entirely by an increased frequency of **movie cuts** (where the visual frame changes abruptly due to an edit).



To evaluate a truly accelerated information flow, researchers required a sensory stimulus that retained realistic spatial and temporal correlations but moved at a velocity far exceeding Hollywood scene cuts.

To generate a highly naturalistic yet high-velocity visual asset, researchers implemented a unique experimental solution:

1. A compact **GoPro camera was mounted directly onto the head of a cat**.
2. The cat was permitted to roam freely outdoors in a natural environment.
3. Once the cat returned home to receive food, the raw footage was recovered.

The resulting first-person feline video provided a continuous, highly natural visual stream characterized by incredibly rapid environmental adjustments, yielding a significantly faster flow of information than artificial movie edits.

Using the gathered data, researchers mapped out three distinct informational speed regimes to compare how temporal resolution shifts when discriminating scenes:

Information Regime	Visual Input Characteristics	Cortical Saturation Point
Slow Regime	Low-velocity environmental movement; high temporal continuity.	Coarser sub-frame saturation knee.
Intermediate Regime	Standard Hollywood cinema tracking; dominated by scene cuts.	Moderate temporal resolution requirements.
Very Fast Regime	Head-mounted cat footage; high-velocity continuous spatial adjustments.	Sharper, finer sub-frame saturation point.

The comparative analysis revealed that while the underlying neural mechanics remain generally consistent, there is a clear, visible shift in the **saturation point** across the three scene velocities. This confirms that the primary visual cortex (V1) of the primate brain utilizes sub-frame temporal coding, and the exact quality and sharpness of this temporal precision are tuned dynamically to match the incoming **bit-rate flow** of the environment.

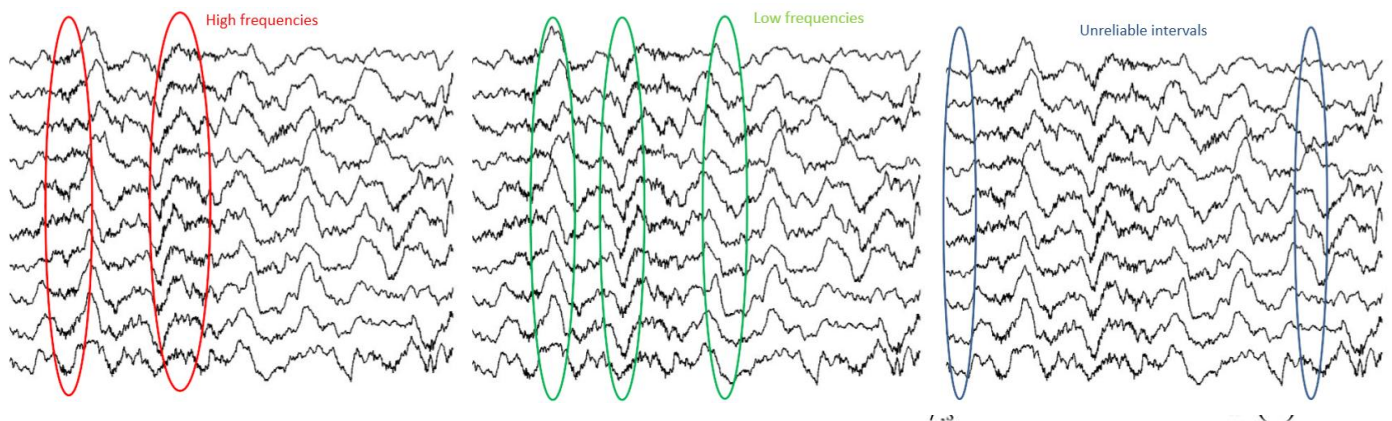
Information Extraction from Continuous Brain Signals (LFP & EEG)

When analyzing continuous brain signals—such as **Local Field Potentials (LFPs)** or **Electroencephalogram (EEG)** fluctuations across different frequency bands—the analytical goal remains the same as with spike trains: identifying features that vary significantly across different stimulus scenes but remain highly consistent and reproducible across multiple experimental trials.

When extracting information from continuous neural waveforms, researchers consistently find **more information within the spectral (frequency) domain than in the purely temporal domain**.

Continuous signals can be broken down into several structural components observed across trials:

- **High-Frequency Oscillations:** Represented by thicker trace lines in signal plots. These fast oscillations are transient; sometimes they are present, and sometimes they are absent. However, when they do appear, they are highly reliable and appear across all trials, exhibiting matching peaks and valleys independent of the trial number.
- **Low-Frequency Components:** This underlying activity can vary from trial to trial, but it consistently acts as the driver that generates the high-frequency oscillations.
- **Very Slow Fluctuations:** These slow trends are found reliably across different trials and vary distinctly depending on the visual scene presented.
- **Unreliable Intervals:** Sections of the recording where activity is inconsistent across trials, making it impossible to determine exactly what the system is encoding.

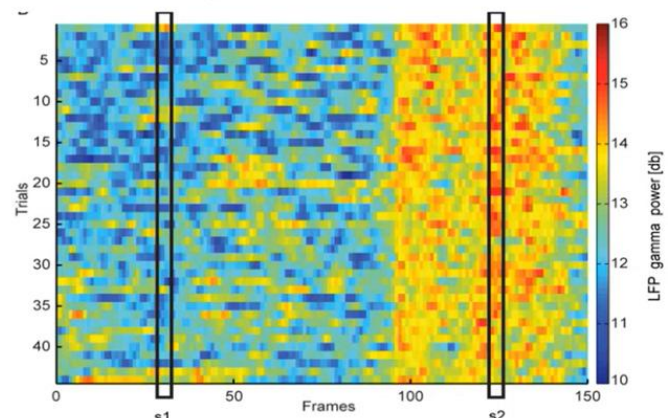


Spectrogram-Like Processing Framework

To calculate the mutual information of a continuous signal, the data must be transformed into an architecture that mimics a spectrogram. Instead of processing a discrete raster plot, the continuous amplitude is mapped across time and trials.

For a pre-selected target frequency band (which represents a single frequency with a specific bandwidth), the data is organized into a multi-dimensional space:

- **X-Axis:** Represents time (and reflects the power of the pre-selected band across the timeline).
- **Y-Axis:** Replaces the standard frequency axis of a traditional spectrogram with **individual trial repetitions**.
- **Z-Axis (Color Scale):** Represents the quantified **power** of the chosen frequency band.



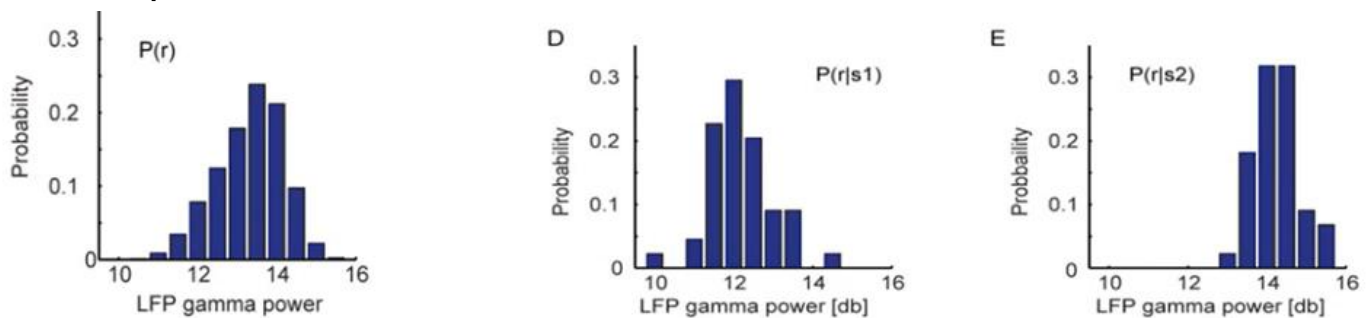
Discretizing Continuous Signals

Unlike spike trains, which inherently consist of discrete, countable events, a signal like gamma power is a **continuous mathematical function**. This requires researchers to apply an intentional discretization (binning) strategy.

The choice of how finely to slice the power axis (the x-axis of the probability distribution histogram) is entirely arbitrary and left to the investigator's discretion:

- **Too Fine:** If the power bins are made too narrow, the histogram empties out, leaving a sparse distribution containing only zeros and ones.
- **Too Coarse:** Binning the power into too few levels (for instance, using only 2 levels) is insufficient to capture structural variations.

The Binning Trade-Off: Selecting the optimal number of power bins (such as an exemplary 10-level quantization) depends strictly on a balance between the total **entropy of the system** you wish to decode and the **number of available experimental trials**.



Applying Info Equations to Spectral Bands

Once an appropriate binning resolution is established, the exact same information-theoretic equations used for discrete spike counts can be applied directly to continuous frequencies.

In this framework, the response variable R no longer represents the raw continuous power of the band. Instead, **R represents the specific discrete bin into which the recorded power falls**. Researchers calculate:

1. **The Overall Distribution of Binned Power:** This yields the overall probability of a response, $P(r)$, evaluated across all trials and all frames.
2. **The Conditional Distribution:** Represented as vertical strips across the stimulus timeline, this tracks the probability of a specific power bin occurring given a particular stimulus frame, $P(r|s)$. For example, what is the probability that when presenting frame 25, the power falls into bin 13 versus bin 11?

Examining these conditional probability strips via the plot's color code reveals distinct tuning profiles. Some stimuli fail to elicit any noticeable gamma power, whereas other specific stimuli elicit intense gamma power.

Applying this uniform mathematical approach across all individual frequencies yields complete **spectral information**. This exact methodology was used to discover key organizational principles of cortical coding:

- **Delta and Gamma Bands:** Significant amounts of descriptive stimulus information are actively carried within both the Delta band and the Gamma band.
- **Beta Band:** No significant stimulus information is carried within the Beta band.
- **Redundancy:** The information found within the Gamma band is highly redundant.

This standardized information-theoretic framework is fully universal: it functions identically whether a researcher is extracting information from local field potentials (LFPs) or macro-scale electroencephalogram (EEG) fluctuations.

